Lecture Notes on Advanced Gyroscopes

Lorenzo Vergata

July 5, 2026

Contents

1 Gyroscopes Applications and Specifications **13**

1.1 Introduction to MEMS Gyroscopes . . . 13
1.2 Units of Angular Velocity . . . 13
1.3 Applications of MEMS Gyroscopes . . . 13
1.3.1 Automotive Applications . . . 13
1.3.2 Consumer Electronics Applications . . . 14
1.3.3 Navigation Applications . . . 14
1.3.4 Medical Applications . . . 14
1.3.5 Additional Applications . . . 14
1.4 Gyroscope Specifications . . . 14
1.4.1 Key Performance Parameters . . . 14
1.5 Performance Parameters of Gyroscopes . . . 15
1.5.1 Scale Factor and Linearity . . . 15
1.5.2 Zero-Rate Output (Offset) and Bias Stability . . . 16
1.5.3 Noise and Minimum Detectable Signal . . . 16
1.5.4 Dynamic Range . . . 17
1.5.5 Bandwidth . . . 17
1.5.6 Temperature Dependence . . . 17
1.5.7 Performance Grades and Application Classes . . . 19
1.5.8 Summary of Key Parameters . . . 19
1.6 Designing Gyroscopes: From Specifications to System . . . 19
1.7 Data Sheet Examples: From Consumer to High-Performance Gyroscopes . . . 21
1.7.1 Consumer-Plus MEMS Gyroscope . . . 21
1.7.2 High-Performance MEMS Gyroscope (Example: Sensonor SAR300) . . . 21
1.8 Allan Deviation: Characterizing Noise and Stability . . . 23
1.8.1 Calculation of Allan Variance . . . 23
1.8.2 Interpretation of the Allan Deviation Plot . . . 24
1.8.3 Relation to Navigation Performance . . . 24
1.8.4 Case Study: Satellite Orientation During Eclipse . . . 25
1.9 Summary: Key Takeaways . . . 25
1.10 Non-MEMS High-Performance Gyroscopes . . . 26
1.10.1 Hemispherical Resonator Gyroscopes (HRGs) . . . 26
1.10.2 Fiber Optic Gyroscopes (FOGs) . . . 27
1.10.3 Miniaturization Attempts: Micro-HRGs and Chip-Scale FOGs . . . 29
1.11 MEMS Transduction Principles and Basic Fabrication . . . 29
1.11.1 General Sensor Working Principle . . . 29
1.11.2 A Basic MEMS Fabrication Process . . . 29
1.11.3 Resulting Structures and Sensing Examples . . . 30
1.11.4 Typical Dimensions and Scale . . . 31

2 Electrostatic Forces in MEMS Gyroscopes **33**
2.1 Introduction . . . 33
2.2 Motion in Non-Inertial Frames . . . 34
2.3 Spring-Mass-Damper Model . . . 34
2.4 Frequency Domain Analysis . . . 35
2.5 Design Considerations for Gyroscopes . . . 36
2.6 Comb Finger Capacitors and Electrostatic Actuation . . . 37
2.7 Electronic Oscillator for Sustaining Oscillation . . . 38
2.7.1 Damping and Quality Factor in Practice . . . 39
2.8 Quality Factor and Damping Sources . . . 39
2.9 Sustaining Oscillation at Resonance . . . 40
2.9.1 Transduction Factors and Motional Current . . . 40
2.9.2 Complete Electromechanical Transfer Function . . . 42
2.10 Performance Comparison and Applications . . . 43
2.11 Recap and Roadmap . . . 43
2.12 Combined Electromechanical Transfer Function . . . 44
2.13 Equivalent Electrical Model . . . 45
2.14 Equivalent Electrical Model (Continued) . . . 45
2.15 Influence of Parasitics: Feedthrough Current . . . 46
2.16 Toward Differential (Push-Pull) Operation . . . 47
2.17 Push-Pull (Differential) Actuation and Sensing . . . 48
2.17.1 Configuration and Force Analysis . . . 48
2.17.2 Key Advantages of Push-Pull . . . 50
2.17.3 Equivalent Electrical Model for Push-Pull Configuration . . . 50
2.17.4 Feedthrough Cancellation in Differential Sensing . . . 51
2.17.5 Design Flexibility . . . 52
2.18 Push-Pull (Differential) Actuation and Sensing . . . 52
2.18.1 Force Analysis for Push-Pull Actuation . . . 52
2.18.2 Key Advantages of Push-Pull . . . 53
2.18.3 Equivalent Electrical Model for Push-Pull Configuration . . . 54
2.18.4 Feedthrough Cancellation in Differential Sensing . . . 55
2.18.5 Summary: Single-Ended vs. Push-Pull . . . 55
2.19 Optimizing the Number of Fingers . . . 55
2.19.1 Trade-Offs in Finger Allocation . . . 56
2.19.2 Lateral Pull-In Instability . . . 56
2.19.3 Design Guideline for Rotor Bias . . . 57
2.20 Layout and Routing Considerations . . . 57
2.20.1 Differential Pair Routing (Option 1) . . . 57
2.20.2 Fully Symmetric Routing (Option 2) . . . 57
2.21 Summary and Outlook . . . 58

3 Amplitude-Modulated Capacitive MEMS Gyroscopes **59**
3.1 Introduction to Gyroscope Modes . . . 59
3.2 Effect of Rotation and Coriolis Force . . . 59
3.3 Sensing the Motion . . . 60
3.4 Typical Dimensions and Fabrication . . . 60
3.5 Subsystem Combination . . . 60
3.6 Capacitance Configurations . . . 60
3.6.1 Comb-Finger (Area-Varying) Capacitors for Drive Mode . . . 60
3.6.2 Parallel-Plate (Gap-Varying) Capacitors for Sense Mode . . . 61
3.7 Readout Mechanism . . . 61
3.8 Current in Sensing Electrodes . . . 61
3.9 Working Principle Summary . . . 62
3.10 Subsystems Review . . . 62

3.11 Advanced Structure: Decoupled Drive and Sense Modes . . . 62
3.11.1 Structural Elements . . . 62
3.11.2 Operational Behavior . . . 63
3.11.3 Benefits . . . 63
3.12 Distinguishing Coriolis from Linear Acceleration . . . 63
3.13 Separating Contributions from Linear Acceleration and Coriolis Acceleration . . . 64
3.13.1 Tuning Fork Principle . . . 64
3.13.2 Response to Angular Rate vs. Linear Acceleration . . . 64
3.13.3 From Conceptual to Practical Implementation . . . 65
3.14 Transition to Detailed Structure Analysis . . . 65
3.15 Detailed Analysis of Yaw-Axis Gyroscope . . . 66
3.15.1 Identification of the Three Frames . . . 66
3.15.2 Anchor Points and Suspension Springs . . . 67
3.15.3 Drive Mode Actuation and Detection . . . 67
3.15.4 Sense Mode Operation under Rotation . . . 67
3.15.5 Differential Sensing and Common-Mode Rejection . . . 67
3.15.6 Additional Electrodes . . . 68
3.16 Summary of Operating Principles . . . 68
3.17 Sensitivity to Different Axes . . . 68
3.17.1 Response to Rotation About Different Axes . . . 68
3.18 Designing for Other Axes: The Roll/Pitch Gyroscope . . . 69
3.18.1 Structural Modifications for Out-of-Plane Sensing . . . 69
3.18.2 Operating Principle for Roll/Pitch Sensing . . . 70
3.18.3 Common-Mode Rejection for Out-of-Plane Acceleration . . . 70
3.19 Design Guidelines for Advanced Gyroscopes . . . 70
3.19.1 Tuning Fork Architecture: Highly Recommended . . . 70
3.19.2 Double Decoupling: Benefits Beyond Parasitic Reduction . . . 71
3.19.3 Mass Distribution: Minimize Drive Frame Mass . . . 71
3.19.4 Framework Structures for Optimal Stiffness-to-Mass Ratio . . . 71
3.20 Integration of Multiple Sensing Axes . . . 72
3.21 Summary . . . 72
3.22 Design Guidelines for Mass Distribution . . . 73
3.22.1 Sense Frame Mass: Minimize for Acceleration Rejection . . . 73
3.22.2 Coriolis Frame Mass: Maximize for Sensitivity . . . 73
3.22.3 Drive Frame Mass: Minimize for Rejection of Unwanted Forces . . . 73
3.22.4 Optimal Mass Distribution Summary . . . 74
3.23 Spring Design Considerations . . . 74
3.23.1 Benefits of Folded Springs . . . 74
3.24 Quality Factor Considerations . . . 74
3.24.1 Drive Mode Quality Factor: Maximize . . . 74
3.24.2 Sense Mode Quality Factor: Trade-offs . . . 75
3.24.3 Natural Q Differentiation via Electrode Geometry . . . 75
3.24.4 Operating Frequency Selection . . . 75
3.25 Drive and Sense Frequency Relationship . . . 75
3.26 Mechanical Stoppers (Travel Limiters) . . . 76
3.27 Quadrature Compensation Electrodes . . . 76
3.28 Monolithic Three-Axis Gyroscopes . . . 76
3.28.1 Operating Principle . . . 76
3.28.2 Advantages . . . 77
3.28.3 Challenges . . . 77
3.28.4 Current Status . . . 77
3.29 Transition to Performance Analysis . . . 78
3.30 General Sensitivity for Accelerometer Subsystem . . . 78
3.30.1 Transfer Function from Acceleration to Displacement . . . 78

3.30.2 Operating Regimes . . . 79
3.31 Application to Gyroscope Sensitivity . . . 80
3.31.1 Quasi-Static Operation ($\omega_d \ll \omega_s$) . . . 80
3.31.2 Mode-Matched Operation ($\omega_d = \omega_s$) . . . 80
3.31.3 Split-Mode Operation ($\omega_d \approx \omega_s$) . . . 80
3.31.4 Post-Resonance Operation ($\omega_d \gg \omega_s$) . . . 81
3.32 Practical Operating Modes in MEMS Gyroscopes . . . 81
3.32.1 Mode-Matched Operation . . . 81
3.32.2 Split-Mode Operation . . . 81
3.32.3 Industry Practice . . . 81
3.33 Mode-Matched Operation . . . 82
3.33.1 Advantages of Mode Matching . . . 82
3.33.2 Disadvantages and Challenges . . . 82
3.34 Mode-Split Operation . . . 82
3.34.1 Advantages of Mode Splitting . . . 83
3.34.2 Disadvantages . . . 83
3.34.3 Comparison Summary . . . 83
3.35 Summary of Gyroscope Subsystems . . . 83
3.36 Electronics Overview . . . 84
3.36.1 Focus: Drive Oscillator Electronics . . . 85
3.37 Loop Gain in Oscillators . . . 85
3.37.1 Contrast with Negative Feedback . . . 85
3.37.2 Applying to the Drive Resonator . . . 85
3.38 Loop Gain Analysis for Drive Oscillator . . . 86
3.38.1 Barkhausen Criterion in the Context of the Drive Loop . . . 86
3.38.2 Start-Up Condition vs. Steady-State Condition . . . 86
3.38.3 Oscillation Buildup in Time Domain . . . 86
3.39 Circuit Implementation for Push-Pull Drive and Differential Sensing . . . 86
3.39.1 System Configuration . . . 87
3.39.2 Differential Charge Amplifier . . . 87
3.39.3 Phase Analysis Around the Loop . . . 88
3.39.4 90° Phase Shifter . . . 88
3.39.5 Amplitude Control and Nonlinearity . . . 89
3.39.6 Complete Loop Operation . . . 89
3.39.7 Summary of Drive Oscillator Design . . . 90
3.40 Why Control the Drive Voltage? . . . 90
3.41 Overall Loop Gain Analysis . . . 90
3.41.1 Stage Transfer Functions . . . 91
3.41.2 Combined Loop Gain . . . 91
3.41.3 Design Rule: Pole Placement . . . 92
3.42 Temperature Dependence of Quality Factor . . . 92
3.43 Automatic Gain Control (AGC) Loop . . . 93
3.43.1 Conceptual Operation . . . 93
3.43.2 Implementation Details . . . 94
3.43.3 Negative Feedback Action . . . 94
3.43.4 AGC Loop Stability . . . 94
3.43.5 Importance for Sensitivity Stability . . . 94
3.44 Summary and Preview . . . 95
3.45 Q&A: AGC Bandwidth Constraints . . . 95

4 Noise and Nonidealities in AM Gyroscopes **97**
4.1 Introduction to Parallel Plate Topology . . . 97
4.2 Sense Mode Capacitance Readout . . . 97
4.3 Differential Output Voltage . . . 97
4.4 System-Level Architecture . . . 98
4.5 Sensitivity as a Design Parameter . . . 98
4.6 Noise Sources in MEMS Gyroscopes . . . 100
4.6.1 Mechanical-Thermal Noise . . . 100
4.6.2 Noise Equivalent Angular Rate Density . . . 101
4.6.3 Noise Optimization and Numerical Example . . . 101
4.7 Electronic Noise Sources . . . 102
4.7.1 Noise Sources in the Sense Charge Amplifier . . . 102
4.7.2 Input-Referred Angular Rate Noise from Electronics . . . 102
4.7.3 Total Electronic Noise and Comparison . . . 103
4.7.4 Design Considerations . . . 103
4.8 Dominance of Noise Terms in Applications . . . 104
4.8.1 Application-Dependent Trade-offs . . . 104
4.8.2 Noise Reduction Guidelines . . . 104
4.9 Numerical Example and Noise Budget Analysis . . . 104
4.9.1 Achieving Low Op-Amp Noise . . . 105
4.9.2 Noise Budget Analysis . . . 105
4.9.3 Current Required for Noise Matching . . . 106
4.10 Noise in the Drive Loop . . . 106
4.11 Q&A Session: White Noise vs. Bias Instability . . . 106
4.12 ADC and Bandwidth Requirements . . . 107
4.12.1 Bandwidth Determination in Mode-Split Operation . . . 107
4.12.2 Bandwidth Limitation and Filtering . . . 107
4.12.3 From Noise Density to RMS Noise . . . 108
4.12.4 Dynamic Range and ADC Resolution . . . 108
4.12.5 Quantization Noise in the Overall Budget . . . 109
4.12.6 Programmable Full-Scale Range . . . 109
4.13 ADC Requirements and Oversampling . . . 110
4.13.1 Quantization Noise Fundamentals . . . 110
4.13.2 Oversampling Principle . . . 110
4.13.3 Decimation . . . 110
4.14 Noise Shaping and Sigma-Delta Converters . . . 111
4.14.1 Principle of Noise Shaping . . . 111
4.14.2 First-Order Sigma-Delta Architecture . . . 111
4.14.3 Operation and Waveform Interpretation . . . 112
4.14.4 Noise Shaping Analysis . . . 112
4.14.5 Higher-Order Modulators . . . 112
4.14.6 Trade-offs and Applicability . . . 113
4.15 Non-Idealities: Fabrication Process Variations . . . 113
4.15.1 Over-etch and Under-etch . . . 113
4.15.2 Consequences for Calibration . . . 113
4.15.3 Design Strategies for Robustness . . . 114
4.16 Calibration Strategies . . . 114
4.16.1 One-Time Calibration . . . 114
4.16.2 Temperature Calibration . . . 114
4.16.3 Design for Robustness . . . 114
4.17 Part-to-Part Variations: Over-Etch and Under-Etch . . . 115
4.17.1 Origin of CD Loss . . . 115
4.17.2 Magnitude and Variability . . . 115
4.17.3 Affected Parameters . . . 115

4.17.4 Design Strategy . . . 116
4.18 Impact on Drive Loop and Scale Factor Stability . . . 116
4.18.1 Drive Displacement vs. AGC Reference . . . 116
4.18.2 Scale Factor Sensitivity to Geometry . . . 116
4.18.3 Numerical Example of Variation . . . 117
4.18.4 Design Freedom: Different Gaps . . . 117
4.18.5 Rule of Thumb . . . 117
4.18.6 Additional Considerations . . . 118
4.19 Trade-offs in Design for Robustness . . . 118
4.20 Frequency Split Stability . . . 119
4.20.1 Resonance Frequency Dependence on Geometry . . . 119
4.20.2 Effect of CD Loss on Frequency Split . . . 119
4.20.3 Design Principle . . . 120
4.21 Temperature Effects . . . 120
4.21.1 Young's Modulus Temperature Dependence . . . 120
4.21.2 Quality Factor Temperature Dependence . . . 120
4.21.3 Impact on Scale Factor . . . 121
4.21.4 Temperature Calibration Requirements . . . 121
4.22 Mechanical Stress and Package Effects . . . 121
4.23 Summary and Key Takeaways . . . 121

5 Quadrature Error and Compensation in AM Gyroscopes 123
5.1 Resolution and Bandwidth in Consumer Applications . . . 123
5.2 Displacement Scales: From Drive Motion to Earth's Rotation . . . 123
5.3 The Challenge of Spurious Motion: Quadrature Error . . . 124
5.3.1 Phase Relationship: The Key to Quadrature . . . 124
5.3.2 Sources of Quadrature Error . . . 124
5.4 Modeling Quadrature in the Signal Chain . . . 125
5.5 Synchronous Demodulation: The Ideal Case . . . 125
5.5.1 Demodulation of the Rate Signal . . . 125
5.5.2 Rejection of the Quadrature Signal . . . 125
5.6 A Critical Caveat: Dynamic Range and Quadrature . . . 126
5.7 Non-Ideal Demodulation: The Impact of Phase Errors . . . 126
5.8 Visualizing Phase Error and Phase Noise . . . 127
5.9 Electronic Implementation: The Passive Mixer Demodulator . . . 127
5.9.1 Operating Principle . . . 127
5.9.2 Key Design Considerations . . . 128
5.10 Modeling the Origin of Quadrature: Cross-Axis Stiffness . . . 128
5.10.1 Equation of Motion in the Sense Axis . . . 128
5.10.2 Why Neglect the Reverse Coupling? . . . 129
5.10.3 Input-Referred Quadrature Rate . . . 129
5.11 Relating Quadrature to a Physical Misalignment Angle . . . 129
5.11.1 A Simple Expression for Input-Referred Quadrature . . . 130
5.11.2 Interpretation and Units . . . 130
5.12 The Frequency Trade-Off: Why Not Operate at Higher Frequencies? . . . 130
5.13 Physical Origins of the Misalignment Angle α . . . 131
5.13.1 In-Plane Non-Uniformities: Spring Asymmetries . . . 131
5.13.2 Out-of-Plane Effects: Wafer Bow and Tilted Etching . . . 131
5.14 Quantitative Impact of a Small Tilt . . . 132
5.15 Minimizing Quadrature: A Mechanical Design Perspective . . . 133
5.15.1 Basic Strategies . . . 133
5.15.2 Advanced Design Strategies . . . 133
5.16 The Challenge of High-Performance Multi-Axis Gyroscopes . . . 133
5.17 Electrostatic Sources of Quadrature: Comb-Drive Asymmetries . . . 135

5.17.1 Modeling the Electrostatic Quadrature Force . . . 135
5.18 Electrostatic Sources of Quadrature and Offset: Summary . . . 136
5.18.1 Statistical Averaging and Typical Magnitudes . . . 137
5.19 The Path to Compensation: Understanding Phase Errors . . . 137
5.19.1 Sources of Phase Error . . . 138
5.19.2 Consequences of Phase Error . . . 139
5.20 Mechanical Phase Error: A Quantitative Example . . . 139
5.20.1 Impact on Output Offset . . . 140
5.20.2 Temperature Dependence of Mechanical Phase Error . . . 140
5.21 Electronic Phase Errors . . . 141
5.21.1 Matched Charge Amplifiers . . . 141
5.21.2 The 90° Phase Shifter and Amplifier Non-Idealities . . . 141
5.21.3 Temperature Dependence of Electronic Phase Error . . . 141
5.22 Summary of Phase Error Contributions . . . 142
5.23 Meeting System-Level Stability Targets . . . 142
5.24 The Impossibility of Reaching High Performance without Quadrature Cancellation . . . 142
5.24.1 Why Calibration and Trimming Are Insufficient . . . 143
5.25 A Simple Partial Solution: Capacitive Quadrature Cancellation . . . 143
5.25.1 Practical Implementation . . . 144
5.25.2 Pros and Cons . . . 144
5.26 Electromechanical Quadrature Cancellation at the Origin . . . 144
5.26.1 Implementation for Yaw (In-Plane) Gyroscopes . . . 145
5.26.2 Extension to Pitch/Roll (Out-of-Plane) Gyroscopes . . . 146
5.27 Electromechanical Quadrature Cancellation: The Principle . . . 146
5.27.1 Force Derivation for In-Plane (Yaw) Gyroscopes . . . 146
5.27.2 Practical Implementation . . . 148
5.28 Extension to Out-of-Plane (Pitch/Roll) Gyroscopes . . . 149
5.28.1 Operating Principle . . . 149
5.28.2 Layout Example . . . 149
5.29 Performance Impact: A Historical Example . . . 150
5.30 Summary . . . 150
5.31 Automatic Quadrature Cancellation: A Closed-Loop Approach . . . 150
5.31.1 System Architecture . . . 151
5.31.2 Operation . . . 151
5.31.3 Why Is This Only for High-End Applications? . . . 151
5.31.4 Interaction Between Loops . . . 152
5.32 Summary of Quadrature Compensation Strategies . . . 152
5.33 Complete Gyroscope Structures: A Review . . . 152
5.33.1 Yaw Gyroscope (In-Plane Sense Mode) . . . 152
5.33.2 Pitch/Roll Gyroscope (Out-of-Plane Sense Mode) . . . 153
5.34 Conclusion . . . 154

6 AM Resistive NEMS Gyroscopes 157
6.1 Introduction to the Alternative Technology . . . 157
6.2 Quick Review of Capacitive Gyroscopes . . . 157
6.2.1 Identified Limitations and Trade-offs . . . 157
6.3 Alternative Approaches . . . 158
6.4 Today's Focus: Piezoresistive NEMS Sensing . . . 158
6.5 The Fabrication Process . . . 158
6.5.1 Key Process Steps (Simplified) . . . 160
6.6 Key Advantages of the Process . . . 160
6.7 Piezoresistive Effect in Heavily Doped Monocrystalline Silicon . . . 161
6.7.1 Simplified Physical Origin of Piezoresistivity . . . 161
6.8 Design Considerations for Scale Factor and Linearity . . . 161

6.8.1 Example Gauge Parameters . . . 161
6.9 Actuation and Drive-Mode Detection Strategy . . . 162
6.10 Incorporation of NEMS Gauges into the Mechanical Structure . . . 162
6.10.1 In-Plane Lever Configuration (for Small Displacements) . . . 163
6.10.2 Drive-Mode Detection (Large Displacements) . . . 163
6.10.3 Out-of-Plane Sensing (Pitch/Roll Gyroscopes) . . . 164
6.11 Overall Gyroscope Architecture . . . 164
6.12 Complete Gyroscope Layout and Symmetry Considerations . . . 165
6.12.1 Pitch / Roll Gyroscopes . . . 166
6.13 Examples of Realised Devices . . . 166
6.14 Mechanical Advantages . . . 167
6.15 Process Summary and Sensitivity Promise . . . 167
6.16 Question from the Audience . . . 167
6.17 Scale-Factor Derivation (Resistive Topology) . . . 167
6.18 Numerical Evaluation of the Scale Factor . . . 168
6.19 Readout Electronics Architecture . . . 169
6.19.1 Wheatstone Bridge Output (for non-electronics students) . . . 169
6.20 Temperature Sensitivity of the Scale Factor . . . 170
6.21 First Experimental Results . . . 170
6.22 Noise Sources . . . 170
6.23 Thermo-Mechanical Noise . . . 171
6.24 Electronic Noise Contributions . . . 171
6.24.1 Amplifier Noise . . . 171
6.24.2 Resistor Thermal Noise (New Term) . . . 171
6.25 Alternative Low-Power Readout Topology . . . 172
6.26 Pros, Cons and Power-Noise Trade-off . . . 172
6.27 Numerical Noise Example . . . 172
6.28 Reliability Considerations (Opening) . . . 173
6.29 Reliability Assessment . . . 174
6.29.1 Why the NEMS Gauges Do Not Compromise Reliability . . . 174
6.29.2 Standard Reliability Test Procedures . . . 174
6.29.3 Experimental Results . . . 175
6.30 Transition to Stability Analysis . . . 175
6.30.1 Biasing Strategy and Voltage Levels . . . 175
6.31 Low-Voltage Operation Advantage . . . 176
6.32 Scale-Factor Derivation with Drive AGC . . . 176
6.32.1 Drive Wheatstone-Bridge Output . . . 176
6.32.2 Final Scale Factor . . . 176
6.33 Feed-Through Capacitance and Parasitic Immunity . . . 177
6.34 Measured Noise and Stability Performance . . . 178
6.34.1 Capacitive Reference Gyroscope . . . 178
6.34.2 Resistive NEMS Gyroscope . . . 178
6.35 Measuring Earth's Rotation with MEMS Gyrocompassing . . . 179
6.35.1 Experimental Setup . . . 179
6.35.2 Results . . . 180
6.36 Temperature Stability of the Scale Factor . . . 180
6.37 Multi-Axis Monolithic Gyroscope Concept . . . 181
6.38 Conclusions and Commercial Perspective . . . 181
6.39 Overview of the Remaining Course . . . 183
6.40 Question & Answer – Vibration Sensitivity (Thierry) . . . 183

7 FM and RIG MEMS Gyroscopes **185**
7.1 Introduction . . . 185
7.2 Review of the Conventional Amplitude-Modulated Capacitive Working Principle . . . 185
7.2.1 Temperature Dependence of the Scale Factor . . . 185
7.2.2 Zero-Rate Offset (ZRO) Behaviour . . . 186
7.3 Motivation for Alternative Working Principles . . . 186
7.4 First Comments on Frequency-Modulated Gyroscopes . . . 187
7.4.1 Linearity and Full-Scale Range . . . 187
7.4.2 Temperature Immunity Claim . . . 187
7.4.3 "Quasi-Digital" Output Claim . . . 188
7.4.4 Intuitive Cartoon Comparison: AM vs. FM Gyroscopes . . . 188
7.5 Equations for the Frequency-Modulated Gyroscope . . . 189
7.5.1 Physics of the Sign Convention . . . 189
7.5.2 Equations of Motion . . . 190
7.5.3 Phasor Analysis (High-Q Approximation) . . . 190
7.5.4 Phasor Diagram . . . 190
7.5.5 In-Phase Matched Modes ($\phi_y - \phi_x = 0°$) . . . 190
7.5.6 Scale Factor and Full-Scale Range in Practice . . . 191
7.5.7 Pros and Cons of In-Phase Operation . . . 191
7.5.8 Quadrature Operation ($\phi_y - \phi_x = \pm 90°$) . . . 191
7.5.9 Lissajous (Split-Frequency Modulated) Frequency-Modulated Gyroscope . . . 192
7.6 Geometrical Factor, Real Implementation and Performance . . . 195
7.6.1 Geometrical Factor O . . . 195
7.6.2 Real Device Example . . . 195
7.6.3 Extension to Pitch and Roll (Out-of-Plane) Axes . . . 195
7.6.4 Experimental Results . . . 196
7.6.5 Electronics Architecture . . . 196
7.6.6 Noise and Long-Term Stability . . . 196
7.7 Rate-Integrating (Whole-Angle / Full-Angle) Gyroscopes . . . 197
7.7.1 Summary of All Architectures Studied . . . 197
7.7.2 Motivation in Inertial Navigation . . . 198
7.7.3 Historical Example: Hemispherical Resonator Gyroscope (HRG) . . . 198
7.7.4 Goal: Silicon MEMS Implementation . . . 199
7.7.5 Requirements and Possible Designs . . . 199
7.7.6 Equations of Motion and Ideal Behaviour . . . 200
7.7.7 Non-Idealities in Real MEMS . . . 201
7.7.8 Control Architecture . . . 202
7.7.9 Conclusion: Two Paths Beyond Amplitude Modulation . . . 205

Chapter 1

Gyroscopes Applications and Specifications

1.1 Introduction to MEMS Gyroscopes

MEMS gyroscopes are angular rate sensors that measure rotational velocity. They naturally complement MEMS accelerometers, together forming inertial measurement units (IMUs). These compact, integrated chips provide comprehensive motion information by combining a three-axis accelerometer for linear acceleration and a three-axis gyroscope for angular rate about three orthogonal axes. Unlike accelerometers, which measure linear acceleration, gyroscopes directly sense angular velocity. The rationale for measuring angular speed rather than angular acceleration will be addressed in Lecture 6.

Gyroscopes are sophisticated devices that integrate physical operating principles, microfabrication challenges, and complex sensor-electronics interactions. Their design requires careful optimization across these domains, demanding a more extensive treatment than accelerometers. This initial lecture establishes context through applications, performance specifications, and an introduction to the operating principle, culminating in a partial derivation of the scale factor (sensitivity), to be completed in the following lecture.

1.2 Units of Angular Velocity

Angular velocity is fundamentally expressed in radians per second (rad/s). In MEMS gyroscope applications, degrees per second (DPS) is more commonly used, with the conversion factor $\pi/180$. Another convenient unit is Hertz (Hz), where 360 DPS corresponds to one complete revolution per second (1 Hz), yielding a conversion factor of 360 DPS/Hz.

1.3 Applications of MEMS Gyroscopes

Different applications impose distinct performance requirements on gyroscope design. While automotive applications drove early development, the application space has expanded considerably.

1.3.1 Automotive Applications

- **Electronic Stability Control (ESC)**: A z-axis gyroscope measures vehicle rotation about its vertical axis to detect skidding on low-friction surfaces. By comparing chassis rotation with steering angle, the system selectively applies brakes to restore stability.
- **Rollover Detection**: An x-axis gyroscope monitors rotation about the longitudinal axis to predict impending rollover, triggering passive safety systems such as airbags.
- **Smart Tires**: Individual wheel rotation monitoring (currently developmental).

1.3.2 Consumer Electronics Applications

- **Gaming**: Enables intuitive motion-based control in handheld devices.
- **Augmented Reality (e.g., Pokémon Go)**: Aligns virtual content with camera perspective through precise orientation tracking, making gyroscopes standard smartphone components.
- **Virtual Reality**: Delivers immersive experiences by accurately translating head and hand movements into corresponding virtual world rotations.
- **Image Stabilization**: Compensates for unintentional hand rotations during image capture, reducing blur through either optical (active) or software-based (passive) correction. Modern smartphones often incorporate multiple gyroscopes for this purpose.
- **Platform Stabilization**: Enables stable flight in drones by measuring tilt and rotation for continuous attitude adjustment.

1.3.3 Navigation Applications

- **Indoor Navigation**: Provides positioning capability in GPS-denied environments (e.g., shopping malls) by integrating accelerometer data for translation with gyroscope data for rotation.
- **Inertial Navigation**: Enables position tracking in environments without GPS coverage, such as submarines or underwater applications where GPS signals are attenuated beyond 5–10 m depth.

1.3.4 Medical Applications

- **Rehabilitation Monitoring**: Tracks limb rotation during post-surgical recovery, enabling remote progress assessment through telemedicine.
- **Vestibular Rehabilitation**: The human vestibular system naturally contains inertial sensors analogous to three-axis accelerometers and gyroscopes (semicircular canals). MEMS gyroscopes can potentially restore function for patients with vestibular damage causing vertigo or imbalance, similar to cochlear implants for hearing loss.

1.3.5 Additional Applications

- **Transportation**: Monitors angular vibrations and stability in trains, aircraft, and other vehicles.
- **Guidance Systems**: Provides rotation measurements for missile navigation and control.

1.4 Gyroscope Specifications

Performance requirements vary significantly across applications. Consumer applications (e.g., smartphones) target specific parameter sets that will be illustrated through datasheet examples.

1.4.1 Key Performance Parameters

Full-Scale Range: The maximum angular rate the device can measure, determined by application requirements. Human hand rotations typically reach 5–6 revolutions per second, corresponding to approximately 2000 DPS. Consequently, consumer gyroscopes commonly feature full-scale ranges of 2000–3000 DPS.

Sensitivity: Derived from the full-scale range and the device's output dynamic range.

Zero-Rate Offset (ZRO): The output when no angular rate is applied, analogous to zero-g offset in accelerometers. ZRO can be significant and requires compensation strategies, including temperature drift correction (to be covered in subsequent lectures).

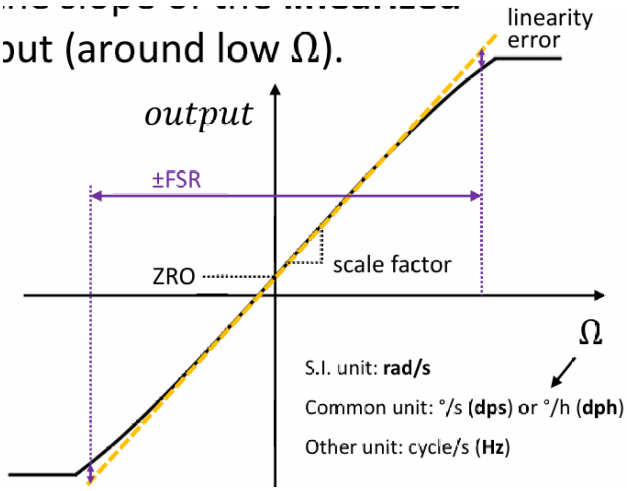

Figure 1.1: Output-input plot.

Resolution: The minimum detectable signal. For hand-motion tracking, noise levels of a few mdps/$\sqrt{\text{Hz}}$ over approximately 100 Hz bandwidth (typically half the output data rate) are sufficient, as hand movements contain negligible energy above 50 Hz.

Power Consumption: Gyroscopes consume more power than accelerometers due to more complex interface electronics. While accelerometers draw tens to hundreds of microamps, gyroscopes in integrated IMUs typically consume approximately 1 mA (recent designs achieving 0.5 mA or less). At supply voltages of a few volts, this corresponds to 1–2 mW power dissipation.

Each specification directly reflects application-specific requirements.

1.5 Performance Parameters of Gyroscopes

We now turn our attention to the key performance parameters that characterize devices of our interest. Understanding these specifications is essential for selecting appropriate gyroscopes for different applications and for interpreting their data sheets.

1.5.1 Scale Factor and Linearity

The fundamental input-output relationship of a gyroscope relates the applied angular rate Ω to the output signal (typically a voltage or digital value). Over its operating range, this relationship should ideally be linear:

$$V_{\text{out}} = S \cdot \Omega + V_{\text{offset}}, \tag{1.1}$$

where S is the **scale factor** (also called sensitivity), typically expressed in mV/(deg/s) or LSB/(deg/s) for digital outputs, and V_{offset} is the zero-rate output.

In practice, due to various nonlinearities in the transduction mechanisms and electronics, the actual response deviates from this ideal straight line. The **non-linearity** is specified as the maximum deviation from a best-fit straight line, expressed as a percentage of the full-scale range. For example, a nonlinearity of 1% means that at any point within the specified range, the actual output differs from the ideal linear response by no more than 1% of the maximum output value.

The **full-scale range** (FSR) is the maximum angular rate (positive or negative) that can be measured while maintaining the specified linearity. This range is typically symmetric around zero and is determined by the application requirements—for instance, a smartphone gyroscope might need ±2000 deg/s to capture rapid gestures, while a navigation system might only require ±300 deg/s but with much higher linearity.

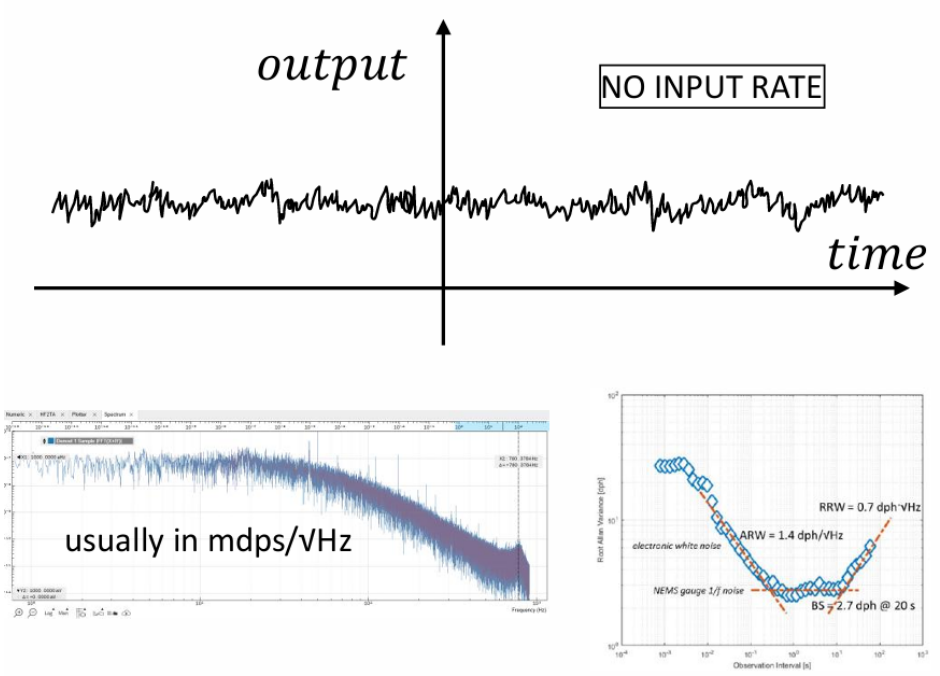

Figure 1.2: Noise representation.

1.5.2 Zero-Rate Output (Offset) and Bias Stability

The **zero-rate output** (ZRO), or offset, is the output signal when no angular rate is applied. In an ideal device, this would be zero, but in reality, various imperfections—mechanical asymmetries, residual stresses, electronic offsets, and parasitic couplings—produce a non-zero output even at rest.

The offset itself can often be calibrated out by measuring it and subtracting it in software. However, the critical issue is that the offset is not perfectly stable; it varies over time and with environmental conditions. This variation is characterized by **bias stability** or **offset drift**.

Bias stability is typically specified in degrees per hour (deg/h) or degrees per second (deg/s) over a specified time interval (e.g., over 10 seconds, 100 seconds, or 1 hour). A bias stability of 10 deg/h means that if you leave the gyroscope stationary for an hour, the integrated angular rate (the inferred angle) would drift by up to 10 degrees due to offset variations alone.

This parameter is arguably the most important for navigation applications, where gyroscope outputs are integrated to determine orientation. Even small offset drifts accumulate over time, leading to rapidly growing position errors.

1.5.3 Noise and Minimum Detectable Signal

Noise represents the random fluctuations in the output signal that are always present, even with a constant input. Unlike offset, noise cannot be calibrated out—it fundamentally limits the smallest signal that can be resolved.

Noise is typically characterized in the frequency domain using **power spectral density** (PSD), denoted as $S_\Omega(f)$ and expressed in units of $(\text{deg/s})/\sqrt{\text{Hz}}$. For many gyroscopes, the noise spectrum is flat (white noise) over a wide frequency range, up to the sensor bandwidth, after which it rolls off due to filtering.

The **minimum detectable signal** (or noise-equivalent rate) is the input angular rate that produces an output equal to the root-mean-square (RMS) noise level. This is found by integrating the noise PSD over the system bandwidth B:

$$\Omega_{\min} = \sqrt{\int_0^B S_\Omega(f)\, df} \approx \sqrt{S_0 \cdot B}, \tag{1.2}$$

where S_0 is the white noise level. For example, with a noise density of 0.01 deg/s/$\sqrt{\text{Hz}}$ and a bandwidth of 100 Hz, the minimum detectable signal is approximately $0.01 \times \sqrt{100} = 0.1$ deg/s.

Another important noise characterization tool for inertial sensors is the **Allan deviation**, which plots the RMS random drift as a function of averaging time. This reveals different noise processes (white noise, random walk, bias instability, rate ramp) operating on different timescales.

1.5.4 Dynamic Range

The **dynamic range** quantifies the span between the smallest and largest signals the gyroscope can measure. It is defined as the ratio of the full-scale range to the noise floor (minimum detectable signal), usually expressed in decibels:

$$\text{DR} = 20 \log_{10} \left(\frac{\text{FSR}}{\Omega_{\min}} \right) \quad \text{(in dB)}. \tag{1.3}$$

A large dynamic range indicates that the sensor can simultaneously measure very small and very large angular rates—a desirable but challenging property to achieve. Trade-offs are inherent: increasing the full-scale range often increases noise, while reducing noise may limit the maximum measurable rate.

1.5.5 Bandwidth

Bandwidth refers to the range of frequencies over which the gyroscope can accurately measure time-varying angular rates. It is determined by the mechanical response of the sense mode and the electrical filtering in the readout electronics.

The required bandwidth varies dramatically by application:

- Consumer electronics (gaming, image stabilization): 50–100 Hz
- Automotive (electronic stability control): 50–100 Hz
- Robotics and drones: 100–400 Hz
- Aerospace (attitude control): 10–100 Hz
- High-shock and vibration measurements: up to 1 kHz or more

The bandwidth is typically set by a low-pass filter in the signal chain, which also limits the noise by restricting the integration bandwidth.

1.5.6 Temperature Dependence

Virtually all gyroscope parameters exhibit some dependence on temperature. Both the scale factor and the zero-rate output change as the device heats up or cools down due to:

- Temperature-dependent material properties (Young's modulus of silicon changes with temperature, affecting resonance frequencies and stiffnesses).
- Thermal expansion mismatches between materials, inducing mechanical stresses.
- Temperature sensitivity of electronics (amplifier gains, offsets).

Scale factor drift with temperature means that the same angular rate produces different output voltages at different temperatures. If uncorrected, this leads to errors in the inferred rate.

Offset drift with temperature is often even more problematic. Even if the zero-rate output is calibrated at room temperature, a temperature change of a few degrees can shift the offset significantly, mimicking a real angular rate.

Compensation techniques include:

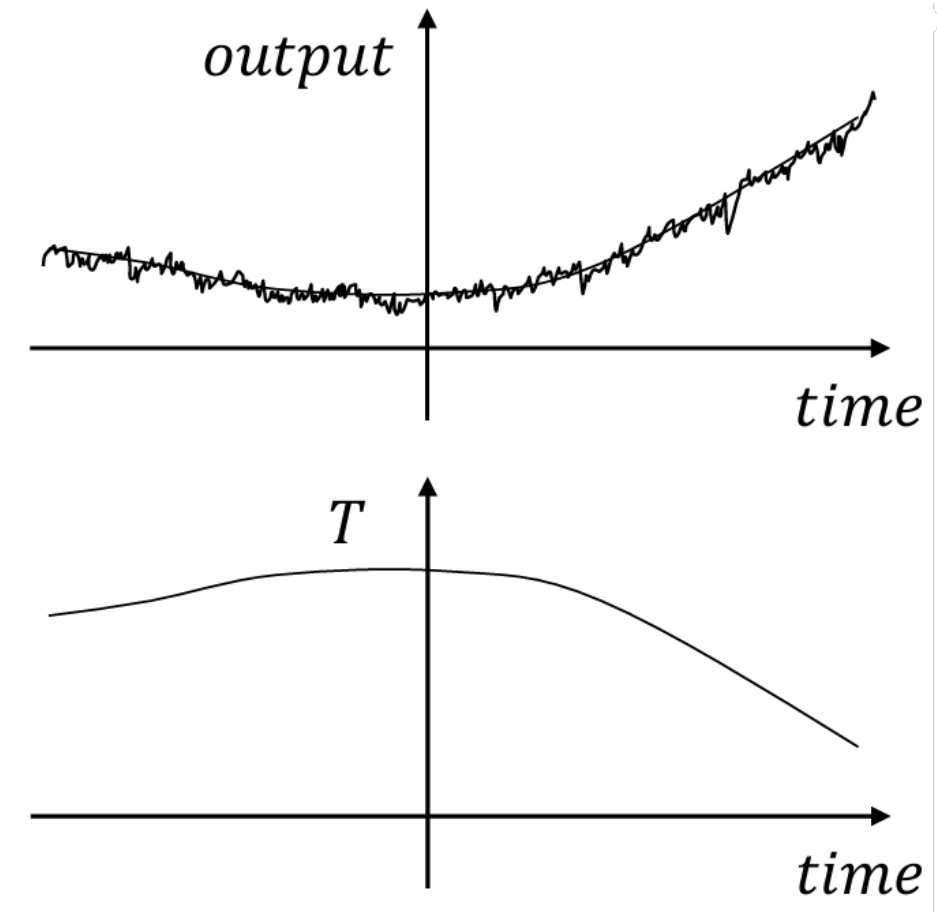

Figure 1.3: ZRO drift in temperature over time.

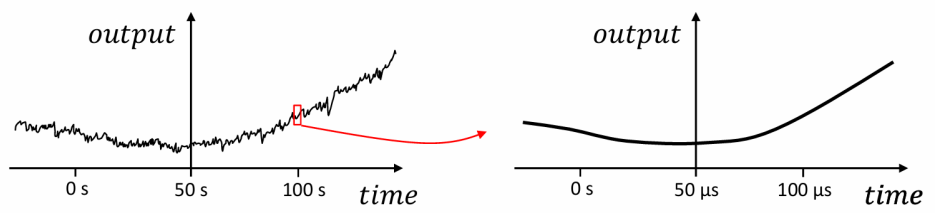

Figure 1.4: Noise and ZRO comparison plot.

- Characterizing the gyroscope over temperature and storing correction coefficients in memory.
- Integrating a temperature sensor on-chip and applying real-time corrections.
- Designing the mechanical structure to be inherently temperature-stable (e.g., using specific crystal orientations or compensation structures).

However, perfect compensation is impossible, and residual drift remains a key performance limiter.

1.5.7 Performance Grades and Application Classes

Based on bias stability, gyroscopes are broadly classified into performance grades:

- **Consumer-grade** (bias stability: 10–100 deg/h): Found in smartphones, gaming controllers, low-cost drones. Small size (1–3 mm^2), low power (milliwatts), low cost ($1–$5). Suitable for short-term orientation tracking (seconds to minutes).
- **Industrial/automotive-grade** (bias stability: 1–10 deg/h): Used in vehicle stability control, platform stabilization, industrial robotics. Moderate size and cost.
- **Tactical-grade** (bias stability: 0.1–1 deg/h): Employed in guidance systems for missiles, unmanned aerial vehicles, and precision agriculture. Larger packages, often vacuum-sealed ceramic or metal, higher power consumption (hundreds of milliwatts to watts).
- **Navigation-grade** (bias stability: 0.001–0.01 deg/h): Required for submarines, aircraft inertial navigation systems, and space applications. Typically large, expensive ($10,000–$100,000), high power (watts to tens of watts), and often based on non-MEMS technologies (fiber optic or ring laser gyroscopes), though high-end MEMS are approaching this class.

As a rule of thumb: better performance demands larger size, higher power consumption, and greater cost. The challenge and opportunity for MEMS gyroscopes lie in pushing high performance into smaller, cheaper, lower-power packages.

1.5.8 Summary of Key Parameters

In the next class, we will see how the drive-mode oscillator design influences many of these parameters, particularly noise and stability, and we will begin our exploration of the sense-mode dynamics where the Coriolis force is detected.

1.6 Designing Gyroscopes: From Specifications to System

The ultimate goal of gyroscope design is to meet the requirements of a specific application. The end user—whether an automotive engineer, a smartphone manufacturer, or a space systems integrator—typically knows little about the internal workings of the gyroscope. They provide a set of performance specifications that their system demands. If an existing device already meets these specifications, they would simply

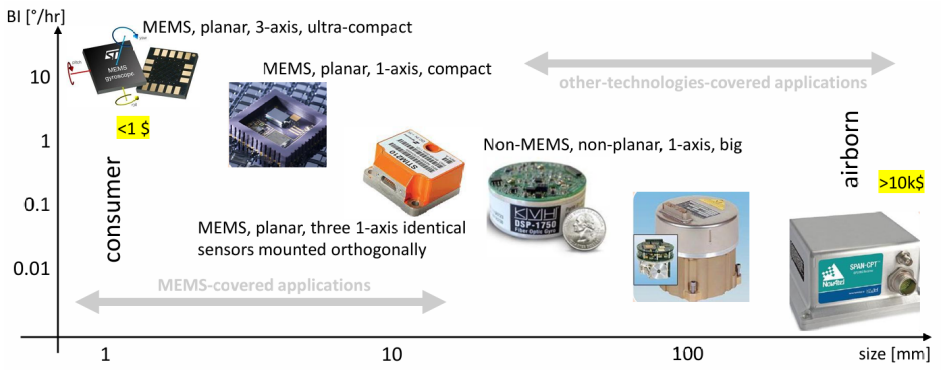

Figure 1.5: Bias Instability over size.

Parameter	**Typical Units**	**Description**
Full-scale range	deg/s, rad/s	Maximum measurable angular rate
Scale factor	mV/(deg/s), LSB/(deg/s)	Sensitivity: output change per unit input
Non-linearity	% of FSR	Deviation from ideal linear response
Zero-rate output (offset)	deg/s, mV	Output with no input rotation
Bias stability	deg/h, deg/s	Long-term variation of offset
Noise density	$(\text{deg/s})/\sqrt{\text{Hz}}$	Random output fluctuations per $\sqrt{\text{Hz}}$
Bandwidth	Hz	Frequency range of accurate measurement
Dynamic range	dB	Ratio of FSR to minimum detectable signal
Temperature drift	%/°C, (deg/s)/°C	Change of parameters with temperature
Power consumption	mW, W	Electrical power required
Size	mm^3, cm^3	Physical footprint

Table 1.1: Summary of key gyroscope performance parameters.

purchase it. Thus, as gyroscope designers, we are tasked with creating a device that surpasses or uniquely satisfies those requirements.

Translating high-level specifications into a functioning system requires a systematic design methodology. A gyroscope is not merely a MEMS structure; it is an integrated system comprising:

- The MEMS sensor element (the proof mass, springs, electrodes)
- The interface and readout electronics
- The transduction mechanisms (electrostatic actuation, capacitive sensing)
- The package (providing mechanical support, electrical interconnection, and often a controlled atmosphere)
- Environmental dependencies (temperature, humidity, stress)

To manage this complexity, designers employ a hierarchy of modeling and simulation tools:

1. **Bottom-up behavioral modeling:** Starting from fundamental physical equations, we build models of individual components (mechanical resonators, capacitive transducers, amplifiers). These are often implemented in system-level simulation environments such as MATLAB/Simulink, which allow us to combine electrical, mechanical, and thermal domains and simulate the complete system response.
2. **Multi-physics finite element analysis (FEA):** Tools like ANSYS or COMSOL are used to analyze the detailed physical behavior of the MEMS structure under coupled stimuli—mechanical deformation, electrostatic forces, fluid damping, thermal expansion. These simulations validate the behavioral models and reveal effects (e.g., stress concentrations, mode shapes, thermoelastic damping) that simplified models might miss.

3. **Iterative refinement:** The process is iterative. Initial designs are simulated, performance is compared to specifications, and parameters are adjusted until requirements are met—or until fundamental trade-offs reveal that the specifications must be revisited.

In this course, we will not delve into the details of these simulation tools, but it is essential to recognize that they are the bridge between abstract customer specifications and a concrete, manufacturable design.

1.7 Data Sheet Examples: From Consumer to High-Performance Gyroscopes

To understand how specifications translate into real devices, let us examine two contrasting data sheets. The differences illustrate the trade-offs between cost, size, power, and performance.

1.7.1 Consumer-Plus MEMS Gyroscope

This example represents the class of gyroscopes found in high-end smartphones, virtual reality headsets, and devices that augment GPS during short outages. These applications demand better-than-basic performance but still operate under tight cost and size constraints.

Key specifications:

- **Full-scale range:** ± 2000 deg/s. This accommodates rapid human gestures (e.g., turning a phone quickly). For context, 2000 deg/s is about 5.5 revolutions per second.
- **Scale factor sensitivity:** Typically expressed in LSB/(deg/s) or mV/(deg/s). The scale factor varies with temperature; here, the data sheet might specify a temperature coefficient of $\pm 2\%$ over the operating range. At 2000 deg/s, a 2% error is 40 deg/s—significant but acceptable for many consumer applications after basic calibration.
- **Zero-rate output (offset):** ± 25 deg/s across samples. This means a randomly selected device, at rest, might indicate a rotation of up to 25 deg/s. This large offset is easily calibrated out by measuring the output at rest, but it consumes dynamic range and requires calibration steps in production.
- **Offset temperature drift:** ± 0.04 deg/s/°C. Over a typical operating temperature range of -40°C to 85°C, the offset can shift by several deg/s. More sophisticated calibration (e.g., polynomial compensation using an on-chip temperature sensor) can reduce this error, but at increased cost.
- **Non-linearity:** 0.2% of full scale. At 2000 deg/s, this corresponds to an error of 4 deg/s—again, correctable with calibration, but each calibration step adds time and cost.
- **Noise density:** 11 mdeg/s/$\sqrt{\text{Hz}}$. Integrating over a 100 Hz bandwidth yields an RMS noise of approximately $0.011 \times \sqrt{100} = 0.11$ deg/s. This sets the short-term resolution limit.
- **Power consumption:** 5 mA at 3 V (15 mW). This is acceptable for battery-powered devices.
- **Package size:** Approximately 3 mm $\times$ 3 mm $\times$ 1 mm (9 mm^3).

This device achieves its low cost (perhaps \$1–\$3) through minimal factory calibration, small size, and high-volume manufacturing. The trade-off is significant residual errors that must be managed at the system level.

1.7.2 High-Performance MEMS Gyroscope (Example: Sensonor SAR300)

This device targets applications such as platform stabilization, surveying, and aerospace, where accuracy and stability are paramount. The dramatic increase in size and cost reflects the engineering invested in performance.

Key specifications:

Table 3. Mechanical characteristics

Symbol	Parameter	Test condition	Min.	Typ.(1)	Max.	Unit
FS	Measurement range	User selectable		±245 ±500 ±2000		dps
So	Sensitivity			114.3 57.2 14.3		digits/dps
SoDr	Sensitivity change vs. temperature(2)	From -40 °C to +85 °C Delta from T = 25 °C		±2		%
DVoff	Digital Zero-rate level	FS = 2000 dps		±25		dps
OffDr	Zero-rate level change vs temperature(3)	FS = 2000 dps		±0.04		dps/°C
NL	Non linearity(3)	Best fit straight line		0.2		% FS
Rn	Rate noise density(3)	BW = 50 Hz		0.011		dps/(√Hz)
ODR	Digital output data rate(3)			11.9/23.7/ 47.3/94.7/ 189.4/ 378.8/ 757.6		Hz
Top	Operating temperature range		-40		+85	°C

25 dps
5 dps
4 dps
0.1 dps
can be calibrated by the user

Table 4. Electrical characteristics

Symbol	Parameter	Test condition	Min.	Typ.(1)	Max.	Unit
Vdd	Supply voltage		2.2	3.0	3.6	V
Vdd_IO	I/O pins supply voltage(2)		1.71		Vdd+0.1	V
Idd	Supply current			5.0		mA
IddSL	Supply current in sleep mode(3)	Selectable by digital interface		2.5		mA
IddPdn	Supply current in power-down mode	Selectable by digital interface		1		µA
VIH	Digital high level input voltage		0.8*Vdd_IO			V
VIL	Digital low level input voltage				0.2*Vdd_IO	V
Ton	Turn-on time(4)	LPF2 disabled ODR = 190 Hz		50		ms

15 mW

Figure 1.6: Sheet table.

- **Package size:** 24 mm × 24 mm × 10 mm (approximately 37,000 mm^3)—over 4000 times larger in volume than the consumer device. The package is ceramic or metal, providing a hermetic seal, low stress, and a stable internal atmosphere (often vacuum). This device is actually a three-axis gyroscope, achieved by mounting three separate sensor chips orthogonally within the package.
- **Full-scale range:** ±400 deg/s. This is sufficient for most navigation and stabilization tasks; extreme rates are not required.
- **Scale factor accuracy:** 500 ppm (0.05%). This is approximately 100 times better than the consumer device, achieved through extensive calibration and stable materials.
- **Non-linearity:** 15 ppm of full scale. At 400 deg/s, this corresponds to an error of only 0.006 deg/s—orders of magnitude better.
- **Offset (bias):** ±250 deg/h (approximately ±0.07 deg/s). This is already far better than the consumer device's raw offset, and the device has likely been factory-calibrated.
- **Offset temperature drift:** 9 deg/h/°C (0.0025 deg/s/°C). This is an order of magnitude lower than the consumer device.
- **Bandwidth:** 200 Hz, comparable to consumer devices.

The superior performance of this device stems not from a fundamentally different operating principle, but from:

- Advanced, stable packaging.
- Extensive calibration across temperature and other environmental factors.
- Higher-quality electronics and design margins.
- Larger physical scale, which reduces certain noise mechanisms and sensitivity to dimensional variations.

The cost of such a device can range from hundreds to thousands of dollars, reflecting the smaller production volumes, larger package, and significant calibration effort.

1.8 Allan Deviation: Characterizing Noise and Stability

A particularly important tool for characterizing gyroscope performance, especially for navigation applications, is the **Allan deviation**. While power spectral density (PSD) describes noise in the frequency domain, the Allan deviation reveals how the sensor's output variance depends on the averaging time, simultaneously capturing short-term noise and long-term drift.

1.8.1 Calculation of Allan Variance

Consider a long data record of the gyroscope output at rest, sampled at a constant rate. Divide this record into M segments, each of duration τ (the averaging time). For each segment k, compute the average output $\bar{\Omega}_k(\tau)$. The Allan variance for that τ is defined as half the mean square difference between consecutive segment averages:

$$\sigma^2_{\text{Allan}}(\tau) = \frac{1}{2}\langle(\bar{\Omega}_{k+1}(\tau) - \bar{\Omega}_k(\tau))^2\rangle, \tag{1.4}$$

where the angle brackets denote averaging over all k. The Allan deviation is the square root: $\sigma_{\text{Allan}}(\tau)$.

The procedure is then repeated for different τ values (longer segments), yielding a plot of $\sigma_{\text{Allan}}(\tau)$ vs. τ.

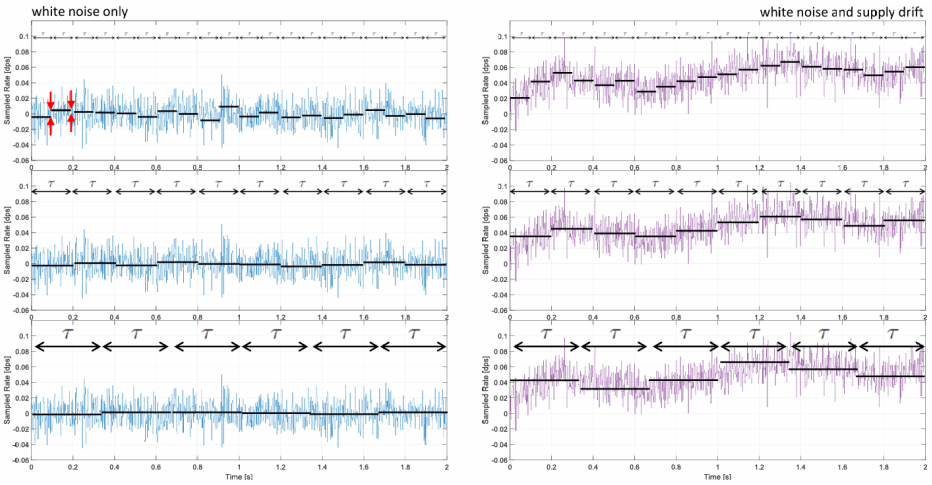

Figure 1.7: Noise intervals.

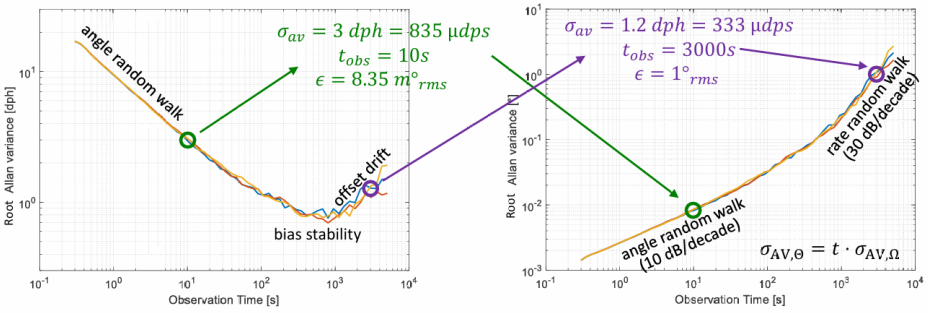

Figure 1.8: Typical Allan deviation plot for a gyroscope, showing characteristic regions.

1.8.2 Interpretation of the Allan Deviation Plot

The shape of the Allan deviation curve reveals different noise processes operating on different timescales:

1. **Angle Random Walk (ARW) region (slope $-1/2$):** At short τ (high frequencies), white noise (rate noise) dominates. As τ increases, the averaging reduces the fluctuation, and $\sigma_{\text{Allan}}(\tau)$ decreases proportionally to $1/\sqrt{\tau}$. The value at $\tau = 1$ s is often quoted as the ARW coefficient, in units of $\text{deg}/\sqrt{\text{h}}$ or $\text{deg}/\sqrt{\text{s}}$. This parameter characterizes the noise floor and limits short-term navigation accuracy.

2. **Bias Instability region (minimum of the curve):** At intermediate τ, the curve reaches a minimum. This minimum value is the **bias stability**—the best achievable stability if the output is averaged over an optimal time. It represents the fundamental limit due to low-frequency noise (e.g., $1/f$ noise) and sets the performance grade of the gyroscope.

3. **Rate Random Walk (RRW) region (slope $+1/2$):** At long τ, the curve increases. This indicates the presence of drift (e.g., due to temperature ramps or very low-frequency noise). The increasing variance means that longer averaging actually degrades the estimate, as the underlying offset is changing.

The Allan deviation thus provides a concise summary of both short-term and long-term behavior in a single graph.

1.8.3 Relation to Navigation Performance

The impact of these parameters on navigation can be understood by considering the integration of angular rate to obtain angle. The angle error variance grows differently depending on the dominant noise process:

- **Angle Random Walk:** The angle error variance grows linearly with time: $\sigma_\theta^2 = \text{ARW}^2 \cdot t$. Thus, the RMS angle error grows as $\sqrt{t}$ (slope +10 dB/decade on a log-log plot).
- **Rate Random Walk (offset drift):** The angle error variance grows as t^3 (since drift in rate integrates to a ramp in angle). The RMS angle error grows as $t^{3/2}$ (slope +30 dB/decade). This is far more damaging over long times.

1.8.4 Case Study: Satellite Orientation During Eclipse

A concrete example illustrates the use of Allan deviation in system design.

Consider a satellite in low Earth orbit with a period of 90 minutes. For 45 minutes, it is in sunlight and can use Sun sensors to determine its orientation and recalibrate its gyroscopes. For the remaining 45 minutes, it is in eclipse (Earth between the satellite and the Sun) and must rely solely on its gyroscopes to maintain attitude knowledge within a 1° error budget.

Given the Allan deviation plot for a candidate gyroscope (as in the Sensonor example), we can evaluate its suitability.

- Recalibration interval: $t = 45$ minutes $= 2700$ seconds.
- From the Allan deviation plot at $\tau = 2700$ s, we read $\sigma_{\text{Allan}} \approx 1.2$ deg/h.
- Convert to deg/s: 1.2 deg/h $= 1.2/3600 = 3.33 \times 10^{-4}$ deg/s.
- The RMS angle error after 2700 s, if the dominant error is rate random walk (drift), is approximately $\sigma_\theta \approx \sigma_{\text{Allan}} \times t = (3.33 \times 10^{-4}$ deg/s$) \times 2700$ s ≈ 0.9 deg.

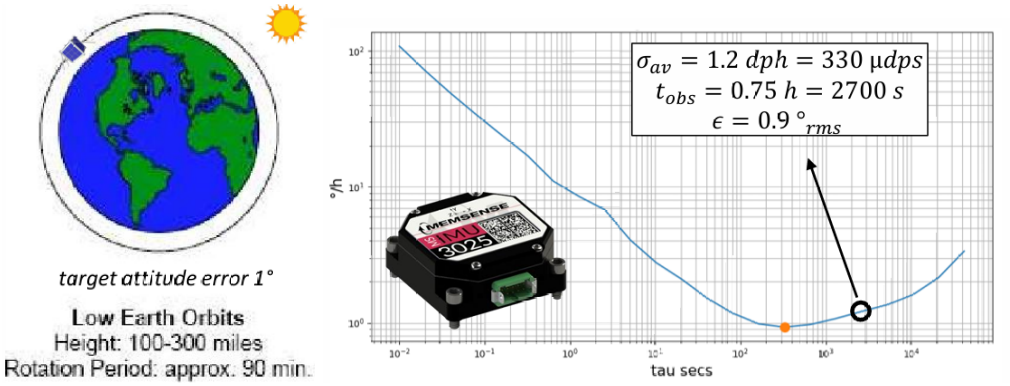

Figure 1.9: Calibration using the Sun.

This is within the 1° requirement, though with little margin. The engineer might accept this gyroscope or seek a slightly better one. At the end of the eclipse period, the Sun sensors provide a fresh orientation measurement, resetting the accumulated error.

This example demonstrates how the Allan deviation directly informs system-level decisions about calibration intervals and expected navigation accuracy.

1.9 Summary: Key Takeaways

- Gyroscope design begins with customer specifications; the designer must translate these into a complete system using hierarchical modeling and simulation tools.
- Data sheets reveal the performance grades: consumer devices achieve small size and low cost at the expense of accuracy and stability, while high-end devices use larger packages, extensive calibration, and stable materials to achieve orders-of-magnitude better performance.

- The Allan deviation is an essential tool for gyroscope characterization, simultaneously displaying the effects of white noise (ARW) and long-term drift (bias instability, RRW) on different timescales.
- The Allan deviation directly impacts navigation system design, determining how often a gyroscope must be recalibrated to maintain a given orientation accuracy.

In the following classes, we will return to the detailed design of the sense mode, exploring how the Coriolis force is detected and how the readout electronics influence key parameters like noise and bandwidth.

1.10 Non-MEMS High-Performance Gyroscopes

Before diving deeper into MEMS gyroscopes, it is instructive to briefly examine the state-of-the-art in high-end, non-MEMS gyroscopes. These devices dominate applications where ultimate performance is required, such as strategic-grade inertial navigation, spacecraft attitude control, and precision stabilization. Understanding their working principles and trade-offs provides context for the ambitious goals of MEMS gyroscope development.

1.10.1 Hemispherical Resonator Gyroscopes (HRGs)

The operating principle of the HRG traces back to a discovery by G.H. Bryan in the 1890s. He observed that if a wine glass is struck, it vibrates in a characteristic resonant mode. If the glass is then rotated, the vibration pattern does not rotate with the glass; instead, it lags behind due to the inertia of the standing wave. This phenomenon, now understood as a manifestation of the Coriolis effect, is the foundation of the HRG.

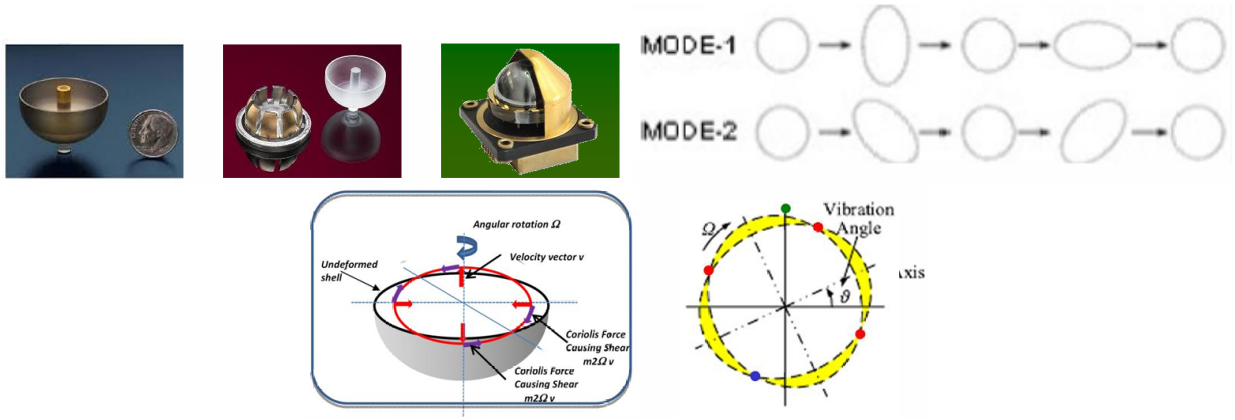

Figure 1.10: Hemispherical resonator (top) with its characteristic vibration mode (bottom). When the resonator rotates, the vibration pattern lags, and the angular displacement of the nodes is proportional to the rotation angle.

A typical HRG consists of a fused quartz or other low-loss material machined into a thin-walled hemisphere (or similar axisymmetric shape such as a cylinder or ring). Electrodes surrounding the resonator electrostatically excite and sustain a specific resonant mode—typically the $n = 2$ flexural mode, where the rim deforms into an elliptical shape with four nodes and antinodes. In this mode, points on the rim oscillate radially, giving them a velocity.

When the entire assembly is rotated about its symmetry axis, the Coriolis effect couples energy into a second, orthogonal vibration mode. The result is that the standing wave pattern precesses relative to the resonator. For an ideal HRG, the angle of precession θ_p is related to the rotation angle θ_r by:

$$\theta_p = -k\theta_r, \tag{1.5}$$

where k is a constant (approximately 0.3 for a thin shell, known as the Bryan factor). Thus, the HRG is inherently an **angle-measuring gyroscope** (or "whole-angle" gyroscope), directly outputting rotation angle without requiring integration of rate. This is a significant advantage over rate gyroscopes, as integration amplifies noise and drift.

Key advantages of HRGs include:

- **Extremely high quality factor (Q):** Quartz resonators can achieve Q values in the millions. This leads to very low noise and exceptional bias stability (on the order of 0.001 deg/h or better—navigation grade).
- **Material stability:** Fused quartz has excellent dimensional and elastic stability over temperature, minimizing drift.
- **Direct angle output:** The inherent precession measurement eliminates integration drift.

Disadvantages include:

- **Large size:** Even the smallest commercial HRGs are several centimeters in diameter. A three-axis system requires three such units, resulting in a bulky assembly.
- **Complexity:** Sustaining the oscillation, controlling amplitude, and reading out the precession angle require sophisticated multi-channel control electronics.
- **Cost:** The combination of precision machining, assembly, and electronics makes HRGs very expensive (tens of thousands of dollars).

Despite these drawbacks, HRGs are the gyroscopes of choice for the most demanding space missions. For example, the James Webb Space Telescope carries six HRGs (two active, four backups) to provide the ultra-stable pointing required for its decade-long mission.

1.10.2 Fiber Optic Gyroscopes (FOGs)

Fiber optic gyroscopes are based on the Sagnac effect, a relativistic phenomenon observed in interferometry. In essence, two beams of light traveling in opposite directions around a closed loop experience a phase difference if the loop is rotating.

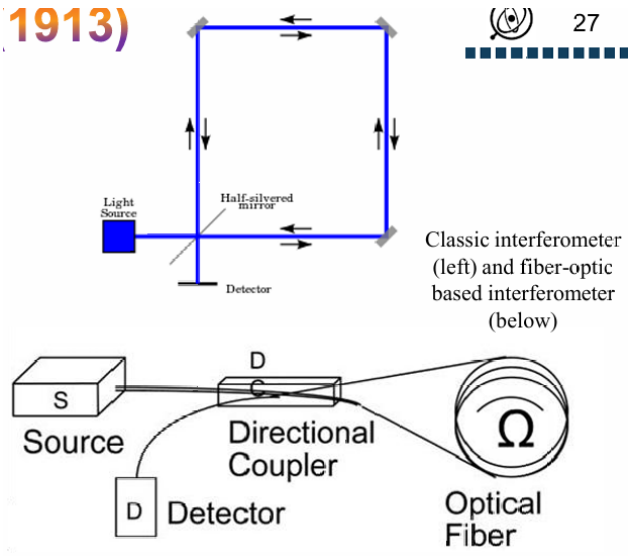

Figure 1.11: Schematic of a fiber optic gyroscope: light from a laser is split, sent clockwise and counterclockwise through a long fiber coil, recombined, and the phase difference is measured.

A basic FOG consists of:

- A coherent light source (laser diode).
- A beam splitter (directional coupler) that divides the light into two paths.
- A long optical fiber coiled around a spool (the sensing loop).
- A photodetector to measure the interference pattern when the two beams recombine.

When the coil is at rest, the two optical paths are identical, and the beams interfere constructively. When the coil rotates about its axis, the effective path length for the beam traveling in the direction of rotation is slightly longer, while the counter-rotating beam travels a slightly shorter path. This path difference induces a phase shift $\Delta\phi$ given by:

$$\Delta\phi = \frac{4\pi LR}{\lambda c}\Omega, \tag{1.6}$$

where L is the fiber length, R is the coil radius, λ is the wavelength, c is the speed of light, and Ω is the angular rate. The phase shift is proportional to Ω, making the FOG a rate gyroscope.

Practical FOGs are more complex, often incorporating:

- Reciprocal configurations to cancel common-mode effects.
- Phase modulators to bias the operating point away from zero (where sensitivity is low) and to enable synchronous detection, reducing $1/f$ noise.
- Multiple turns of fiber (often kilometers in length) wound in a compact coil to maximize sensitivity without increasing physical diameter.

The telecommunications boom of the 1980s and 1990s made high-quality, low-loss optical fiber widely available and inexpensive, catalyzing FOG development.

Advantages of FOGs:

- **No moving parts:** Inherently rugged and immune to mechanical shock and vibration (unlike HRGs or MEMS).
- **Wide dynamic range:** Can measure from very slow rates to high rates, limited only by phase unwrapping.
- **Excellent bias stability:** Typically 0.01–0.1 deg/h, with navigation-grade units achieving better than 0.001 deg/h.
- **Scalability:** Performance can be traded against size and cost by adjusting fiber length and coil diameter.

Disadvantages:

- **Size and weight:** Navigation-grade FOGs require long fiber lengths (hundreds of meters to kilometers) and relatively large coil diameters (tens of centimeters). Three-axis assemblies are bulky.
- **Cost:** While cheaper than HRGs, high-performance FOGs are still expensive (thousands to tens of thousands of dollars).
- **Temperature sensitivity:** The refractive index of fiber changes with temperature, causing drift; compensation is required.

1.10.3 Miniaturization Attempts: Micro-HRGs and Chip-Scale FOGs

The exceptional performance of HRGs and FOGs has inspired efforts to miniaturize them using MEMS and silicon photonics technologies.

Micro-HRGs: Research groups (e.g., at UC Irvine, University of Michigan) have fabricated miniature versions of HRGs using micromachined fused quartz or silicon. These devices, often called micro-shell or micro-ring resonators, are typically 1–5 mm in diameter. They leverage the same high-Q material properties but at a much smaller scale. While performance has improved dramatically, it has not yet reached navigation-grade levels, though tactical-grade performance is within reach.

Chip-scale FOGs: The rise of silicon photonics has enabled the integration of optical waveguides, couplers, modulators, and detectors on a single chip. Companies like Anello (formerly Honeywell) have demonstrated FOGs where the sensing coil is replaced by a spiral waveguide etched into a silicon chip. By using a high-index-contrast material system (e.g., silicon nitride on silicon dioxide), very long optical path lengths can be achieved in a compact footprint. Performance of such chip-scale FOGs is currently in the 0.1 deg/h range, with potential for improvement.

These miniaturized versions represent a convergence of non-MEMS principles with MEMS fabrication, blurring the lines between traditional categories.

1.11 MEMS Transduction Principles and Basic Fabrication

Having surveyed the high-performance landscape, we now turn to the core of this course: MEMS gyroscopes. While many transduction principles exist, we will focus on capacitive transduction, the most widely adopted in commercial devices due to its simplicity, low power consumption, and compatibility with standard integrated circuit processes.

1.11.1 General Sensor Working Principle

The operating principle of a MEMS gyroscope (and many other sensors) follows a generic signal chain:

1. **Input:** The physical quantity to be measured—in our case, angular rotation Ω—generates a force. For a gyroscope, this is the Coriolis force, which is proportional to Ω and the driven velocity.

2. **Mechanical domain:** This force acts on a mechanical element (the proof mass, suspended by springs), causing a displacement or deformation.

3. **Transduction:** The displacement is converted into an electrical signal through a transduction mechanism. Capacitive sensing is most common: motion changes the capacitance between the moving mass and fixed electrodes.

4. **Electronics:** The tiny capacitance change is amplified, filtered, and converted to a usable output (analog voltage or digital word).

The beauty of MEMS technology is that it allows the mechanical element and the transduction electrodes to be fabricated simultaneously from the same material (typically silicon) using processes derived from the integrated circuit industry. This co-fabrication enables extreme miniaturization, low cost, and the potential for on-chip integration of electronics.

1.11.2 A Basic MEMS Fabrication Process

The following outlines a simplified surface micromachining process, similar to what we introduced earlier but now with more detail. This process yields suspended structures capable of in-plane and out-of-plane motion.

1. **Substrate:** Start with a silicon wafer (200–300 mm diameter), which provides mechanical support.

2. **Sacrificial layer deposition:** Deposit a thin layer of silicon dioxide (SiO_2) by thermal oxidation or chemical vapor deposition. This layer will later be removed to free the moving parts.

3. **First structural layer deposition and patterning:** Deposit a layer of polysilicon (poly-Si) by low-pressure chemical vapor deposition (LPCVD). This layer forms electrical interconnects and routing lines. Pattern it using photolithography and plasma etching to define the desired interconnect paths.
4. **Second sacrificial layer deposition and patterning:** Deposit another SiO_2 layer. Etch contact holes (vias) through this oxide down to the underlying polysilicon interconnects where electrical connections are needed.
5. **Second structural layer deposition and patterning:** Deposit the main structural polysilicon layer (typically 2–10 μm thick). This layer will form the proof mass, springs, and electrodes. Pattern it using deep reactive ion etching (DRIE) to define the intricate geometry: comb fingers, beams, anchors, and etch access holes.
6. **Release:** Expose the wafer to a hydrofluoric acid (HF) vapor or liquid etchant. HF selectively etches the sacrificial SiO_2 layers without attacking the silicon. The etchant enters through the access holes, removes the oxide under the moving parts, and leaves them suspended. Regions where the structural layer is anchored directly to the substrate (through the vias) remain fixed.
7. **Packaging (wafer bonding):** Bond a second silicon wafer (capping wafer) to the device wafer in a vacuum or controlled atmosphere. This creates a hermetically sealed cavity that protects the delicate MEMS structure from contaminants, moisture, and mechanical damage, while also providing a stable environment for damping control. Electrical feedthroughs connect the internal interconnects to external bond pads.

This process requires only 5–6 main mask layers, making it relatively simple and inexpensive. The materials involved (silicon, silicon dioxide, and metal for contacts) are well-understood and reliable.

1.11.3 Resulting Structures and Sensing Examples

The process yields structures with distinct features:

- **Anchors:** Regions where the structural layer is fixed to the substrate via the oxide vias.
- **Springs:** Thin, flexible beams (often serpentine or folded) that connect the movable mass to the anchors, providing the restoring force.
- **Movable mass (rotor):** A large plate with etch access holes (required for release) and attached comb fingers or electrodes.
- **Fixed electrodes (stators):** Interdigitated comb fingers or parallel plates anchored to the substrate, forming capacitors with the movable electrodes.

In-Plane Sensing

Consider an acceleration along the x-axis (in the plane of the chip). In the non-inertial reference frame of the accelerating substrate, the proof mass experiences an inertial force $F = -ma_x$ opposite the acceleration direction. This force displaces the mass relative to the substrate, changing the capacitance of the comb fingers. For a differential comb drive, one set of gaps increases while the opposite set decreases, producing a differential capacitance signal proportional to displacement, and hence to acceleration.

This principle can be extended to two orthogonal in-plane axes (x and y) by designing appropriate suspension and electrode arrangements, all within the same planar process.

Out-of-Plane Sensing

By modifying the spring design (e.g., using torsional or folded suspensions), the proof mass can be made sensitive to forces perpendicular to the chip plane (z-axis). An out-of-plane acceleration causes the mass to tilt or move vertically. Electrodes placed beneath (or above) the mass, often on a separate conductive layer or the substrate itself, form parallel-plate capacitors. The vertical displacement changes the gap, altering the capacitance. Differential sensing can be achieved with electrodes on both sides of the mass.

Thus, with a single planar fabrication process, we can realize inertial sensors for all three axes.

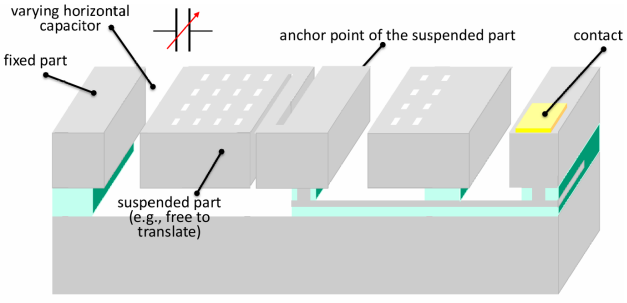

Figure 1.12: In-plane MEMS.

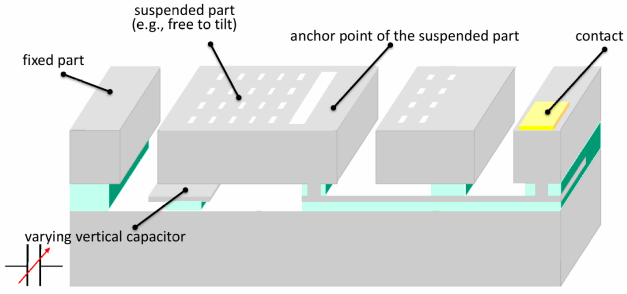

Figure 1.13: Out-of-plain MEMS.

1.11.4 Typical Dimensions and Scale

- Minimum feature size (gap between fingers, beam width): 1–2 μm.
- Structural layer thickness: 2–50 μm, depending on process.
- Proof mass dimensions: 100–1000 μm per side.
- Overall die size: 0.5–5 mm per side.

These tiny dimensions are what enable the integration of complete motion-sensing systems into handheld devices.

In the following classes, we will build upon this foundation, exploring how to design the drive and sense modes of a MEMS gyroscope, model their dynamics, and interface them with sustaining and readout electronics. We will see how the simple structures described here can be configured to measure angular rate with remarkable precision.

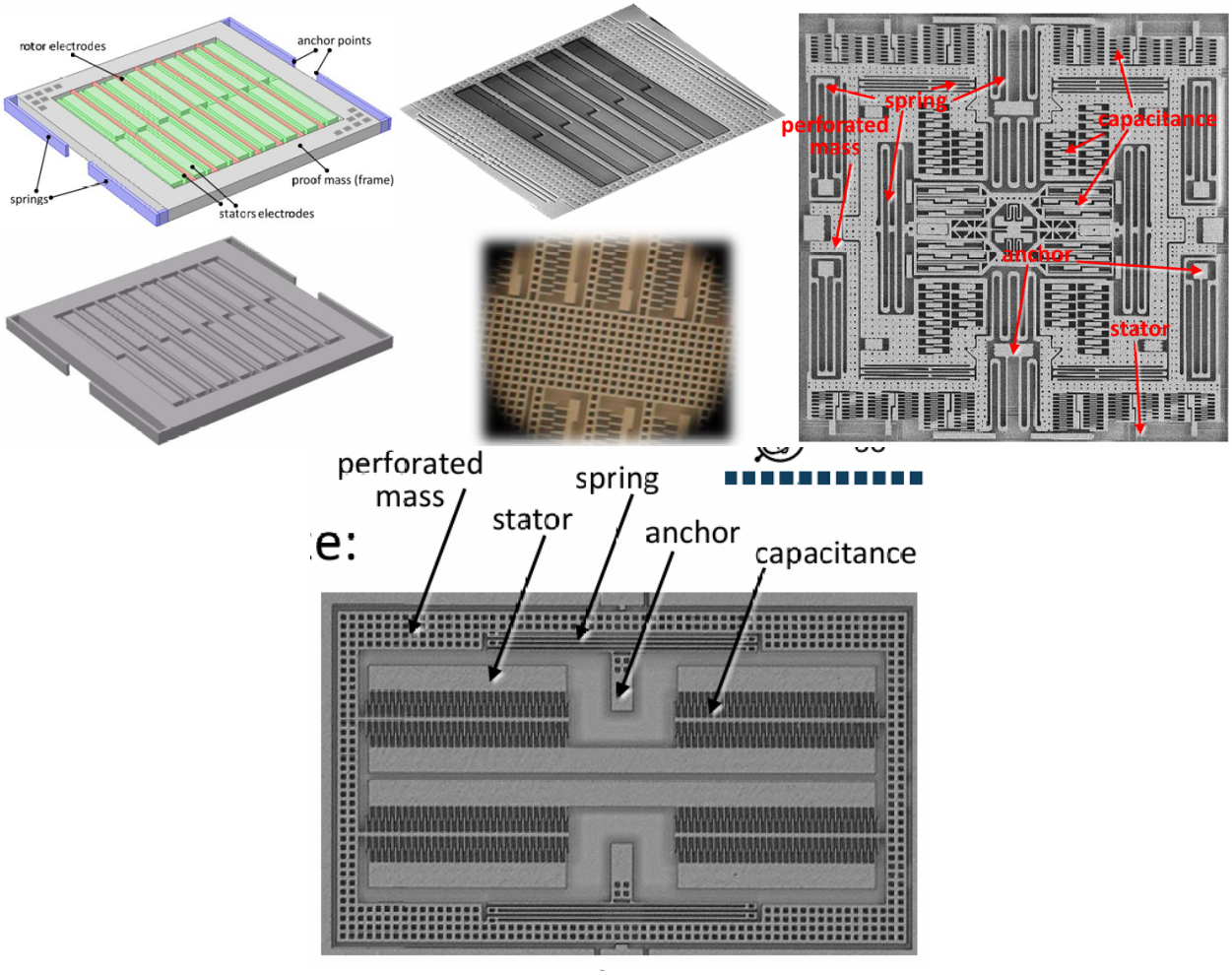

Figure 1.14: MEMS components.

Chapter 2

Electrostatic Forces in MEMS Gyroscopes

2.1 Introduction

In the previous class, we introduced the key performance parameters relevant to gyroscope design and briefly surveyed the state-of-the-art in high-end non-MEMS gyroscopes, including experimental resonant and fiber optic gyroscopes. We also laid the groundwork by reviewing the fundamentals of a MEMS fabrication process.

This brings us to the central question: How can we realize a gyroscope using a MEMS process? We will begin addressing this question today, with a complete answer to follow in the next lecture. As mentioned in the first class, some of this material may be familiar. However, given that this is an advanced MEMS course, we will proceed at a measured pace through the foundational concepts to allow sufficient time for the more advanced, yet critical, topics. Each topic we revisit will be explored in greater depth than before.

To begin, let us recall where we ended the last class. We established that a relatively simple—and consequently, inexpensive—fabrication process can be used to create suspended structures. These structures are held in place by compliant mechanisms, which we will refer to as springs, and are free to displace, for example, in-plane or out-of-plane.

To understand how such a process can yield a functional gyroscope, we must first learn to model these devices. Therefore, the initial part of this class will focus on the motion modeling of these suspended elements. This modeling is essential for conceiving a gyroscope design based on the photolithographic steps of the process. As we will see, this type of gyroscope is composed of distinct sub-elements. We will begin by analyzing and modeling the first of these sub-elements—the drive—in detail, including its electrical characteristics.

The lecture is structured into three main parts. First, we will review the fundamentals of MEMS modeling, which will then be directly applied to gyroscopes. We will see that designing a gyroscope requires sustaining a specific mode of oscillation. To achieve this, we must apply electrostatic forces to the structure. We will examine how this is typically done, focusing on the most common method: push-pull actuation. Its advantages over single-ended actuation will be discussed. Finally, we will develop an electrical equivalent circuit for this entire concept, culminating in a detailed and advanced model for the push-pull drive configuration.

Let us now outline the simplest conceptual vision of a MEMS gyroscope. The device consists of suspended microstructures. As you are likely aware, the Coriolis effect enables the measurement of angular rotation rate. Consider a body moving with a velocity along one axis (e.g., the x-axis). If this body is also subjected to an angular rotation, it experiences an apparent acceleration in a non-inertial reference frame—the Coriolis acceleration.

It is therefore intuitive that if we can sustain the motion of a suspended mass with a known velocity, we can then measure the effects of the Coriolis acceleration. The velocity cannot be constant, as the mass would eventually hit a physical boundary; instead, it is an oscillatory motion, typically on the order of a few

microns. This results in a sinusoidal velocity. By sustaining this velocity and measuring the Coriolis-induced displacement, the only unknown in the equation becomes the angular velocity we wish to measure. This is the fundamental operating principle of a vibratory gyroscope.

2.2 Motion in Non-Inertial Frames

To proceed, we need a quantitative understanding of how apparent accelerations—and thus apparent forces—act on a suspended mass. This requires modeling the effects within a non-inertial reference frame.
We can consider ourselves on Earth, which for many purposes is a sufficiently accurate inertial reference frame. The gyroscope is mounted on a moving system (e.g., a car or a mobile phone), which constitutes our non-inertial reference frame. This frame can accelerate and rotate. Within this non-inertial frame, we have a suspended mass, which we will initially assume is completely free, though we will later introduce the constraints of springs and dampers.
The relationship between the position, velocity, and acceleration of the mass as seen from the inertial (absolute) and non-inertial (relative) frames is derived from vector kinematics. The position vector is a simple sum. However, velocities and accelerations include additional terms due to the rotation of the non-inertial frame.
If we sit in the non-inertial frame and wish to apply Newton's laws, we must introduce apparent forces to account for the frame's acceleration and rotation. The general expression for the apparent acceleration of a point P in a rotating frame includes terms due to:

- Linear acceleration of the rotating frame origin;
- Angular acceleration;
- Centripetal acceleration;
- Coriolis acceleration.

In MEMS gyroscopes, the displacements are minuscule (on the order of micrometers). Consequently, the terms proportional to the position vector $\vec{r}$ (centripetal and angular acceleration terms) are negligible compared to the Coriolis term. The Coriolis acceleration is given by:

$$\vec{a}_c = -2\vec{\omega} \times \vec{v}_r, \tag{2.1}$$

where $\vec{\omega}$ is the angular velocity vector of the non-inertial frame (the quantity we aim to measure) and $\vec{v}_r$ is the velocity vector of the mass as observed within that rotating frame (the velocity we sustain). In the context of our gyroscope, $\vec{\omega}$ is the rotation of the chip, and $\vec{v}_r$ is the oscillatory velocity of the suspended mass.

2.3 Spring-Mass-Damper Model

In a realistic MEMS device, the suspended mass is not free-floating. As we saw in the last class, it is anchored to a substrate by elastic beams, which we model as a spring with stiffness k. Furthermore, the mass operates within an environment containing various energy dissipation mechanisms, which we collectively model as a damper with coefficient b. The damping term accounts for the decay of oscillations if energy is not continuously supplied.

The complete equation of motion for the mass m within the non-inertial frame must therefore include the forces from the spring and the damper, in addition to any external or inertial forces. The spring force is restoring and proportional to displacement ($-kx$), while the damping force is proportional to velocity ($-b\dot{x}$).

Considering the inertial forces, we can design the gyroscope to be insensitive to linear accelerations (a topic for a later lecture) and primarily responsive to the Coriolis force. Applying Newton's second law ($m\vec{a} = \sum \vec{F}$) in the non-inertial frame, and including the Coriolis force as an external apparent force, we obtain the familiar second-order differential equation for a spring-mass-damper system:

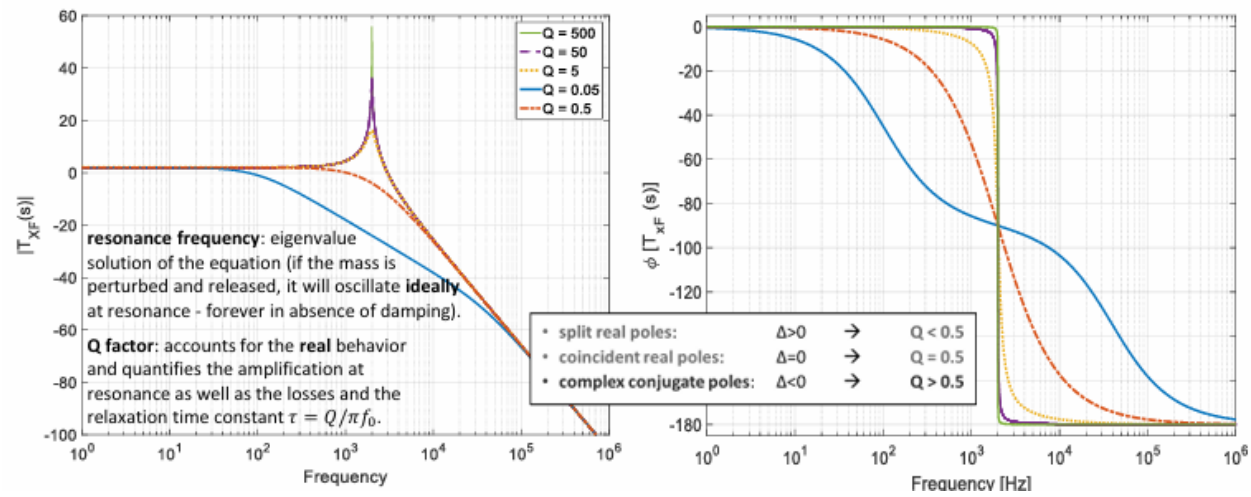

Figure 2.1: Transfer function plot.

$$m\ddot{x} + b\dot{x} + kx = F, \tag{2.2}$$

where x is the displacement of the mass from its rest position within the non-inertial frame. The force term F on the right-hand side encompasses all forces acting on the mass, including the Coriolis force and any intentionally applied forces, such as electrostatic actuation forces. Equation (2.2) is the fundamental equation we will use to model the dynamics of the gyroscope's sensitive element.

2.4 Frequency Domain Analysis

A key performance parameter introduced in the previous class is the bandwidth, typically around 100 Hz for many gyroscope applications. To understand how the system responds to forces at different frequencies, it is instructive to examine the relationship between force and displacement in the frequency domain rather than solely in the time domain.

This is readily accomplished by applying the Laplace transform to the equation of motion we derived earlier. Transforming the time derivatives ($d^2/dt^2 \to s^2$, $d/dt \to s$) yields the transfer function $H(s)$ between an applied force $F(s)$ (which may be inertial, electrostatic, or any other external force) and the resulting displacement $X(s)$:

$$H(s) = \frac{X(s)}{F(s)} = \frac{1}{ms^2 + bs + k}. \tag{2.3}$$

This second-order system has two poles, which may be real and distinct, real and coincident, or complex conjugates, depending on the damping ratio. In gyroscope applications, the devices typically operate at low pressures, resulting in light damping. Consequently, the poles are almost always complex conjugates, leading to the characteristic frequency response shown in Figure 2.1. The response is flat at low frequencies, exhibits a resonant peak, and then rolls off at -40 dB/decade due to the double pole.

To better interpret this behaviour, it is convenient to re-express the transfer function in terms of two key parameters: the natural (or resonance) frequency ω_n and the quality factor Q. These are defined as:

$$\omega_n = \sqrt{\frac{k}{m}}, \qquad Q = \frac{\sqrt{mk}}{b}. \tag{2.4}$$

The natural frequency ω_n (in rad/s) is the undamped oscillation frequency of the system. The quality factor Q is inversely proportional to the damping coefficient b and quantifies the system's ability to store energy. A high Q means that an impulse will cause the mass to oscillate for many cycles before decaying; a low Q results in rapid damping.

In terms of ω_n and Q, the transfer function magnitude can be examined in three distinct frequency regions:

- **Well below resonance** ($\omega \ll \omega_n$): The response is stiffness-dominated, and $|H(j\omega)| \approx 1/k$. This is the quasi-static regime, where Hooke's law prevails.

- **At resonance** ($\omega = \omega_n$): For an underdamped system with high Q, the response is amplified by a factor of Q compared to the quasi-static value: $|H(j\omega_n)| = Q/k = 1/(b\,\omega_n)$.

- **Well above resonance** ($\omega \gg \omega_n$): The response is mass-dominated and rolls off as $|H(j\omega)| \approx 1/(m\omega^2)$, decreasing by a factor of 100 for each decade increase in frequency (-40 dB/decade).

The phase response is equally important. At frequencies well below resonance, the phase shift is near $0°$. As frequency increases through resonance, the phase lags by $-90°$ exactly at $\omega = \omega_n$. Well above resonance, the phase approaches $-180°$. The steepness of this phase transition is directly related to the quality factor: a higher Q results in a sharper, more abrupt phase change near resonance.

While these concepts are foundational, they are essential for understanding the design and operation of resonant sensors like gyroscopes.

2.5 Design Considerations for Gyroscopes

Let us now return to the fundamental operating principle. To exploit the Coriolis effect, we must sustain the suspended mass with a controlled velocity and then measure the displacement caused by the Coriolis acceleration, from which the angular rate $\vec{\omega}$ can be inferred.

The Coriolis acceleration is given by the cross product $\vec{a}_c = -2\vec{\omega} \times \vec{v}$. This vector relationship imposes a critical geometric constraint. Suppose we sustain the mass with a velocity $\vec{v}$ along the x-direction. If an angular rate $\vec{\omega}$ is applied along an orthogonal axis, for example the z-direction, then the resulting Coriolis acceleration $\vec{a}_c$ will appear along the third orthogonal axis, the y-direction.

Consequently, the suspended mass must be capable of moving along two distinct, orthogonal directions:

- **Drive direction** (e.g., x-axis): Along this axis, we intentionally apply an electrostatic force to sustain the mass in continuous oscillation. This establishes the required velocity $\vec{v}$.

- **Sense direction** (e.g., y-axis): Along this axis, the mass responds to the Coriolis force when an external angular rate is present.

Thus, the system exhibits two distinct vibrational modes. The parameters governing each mode—effective mass, stiffness, and damping—may differ if the mechanical structure is not perfectly symmetric. In the drive mode, the equation of motion is driven by an applied electrostatic force. In the sense mode, the governing equation includes the Coriolis force as the input. Without a sustained velocity in the drive mode, there is no Coriolis force to detect. Therefore, our first task is to understand how to establish and maintain this oscillation.

A useful conceptual model is to view a gyroscope as a combination of two subsystems: a **resonator** along the drive direction and an **accelerometer** along the sense direction. The resonator is responsible for generating a constant, known velocity. The accelerometer then measures the Coriolis acceleration, which is proportional to the product of that velocity and the unknown angular rate.

In the following sections, we will first study the capacitively driven resonator (the drive mode). We will examine how electrostatic forces can sustain continuous oscillation, a function typically implemented by an electronic oscillator circuit built around the MEMS resonator. Once this is established, we will then turn our attention to the sense mode. Here, the challenge is to read out the small displacements induced by the Coriolis force, essentially treating it as a specialized accelerometer. This two-part analysis will provide a complete understanding of the gyroscope's operation.

2.6 Comb Finger Capacitors and Electrostatic Actuation

To apply forces to our suspended microstructures, we need a practical and efficient transduction mechanism. In MEMS, electrostatic actuation is the most widely adopted method due to its simplicity, low power consumption, and compatibility with standard fabrication processes. A common implementation is the **comb finger capacitor** (or comb drive).

Consider a top view of a MEMS structure, as illustrated in typical process flows. A fixed part (stator) is shown in green, and a moving part (rotor) in blue. The moving part is suspended by springs (not shown for clarity) and can displace along a defined axis, for instance the x-direction. In the region of interest, the structure consists of interdigitated fingers—hence the name "comb fingers." Multiple fingers are used in parallel to increase the total force.

Let us define the relevant geometric parameters:

- g: the gap between a fixed finger and a moving finger,
- l_0: the initial overlap length between the fingers,
- h: the height (thickness) of the structure, determined by the process,
- n: the number of finger pairs.

Both the fixed and moving parts are fabricated from conductive silicon, with air (or vacuum) as the dielectric. Thus, each pair of opposing fingers forms a capacitor. The total capacitance between the fixed and moving sets can be approximated by considering the parallel-plate contributions from both sides of each finger. For small displacements x along the direction that increases overlap, the capacitance is:

$$C(x) = 2n\epsilon_0 \frac{h(l_0 + x)}{g}, \tag{2.5}$$

where ϵ_0 is the permittivity of free space. The factor of 2 accounts for the two capacitive gaps per finger (both sidewalls). As the moving part displaces, the overlap area changes, and so does the capacitance.

When a voltage V is applied across the capacitor, electrostatic energy is stored. The system tends to minimize its potential energy. Since the electrostatic energy is $U_e = -\frac{1}{2}CV^2$ (considering the work done by the voltage source), minimizing energy corresponds to maximizing capacitance. Hence, an attractive force arises that pulls the moving fingers further into the fixed fingers, increasing the overlap x.

The electrostatic force can be derived from the principle of virtual work. The change in electrostatic energy dU_e must equal the mechanical work done by the force F_e in displacing the structure, $F_e dx$. For a capacitor with a constant voltage source, the force is:

$$F_e = \frac{1}{2}\frac{dC}{dx}V^2. \tag{2.6}$$

This force is always attractive and depends on the square of the applied voltage. For the comb drive, the capacitance derivative is:

$$\frac{dC}{dx} = 2n\epsilon_0 \frac{h}{g}, \tag{2.7}$$

which is constant with respect to x (to first order). This is a key advantage of comb drives: the electrostatic force is independent of displacement, avoiding the strong nonlinearities and pull-in instability seen in parallel-plate actuators. The resulting force is:

$$F_e = n\epsilon_0 \frac{h}{g}V^2. \tag{2.8}$$

Thus, by applying a voltage V across the comb fingers, we can exert a controlled electrostatic force on the suspended mass. If we apply an alternating voltage (AC), we can excite oscillatory motion. In a gyroscope, this force is used to sustain the drive-mode oscillation along the x-axis, providing the necessary velocity $\vec{v}$ for the Coriolis effect.

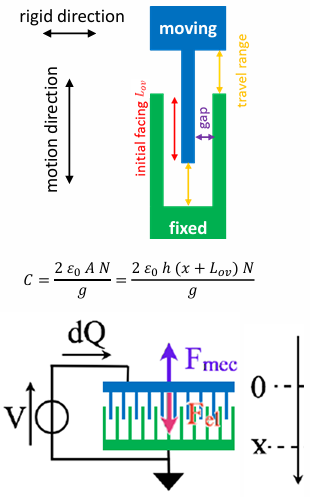

Figure 2.2: Parallel plate representation.

Audience question: Does the vacuum level (pressure) affect the permittivity ϵ_0 and hence the force?
The permittivity of free space ϵ_0 is a fundamental constant and does not change with pressure. However, in a non-perfect vacuum, the dielectric medium is a rarefied gas, whose relative permittivity differs from unity by a tiny amount proportional to pressure. For practical pressures in MEMS packages (e.g., 0.1 mbar to atmospheric pressure), this effect is completely negligible compared to other factors like damping.

2.7 Electronic Oscillator for Sustaining Oscillation

Having established that we can apply an electrostatic force, the next question is: how do we sustain a continuous, stable oscillation of the suspended mass without manually applying an external driving signal? The answer lies in building an **electronic oscillator** around the MEMS resonator.

An oscillator is a system that generates a periodic signal from a DC power supply. The basic architecture consists of:

- A **resonant element** (here, our MEMS spring-mass-damper system),
- An **amplifier** to provide gain,
- A **feedback path** that returns a fraction of the output to the input,
- A **nonlinearity** (often the amplifier's saturation or an automatic gain control) to stabilize the amplitude.

The principle is positive feedback: a small perturbation (e.g., due to noise) causes the mass to move. This motion is transduced into an electrical signal (e.g., via a sense capacitor). The signal is amplified and fed back to the actuation port, applying a force that reinforces the motion. If the loop gain is greater than unity and the phase shift around the loop is an integer multiple of 360° (i.e., 0° effectively), the oscillation grows. The nonlinearity then limits the amplitude to a stable value.

The conditions for sustained oscillation are known as the **Barkhausen criteria**:

$$|G_{\text{loop}}(j\omega)| \geq 1, \tag{2.9}$$

$$\angle G_{\text{loop}}(j\omega) = 0^\circ \ (\text{mod } 360^\circ), \tag{2.10}$$

where G_{loop} is the total complex gain around the loop, including the MEMS resonator, amplifier, and any other components.

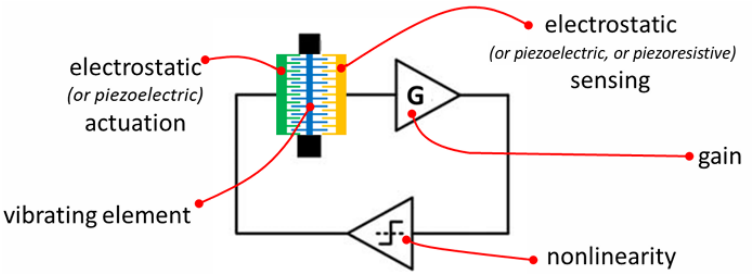

Figure 2.3: Basic oscillator architecture.

The MEMS resonator, especially when operated at low pressure with a high quality factor Q, has a very sharp phase transition near its resonance frequency ω_n. Away from resonance, the phase shift is either near $0°$ (well below resonance) or near $-180°$ (well above). Exactly at resonance, the phase shift through the mechanical element is $-90°$. By designing the rest of the electronic circuit to contribute an additional $+90°$ phase shift, we can satisfy the $0°$ loop phase condition precisely at ω_n. Because the phase changes very rapidly near resonance for a high-Q resonator, only frequencies extremely close to ω_n will meet the phase criterion. Thus, the oscillation frequency is locked to the mechanical resonance frequency.

In summary, the MEMS resonator, with its high Q, acts as a very selective frequency filter, ensuring that the oscillator operates at the desired drive-mode frequency. This is essential for the gyroscope, as the velocity amplitude and frequency must be well-controlled to accurately interpret the Coriolis signal.

2.7.1 Damping and Quality Factor in Practice

The quality factor Q is a critical parameter. It is inversely proportional to the damping coefficient b: $Q = \sqrt{mk}/b$. Damping in MEMS originates from several sources, but for comb-drive resonators, the dominant mechanism is often **gas damping** (squeeze-film and slide-film damping) due to interaction with the surrounding gas molecules.

The dependence of Q on pressure is illustrated by experimental data from test structures (similar to our comb-drive model). At higher pressures (e.g., near atmospheric), the quality factor is low (e.g., $Q \sim 10$–100) because many gas molecules collide with the moving structure, dissipating energy. As the pressure is reduced (by vacuum packaging), the mean free path of molecules increases, and gas damping decreases. Consequently, Q rises, often reaching values of 10^4–10^5 or more at pressures below 1 mbar. This high Q is desirable for oscillator stability and low phase noise, which directly impacts the gyroscope's sensitivity and noise floor. Therefore, gyroscopes are typically operated in a moderate vacuum.

2.8 Quality Factor and Damping Sources

As shown in the experimental data, the quality factor Q exhibits a strong dependence on the ambient pressure. At higher pressures (e.g., above a few millibar), Q increases as pressure decreases, typically ranging from about 100 to over 10,000. This behavior is attributed to gas damping: fewer gas molecules surround the structure, reducing energy dissipation through collisions.

However, below a certain pressure threshold (around 1 mbar in this example), Q plateaus and no longer increases with further pressure reduction. The saturation value depends on the specific device geometry and design. This indicates that once gas damping is sufficiently suppressed, other energy dissipation mechanisms become dominant and limit the maximum achievable Q to values on the order of 10^5 in this case.

The practical implication of high Q can be visualized through the ring-down time. When a short impulse is applied to the structure, a high-Q system will continue oscillating for many cycles—potentially for seconds—before the amplitude decays significantly. This underscores the ability of high-Q resonators to store mechanical energy.

What are these limiting dissipation mechanisms in the low-pressure regime? One important source is **thermoelastic damping** (TED). When the suspension springs deform during motion, some portions are

compressed while others are stretched. In most materials, compression induces local heating, while tension induces local cooling. This creates microscopic temperature gradients within the spring. Heat flows from hotter to cooler regions, and this irreversible heat conduction represents a loss of energy from the mechanical system, manifesting as damping. TED is geometry- and material-dependent and sets a fundamental upper limit on Q for many MEMS resonators, even in perfect vacuum.

2.9 Sustaining Oscillation at Resonance

For efficient operation, we wish to sustain the drive-mode oscillation at the mechanical resonance frequency ω_n. The reason is clear: at resonance, the displacement amplitude is amplified by the quality factor Q compared to the quasi-static response. Thus, a given actuation force produces a much larger motion, reducing power consumption and improving signal-to-noise ratio.

However, we cannot simply apply an AC voltage at ω_n directly to the actuation port. The electrostatic force is proportional to the square of the applied voltage ($F_e \propto V^2$). If we apply $v_a \sin(\omega_n t)$, the force would contain a component at $2\omega_n$ (from the $\sin^2$ term) and a DC component, but no component at ω_n itself. This would not efficiently excite the resonance.

The standard solution is to linearize the force-voltage relationship using a **small-signal approximation** combined with a large DC bias. We apply a voltage difference consisting of a large DC component V_{DC} and a small AC signal v_a at the desired frequency ω:

$$V(t) = V_{DC} - v_a \sin(\omega t), \tag{2.11}$$

where the minus sign is a convention depending on which port is driven. The resulting electrostatic force is:

$$F_e = \frac{1}{2}\frac{dC}{dx}(V_{DC} - v_a \sin(\omega t))^2. \tag{2.12}$$

Expanding the square:

$$F_e = \frac{1}{2}\frac{dC}{dx}\left(V_{DC}^2 - 2V_{DC}v_a \sin(\omega t) + v_a^2 \sin^2(\omega t)\right). \tag{2.13}$$

The terms are:

- A DC force term: $\frac{1}{2}\frac{dC}{dx}V_{DC}^2$, which causes a static displacement but does not contribute to oscillation.
- A force at frequency ω: $-\frac{dC}{dx}V_{DC}v_a \sin(\omega t)$, which is the desired component for resonant driving.
- A force at frequency 2ω: $\frac{1}{2}\frac{dC}{dx}v_a^2 \sin^2(\omega t) = \frac{1}{4}\frac{dC}{dx}v_a^2(1 - \cos(2\omega t))$, which includes a DC term and a component at twice the drive frequency.

If we ensure that $v_a \ll V_{DC}$ (small-signal condition), the v_a^2 term becomes negligible compared to the $2V_{DC}v_a$ term. Thus, to first order, the force at the drive frequency ω is:

$$F_e(\omega) \approx -\frac{dC}{dx}V_{DC}v_a \sin(\omega t). \tag{2.14}$$

By setting $\omega = \omega_n$, we can resonantly excite the structure with an amplified displacement response.

Typical values: V_{DC} may range from 5 V to 40 V, while v_a is kept smaller, e.g., hundreds of millivolts to a few volts, to satisfy the small-signal approximation.

2.9.1 Transduction Factors and Motional Current

To model the MEMS device as part of an electronic oscillator loop, we need to relate the electrical input (voltage v_a) to the electrical output (current from the sense port). We define two transduction factors.

For the **actuation port**, the force generated per unit AC voltage is:

$$\eta_a = V_{DC}\frac{dC_a}{dx}, \tag{2.15}$$

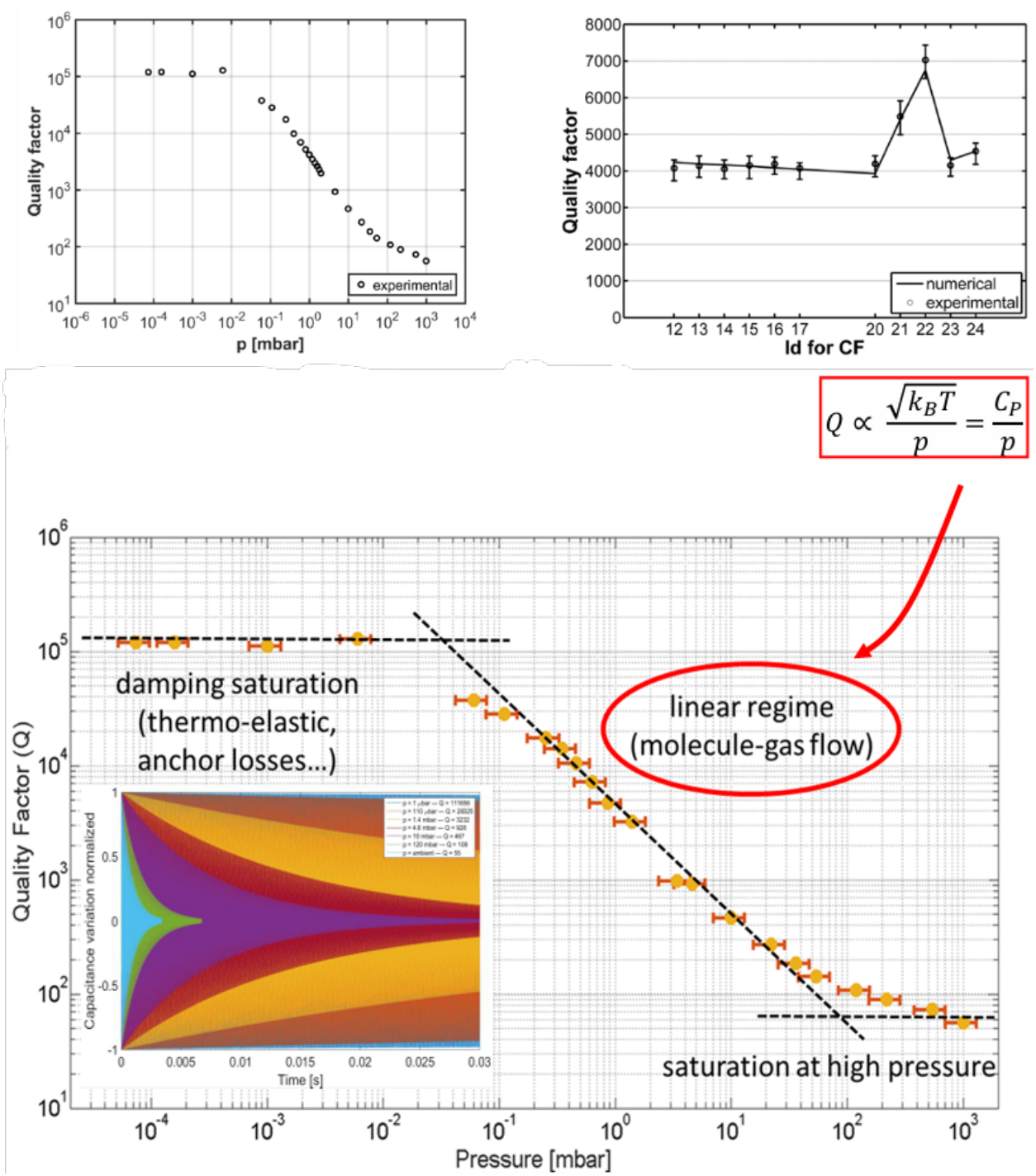

Figure 2.4: Q-factor plots.

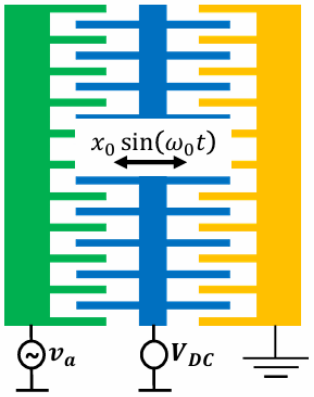

Figure 2.5: Rotor and stators voltage representation.

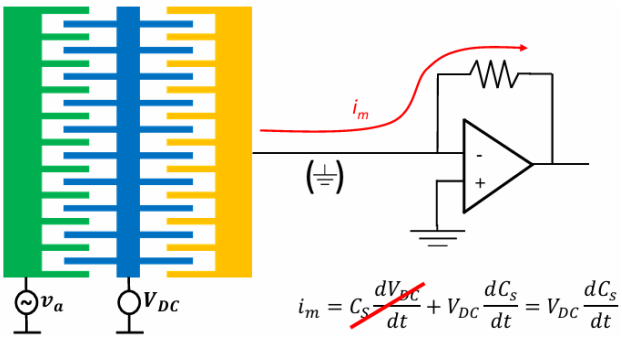

Figure 2.6: Sense port transduction system.

where C_a is the actuation-port capacitance. In the phasor domain (for sinusoidal signals at frequency ω), the force phasor $\tilde{F}_e$ is related to the input voltage phasor $\tilde{v}_a$ by:

$$\tilde{F}_e = \eta_a \tilde{v}_a. \tag{2.16}$$

For the **sense port**, we consider how motion is converted into an electrical signal. The sense port is typically held at a virtual ground potential (e.g., by a transimpedance amplifier). The voltage across the sense capacitor is constant (V_{DC}), but the capacitance $C_s(x)$ varies with displacement x. The charge on the sense port is $Q_s = C_s(x)V_{DC}$. When the mass moves, the charge changes, producing a current:

$$i_s = \frac{dQ_s}{dt} = V_{DC}\frac{dC_s}{dt} = V_{DC}\frac{dC_s}{dx}\frac{dx}{dt}. \tag{2.17}$$

Defining the sense-port transduction factor:

$$\eta_s = V_{DC}\frac{dC_s}{dx}, \tag{2.18}$$

the motional current becomes:

$$i_s = \eta_s \dot{x}. \tag{2.19}$$

In the phasor domain, $\tilde{i}_s = \eta_s(j\omega\tilde{x})$, where $\tilde{x}$ is the displacement phasor.

For a symmetric design, $dC_a/dx = dC_s/dx$, and thus $\eta_a = \eta_s = \eta$.

2.9.2 Complete Electromechanical Transfer Function

We now have all the elements to derive the transfer function from input voltage $\tilde{v}_a$ to output motional current $\tilde{i}_s$. The force $\tilde{F}_e = \eta\tilde{v}_a$ drives the mechanical system, whose displacement is given by the mechanical

transfer function $H_m(s) = 1/(ms^2 + bs + k)$. Thus:

$$\tilde{x} = H_m(s)\,\eta\tilde{v}_a. \tag{2.20}$$

The velocity is $s\tilde{x}$, so the motional current is:

$$\tilde{i}_s = \eta\, s\tilde{x} = \eta^2\, sH_m(s)\,\tilde{v}_a. \tag{2.21}$$

Therefore, the electromechanical transfer function of the MEMS device (from actuation voltage to sense current) is:

$$\frac{\tilde{i}_s}{\tilde{v}_a} = \eta^2\,\frac{s}{ms^2 + bs + k}. \tag{2.22}$$

At resonance ($s = j\omega_n$), this simplifies to:

$$\left|\frac{\tilde{i}_s}{\tilde{v}_a}\right|_{\omega=\omega_n} = \eta^2\,\frac{\omega_n}{b} = \eta^2\,\frac{Q}{k}\omega_n, \tag{2.23}$$

since $Q = \sqrt{mk}/b$ and $\omega_n = \sqrt{k/m}$. This transfer function is a key building block for designing the oscillator loop that sustains the drive-mode vibration.

2.10 Performance Comparison and Applications

The performance landscape of gyroscopes spans several orders of magnitude. At one end, we have low-cost, miniaturized MEMS gyroscopes—ubiquitous in mobile phones, automotive systems, and consumer electronics—which offer adequate performance for their intended applications but fall short of the highest precision requirements. At the other end, we find high-end non-MEMS gyroscopes such as fiber optic gyroscopes (FOGs) and hemispherical resonator gyroscopes (HRGs), which deliver exceptional accuracy and stability but are bulky, expensive, and power-hungry.

Between these extremes lies a vast application space that would greatly benefit from high-performance gyroscopes if they could be miniaturized. Inertial navigation systems for autonomous vehicles, drones, robotics, and personal localization demand the precision of high-end gyroscopes in a compact, low-cost, low-power form factor. This gap motivates the development of advanced MEMS gyroscopes that approach the performance of their macroscopic counterparts.

2.11 Recap and Roadmap

We began by establishing that MEMS processes can create suspended microstructures. By sustaining one of these structures in continuous oscillation, we impart a velocity $\vec{v}$. When the device undergoes an external rotation $\vec{\omega}$, the Coriolis effect generates an apparent acceleration $\vec{a}_c = -2\vec{\omega} \times \vec{v}$, producing a force on the suspended mass. Measuring this force allows us to determine the angular rate.

The dynamics of the suspended mass are described by a spring-mass-damper system:

$$m\ddot{x} + b\dot{x} + kx = F. \tag{2.24}$$

However, a gyroscope requires motion in at least two orthogonal directions: a drive direction (along which we sustain oscillation) and a sense direction (along which the Coriolis force appears). We therefore decomposed the problem into two coupled subsystems.

Focusing on the drive subsystem, we introduced the comb-finger capacitor as a means of electrostatic actuation. The force generated is quadratic in voltage: $F_e \propto V^2$. To excite the structure at its resonance frequency ω_n—where the displacement is amplified by the quality factor Q—we linearized this relationship using a large DC bias V_{DC} and a small AC signal v_a:

$$F_e(\omega) \approx \eta\, v_a, \quad \text{with } \eta = V_{DC}\frac{dC}{dx}. \tag{2.25}$$

This yields a force component at the drive frequency ω. On the sense side, motion induces a motional current:

$$i_s = \eta\,\dot{x}, \tag{2.26}$$

assuming symmetric actuation and sense ports ($\eta_a = \eta_s = \eta$).

2.12 Combined Electromechanical Transfer Function

We now assemble the complete relationship from input voltage $\tilde{v}_a$ to output motional current $\tilde{i}_s$. The mechanical displacement in response to the electrostatic force is given by the mechanical transfer function:

$$\tilde{x} = \frac{1}{ms^2 + bs + k}\,\tilde{F}_e = \frac{1}{ms^2 + bs + k}\,\eta\tilde{v}_a. \tag{2.27}$$

The velocity is $s\tilde{x}$, and the motional current is $\tilde{i}_s = \eta s\tilde{x}$. Combining these:

$$\tilde{i}_s = \eta^2\,\frac{s}{ms^2 + bs + k}\,\tilde{v}_a. \tag{2.28}$$

Thus, the electromechanical transfer function—an electrical admittance $Y_m(s)$—is:

$$Y_m(s) = \frac{\tilde{i}_s}{\tilde{v}_a} = \eta^2\,\frac{s}{ms^2 + bs + k}. \tag{2.29}$$

This transfer function has several notable features:

- The numerator contains an s term, reflecting that the output current is proportional to velocity (derivative of displacement).
- The denominator is the characteristic polynomial of the second-order mechanical system.
- The transduction factor η appears squared, meaning the response scales with V_{DC}^2.

The frequency response of $Y_m(s)$ (for $s = j\omega$) exhibits:

- **Below resonance** ($\omega \ll \omega_n$): The response rises with frequency at +20 dB/decade due to the s term, as the velocity increases with frequency for a given force.
- **At resonance** ($\omega = \omega_n$): The displacement is amplified by Q, leading to a peak in the admittance:

$$|Y_m(j\omega_n)| = \eta^2\frac{\omega_n}{b} = \eta^2\frac{Q}{k}\omega_n. \tag{2.30}$$

- **Above resonance** ($\omega \gg \omega_n$): The response rolls off at -20 dB/decade, as the mass-dominated behavior reduces displacement and thus velocity.

The phase starts at $+90°$ (due to the s term), crosses $0°$ at resonance (since the mechanical system contributes $-90°$, net $0°$), and approaches $-90°$ at high frequencies.

Crucially, the MEMS resonator acts as an extremely sharp frequency filter. Quality factors in MEMS can reach tens of thousands to hundreds of thousands—far exceeding what is practical with electronic RLC filters (typically $Q < 100$). This high-Q behavior ensures that the phase and gain conditions for oscillation are satisfied only in a very narrow band around ω_n, making the MEMS device the ideal frequency-selective element in an oscillator loop.

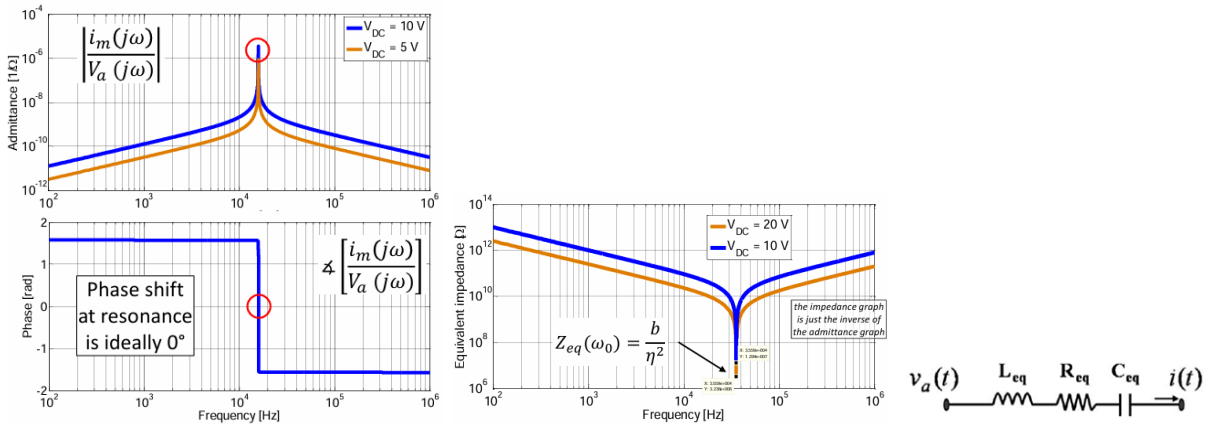

Figure 2.7: Transfer function and equivalent impedance plots. RLC equivalent circuit.

2.13 Equivalent Electrical Model

The admittance $Y_m(s)$ in Equation (2.29) has a form that suggests an equivalent electrical circuit. Rewriting it as:

$$Y_m(s) = \frac{1}{\frac{m}{\eta^2}s + \frac{b}{\eta^2} + \frac{k}{\eta^2}\frac{1}{s}}, \tag{2.31}$$

we recognize that this is the admittance of a series RLC circuit. Specifically, the impedance $Z_m(s) = 1/Y_m(s)$ is:

$$Z_m(s) = \frac{m}{\eta^2}s + \frac{b}{\eta^2} + \frac{k}{\eta^2}\frac{1}{s}. \tag{2.32}$$

Comparing with the impedance of a series RLC network, $Z_{RLC}(s) = Ls + R + 1/(Cs)$, we obtain the equivalent electrical parameters:

$$L_{eq} = \frac{m}{\eta^2}, \tag{2.33}$$

$$R_{eq} = \frac{b}{\eta^2}, \tag{2.34}$$

$$C_{eq} = \frac{\eta^2}{k}. \tag{2.35}$$

Thus, from an electrical perspective, the MEMS resonator (with its actuation and sensing ports biased appropriately) behaves exactly like a series RLC circuit. The motional current i_s is the current flowing through this equivalent branch, and the applied voltage v_a appears across it. This **LCR equivalent model** is immensely powerful: it allows us to simulate and design the oscillator electronics using familiar circuit analysis tools, with the MEMS device represented by simple lumped electrical components. The model captures the resonant behavior, the quality factor ($Q = \frac{1}{R_{eq}}\sqrt{\frac{L_{eq}}{C_{eq}}}$), and the frequency selectivity essential for oscillator design.

2.14 Equivalent Electrical Model (Continued)

The equivalent electrical parameters we derived provide deep insight into the system behavior. The equivalent resistance,

$$R_{eq} = \frac{b}{\eta^2}, \tag{2.36}$$

represents energy dissipation in the electrical domain, directly corresponding to the mechanical damping coefficient b. Similarly, the equivalent inductance,

$$L_{eq} = \frac{m}{\eta^2}, \tag{2.37}$$

models energy storage via kinetic energy (mass), while the equivalent capacitance,

$$C_{eq} = \frac{\eta^2}{k}, \tag{2.38}$$

represents potential energy storage in the springs (stiffness). Thus, the MEMS resonator, including its electrostatic actuation and sensing, behaves exactly as a series RLC circuit from its electrical terminals.

This equivalence is tremendously powerful. It allows us to:

- Simulate the combined MEMS+electronics system using standard circuit simulators (SPICE, etc.) with simple passive components.
- Easily incorporate more complex electrode configurations (e.g., differential drives and senses) by extending the equivalent circuit.
- Analyze oscillator loop dynamics using familiar impedance concepts.

At the resonance frequency $\omega_n = 1/\sqrt{L_{eq}C_{eq}}$, the inductive and capacitive reactances cancel, leaving only the equivalent resistance R_{eq}. Therefore, at resonance, the MEMS resonator simplifies to a pure resistance R_{eq} between the actuation input and the virtual-ground sense node. This makes the design of sustaining amplifiers particularly straightforward: we essentially need to provide gain around this resistor to satisfy oscillation conditions.

Typical values for R_{eq} in MEMS gyroscopes are in the megaohm range, depending on design parameters and bias voltage. Because $\eta \propto V_{DC}$, $R_{eq} \propto 1/V_{DC}^2$; doubling V_{DC} reduces R_{eq} by a factor of four. This tunability is useful in practice.

2.15 Influence of Parasitics: Feedthrough Current

In any real implementation, the idealized two-port MEMS resonator is plagued by parasitic capacitances. Consider a parasitic capacitance C_f directly coupling the actuation port to the sense port. Such a capacitance arises from:

- Electric field lines between the actuation and sense electrodes (through the substrate or air),
- Overlapping bond pads and interconnect lines,
- Substrate coupling.

With C_f present, the total current measured at the sense port becomes the sum of the motional current i_m (from the equivalent RLC branch) and a feedthrough current i_f:

$$i_{\text{total}} = i_m + i_f = \frac{\eta^2 s}{ms^2 + bs + k} v_a + sC_f v_a. \tag{2.39}$$

The admittance is now:

$$Y_{\text{total}}(s) = \frac{\eta^2 s}{ms^2 + bs + k} + sC_f. \tag{2.40}$$

The impact on the frequency response is significant:

- At low frequencies, the sC_f term dominates, giving a +20 dB/decade slope (capacitive feedthrough).
- Near resonance, the motional current peaks, but the feedthrough current also contributes, potentially obscuring the resonance if C_f is large.

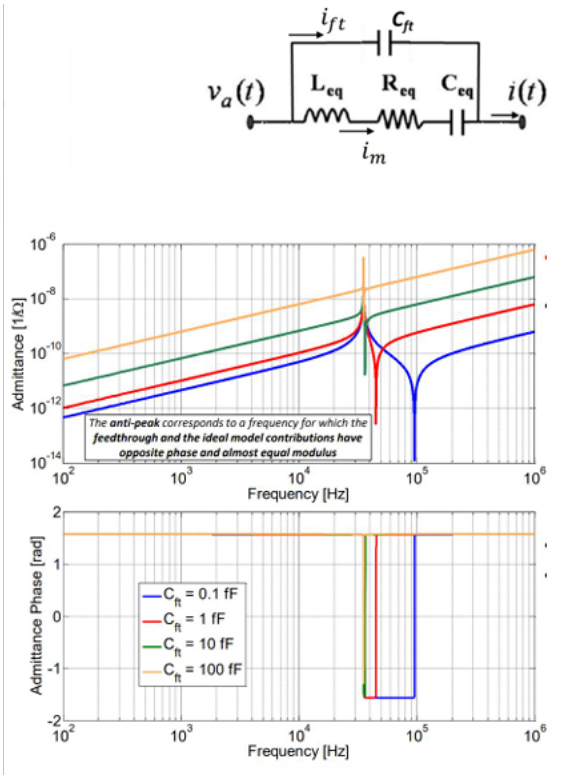

Figure 2.8: Parasitic capacitance influence.

- At high frequencies, the motional current rolls off (−20 dB/decade), but the feedthrough current continues increasing (+20 dB/decade), so the total admittance eventually rises again.

The phase response is similarly corrupted: instead of a clean transition from +90° to −90° through resonance, the phase may pull back toward +90° at high frequencies due to the feedthrough capacitance. This compromises the resonator's effectiveness as a sharp frequency-selective element in an oscillator loop, as multiple frequencies may satisfy the gain condition, and the phase condition becomes ambiguous.

Various techniques exist to mitigate feedthrough, including shielding, layout optimization, and differential measurement. The latter leads us to a more robust configuration.

2.16 Toward Differential (Push-Pull) Operation

The single-ended actuation and sensing scheme we have analyzed so far has several limitations:

1. **Small-signal constraint:** The linearization requires $v_a \ll V_{DC}$, limiting the achievable oscillation amplitude or demanding large, inconvenient DC voltages.

2. **Residual nonlinearity:** Even with small signals, the v_a^2 term generates force components at 2ω, which can excite spurious mechanical modes (undesired resonances) in the gyroscope structure.

3. **Common-mode susceptibility:** Single-ended sensing is vulnerable to noise and interference coupled from the substrate, power supplies, or other sources, as these disturbances appear directly in the output current.

To overcome these issues, we introduce a **differential (push-pull) configuration**, as illustrated conceptually in Fig. 2.9. The key features are:

- Two actuation ports (D1 and D2) placed symmetrically on opposite sides of the rotor.

- Two sensing ports (S1 and S2) similarly placed symmetrically.
- The rotor is biased at a constant DC voltage V_{DC}.
- Actuation signals are applied differentially: $v_{D1} = +v_a$ and $v_{D2} = -v_a$ (opposite phases).
- The sensing ports are typically shorted together or connected to differential amplifiers.

Let us analyze the electrostatic force under this differential drive. The force on the rotor is the sum of forces from both sides. For small displacements, the capacitance derivatives are equal in magnitude but opposite in sign for the two sides (moving toward one side increases its capacitance while decreasing the other). The force from each side is:

$$F_{e1} = \frac{1}{2}\frac{dC_1}{dx}(V_{DC} - v_a)^2, \quad F_{e2} = \frac{1}{2}\frac{dC_2}{dx}(V_{DC} + v_a)^2, \tag{2.41}$$

where we assume $dC_2/dx = -dC_1/dx = -dC/dx$ for symmetry. Expanding and summing:

$$\begin{aligned} F_e &= F_{e1} + F_{e2} && (2.42)\\ &= \frac{1}{2}\frac{dC}{dx}\left[(V_{DC}^2 - 2V_{DC}v_a + v_a^2) - (V_{DC}^2 + 2V_{DC}v_a + v_a^2)\right] && (2.43)\\ &= \frac{1}{2}\frac{dC}{dx}\left[-4V_{DC}v_a\right] && (2.44)\\ &= -2\frac{dC}{dx}V_{DC}v_a. && (2.45) \end{aligned}$$

Several important observations emerge:

- The DC force terms (V_{DC}^2) cancel completely.
- The second-harmonic terms (v_a^2) also cancel.
- The net force is proportional to v_a (linear) without requiring $v_a \ll V_{DC}$; the linearization is exact for this symmetric configuration.
- The force amplitude is doubled compared to the single-ended case for the same v_a and V_{DC}.

Thus, differential actuation eliminates the small-signal constraint and suppresses unwanted frequency components. Similarly, differential sensing (taking the difference of currents from S1 and S2) cancels common-mode feedthrough currents and other disturbances, leaving only the differential motional signal. This makes the push-pull architecture vastly superior for precision gyroscope applications. We will explore its full implications, including the equivalent circuit and oscillator design, in the next class.

2.17 Push-Pull (Differential) Actuation and Sensing

The limitations of single-ended actuation and sensing motivate a more symmetric approach: push-pull (differential) actuation combined with differential sensing. This configuration, illustrated conceptually in Figure 2.9 (not shown), offers numerous advantages that are critical for high-performance gyroscopes.

2.17.1 Configuration and Force Analysis

In the push-pull scheme:

- The rotor (moving mass) is biased at a constant DC voltage V_{DC}.
- Two actuation electrodes are placed symmetrically on opposite sides of the rotor: Drive 1 (D1) and Drive 2 (D2).
- Two sensing electrodes are similarly placed symmetrically: Drive Detection 1 (DD1) and Drive Detection 2 (DD2).

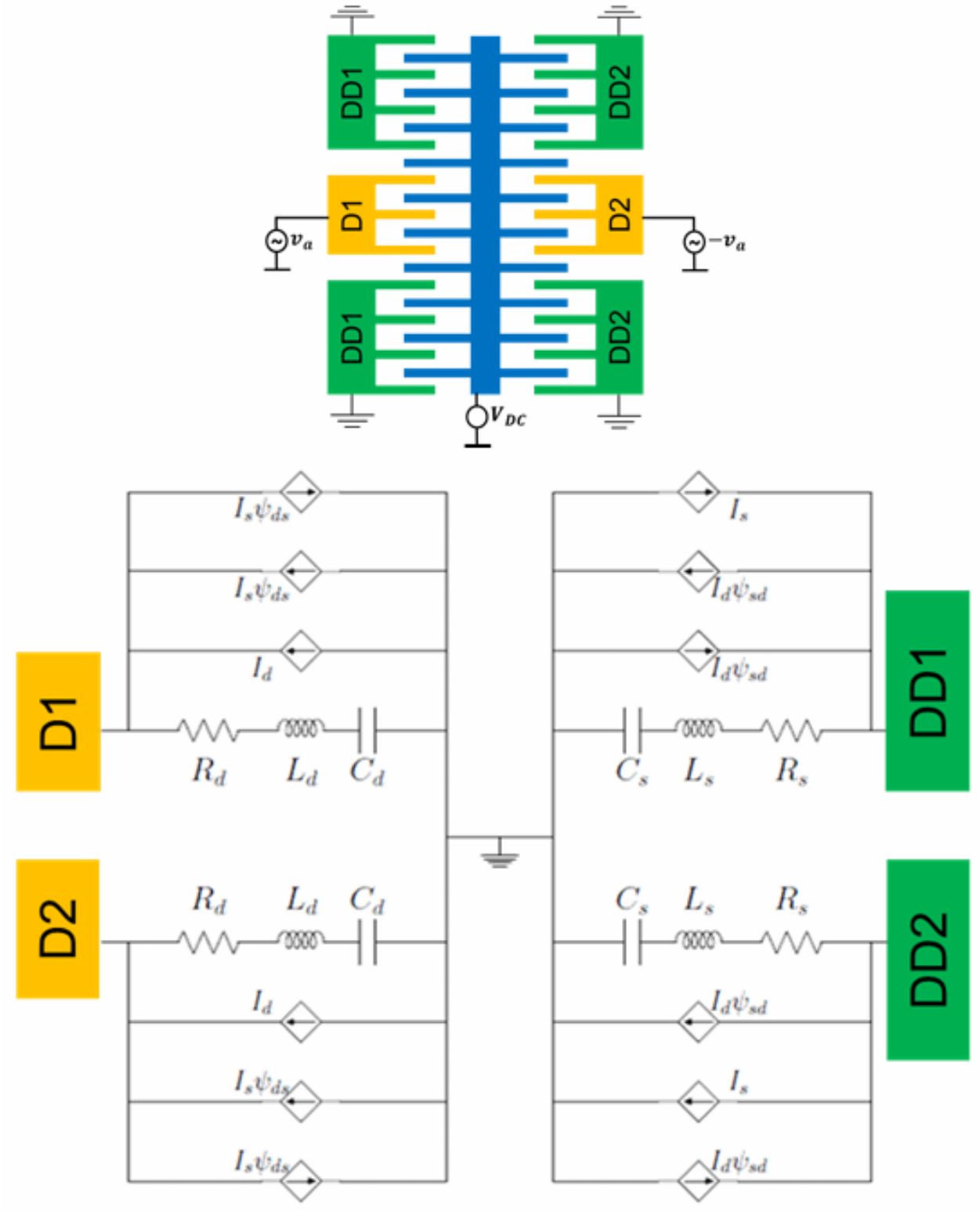

Figure 2.9: Differential push-pull scheme.

- AC actuation voltages are applied in anti-phase: $v_{D1} = +v_a$ and $v_{D2} = -v_a$.
- The sensing electrodes are typically connected to a differential transimpedance amplifier, effectively holding them at virtual ground.

Let us derive the net electrostatic force acting on the rotor. For small displacements, the capacitance derivatives for the two sides are equal in magnitude but opposite in sign: moving toward D1 increases C_{D1} while decreasing C_{D2}. Thus, $dC_{D1}/dx = -dC_{D2}/dx = dC/dx$. The force from each side is:

$$F_{e1} = \frac{1}{2}\frac{dC}{dx}(V_{DC} - v_a)^2, \tag{2.46}$$

$$F_{e2} = \frac{1}{2}\left(-\frac{dC}{dx}\right)(V_{DC} + v_a)^2 = -\frac{1}{2}\frac{dC}{dx}(V_{DC} + v_a)^2. \tag{2.47}$$

Expanding the squares:

$$F_{e1} = \frac{1}{2}\frac{dC}{dx}\left(V_{DC}^2 - 2V_{DC}v_a + v_a^2\right), \tag{2.48}$$

$$F_{e2} = -\frac{1}{2}\frac{dC}{dx}\left(V_{DC}^2 + 2V_{DC}v_a + v_a^2\right). \tag{2.49}$$

The total force is the sum:

$$F_e = F_{e1} + F_{e2} \tag{2.50}$$

$$= \frac{1}{2}\frac{dC}{dx}\left[(V_{DC}^2 - 2V_{DC}v_a + v_a^2) - (V_{DC}^2 + 2V_{DC}v_a + v_a^2)\right] \tag{2.51}$$

$$= \frac{1}{2}\frac{dC}{dx}(-4V_{DC}v_a) \tag{2.52}$$

$$= -2\frac{dC}{dx}V_{DC}v_a. \tag{2.53}$$

2.17.2 Key Advantages of Push-Pull

Several profound benefits emerge from this analysis:

1. **Exact linearization:** The force is proportional to v_a with *no* small-signal approximation. The DC terms (V_{DC}^2) and second-harmonic terms (v_a^2) cancel exactly due to symmetry. This eliminates distortion and spurious excitation of mechanical modes at 2ω.
2. **Increased force efficiency:** For the same V_{DC} and v_a, the force amplitude is doubled compared to the single-ended case (where $F_e = -\frac{dC}{dx}V_{DC}v_a$).
3. **No DC force component:** The static deflection caused by V_{DC}^2 terms cancels, so the equilibrium position remains centered.
4. **Common-mode rejection:** Differential sensing cancels common-mode disturbances, including feedthrough currents, substrate noise, and electromagnetic interference.

The only practical drawback is the increased number of electrical interconnects (pads), which adds marginal cost but is readily justified by the performance gains.

2.17.3 Equivalent Electrical Model for Push-Pull Configuration

The equivalent circuit for the four-port push-pull resonator is more complex than the simple series *RLC* of the single-ended case. The increased complexity arises from cross-coupling: a voltage applied to any port induces motion, which in turn generates currents at *all* other ports via the motional mechanism.

Consider the current i_{D1} flowing into actuation port D1. This current has four contributions:

- Self-induced: from the voltage v_{D1} applied to D1 itself, producing motion and thus a current back into D1. This is analogous to the single-ended case and can be represented by an equivalent RLC series branch between D1 and the rotor.
- Cross-induced from D2: voltage v_{D2} causes motion, which generates a current at D1. This is represented by a current-controlled current source (or a transadmittance) proportional to v_{D2}.
- Cross-induced from DD1 and DD2: similarly, voltages at the sensing ports (though these are typically held at virtual ground and thus have zero AC voltage) would also contribute.

Generalizing, each port i experiences a motional current:

$$i_{m,i} = \sum_j Y_{ij} v_j, \tag{2.54}$$

where Y_{ij} is the transadmittance from port j to port i. For a symmetric, perfectly balanced structure:

- $Y_{ii} = \frac{\eta_i^2 s}{ms^2+bs+k}$ (self-admittance, equivalent to series RLC).
- $Y_{ij} = \frac{\eta_i \eta_j s}{ms^2+bs+k}$ for $i \neq j$ (mutual admittance).

If actuation and sense ports have different transduction factors (e.g., different numbers of fingers), we define $\psi = \eta_{\text{sense}}/\eta_{\text{drive}}$; then mutual terms scale accordingly.

In addition to motional currents, each port has a static parasitic capacitance $C_{p,i}$ to the rotor (and to other ports). The full equivalent circuit thus consists of:

- A network of coupled RLC branches representing motional coupling between all ports.
- Lumped parasitic capacitors from each port to ground (rotor) and between ports.

While this appears daunting, in normal operation, the sensing ports are held at virtual ground (AC ground), so their voltages are zero. This simplifies the model considerably: only the actuation ports (D1, D2) have applied voltages, and the sensing ports (DD1, DD2) are outputs. The currents at DD1 and DD2 become:

$$i_{DD1} = Y_{\text{DD1,D1}} v_{D1} + Y_{\text{DD1,D2}} v_{D2}, \tag{2.55}$$

$$i_{DD2} = Y_{\text{DD2,D1}} v_{D1} + Y_{\text{DD2,D2}} v_{D2}. \tag{2.56}$$

With symmetry, $Y_{\text{DD1,D1}} = Y_{\text{DD2,D2}}$ (self-like) and $Y_{\text{DD1,D2}} = Y_{\text{DD2,D1}}$ (mutual). Since $v_{D2} = -v_{D1}$, the differential output current $i_{DD1} - i_{DD2}$ becomes:

$$i_{\text{diff}} = (Y_{\text{DD1,D1}} - Y_{\text{DD1,D2}} - Y_{\text{DD2,D1}} + Y_{\text{DD2,D2}}) v_{D1} = 2(Y_{\text{self}} - Y_{\text{mutual}}) v_{D1}, \tag{2.57}$$

which is proportional to the motional current and retains the desired resonant behavior.

2.17.4 Feedthrough Cancellation in Differential Sensing

Parasitic feedthrough capacitances exist between each actuation port and each sensing port. In a four-port system, there are four such capacitors: $C_{f,11}$ (D1 to DD1), $C_{f,12}$ (D1 to DD2), $C_{f,21}$ (D2 to DD1), and $C_{f,22}$ (D2 to DD2).

If the layout is perfectly symmetric, $C_{f,11} = C_{f,22}$ and $C_{f,12} = C_{f,21}$. Now consider the currents injected into the sensing ports via these feedthrough capacitors due to the anti-phase drive signals $v_{D1} = +v_a$ and $v_{D2} = -v_a$:

$$i_{ft,DD1} = sC_{f,11} v_a + sC_{f,21}(-v_a) = s(C_{f,11} - C_{f,21}) v_a, \tag{2.58}$$

$$i_{ft,DD2} = sC_{f,12} v_a + sC_{f,22}(-v_a) = s(C_{f,12} - C_{f,22}) v_a. \tag{2.59}$$

If all four capacitors are equal ($C_{f,11} = C_{f,12} = C_{f,21} = C_{f,22}$), then $i_{ft,DD1} = i_{ft,DD2} = 0$. Even if they are not perfectly equal but symmetric, the differential output $i_{DD1} - i_{DD2}$ cancels the common-mode feedthrough components. This provides a twofold rejection:

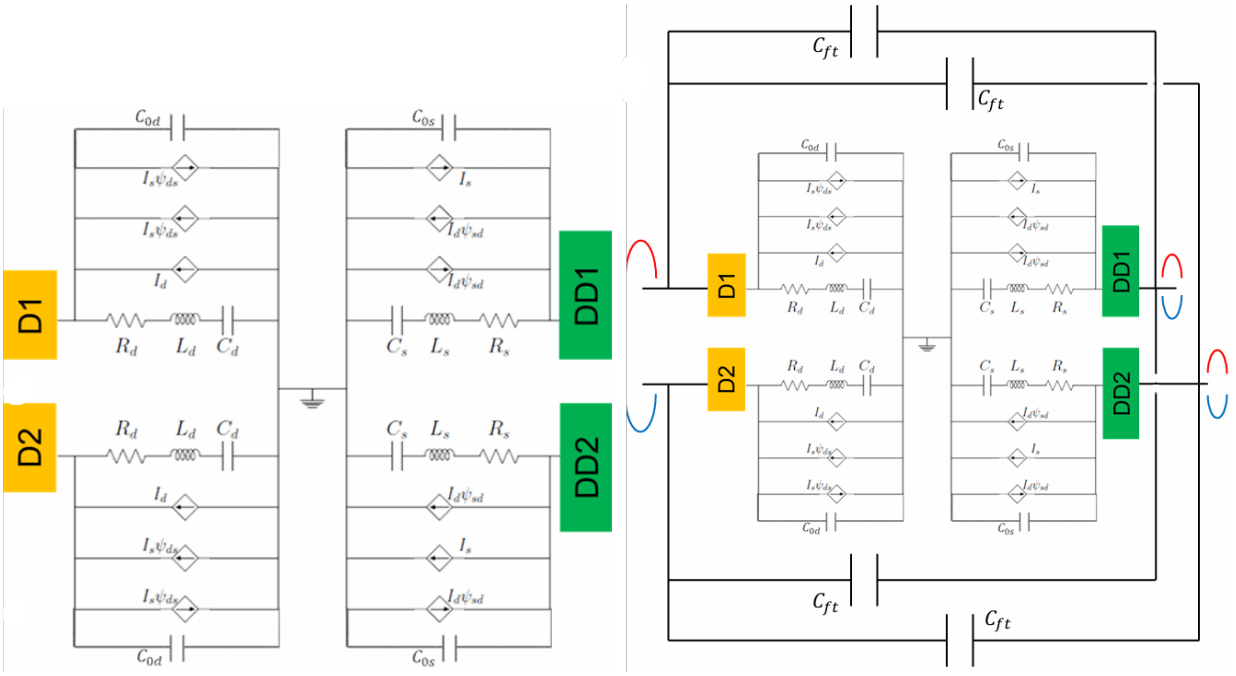

Figure 2.10: Complete differential push-pull model without (left) and with (right) parasitic feedthrough capacitances.

1. Cancellation due to anti-phase drives on symmetric capacitors.
2. Cancellation due to differential sensing (subtracting the two sense currents).

Thus, push-pull operation with differential sensing dramatically reduces feedthrough contamination, preserving the clean resonant peak and phase characteristics essential for oscillator stability.

2.17.5 Design Flexibility

An additional benefit of the push-pull architecture is design freedom. In the single-ended case, maintaining force balance often required equal numbers of actuation and sensing fingers. With differential drive and sense, the number of fingers on each side can be optimized independently for actuation and sensing performance. For example, one might use fewer actuation fingers (to reduce capacitive loading on the drive electronics) and more sensing fingers (to increase the motional current signal). This flexibility allows the designer to trade off parameters such as force generation, signal strength, and area consumption to meet specific system requirements.

2.18 Push-Pull (Differential) Actuation and Sensing

The limitations of single-ended actuation and sensing motivate a more symmetric approach: push-pull (differential) actuation combined with differential sensing. This configuration, illustrated in Figure 2.10, offers numerous advantages that are critical for high-performance gyroscopes. The rotor is biased at V_{DC}. Drive signals are applied in anti-phase: $v_{D1} = +v_a$, $v_{D2} = -v_a$. Sense electrodes are held at virtual ground, and their currents are subtracted.

2.18.1 Force Analysis for Push-Pull Actuation

In the push-pull scheme:

- The rotor (moving mass) is biased at a constant DC voltage V_{DC}.
- Two actuation electrodes are placed symmetrically on opposite sides of the rotor: Drive 1 (D1) and Drive 2 (D2).
- Two sensing electrodes are similarly placed symmetrically: Sense 1 (S1) and Sense 2 (S2).
- AC actuation voltages are applied in anti-phase: $v_{D1} = +v_a$ and $v_{D2} = -v_a$ (with respect to the rotor bias, the actual voltages are $V_{DC} + v_a$ on D1 and $V_{DC} - v_a$ on D2).
- The sensing electrodes are typically connected to a differential transimpedance amplifier, effectively holding them at virtual ground.

Let us derive the net electrostatic force acting on the rotor. For small displacements, the capacitance derivatives for the two sides are equal in magnitude but opposite in sign: moving toward D1 increases C_{D1} while decreasing C_{D2}. Thus,

$$\frac{dC_{D1}}{dx} = +\frac{dC}{dx}, \quad \frac{dC_{D2}}{dx} = -\frac{dC}{dx}. \tag{2.60}$$

The electrostatic force from each side is given by the general expression $F_e = \frac{1}{2}\frac{dC}{dx}V^2$. For D1, the voltage across the capacitor is $(V_{DC} + v_a)$, so:

$$F_{e1} = \frac{1}{2}\frac{dC}{dx}(V_{DC} + v_a)^2. \tag{2.61}$$

For D2, the voltage is $(V_{DC} - v_a)$, but note the sign of the derivative:

$$F_{e2} = \frac{1}{2}\left(-\frac{dC}{dx}\right)(V_{DC} - v_a)^2 = -\frac{1}{2}\frac{dC}{dx}(V_{DC} - v_a)^2. \tag{2.62}$$

Expanding both squares:

$$F_{e1} = \frac{1}{2}\frac{dC}{dx}\left(V_{DC}^2 + 2V_{DC}v_a + v_a^2\right), \tag{2.63}$$

$$F_{e2} = -\frac{1}{2}\frac{dC}{dx}\left(V_{DC}^2 - 2V_{DC}v_a + v_a^2\right). \tag{2.64}$$

The total force is the sum:

$$\begin{aligned} F_e &= F_{e1} + F_{e2} \\ &= \frac{1}{2}\frac{dC}{dx}\left[(V_{DC}^2 + 2V_{DC}v_a + v_a^2) - (V_{DC}^2 - 2V_{DC}v_a + v_a^2)\right] \\ &= \frac{1}{2}\frac{dC}{dx}(4V_{DC}v_a) \\ &= 2\frac{dC}{dx}V_{DC}v_a. \end{aligned} \tag{2.65}$$

2.18.2 Key Advantages of Push-Pull

Equation (2.65) reveals several profound benefits:

1. **Exact linearization:** The force is proportional to v_a with *no* small-signal approximation. The DC terms (V_{DC}^2) and second-harmonic terms (v_a^2) cancel exactly due to symmetry. This eliminates distortion and spurious excitation of mechanical modes at 2ω.
2. **Increased force efficiency:** For the same V_{DC} and v_a, the force amplitude is doubled compared to the single-ended case (where $F_e = \frac{dC}{dx}V_{DC}v_a$).
3. **No DC force component:** The static deflection caused by V_{DC}^2 terms cancels, so the equilibrium position remains centered.

4. **Common-mode rejection:** Differential sensing cancels common-mode disturbances, including feedthrough currents, substrate noise, and electromagnetic interference.

The only practical drawback is the increased number of electrical interconnects (pads), which adds marginal cost but is readily justified by the performance gains.

2.18.3 Equivalent Electrical Model for Push-Pull Configuration

The equivalent circuit for the four-port push-pull resonator is more complex than the simple series RLC of the single-ended case. The increased complexity arises from cross-coupling: a voltage applied to any port induces motion, which in turn generates currents at *all* other ports via the motional mechanism.

Consider the current i_{D1} flowing into actuation port D1. This current has four contributions:

- **Self-induced:** from the voltage v_{D1} applied to D1 itself, producing motion and thus a current back into D1. This is analogous to the single-ended case and can be represented by an equivalent RLC series branch between D1 and the rotor.
- **Cross-induced from D2:** voltage v_{D2} causes motion, which generates a current at D1.
- **Cross-induced from S1 and S2:** similarly, voltages at the sensing ports (though these are typically held at virtual ground and thus have zero AC voltage) would also contribute.

Generalizing, each port i experiences a motional current:

$$i_{m,i} = \sum_j Y_{ij} v_j, \tag{2.66}$$

where Y_{ij} is the transadmittance from port j to port i. For a symmetric, perfectly balanced structure:

- $Y_{ii} = \dfrac{\eta_i^2 s}{ms^2 + bs + k}$ (self-admittance, equivalent to series RLC).
- $Y_{ij} = \dfrac{\eta_i \eta_j s}{ms^2 + bs + k}$ for $i \neq j$ (mutual admittance).

If actuation and sense ports have different transduction factors (e.g., different numbers of fingers), we define $\psi = \eta_{\text{sense}}/\eta_{\text{drive}}$; then mutual terms scale accordingly.

In addition to motional currents, each port has a static parasitic capacitance $C_{p,i}$ to the rotor (and to other ports). The full equivalent circuit thus consists of:

- A network of coupled RLC branches representing motional coupling between all ports.
- Lumped parasitic capacitors from each port to ground (rotor) and between ports.

While this appears daunting, in normal operation, the sensing ports are held at virtual ground (AC ground), so their voltages are zero. This simplifies the model considerably: only the actuation ports (D1, D2) have applied voltages, and the sensing ports (S1, S2) are outputs. The currents at S1 and S2 become:

$$i_{S1} = Y_{\text{S1,D1}} v_{D1} + Y_{\text{S1,D2}} v_{D2}, \tag{2.67}$$

$$i_{S2} = Y_{\text{S2,D1}} v_{D1} + Y_{\text{S2,D2}} v_{D2}. \tag{2.68}$$

With symmetry, $Y_{\text{S1,D1}} = Y_{\text{S2,D2}}$ (self-like) and $Y_{\text{S1,D2}} = Y_{\text{S2,D1}}$ (mutual). Since $v_{D2} = -v_{D1}$, the differential output current $i_{S1} - i_{S2}$ becomes:

$$i_{\text{diff}} = (Y_{\text{S1,D1}} - Y_{\text{S1,D2}} - Y_{\text{S2,D1}} + Y_{\text{S2,D2}}) v_{D1} = 2(Y_{\text{self}} - Y_{\text{mutual}}) v_{D1}, \tag{2.69}$$

which is proportional to the motional current and retains the desired resonant behavior.

2.18.4 Feedthrough Cancellation in Differential Sensing

Parasitic feedthrough capacitances exist between each actuation port and each sensing port. In a four-port system, there are four such capacitors: $C_{f,11}$ (D1 to S1), $C_{f,12}$ (D1 to S2), $C_{f,21}$ (D2 to S1), and $C_{f,22}$ (D2 to S2).

If the layout is perfectly symmetric, $C_{f,11} = C_{f,22}$ and $C_{f,12} = C_{f,21}$. Now consider the currents injected into the sensing ports via these feedthrough capacitors due to the anti-phase drive signals $v_{D1} = +v_a$ and $v_{D2} = -v_a$:

$$i_{ft,S1} = sC_{f,11}v_a + sC_{f,21}(-v_a) = s(C_{f,11} - C_{f,21})v_a, \tag{2.70}$$

$$i_{ft,S2} = sC_{f,12}v_a + sC_{f,22}(-v_a) = s(C_{f,12} - C_{f,22})v_a. \tag{2.71}$$

If all four capacitors are equal ($C_{f,11} = C_{f,12} = C_{f,21} = C_{f,22}$), then $i_{ft,S1} = i_{ft,S2} = 0$. Even if they are not perfectly equal but symmetric, the differential output $i_{S1} - i_{S2}$ cancels the common-mode feedthrough components. This provides a twofold rejection:

1. Cancellation due to anti-phase drives on symmetric capacitors.
2. Cancellation due to differential sensing (subtracting the two sense currents).

Thus, push-pull operation with differential sensing dramatically reduces feedthrough contamination, preserving the clean resonant peak and phase characteristics essential for oscillator stability.

2.18.5 Summary: Single-Ended vs. Push-Pull

The comparison between the two approaches is summarized in Table 2.1.

Feature	**Single-Ended**	**Push-Pull Differential**
Force linearity	Requires $v_a \ll V_{DC}$ (small-signal approx.)	Exact linearization, valid for any v_a
Harmonic distortion	Contains 2ω force components	2ω terms cancel completely
DC force component	Present ($\propto V_{DC}^2$)	Cancels
Feedthrough sensitivity	Single feedthrough path; directly adds to output	Symmetry and differential sensing provide strong rejection
Common-mode noise rejection	None (single-ended output)	Excellent (differential output)
Design flexibility	Constrained (often equal actuation/sense fingers)	Independent optimization of actuation and sense ports
Complexity	Lower (fewer pads, simpler model)	Higher (more pads, more complex equivalent circuit)

Table 2.1: Comparison of single-ended and push-pull differential drive/sense configurations for the drive-mode resonator.

In conclusion, the push-pull differential architecture is overwhelmingly preferred for high-performance MEMS gyroscopes. It provides linear, distortion-free actuation, excellent rejection of parasitics and disturbances, and greater design flexibility—all essential for achieving the low noise, high stability, and accurate angular rate sensing required in advanced applications. We will explore the integration of this drive-mode resonator into a complete oscillator design in the next lecture.

2.19 Optimizing the Number of Fingers

The push-pull configuration offers the freedom to independently choose the number of actuation fingers (on D1 and D2) and the number of sense fingers (on S1 and S2). This choice involves fundamental trade-offs that impact the overall gyroscope performance.

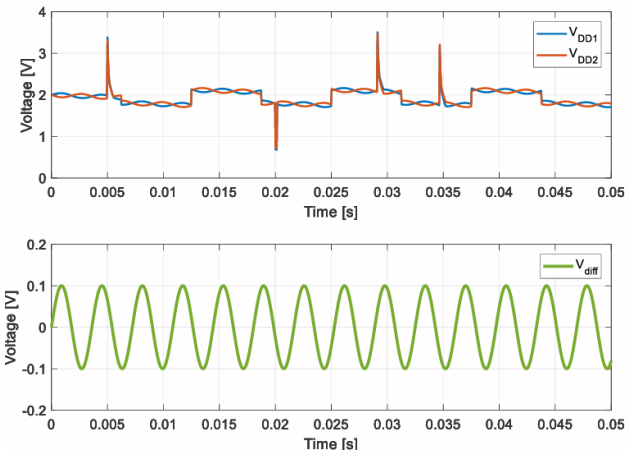

Figure 2.11: Differential readout compensation of spurious common mode terms.

2.19.1 Trade-Offs in Finger Allocation

A larger number of actuation fingers increases the transduction factor $\eta_a = V_{DC} dC_a/dx$, and thus generates a larger electrostatic force for a given actuation voltage v_a. This yields a larger drive displacement amplitude.

However, in high-Q MEMS resonators, achieving a large displacement is usually not the limiting factor; even moderate forces produce significant motion due to resonance amplification. Therefore, a more strategic approach is to minimize the number of actuation fingers, subject to reaching the target displacement with the maximum available v_a (limited by the electronic interface or breakdown constraints).

The area saved by reducing actuation fingers can be reallocated to increase the number of **sense fingers**. A larger number of sense fingers increases the sense transduction factor $\eta_s = V_{DC} dC_s/dx$, which directly boosts the motional current $i_s = \eta_s \dot{x}$. A larger motional current improves the signal-to-noise ratio of the front-end electronics and reduces the phase noise of the oscillator. Lower phase noise in the drive oscillator translates directly to lower noise in the angular rate measurement, as phase fluctuations in the drive velocity appear as noise in the Coriolis signal.

Thus, a typical design rule of thumb is:

> Allocate just enough actuation fingers to achieve the required drive displacement with the maximum available v_a. Use the remaining available area to maximize the number of sense fingers.

2.19.2 Lateral Pull-In Instability

A critical constraint on the rotor bias voltage V_{DC} arises from an effect known as **lateral pull-in**. While our analysis has focused on the intended motion along the x-axis (drive direction), the charged comb fingers also generate electrostatic forces in the orthogonal y-direction (the direction transverse to the intended motion, which may coincide with the sense direction or another off-axis direction).

Consider the symmetric pair of drive fingers on the left and right of a given rotor finger. Ideally, with the rotor perfectly centered, the electrostatic forces in the y-direction from opposing fingers cancel exactly. However, if the rotor experiences a small perturbation in the y-direction, the balance is broken: the capacitor on the side toward which the rotor moves has a smaller gap, increasing its capacitance and attracting force, while the opposite side has a larger gap and a weaker force. The net effect is an electrostatic force that *pulls the rotor further in the y-direction*, acting as a **negative spring stiffness** (electrostatic softening).

This negative stiffness reduces the effective mechanical stiffness in the y-direction:

$$k_{y,\text{eff}} = k_{y,\text{mech}} - k_{y,\text{elec}}(x), \tag{2.72}$$

where $k_{y,\text{elec}}(x)$ is the electrostatic spring softening, which depends on the instantaneous overlap (hence on the drive displacement x). If the electrostatic softening exceeds the mechanical stiffness ($k_{y,\text{elec}} \geq k_{y,\text{mech}}$),

the system becomes unstable, and the rotor will snap laterally into contact with one of the fixed fingers—a phenomenon called lateral pull-in.

The electrostatic softening can be derived from the derivative of the y-directed force with respect to y displacement. For a symmetric comb drive, the softening per side is proportional to the overlap length and the square of the voltage difference. A detailed expression (not required to memorize) is provided in the notes.

2.19.3 Design Guideline for Rotor Bias

To avoid lateral pull-in, the designer must ensure that the electrostatic softening remains a small fraction of the mechanical stiffness throughout the entire drive oscillation cycle. A conservative guideline is to keep the maximum softening (which occurs at maximum overlap) below, say, 10% of $k_{y,\mathrm{mech}}$.

Numerical example:

- Mechanical stiffness in y-direction: $k_y = 3000\,\mathrm{N/m}$
- Rotor bias: $V_{DC} = 20\,\mathrm{V}$
- Finger overlap: $l_0 = 15\,\mu\mathrm{m}$ (varies by $\pm x$ during oscillation)
- Gap: $g = 1\,\mu\mathrm{m}$
- Number of fingers: $n = 50$

At maximum overlap ($l_0 + x_{\max}$), the calculated electrostatic softening might be, e.g., $300\,\mathrm{N/m}$ (10% of k_y), which is acceptable. At minimum overlap, it might drop to $53\,\mathrm{N/m}$. This example shows a safe design.

Thus, the choice of V_{DC} is a trade-off: higher V_{DC} increases transduction (desirable for signal strength) but exacerbates electrostatic softening and pull-in risk. Typical values range from 5 V to 20 V, with careful verification against the specific geometry.

2.20 Layout and Routing Considerations

The benefits of differential drive and sense—particularly feedthrough cancellation—rely on symmetry. The physical layout of the electrical connections (routing) must preserve this symmetry as much as possible.

2.20.1 Differential Pair Routing (Option 1)

One approach is to treat each pair of signals (D1/D2 and S1/S2) as high-quality differential pairs. This means routing the two traces of each pair close together, with identical geometry, so that any external interference couples equally to both. This maximizes common-mode rejection for disturbances originating outside the chip.

However, this approach may not be optimal for internal parasitic coupling between different pairs (e.g., D1 to S1 vs. D1 to S2). The distances between D1 and S1 may differ from those between D1 and S2, breaking the symmetry required for cancellation of feedthrough via D2's opposite phase.

Figure 2.12: Differential pair routing.

2.20.2 Fully Symmetric Routing (Option 2)

A more rigorous approach is to design the entire electrode and routing structure with full geometrical symmetry. For example:

- Place D1 and S1 on one side of the rotor, with their bond pads and traces arranged symmetrically with respect to the rotor's centerline.
- Place D2 and S2 on the opposite side, with mirror symmetry.
- Within each side, the traces for D1 and S1 may not be a tightly coupled differential pair, but the overall structure ensures that parasitic capacitances $C_{f,D1S1} = C_{f,D2S2}$ and $C_{f,D1S2} = C_{f,D2S1}$.

Figure 2.13: Fully symmetric routing.

With this symmetry, the feedthrough cancellation mechanism described earlier (anti-phase drives on symmetric capacitors + differential sensing) works optimally. The trade-off is slightly reduced rejection of certain common-mode external disturbances, but this is often a worthwhile compromise given the strong suppression of on-chip parasitic feedthrough.

2.21 Summary and Outlook

In this lecture, we have covered the foundational concepts for modeling and designing the drive mode of a MEMS gyroscope:

- We reviewed the spring-mass-damper model and its frequency-domain characteristics, highlighting the importance of resonance and quality factor.
- We introduced electrostatic comb-drive actuation and derived the force expression, noting its quadratic dependence on voltage.
- We developed the small-signal linearization technique using a DC bias to enable resonant excitation.
- We derived the electromechanical transfer function and its equivalent series RLC circuit model, a powerful tool for system-level simulation.
- We identified the limitations of single-ended actuation and sensing: small-signal constraint, 2ω force components, susceptibility to common-mode noise, and design constraints.
- We presented the push-pull (differential) drive and differential sense configuration, showing how it provides exact linearization, cancellation of harmonics, common-mode rejection, and design flexibility.
- We developed the more complex equivalent circuit for the four-port differential system and explained how symmetry leads to effective feedthrough cancellation.
- We discussed optimization of finger counts, trading off actuation force against sense signal strength.
- We introduced the important constraint of lateral pull-in instability, which limits the maximum allowable rotor bias voltage.
- We briefly considered layout strategies to preserve symmetry and maximize parasitic rejection.

The drive mode is now well characterized. However, the drive resonator alone does not constitute a gyroscope. To sustain the oscillation autonomously, we must incorporate the MEMS device into a feedback loop—an electronic oscillator. The next class will focus on oscillator design: selecting the amplifier, ensuring the Barkhausen criteria are met, managing amplitude stability, and analyzing phase noise. Subsequently, we will turn our attention to the sense mode, where the Coriolis force is detected, and finally, we will combine both subsystems into a complete gyroscope model.

Chapter 3

Amplitude-Modulated Capacitive MEMS Gyroscopes

3.1 Introduction to Gyroscope Modes

As introduced previously, this discussion will clarify the operational principles of gyroscopes by examining their two primary modes of motion. These principles can be observed in practice using a plastic demonstration model. The first mode is the drive mode, which we established in the drive equation. In practice, we typically employ comb-finger-based actuation to create a resonator. Although a push-pull actuation with differential sensing using multiple electrodes on both sides is standard—allowing us to close an electronic loop around these electrodes—a single electrode is shown here for graphical simplicity.

Each vibrational mode of the structure—both along the x-axis (drive direction) and the y-axis (sense direction)—can be modeled as an independent spring-mass-damper system. For the drive mode, the stiffness k is provided by the four external springs that constrain the motion. The moving mass in this direction is essentially the entire mass of the structure. The damping coefficient, which depends on the fluid-structure interaction, may differ between directions due to the geometry of the moving parts.

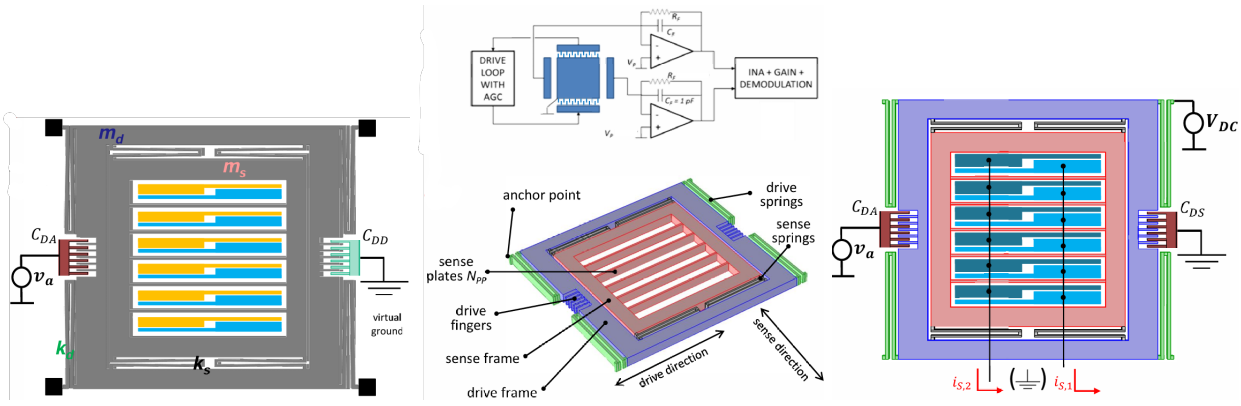

Figure 3.1: Gyroscope representations.

3.2 Effect of Rotation and Coriolis Force

If the structure is oscillating in the drive direction and an external rotation is applied, a non-inertial reference frame reveals a displacement in the y-direction. This apparent motion arises solely from the Coriolis effect and is therefore dependent on the angular velocity. The phase and amplitude of the y-displacement correspond directly to the sign and magnitude of the angular rate. With no angular rate, no y-motion occurs. A

larger angular rate produces a larger y-displacement. This principle defines amplitude-modulated (AM) gyroscopes, where the Coriolis acceleration—proportional to the drive-mode velocity—modulates the sense-mode displacement amplitude at the drive frequency.

In the typical design, the outer frame does not displace in the y-direction despite experiencing the same Coriolis acceleration. This is because the external springs, which are compliant in the x-direction for drive motion, are designed to be very stiff in the y-direction. In contrast, the inner frame is mechanically decoupled from the outer frame by a separate set of springs with stiffness k_s. This allows the inner frame to respond to the Coriolis force by displacing in the y-direction.

3.3 Sensing the Motion

Motion along the y-direction is detected capacitively using fixed stators that form parallel-plate capacitors with the moving arms of the inner frame. These are arranged differentially: as the structure moves, the gap to one set of electrodes (e.g., blue) decreases while the gap to the opposing set (e.g., yellow) increases. This creates a differential capacitance change that is proportional to the displacement. A physical toy model can demonstrate both the drive and sense modes, though it lacks the fixed stators found on a fabricated device due to the absence of a substrate.

In this configuration, the drive velocity is along the x-axis, and the angular rate is about the out-of-plane z-axis (yaw). The resulting Coriolis acceleration, given by the cross product, acts along the perpendicular y-axis.

3.4 Typical Dimensions and Fabrication

A complete MEMS gyroscope structure typically occupies an area of approximately 1 mm $\times$ 1 mm. It is fabricated using processes derived from integrated circuit manufacturing, enabling minimum feature sizes as small as one micron. This combination of micron-scale mechanical elements with electrical interfaces defines the microelectromechanical system (MEMS), which is then integrated with surrounding electronics.

3.5 Subsystem Combination

A gyroscope effectively combines two subsystems:

1. A **drive resonator** that is maintained in oscillation along the x-axis using a positive feedback loop.
2. A **sense accelerometer** that measures the Coriolis acceleration by detecting differential capacitance changes at the sense electrodes.

The output voltage, representing the angular rate, is modulated at the drive frequency and must be demodulated to recover the baseband signal.

3.6 Capacitance Configurations

Two primary types of variable capacitors are used in MEMS gyroscopes, each suited to different functions based on the expected displacement scale.

3.6.1 Comb-Finger (Area-Varying) Capacitors for Drive Mode

Drive-mode actuation typically uses comb-finger capacitors. In this configuration, motion changes the overlapping area between fixed and moving fingers. A key advantage is that the electrostatic force is independent of displacement, allowing for large, linear oscillations without introducing geometric nonlinearities. While "large" in MEMS terms, displacements are on the order of 5–20 microns, given that gaps and features are typically a few microns.

3.6.2 Parallel-Plate (Gap-Varying) Capacitors for Sense Mode

Sense-mode detection uses parallel-plate capacitors where motion changes the gap between electrodes. The Coriolis-induced sense displacement is very small—typically 10–50 nanometers. While the capacitance variation in a gap-varying design is inherently nonlinear, it can be linearized for such small displacements. The primary advantage is its significantly higher transduction efficiency (capacitance change per unit displacement) compared to comb fingers, making it ideal for detecting minute motions even at the cost of slight nonlinearity.

3.7 Readout Mechanism

The capacitance variation at the sense electrodes is read out electronically. All moving (rotor) structures are biased at the same DC voltage, as they are fabricated from conductive silicon. The fixed sense plates are typically connected to a virtual ground, such as the input of an operational amplifier. This configuration allows the current induced by the motion-related capacitance change to flow through a feedback branch, producing an output voltage proportional to the sense-mode displacement. This method is analogous to the readout used in the drive resonator.

3.8 Current in Sensing Electrodes

Consider a single set of sense electrodes, for instance the light blue ones. The current flowing through these electrodes is given by the time derivative of charge:

$$i = \frac{dQ}{dt} = \frac{d(CV)}{dt} = V\frac{dC}{dt} + C\frac{dV}{dt}$$

In our configuration, the voltage across the capacitor is held constant because one plate is biased at V_{dc} and the other is held at virtual ground (0 V). Therefore, the $\frac{dV}{dt}$ term disappears, leaving:

$$i = V_{dc}\frac{dC}{dt}$$

We can expand this using the chain rule:

$$\frac{dC}{dt} = \frac{dC}{dy}\frac{dy}{dt} = \frac{dC}{dy}\dot{y}$$

Since the sense motion is sinusoidal at the drive frequency ω_d, the velocity $\dot{y}$ is proportional to ω_d times the displacement amplitude. Thus, the current amplitude becomes:

$$i = V_{dc}\frac{dC}{dy}\omega_d y$$

For the differential electrode pair (light blue and dark blue), the currents will be equal and opposite. When the rotor moves upward, the gap to the light blue electrode decreases (capacitance increases) while the gap to the dark blue electrode increases (capacitance decreases), producing currents of opposite polarity.

For a parallel-plate capacitor, the capacitance is:

$$C = \epsilon_0\frac{A}{g} = \epsilon_0\frac{L \cdot t}{g}$$

where L is the electrode length, t is the process thickness (constant), and g is the gap. For small displacements Δy around the nominal gap g_0, the capacitance variation can be linearized. The differential capacitance variation per unit displacement for a pair of electrodes is:

$$\frac{dC_{diff}}{dy} \approx 2\frac{C_0}{g_0}$$

where C_0 is the nominal capacitance of one electrode at rest. This factor accounts for the differential configuration where one gap decreases while the other increases. Multiple electrode pairs can be paralleled to increase the overall transduction efficiency.

3.9 Working Principle Summary

The gyroscope operating principle can be summarized as follows:

1. Establish a sustained oscillation along the drive direction (x-axis) using electrostatic actuation and control electronics.
2. An external angular rotation Ω about an orthogonal axis (here, the z-axis) generates a Coriolis acceleration along the third perpendicular axis (y-axis):

$$\vec{a}_c = -2\vec{\Omega} \times \vec{v}$$

3. This acceleration produces a force on the proof mass, causing displacement along the sense direction.
4. The resulting motion is detected by fixed electrodes that form differential parallel-plate capacitors with the moving structure.
5. The capacitance variation generates currents proportional to the angular rate, which are converted to voltage by readout electronics.

Note that all electrodes are fixed to the substrate and therefore share the same non-inertial reference frame as the rotating structure. They observe the apparent motion of the proof mass induced by the Coriolis acceleration, analogous to observing a moving ball from a rotating reference frame.

3.10 Subsystems Review

As introduced previously, the gyroscope comprises two distinct spring-mass-damper subsystems:

- **Drive mode (x-direction)**: Characterized by mass m_x, stiffness k_x, and damping coefficient c_x. Oscillation is sustained by electrostatic forces from comb-drive actuators.
- **Sense mode (y-direction)**: Characterized by mass m_y, stiffness k_y, and damping coefficient c_y. These parameters differ from the drive mode because different springs constrain the sense motion. Excitation comes from the Coriolis force rather than electrostatic actuation.

Geometrical parameters are carefully designed to achieve appropriate resonant frequencies and bandwidth for both modes.

3.11 Advanced Structure: Decoupled Drive and Sense Modes

In the basic gyroscope structure, the sense electrodes experience undesired capacitance variations due to drive-mode motion, caused by fringe effects and imperfect electrode alignment. These parasitic signals can interfere with the desired Coriolis-induced measurements. A more advanced architecture, the **doubly decoupled gyroscope**, addresses this limitation by introducing three nested frames instead of two.

3.11.1 Structural Elements

- **Drive frame (outer)**: Excited by comb-finger actuators to oscillate along the x-direction. It is stiff in the y-direction due to its suspension springs.
- **Coriolis frame (intermediate)**: Nested within the drive frame and connected to it by springs that are compliant in the x-direction (allowing it to follow drive motion) but stiff in the y-direction. This frame carries no sense electrodes.
- **Sense frame (inner)**: Nested within the Coriolis frame and connected to it by springs that are compliant in the y-direction but stiff in the x-direction. This frame contains the parallel-plate sense electrodes and is anchored to the substrate at specific points.

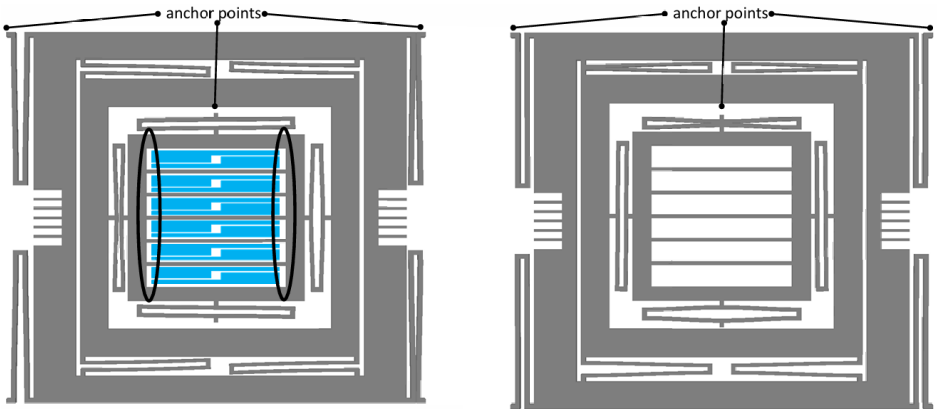

Figure 3.2: Doubly-decoupled gyroscope.

3.11.2 Operational Behavior

During drive motion (no rotation):

- The drive frame oscillates along x.
- The Coriolis frame, connected through x-compliant springs, follows this motion with the same velocity.
- The sense frame, anchored through springs that are stiff in x, remains stationary along the x-direction. It does not participate in drive oscillation.

Under rotation (Ω_z):

- The drive frame experiences Coriolis acceleration but does not displace in y due to its stiff y-suspension.
- The Coriolis frame, moving with drive velocity, experiences Coriolis acceleration and displaces in the y-direction (its suspension is compliant in y).
- This y-displacement of the Coriolis frame drags the sense frame through the y-compliant connecting springs, causing the sense frame to move in y.

The key advantage is that the sense frame remains stationary during pure drive motion, eliminating parasitic capacitance variations from drive displacement. Only when Coriolis force acts does the sense frame move, producing a clean signal proportional to angular rate.

3.11.3 Benefits

- Drive motion does not directly modulate sense capacitances.
- Sense electrodes respond only to Coriolis-induced displacement.
- Reduced feedthrough and improved signal-to-noise ratio.

3.12 Distinguishing Coriolis from Linear Acceleration

A fundamental challenge in gyroscope operation is distinguishing between:

- Displacement caused by Coriolis acceleration (desired angular rate signal)
- Displacement caused by external linear acceleration (undesired disturbance)

Both forces act on the same proof mass and produce motion in the sense direction. The key differentiator is **frequency**.

- The drive oscillation is maintained at a specific frequency ω_d, typically in the range of 20–50 kHz. This frequency is chosen to be well above the bandwidth of typical external accelerations encountered in applications (e.g., human motion, device handling), which rarely exceed a few kilohertz.

- Coriolis force is proportional to drive velocity, which oscillates at ω_d. Therefore, the resulting sense-mode displacement is also modulated at ω_d.

- Linear accelerations produce sense-mode displacement at the frequency of the applied acceleration, which is generally much lower than ω_d.

This frequency separation enables discrimination through **synchronous demodulation**. By multiplying the sense output with a reference signal derived from the drive oscillation (at frequency ω_d), components at ω_d are converted to DC (representing angular rate), while components at other frequencies are shifted to AC and can be filtered out. This technique effectively extracts the Coriolis signal while rejecting low-frequency acceleration disturbances.

3.13 Separating Contributions from Linear Acceleration and Coriolis Acceleration

While frequency-based discrimination through synchronous demodulation effectively separates Coriolis signals from linear acceleration disturbances, practical non-idealities in real devices mean that some acceleration-induced motion may still appear in the sense output. To further enhance this separation, gyroscopes employ tuning fork architectures.

3.13.1 Tuning Fork Principle

In a tuning fork configuration, instead of a single oscillating structure, two identical structures are coupled together and driven to move in anti-phase (opposite directions). To visualize this, imagine taking the structure we have studied, mirroring it, and arranging the two halves such that at any instant, their velocities are equal in magnitude but opposite in direction:

$$v_1 = -v_2$$

3.13.2 Response to Angular Rate vs. Linear Acceleration

This anti-phase motion creates a fundamental distinction between how the structure responds to angular rate and linear acceleration:

- **Under angular rotation** Ω: The Coriolis force $\vec{F}_c = -2m\vec{\Omega} \times \vec{v}$ acts on each mass. Because the velocities are opposite, the Coriolis forces are also opposite:

$$\vec{F}_{c1} = -2m\vec{\Omega} \times \vec{v}_1, \quad \vec{F}_{c2} = -2m\vec{\Omega} \times \vec{v}_2 = -2m\vec{\Omega} \times (-\vec{v}_1) = -\vec{F}_{c1}$$

  Thus, one mass moves upward while the other moves downward—a differential mode response.

- **Under linear acceleration** a: An inertial force $\vec{F} = m\vec{a}$ acts on both masses in the same direction. Both masses displace identically—a common mode response.

By designing the sensing electrodes to detect differential motion while rejecting common motion, the tuning fork architecture provides an additional mechanism to separate angular rate signals from linear acceleration disturbances. The two halves of the device are designed to resonate in anti-phase at the same frequency, typically using coupling springs that enforce this mode shape.

3.13.3 From Conceptual to Practical Implementation

While the classic tuning fork is a three-dimensional structure, MEMS gyroscopes implement this principle in planar (quasi-2D) form. The yaw-axis gyroscope we will now examine in detail employs exactly this concept: the top and bottom portions of the structure oscillate in anti-phase, providing the benefits of differential sensing and common-mode rejection.

3.14 Transition to Detailed Structure Analysis

Having established the working principles, we now adopt a top-down approach: we examine the complete structure of a two-axis gyroscope, then systematically identify its features and understand their roles. This approach reveals how the relatively simple operating principles translate into the complex geometries seen in practical devices.

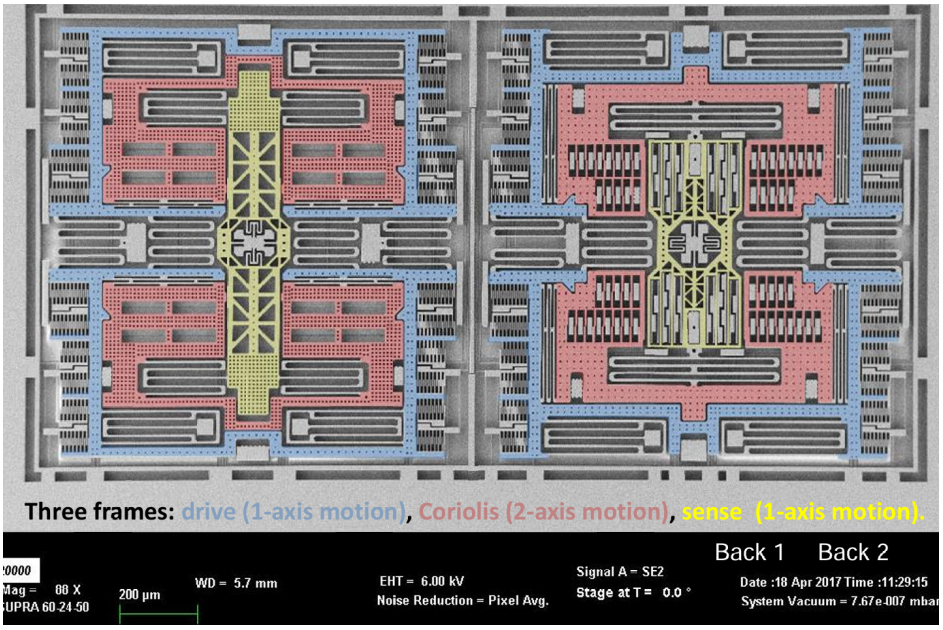

Figure 3.3: Complete two-axis MEMS gyroscope structure. The right-hand side is a yaw-axis gyroscope (sensing out-of-plane rotation), while the left-hand side is a roll-axis gyroscope (sensing in-plane rotation about the x-axis).

Key observable features include:

- Comb finger arrays for electrostatic actuation, oriented along what we identify as the drive direction.
- Perforations (holes) throughout suspended structures, which facilitate release etching and reduce damping.
- Suspension springs of various geometries.
- Parallel-plate electrodes for capacitive sensing.
- Two distinct gyroscope structures on the same chip: one for yaw sensing (out-of-plane rotation) and one for roll sensing (in-plane rotation about the x-axis).

3.15 Detailed Analysis of Yaw-Axis Gyroscope

We now focus on the yaw-axis gyroscope (the right-hand structure in the previous slide). For clarity, we examine only the top half of this structure, understanding that the bottom half is symmetric and operates in anti-phase, realizing the tuning fork configuration.

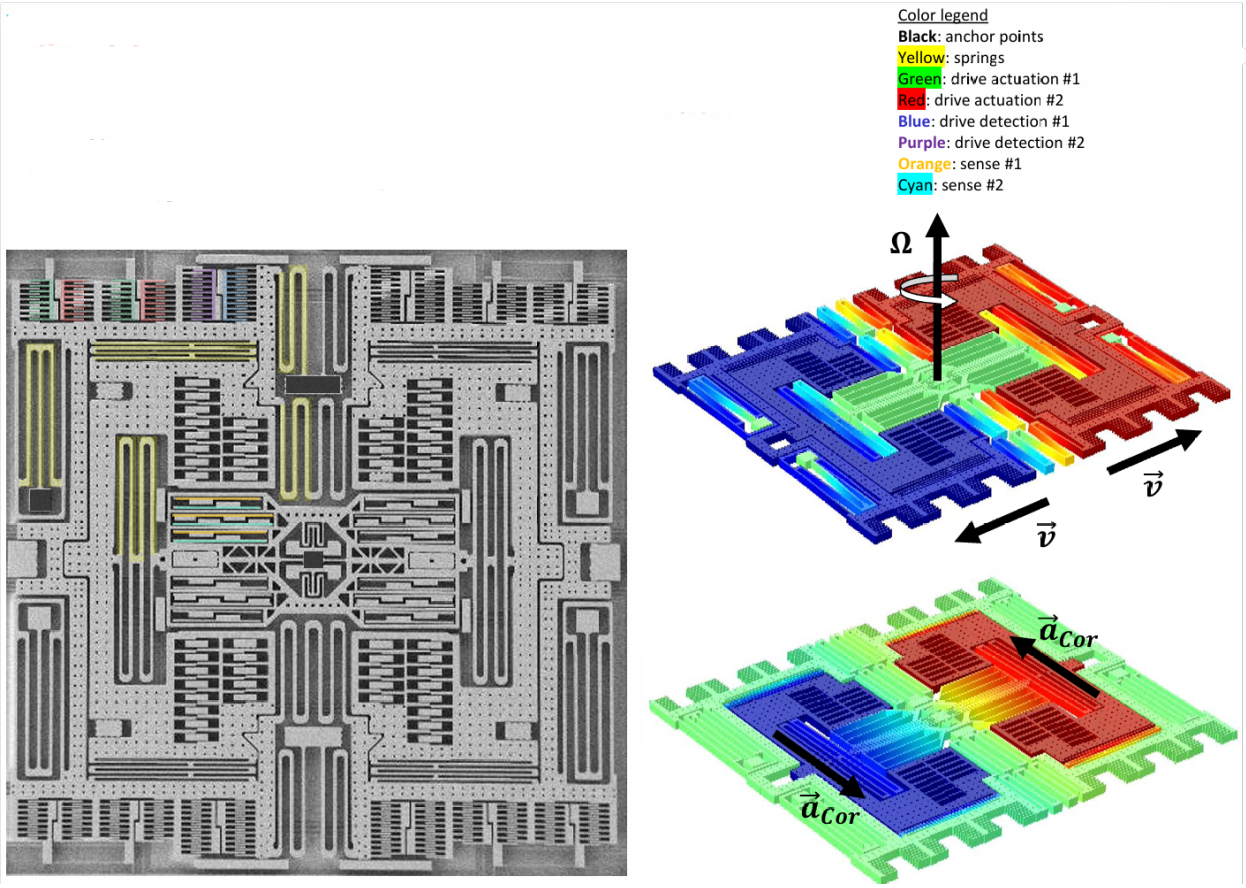

Figure 3.4: Yaw-axis gyroscope detail with key frames highlighted. The structure has been rotated 90° so that the drive direction is horizontal.

3.15.1 Identification of the Three Frames

As introduced in the doubly decoupled architecture, we identify three nested frames:

1. **Drive frame (outer frame)**: The blue-highlighted frame in the figure. This frame is actuated by comb fingers and constrained by suspension springs to move only along the drive direction (x-axis). The four suspension springs (one in each quadrant) are anchored to the substrate and designed for high stiffness in the sense direction (y-axis) while being compliant in the drive direction.

2. **Coriolis frame (intermediate frame)**: The pink-highlighted frame. Connected to the drive frame by springs that are stiff in the drive direction (so it follows drive motion) but compliant in the sense direction. This frame carries no sense electrodes.

3. **Sense frame (inner frame)**: The yellow-highlighted frame. Connected to the Coriolis frame by springs that are stiff in the sense direction (allowing it to be dragged in y) but compliant in the drive direction (so it remains stationary during pure drive motion). This frame carries the parallel-plate sense electrodes.

3.15.2 Anchor Points and Suspension Springs

Anchor points (blacked out in the figure) connect the suspension springs to the substrate, keeping the structure suspended. Key springs include:

- **Drive frame suspension springs**: Four external springs (one in each quadrant) that anchor the drive frame and constrain its motion primarily to the drive direction.
- **Coriolis frame coupling springs**: Connect the drive frame to the Coriolis frame. Stiff in x (transmitting drive motion), compliant in y (allowing Coriolis-induced displacement).
- **Sense frame decoupling springs**: Connect the Coriolis frame to the sense frame. Stiff in y (transmitting Coriolis-induced motion), compliant in x (isolating sense frame from drive motion).
- **Tuning fork coupling spring**: A spring spanning from one drive frame half to the other, coupling the two halves of the device and enforcing anti-phase oscillation.

3.15.3 Drive Mode Actuation and Detection

Push-pull actuation is implemented using two sets of comb fingers:

- **Green electrodes (push)**: Apply electrostatic force attracting the drive frame in one direction.
- **Red electrodes (pull)**: Apply force attracting the drive frame in the opposite direction.

By applying anti-phase sinusoidal signals to these electrode sets, sustained oscillation is established in the drive mode.

Drive motion is detected using differential sensing electrodes:

- **Light blue and purple electrodes**: These form differential capacitive pairs that measure drive displacement. This signal is used in a feedback loop to sustain oscillation at the resonant frequency.

All electrode sets are replicated across the structure and configured to sense the anti-phase drive mode, ensuring that the two halves oscillate in opposite directions as required for tuning fork operation.

3.15.4 Sense Mode Operation under Rotation

When an angular rate Ω_z is applied while the drive oscillation is active:

1. The Coriolis frame, moving with drive velocity, experiences Coriolis acceleration in the y-direction.
2. Being compliant in y, the Coriolis frame displaces along y.
3. This displacement is transmitted through the y-stiff coupling springs to the sense frame.
4. The sense frame moves in y, changing the gaps of the parallel-plate electrodes.
5. The resulting differential capacitance change is converted to a voltage proportional to angular rate.

Crucially, the sense frame remains stationary during pure drive motion (no rotation), eliminating parasitic capacitive changes from drive displacement.

3.15.5 Differential Sensing and Common-Mode Rejection

The tuning fork architecture ensures that under rotation, the two halves of the device move in opposite directions in the sense mode (differential signal). Linear acceleration, however, causes both halves to move in the same direction (common-mode signal). The sensing electronics are designed to amplify the differential component while rejecting the common-mode component, providing additional rejection of acceleration disturbances beyond the frequency-domain separation.

3.15.6 Additional Electrodes

Some electrodes visible in the structure (e.g., certain parallel-plate sets) serve purposes beyond basic sense detection. These are typically used for:

- Quadrature error correction (electrostatic tuning)
- Mode matching
- Self-test and diagnostics

Their function will be addressed in subsequent discussions.

3.16 Summary of Operating Principles

The complete yaw-axis gyroscope operation can be summarized as follows:

1. **Drive mode**: Push-pull electrostatic actuation sustains anti-phase oscillation of the two device halves along the x-axis at frequency ω_d (typically 20-50 kHz).
2. **Coriolis force generation**: External angular rate Ω_z produces Coriolis acceleration on the moving Coriolis frames, generating force along the y-axis.
3. **Sense mode response**: The sense frames displace along y, modulated at ω_d with amplitude proportional to Ω_z.
4. **Differential detection**: Parallel-plate capacitors transduce this motion into differential currents.
5. **Signal processing**: Synchronous demodulation extracts the baseband angular rate signal while rejecting low-frequency acceleration disturbances. The differential sensing further rejects common-mode acceleration inputs.

The complexity of the final structure—with its nested frames, multiple spring types, and electrode configurations—arises from the need to achieve this functionality while minimizing parasitic effects and maximizing rejection of disturbances.

3.17 Sensitivity to Different Axes

Having analyzed the yaw-axis gyroscope in detail, we now examine why this structure responds selectively to angular rate about the z-axis while rejecting rates about the other two axes. Understanding this selectivity is crucial for multi-axis gyroscope design.

3.17.1 Response to Rotation About Different Axes

Consider the drive velocity $\vec{v}$ along the x-axis. The Coriolis acceleration is given by $\vec{a}_c = -2\vec{\Omega} \times \vec{v}$. We examine three cases:

1. **Angular rate about z-axis (yaw)**: $\vec{\Omega} = \Omega_z \hat{k}$, $\vec{v} = v_x \hat{i}$
   $\vec{a}_c = -2\Omega_z v_x (\hat{k} \times \hat{i}) = -2\Omega_z v_x \hat{j}$
   The resulting Coriolis acceleration is along the y-axis, exactly the direction for which the sense mode is designed. The structure responds with measurable displacement.
2. **Angular rate about x-axis (roll)**: $\vec{\Omega} = \Omega_x \hat{i}$, $\vec{v} = v_x \hat{i}$
   $\vec{a}_c = -2\Omega_x v_x (\hat{i} \times \hat{i}) = 0$
   The cross product of parallel vectors is zero, so no Coriolis acceleration occurs regardless of the angular rate magnitude.

3. **Angular rate about y-axis (pitch)**: $\vec{\Omega} = \Omega_y \hat{j}$, $\vec{v} = v_x \hat{i}$
   $\vec{a}_c = -2\Omega_y v_x(\hat{j} \times \hat{i}) = -2\Omega_y v_x(-\hat{k}) = 2\Omega_y v_x \hat{k}$
   Coriolis acceleration acts along the z-axis (out-of-plane). However, the suspension springs are designed to be very stiff in the z-direction, so the resulting displacement is negligible.

Thus, the structure exhibits high sensitivity only to the intended yaw axis, with cross-axis sensitivity effectively suppressed by a combination of zero cross-product (for x-axis rate) and high mechanical stiffness (for z-axis motion from y-axis rate).

3.18 Designing for Other Axes: The Roll/Pitch Gyroscope

The same principles can be extended to measure angular rates about in-plane axes by designing a sense mode that responds to out-of-plane motion. This is exactly the function of the left-hand structure in the two-axis gyroscope shown earlier—a roll-axis gyroscope (sensing rotation about the x-axis) or pitch-axis gyroscope (sensing rotation about the y-axis), depending on orientation.

3.18.1 Structural Modifications for Out-of-Plane Sensing

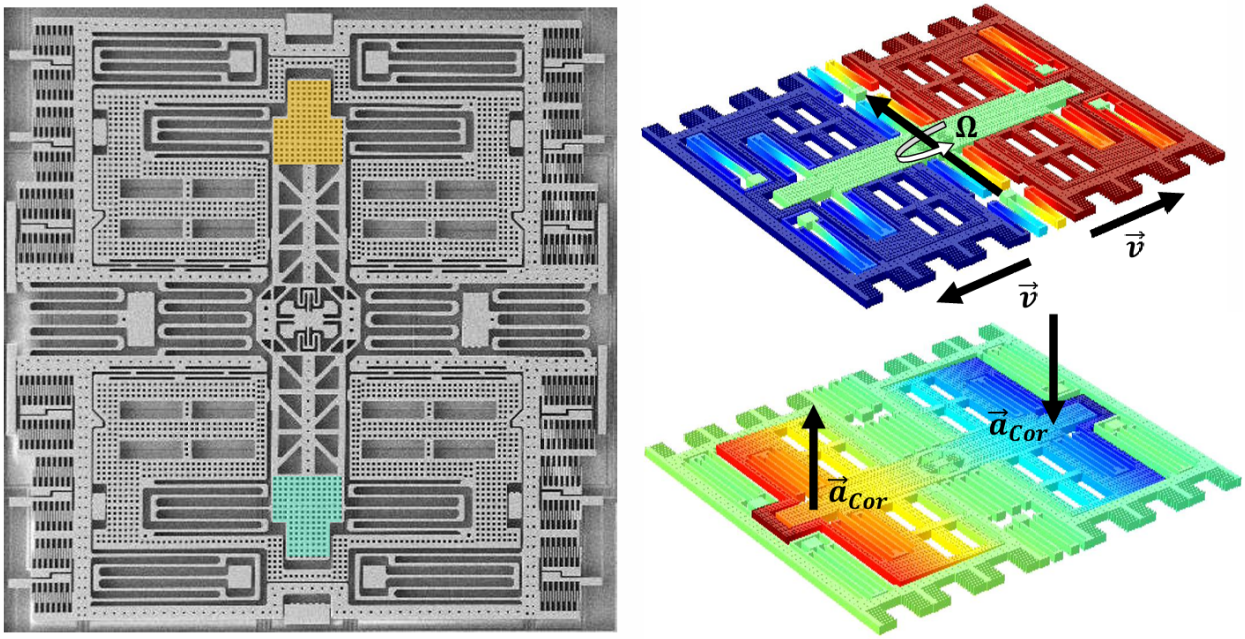

Figure 3.5: Roll/pitch gyroscope structure. The drive mechanism remains similar, but the sense mode uses torsional springs for out-of-plane motion.

Key features of the in-plane rate sensing structure:

- **Drive frame**: Nearly identical to the yaw gyroscope, with comb-finger actuation establishing oscillation along the x-axis. Push-pull actuation and drive detection electrodes function as before.
- **Coriolis frame**: Connected to the drive frame by springs that are stiff in the drive direction (transmitting drive motion) but designed as **torsional springs** that permit rotation about an in-plane axis. These springs enable the Coriolis frame to execute a flapping motion out of the plane.
- **Sense frame**: Coupled to the Coriolis frame through mechanisms that transmit out-of-plane motion while decoupling in-plane drive motion.

- **Sensing electrodes**: Unlike the in-plane parallel plates used for yaw sensing, out-of-plane sensing requires electrodes placed **underneath** the moving structure. These form parallel-plate capacitors with the sense frame, with the gap varying as the structure moves up and down.

3.18.2 Operating Principle for Roll/Pitch Sensing

With drive velocity along x-axis and angular rate Ω_y (pitch) applied:

1. Coriolis acceleration acts along the z-axis: $a_c = 2\Omega_y v_x$ (direction depends on sign conventions).
2. This acceleration produces a torque on the Coriolis frame due to the offset between its center of mass and the torsional spring axis.
3. The Coriolis frame undergoes an anti-phase flapping motion: when one half moves up, the other moves down.
4. This motion is transmitted to the sense frame, which moves out-of-plane relative to fixed underlying electrodes.
5. Differential capacitance changes between the two halves provide a signal proportional to angular rate.

3.18.3 Common-Mode Rejection for Out-of-Plane Acceleration

The tuning fork architecture provides the same benefit for out-of-plane sensing:

- **Under angular rate**: The two halves move in opposite directions (differential mode). One side's capacitance increases while the other decreases.
- **Under linear z-axis acceleration**: Both halves move in the same direction (common mode). Both capacitances change identically.

By taking the differential output, linear acceleration effects are cancelled, leaving only the Coriolis-induced signal.

3.19 Design Guidelines for Advanced Gyroscopes

Based on the structures we have analyzed, several design principles emerge for high-performance MEMS gyroscopes.

3.19.1 Tuning Fork Architecture: Highly Recommended

While frequency-domain separation (modulation/demodulation) theoretically distinguishes Coriolis signals from acceleration disturbances, practical limitations of filtering make complete rejection impossible. The tuning fork architecture provides a superior solution by rejecting accelerations **at the capacitance level**:

- Angular rate produces differential capacitance changes
- Linear acceleration produces common-mode capacitance changes
- Differential readout inherently cancels acceleration effects

This mechanical rejection complements electronic filtering, significantly improving overall performance.

3.19.2 Double Decoupling: Benefits Beyond Parasitic Reduction

The doubly decoupled architecture (drive frame, Coriolis frame, sense frame) offers multiple advantages:

1. **Minimizes parasitic effects**: As discussed, fringe effects from drive motion on sense electrodes are virtually eliminated because the sense frame remains stationary during pure drive oscillation.
2. **Enables quadrature compensation**: As will be discussed in subsequent lectures, electromechanical imperfections (quadrature error) require dedicated compensation electrodes. These are challenging to implement without double decoupling. The structures highlighted in the yaw and roll devices correspond to quadrature compensation electrodes.
3. **Improves robustness to imperfections**: Even if perfect symmetry is not achieved in fabrication, the decoupled architecture reduces the impact of asymmetries.

3.19.3 Mass Distribution: Minimize Drive Frame Mass

A critical design consideration is how mass is distributed among the three frames. Recall that Coriolis force acts on **all** moving masses that have velocity:

$$F_c = -2m\vec{\Omega} \times \vec{v}$$

- The drive frame has velocity and therefore experiences Coriolis force.
- However, we do **not** want the drive frame to displace in the sense direction—it is designed with high stiffness in that direction.
- Any Coriolis force on the drive frame must be absorbed by its suspension without producing displacement.
- Therefore, minimizing drive frame mass minimizes the Coriolis force that must be rejected by its stiff suspension.

Design rule: Make the drive frame as light as possible while maintaining sufficient rigidity to resist deformation under operational forces. This is typically achieved through:

- Minimal width of frame members
- Generous perforation (holes) to remove mass without compromising stiffness
- **Framework structures** that maximize stiffness-to-mass ratio

3.19.4 Framework Structures for Optimal Stiffness-to-Mass Ratio

The concept of framework structures, familiar from civil engineering (e.g., electrical transmission towers), applies directly to MEMS design:

Key principles:

- Triangular truss geometries provide inherent rigidity
- Members can be very slender because loads are carried axially
- Mass is concentrated only where structurally necessary
- The planar nature of MEMS fabrication constrains designs to 2D frameworks, but within that constraint, significant optimization is possible

In the yaw gyroscope example shown, the drive frame could be further hollowed or replaced with a truss structure to reduce mass while maintaining the rigidity needed to preserve mode shapes and resonant frequencies.

Figure 3.6: Example of a lightweight rigid framework structure. Triangular trusses provide high stiffness with minimal material.

3.20 Integration of Multiple Sensing Axes

The complete two-axis gyroscope combines:

- A yaw-axis structure (sensing Ω_z) with in-plane sense motion and parallel-plate electrodes
- A roll/pitch-axis structure (sensing Ω_x or Ω_y) with out-of-plane sense motion and underlying electrodes
- Shared drive electronics (though each axis may have its own drive loop)
- Common signal processing for demodulation and output

This integration on a single chip enables compact multi-axis inertial measurement units (IMUs) for applications ranging from consumer electronics to industrial and automotive systems.

3.21 Summary

We have seen how:

- Gyroscopes are designed to be sensitive only to their intended axis through a combination of cross-product physics and mechanical stiffness engineering.
- Different architectures enable sensing of out-of-plane vs. in-plane angular rates.
- Tuning fork configurations provide essential rejection of linear acceleration disturbances.
- Double decoupling improves performance and enables quadrature error compensation.
- Mass distribution, particularly minimizing drive frame mass, is a key design optimization.
- Framework structures offer an approach to achieve high rigidity with minimal mass.

These principles form the foundation for understanding and designing high-performance MEMS gyroscopes. The next lectures will address specific challenges such as quadrature error and its compensation, as well as the electronic interfaces that complete the system.

3.22 Design Guidelines for Mass Distribution

Building on our analysis of the doubly decoupled architecture, we now examine optimal mass distribution among the three frames. This is a critical design consideration that significantly influences gyroscope performance.

3.22.1 Sense Frame Mass: Minimize for Acceleration Rejection

The sense frame (yellow) raises an important question: should it be heavy or light? Recall its role in the system:

- The sense frame does **not** move with drive velocity. During drive oscillation, the decoupling springs keep it stationary.
- Consequently, it does **not** directly experience Coriolis force. Coriolis force acts only on masses with velocity relative to the rotating frame.
- However, the sense frame **does** directly experience linear acceleration according to $F = ma$.

Therefore, the design guideline is clear: **minimize sense frame mass**. A lighter sense frame:

- Reduces the force generated by linear acceleration disturbances
- Minimizes the motion that must be rejected by differential sensing
- Complements the tuning fork architecture's common-mode rejection

The framework structure visible in some implementations (including the example shown) achieves precisely this: maximum rigidity with minimum mass through truss-like geometries.

3.22.2 Coriolis Frame Mass: Maximize for Sensitivity

The Coriolis frame (pink) plays a fundamentally different role:

- It moves with drive velocity (coupled through stiff-in-x springs)
- Under angular rate, it experiences Coriolis force: $F_c = -2m\vec{\Omega} \times \vec{v}$
- This force drives the sense motion that ultimately produces the output signal

Since Coriolis force is proportional to mass, **maximize Coriolis frame mass** to maximize sensitivity. Given a fixed total area available for the device, mass should be concentrated where it contributes to signal generation.

3.22.3 Drive Frame Mass: Minimize for Rejection of Unwanted Forces

The drive frame (blue) also experiences Coriolis force (it has velocity), but we deliberately prevent it from moving in the sense direction through stiff y-direction suspension. Any Coriolis force on the drive frame must be absorbed by its suspension without producing displacement. Therefore:

- **Minimize drive frame mass** to minimize the Coriolis force that must be rejected
- Maintain sufficient rigidity to preserve mode shapes and resonant frequencies
- Framework structures or generous perforation achieve the stiffness-to-mass ratio objective

3.22.4 Optimal Mass Distribution Summary

For a given total device area:

- Drive frame: as light as possible (thin members, hollowed, framework structures)
- Sense frame: as light as possible (minimizes acceleration response)
- Coriolis frame: as heavy as possible (maximizes Coriolis force)

This distribution ensures that mass is placed where it contributes to desired signal generation and minimized where it would amplify unwanted disturbances.

3.23 Spring Design Considerations

Throughout the structures we have examined, springs exhibit multiple folds and serpentine patterns. This design choice is not arbitrary but serves specific engineering purposes.

3.23.1 Benefits of Folded Springs

- **Fabrication tolerance**: For a given target stiffness, a folded spring can be designed with wider individual beam widths compared to a straight beam. Wider features are less sensitive to process variations (over-etching, under-etching, lithography errors).
- **Stress distribution**: Folded geometries distribute mechanical stress more evenly, reducing risk of fracture or fatigue.
- **Area efficiency**: Long effective spring lengths can be achieved in compact areas through folding.
- **Decoupling**: Multiple-fold springs can be designed with different stiffnesses in orthogonal directions, enabling the directional compliance characteristics essential for decoupled architectures.

The analogy to electronic design is apt: just as minimum-size transistors are more sensitive to process variations than larger devices, minimum-width springs are more affected by fabrication imperfections. Wider features, made possible by folding, provide greater robustness.

3.24 Quality Factor Considerations

Quality factor (Q) characterizes the sharpness of resonance and the energy dissipation in a spring-mass-damper system. The requirements for drive and sense modes differ significantly.

3.24.1 Drive Mode Quality Factor: Maximize

For the drive resonator:

- High Q means less energy required to sustain oscillation (lower power consumption)
- Sharper filtering around drive frequency improves rejection of off-frequency disturbances
- Lower phase noise in the oscillator loop
- Typical values: 10,000–30,000 or even higher in vacuum-packaged devices

The design guideline is unambiguous: **make drive mode Q as high as practically achievable**.

3.24.2 Sense Mode Quality Factor: Trade-offs

The sense mode presents a more complex optimization:

- **High Q advantages**: Lower mechanical noise (Brownian noise), higher signal amplitude at resonance
- **High Q disadvantages**: Longer ring-down time means shock or vibration inputs can cause prolonged oscillations that interfere with measurement; reduced bandwidth

A shock contains energy across a broad frequency spectrum. If the sense mode has very high Q, even a brief shock can excite oscillations that persist for many cycles, corrupting the output. This creates a fundamental trade-off between noise performance and transient response.

3.24.3 Natural Q Differentiation via Electrode Geometry

Fortunately, the electrode configurations themselves provide a natural way to achieve different Q values for drive and sense modes:

- **Drive mode (comb fingers)**: Motion is parallel to electrode fingers (sliding). This generates **squeeze film damping**? Actually, sliding motion produces primarily **viscous drag** with relatively low damping coefficient. The fluid is not significantly compressed; it flows around the fingers.
- **Sense mode (parallel plates)**: Motion is perpendicular to plates (closing/opening gap). This generates **squeeze film damping**—the fluid between plates must be squeezed out or drawn in, creating significant damping forces, especially at small gaps.

Thus, even in the same vacuum package, the sense mode naturally exhibits lower Q than the drive mode due to the different damping mechanisms. This aligns well with design objectives: high Q for efficient drive oscillation, moderate Q for acceptable transient response in the sense mode.

3.24.4 Operating Frequency Selection

Drive frequency selection involves several considerations:

- **Lower bound**: Must be well above the frequency range of expected vibrations and accelerations (typically a few kHz) to enable effective filtering. 20 kHz is a common lower bound.
- **Upper bound**: Limited by increasing quadrature error (an electromechanical imperfection to be discussed) and by power consumption (higher frequency requires more drive power).
- **Typical range**: 20–50 kHz represents a good compromise for most applications.

3.25 Drive and Sense Frequency Relationship

An important design question: should drive and sense resonant frequencies be matched or separated?

This will be addressed in detail in subsequent discussions, but the trade-off can be summarized:

- **Matched frequencies**: Maximum mechanical gain (signal amplitude) because the Coriolis force excites the sense mode at its resonance. However, this also maximizes sensitivity to parameter variations and can exacerbate certain errors.
- **Split frequencies**: Reduced gain but improved stability and bandwidth. The sense mode is operated off-resonance, making the response less sensitive to exact frequency matching.

The optimal choice depends on target application, performance requirements, and the ability to control and match frequencies through design and fabrication.

Figure 3.7: Mechanical stoppers (bumpers) limit maximum displacement to prevent short circuits or structural damage.

3.26 Mechanical Stoppers (Travel Limiters)

Practical devices include **mechanical stoppers** (also called travel limiters or over-travel stops):

Functions:

- **Prevent short circuits**: The moving rotor structure and fixed stator electrodes are typically at different DC bias voltages. If they contact, a short circuit can occur, potentially damaging electronics or the device itself.
- **Prevent structural damage**: Excessive displacement could overstress suspension springs or cause fracture.
- **Limit travel**: Stoppers engage when displacement exceeds the normal operating range, providing a mechanical hard stop.

Stoppers are designed with sufficient clearance to avoid interfering with normal motion but engage before dangerous contact occurs. They are typically placed along both drive and sense directions.

3.27 Quadrature Compensation Electrodes

As noted in the structure analysis, specific electrodes are dedicated to **quadrature error compensation**. Quadrature error is an electromechanical imperfection that will be examined in detail in subsequent lectures. For now, note that:

- Quadrature error produces a signal at the sense output that is in quadrature phase (90° offset) from the Coriolis signal
- It arises from stiffness coupling between drive and sense modes due to fabrication imperfections
- Compensation electrodes apply electrostatic forces to cancel this coupling
- These electrodes are challenging to implement without the double-decoupled architecture

For high-performance devices, quadrature compensation is essentially mandatory.

3.28 Monolithic Three-Axis Gyroscopes

Beyond the discrete yaw and roll/pitch structures examined, there is interest in **monolithic three-axis gyroscopes**—a single mechanical structure sensitive to angular rate about all three axes simultaneously.

3.28.1 Operating Principle

- A single drive motion is established, but it involves motion along **two directions simultaneously** (Lissajous figures or more complex trajectories)
- With velocity components along multiple axes, Coriolis acceleration can occur for angular rates about all three axes
- Sense electrodes are designed to detect displacements along the three corresponding directions
- Three sense modes are implemented within the same structure

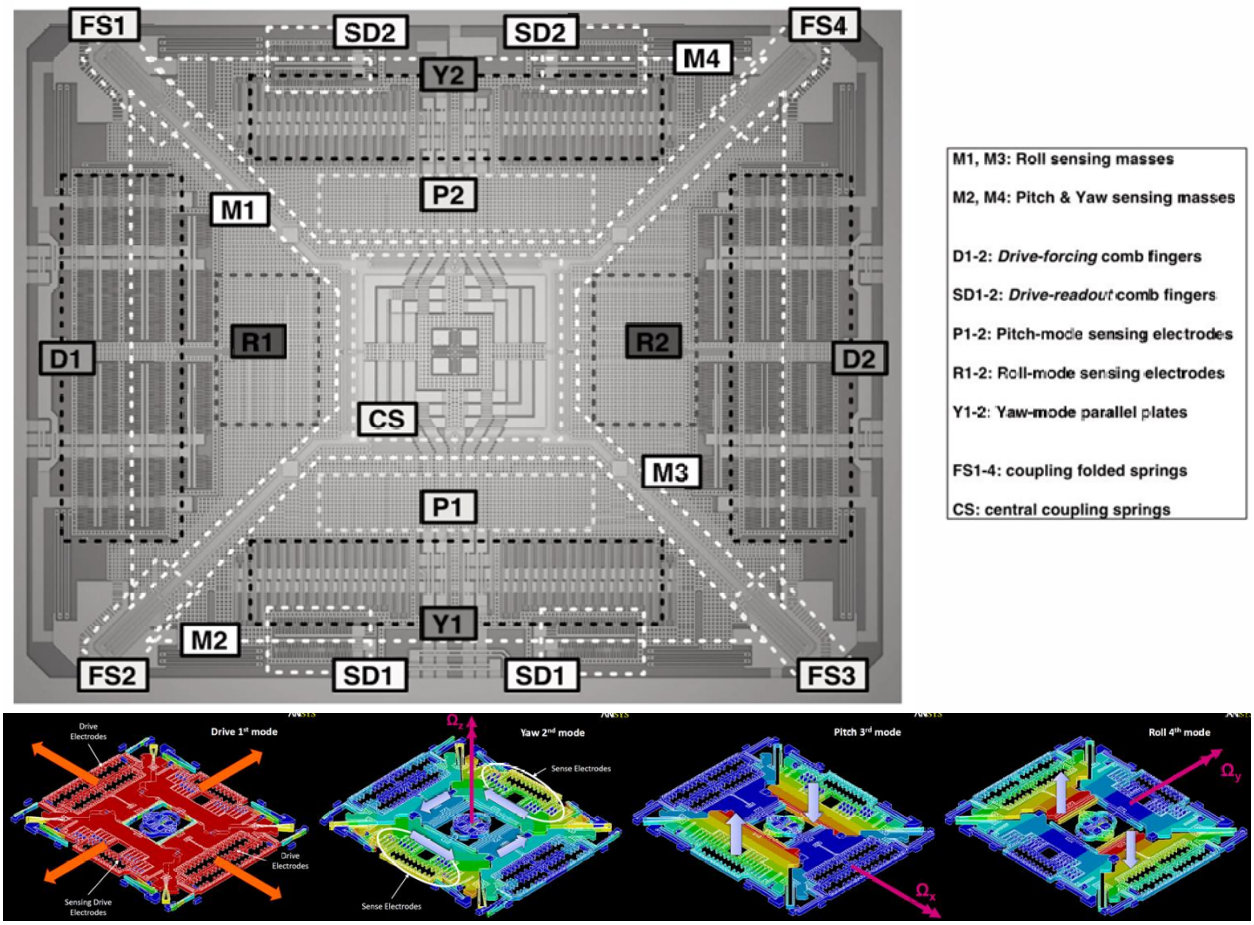

Figure 3.8: 3-axis monolithic gyroscope.

3.28.2 Advantages

- **Single drive oscillator**: Only one drive loop needed, reducing power consumption significantly
- **Area efficiency**: One structure instead of three separate gyroscopes
- **Potential cost reduction**: Smaller die area, simpler assembly

3.28.3 Challenges

- **Quadrature compensation complexity**: With multiple simultaneous motions, implementing compensation electrodes becomes extremely challenging
- **Mode separation**: Ensuring distinct, well-behaved sense modes for each axis is difficult
- **Cross-axis sensitivity**: Parasitic coupling between axes is harder to control

3.28.4 Current Status

- Primarily used in **consumer applications** where low power and small size outweigh ultimate performance requirements
- Some manufacturers (e.g., Murata) have introduced three-axis devices with competitive performance using alternative principles

- High-performance applications still typically use separate structures for each axis
- Ongoing research aims to improve three-axis performance for broader applications

3.29 Transition to Performance Analysis

Having established the structural and design principles of MEMS gyroscopes, we now transition to analyzing their performance in terms of the specifications discussed at the beginning of the course. To understand parameters such as:

- Sensitivity (scale factor)
- Noise equivalent rate
- Bandwidth
- Dynamic range
- Linear acceleration rejection

we first need to develop a complete model of the transduction chain: from input angular rate Ω to output voltage V_{out}. This model will incorporate:

- Mechanical domain: Coriolis force generation, sense mode dynamics
- Electrostatic domain: Capacitance variation with displacement
- Electronic domain: Current generation, transimpedance amplification, demodulation

The following sections will develop this transduction model, providing the foundation for quantitative performance analysis and specification derivation.

3.30 General Sensitivity for Accelerometer Subsystem

To analyze gyroscope performance, we first examine the general displacement sensitivity of an accelerometer to an external acceleration. This approach is logical because, as established, the sense subsystem of a gyroscope is fundamentally an accelerometer that responds to the Coriolis acceleration. Once we understand the displacement response to acceleration, extending the analysis to Coriolis acceleration and ultimately to angular rate becomes straightforward.

3.30.1 Transfer Function from Acceleration to Displacement

Consider the sense mode (y-direction) of our yaw-axis gyroscope. The relationship between displacement y and an applied force F is that of a spring-mass-damper system:

$$\frac{y}{F} = \frac{1/m}{\omega_s^2 - \omega^2 + j\frac{\omega\omega_s}{Q_s}}$$

where ω_s is the sense mode resonant frequency and Q_s is its quality factor. Since inertial force from acceleration is $F = ma$, the displacement per unit acceleration becomes:

$$\frac{y}{a} = \frac{1}{\omega_s^2 - \omega^2 + j\frac{\omega\omega_s}{Q_s}}$$

The magnitude of this transfer function, assuming $Q_s \gg 1$, exhibits the characteristic resonant peak shown schematically in Figure 3.9.

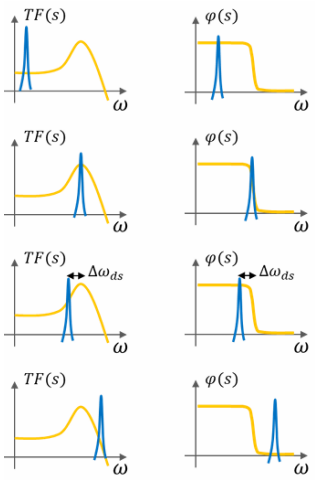

Figure 3.9: Magnitude and phase of the sense mode transfer function y/a. Three operating regions are identified: quasi-static (far below resonance), resonant (at resonance), and split-mode (near but not at resonance).

3.30.2 Operating Regimes

Depending on the frequency ω of the input acceleration relative to the sense mode resonance ω_s, three distinct operating regimes emerge:

Quasi-Static Regime ($\omega \ll \omega_s$)

When the acceleration frequency is much lower than the sense resonance, the inertial and damping terms are negligible compared to the stiffness term. The transfer function simplifies to:

$$\frac{y}{a} \approx \frac{1}{\omega_s^2}$$

This regime offers the simplest relationship: displacement is proportional to acceleration with no frequency-dependent amplification and zero phase shift. However, the sensitivity is relatively low.

Resonant Regime ($\omega = \omega_s$)

When the acceleration frequency exactly matches the sense resonance, the transfer function is dominated by the damping term:

$$\frac{y}{a} = \frac{Q_s}{\omega_s^2} \angle -90^\circ$$

This provides maximum sensitivity, amplified by the quality factor Q_s. The -90° phase shift is characteristic of resonance. This is the regime used for drive mode excitation (though with electrostatic force rather than inertial acceleration).

Split-Mode Regime ($\omega \approx \omega_s$, but not equal)

When the acceleration frequency is near but not exactly at resonance, the response lies between the quasi-static and resonant cases. The displacement per unit acceleration can be expressed as:

$$\frac{y}{a} = \frac{Q_{\text{eff}}}{\omega_s^2}$$

where Q_{eff} is an **effective quality factor** that depends on the frequency separation $\Delta\omega = |\omega - \omega_s|$. For a given offset, $Q_{\text{eff}} < Q_s$ and, importantly, is relatively insensitive to the exact value of Q_s. The phase shift in this region transitions gradually from $0°$ to $-90°$ to $-180°$ as frequency increases through resonance.

Post-Resonance Regime ($\omega \gg \omega_s$)

Far above resonance, the mass term dominates, and sensitivity rolls off rapidly with a $-180°$ phase shift. This regime is not useful for sensing.

3.31 Application to Gyroscope Sensitivity

In a gyroscope, the acceleration of interest is the Coriolis acceleration:

$$a_c = 2\Omega v_d$$

where Ω is the angular rate and v_d is the drive velocity. For sinusoidal drive motion at frequency ω_d with displacement amplitude X_d, the velocity amplitude is $\omega_d X_d$, giving:

$$a_c = 2\Omega\omega_d X_d \sin(\omega_d t)$$

The Coriolis acceleration is therefore a sinusoidal signal at the drive frequency ω_d. Substituting into the accelerometer transfer function yields the displacement in the sense direction per unit angular rate:

$$\frac{y}{\Omega} = 2\omega_d X_d \cdot \frac{y}{a}\bigg|_{\omega=\omega_d}$$

The specific form depends on the relationship between ω_d and the sense mode resonance ω_s.

3.31.1 Quasi-Static Operation ($\omega_d \ll \omega_s$)

If the drive frequency is well below the sense resonance:

$$\frac{y}{\Omega} = \frac{2\omega_d X_d}{\omega_s^2}$$

This configuration is simple and stable but offers low sensitivity. It is rarely used in practice.

3.31.2 Mode-Matched Operation ($\omega_d = \omega_s$)

When the drive frequency exactly matches the sense resonance:

$$\frac{y}{\Omega} = \frac{2\omega_d X_d Q_s}{\omega_s^2} = \frac{2X_d Q_s}{\omega_s}$$

This provides maximum sensitivity, amplified by the sense mode quality factor Q_s. The displacement is in phase quadrature ($-90°$) relative to the drive motion.

3.31.3 Split-Mode Operation ($\omega_d \approx \omega_s$)

When the drive frequency is close to but not exactly at the sense resonance:

$$\frac{y}{\Omega} = \frac{2\omega_d X_d Q_{\text{eff}}}{\omega_s^2}$$

where Q_{eff} depends on the frequency separation $\Delta\omega = |\omega_d - \omega_s|$. For a fixed offset, Q_{eff} is relatively insensitive to variations in Q_s.

3.31.4 Post-Resonance Operation ($\omega_d \gg \omega_s$)

This regime yields very low sensitivity and is not used.

3.32 Practical Operating Modes in MEMS Gyroscopes

The choice of operating mode involves fundamental trade-offs among sensitivity, bandwidth, stability, and fabrication complexity.

3.32.1 Mode-Matched Operation

Advantages:

- Maximum sensitivity (proportional to Q_s)
- Highest signal-to-noise ratio potential

Disadvantages:

- Limited bandwidth (the sense mode's resonant peak is narrow)
- Extreme sensitivity to frequency matching: fabrication tolerances make perfect matching difficult without post-fabrication trimming or active tuning
- Sensitivity to temperature variations, which shift both ω_d and ω_s
- Phase shift of exactly $-90°$ requires careful demodulation alignment

3.32.2 Split-Mode Operation

Advantages:

- Good sensitivity (though lower than mode-matched) with moderate gain from Q_{eff}
- Much less sensitive to exact frequency matching—relaxes fabrication tolerances
- Improved bandwidth
- **Temperature robustness**: The gain in split-mode operation is largely independent of the sense mode quality factor Q_s. Since Q_s typically varies with temperature, this provides inherent stability, unlike the peak gain which varies significantly..

Disadvantages:

- Lower sensitivity than mode-matched operation
- Some residual phase shift that may vary with temperature

3.32.3 Industry Practice

For these reasons, many state-of-the-art MEMS gyroscopes operate in **split-mode** configuration. This choice:

- Relaxes the requirement for perfect mode matching, simplifying design and fabrication
- Provides robust performance across temperature without active compensation
- Offers a good compromise between sensitivity and bandwidth
- Reduces the impact of sense mode Q variations on scale factor stability

High-performance devices may incorporate electrostatic tuning to adjust frequencies dynamically, but split-mode operation remains a common and effective approach. In the following analyses, we will maintain equations for both mode-matched and split-mode operation to illustrate the trade-offs quantitatively.

3.33 Mode-Matched Operation

In mode-matched operation, the drive oscillator maintains the structure in oscillation at the drive resonance frequency ω_d. When an angular rate Ω is applied about the sensitive axis, the resulting Coriolis force excites the sense mode at the same frequency ω_d. Since the modes are matched ($\omega_d = \omega_s$), the sense mode is excited exactly at its resonance, and the displacement amplitude is amplified by the sense mode quality factor Q_s.

The displacement in the sensing direction per unit angular rate can be derived as:

$$\frac{|y|}{\Omega} = \frac{2X_d Q_s}{\omega_s} = \frac{2X_d}{\Delta\omega_{\text{FWHM}}}$$

where X_d is the drive displacement amplitude, ω_s is the sense resonant frequency, and $\Delta\omega_{\text{FWHM}} = \omega_s/Q_s$ is the full width at half maximum (the 3 dB bandwidth) of the sense mode resonance peak. This expression reveals that the sensitivity is inversely proportional to the width of the resonance peak—a higher Q_s gives a narrower peak and higher sensitivity.

3.33.1 Advantages of Mode Matching

- **Maximum sensitivity**: The displacement per unit angular rate is amplified by the sense mode quality factor Q_s, providing the largest possible mechanical gain.
- **Signal-to-noise ratio**: Higher mechanical gain directly improves the signal level relative to electronic noise, potentially enabling better overall performance.

3.33.2 Disadvantages and Challenges

- **Narrow bandwidth**: The sense mode's resonant peak width $\Delta\omega_{\text{FWHM}}$ determines the system bandwidth. High Q_s gives a very narrow peak, limiting the frequency range over which the gyroscope can respond to varying angular rates.
- **Temperature sensitivity**: Both the quality factor Q_s and the resonant frequencies ω_d and ω_s vary with temperature. Since sensitivity in mode-matched operation is proportional to Q_s, any temperature-induced change in Q_s directly affects the scale factor (sensitivity). Similarly, frequency shifts can cause the modes to become mismatched, drastically reducing sensitivity.
- **Fabrication tolerances**: Achieving perfect mode matching despite process variations is extremely challenging. A mismatch of even a few tens of hertz can significantly reduce sensitivity, especially for high-Q sense modes.
- **Mode matching circuitry**: Maintaining mode matching across temperature and lifetime typically requires active tuning, often by applying electrostatic forces to adjust the sense mode frequency. This adds circuit complexity, increases power consumption, and requires additional calibration and control loops.
- **Phase stability**: The exact $-90°$ phase shift at resonance requires precise demodulation alignment; any phase errors directly affect the output signal.

For these reasons, mode-matched operation, while offering the highest theoretical sensitivity, is challenging to implement robustly in practical systems without significant overhead. It is typically reserved for high-end applications where performance justifies the complexity, or combined with sophisticated control loops that continuously maintain matching.

3.34 Mode-Split Operation

In mode-split operation, the drive frequency ω_d is deliberately designed to be offset from the sense mode resonance ω_s by a fixed amount $\Delta\omega = |\omega_d - \omega_s|$. Typical drive frequencies are in the 20–30 kHz range, while the split $\Delta\omega$ is usually chosen between a few hundred hertz and about one kilohertz. This split is

intentionally much larger than the 3 dB bandwidth of the sense mode ($\Delta\omega \gg \omega_s/Q_s$), ensuring that the drive frequency lies on the sloping side of the sense resonance peak rather than at its apex.

3.34.1 Advantages of Mode Splitting

- **Relaxed matching requirements**: The exact value of the frequency split is not critical. If the target split is 600 Hz and fabrication results in 580 Hz or 650 Hz, performance remains essentially unchanged. This dramatically reduces sensitivity to process variations.
- **Temperature robustness**: The gain in split-mode operation is largely independent of the sense mode quality factor Q_s. Since Q_s typically varies with temperature, this provides inherent scale factor stability. The sensitivity depends primarily on the well-controlled frequency split $\Delta\omega$ rather than on Q_s.
- **Extended bandwidth**: The system bandwidth is no longer limited by the narrow resonant peak but is instead determined by the overall sense mode transfer function, which is much wider off-resonance.
- **Simplified electronics**: No active mode matching circuitry is required, reducing power consumption and design complexity.
- **Predictable phase**: The phase shift at the operating point is relatively stable and can be accounted for in demodulation.

3.34.2 Disadvantages

- **Lower sensitivity**: The gain is reduced compared to mode-matched operation. The effective quality factor Q_{eff} in split mode is typically in the range of 10–50, compared to a Q_s that might be several thousand. This represents a sensitivity reduction of one to two orders of magnitude.
- **Still requires drive amplitude stabilization**: The drive displacement X_d may still vary with temperature due to changes in the drive mode quality factor, requiring separate amplitude control circuitry.

3.34.3 Comparison Summary

Feature	Mode-Matched	Mode-Split
Sensitivity	High ($\propto Q_s$)	Moderate ($\propto Q_{\text{eff}}$)
Bandwidth	Narrow ($\approx \omega_s/Q_s$)	Wider
Temperature stability	Poor (gain $\propto Q_s(T)$)	Good (gain independent of Q_s)
Fabrication tolerance	Extremely tight	Relaxed
Circuit complexity	High (active tuning)	Low
Power consumption	Higher	Lower
Phase at operating point	Exactly $-90°$	Varies with split

For these reasons, many state-of-the-art MEMS gyroscopes operate in split-mode configuration, accepting the moderate sensitivity reduction in exchange for robust, stable performance with lower complexity and power consumption. The sensitivity loss can often be compensated through careful electronic design and does not necessarily degrade the overall noise performance if other noise sources dominate.

3.35 Summary of Gyroscope Subsystems

We have now established that a MEMS gyroscope comprises two main electromechanical subsystems:

1. **Drive resonator**: A spring-mass-damper system along the x-direction, maintained in sustained oscillation at its resonant frequency ω_d by electrostatic actuation and feedback electronics.

2. **Sense accelerometer**: A spring-mass-damper system along the y-direction (for yaw sensing) that responds to the Coriolis acceleration $a_c = 2\Omega v_d$.

These two subsystems are designed with slightly split resonant frequencies ($\omega_d \neq \omega_s$) for the reasons discussed above. The remaining discussion will focus on the electronic interfaces required to:

- Sustain the drive oscillation (drive oscillator electronics)
- Read out the sense displacement (capacitive sensing electronics)
- Demodulate the sense signal to recover the angular rate information

Today we will address the drive oscillator; the sense channel and demodulation will be covered in the next class, followed by analysis of noise, bandwidth, and overall system performance.

3.36 Electronics Overview

Figure 3.10 shows a simplified block diagram of the complete gyroscope system.

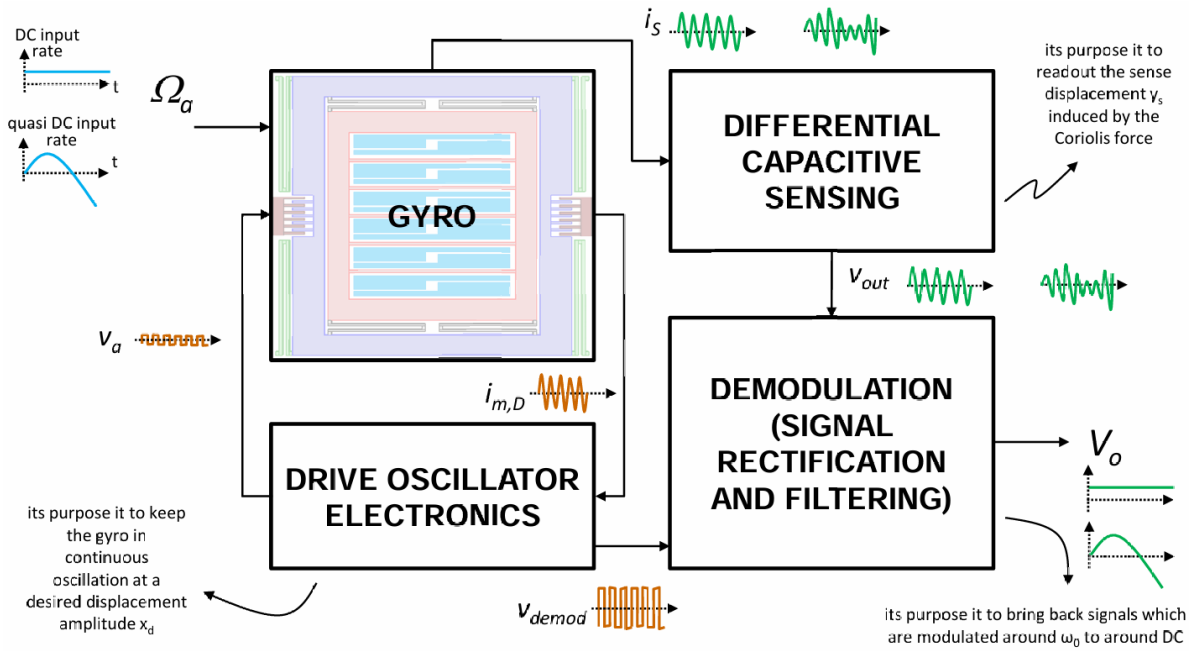

Figure 3.10: Simplified block diagram of a MEMS gyroscope system showing drive oscillator loop and sense channel.

The system operates as follows:

- The **drive resonator** is sustained in oscillation at ω_d by a feedback loop that compensates mechanical losses. The drive motion is detected capacitively (drive sense electrodes), producing a current i_d at frequency ω_d. This signal is amplified, conditioned, and fed back to the drive actuators (drive electrodes) with appropriate phase and amplitude to maintain constant oscillation amplitude.
- When an angular rate Ω is applied, the resulting Coriolis force produces a sense-mode displacement also at frequency ω_d. This displacement is detected by the sense electrodes, producing a current i_s at ω_d whose amplitude is proportional to Ω.
- The sense signal is amplified by a transimpedance amplifier (TIA) and then fed to a **demodulator**. The demodulation reference is derived from the drive oscillator (e.g., the drive sense signal). Synchronous demodulation shifts the signal from ω_d down to baseband.

- A low-pass filter removes the high-frequency components, leaving an output voltage V_{out} that replicates the original angular rate waveform $\Omega(t)$.

For a constant angular rate Ω_0, the sense current is a pure sinusoid at ω_d. For a time-varying rate $\Omega(t)$, the sense current is amplitude-modulated: its envelope follows $\Omega(t)$ while its carrier is at ω_d. Demodulation recovers the envelope.

3.36.1 Focus: Drive Oscillator Electronics

The remainder of this lecture focuses on the drive oscillator loop. Key requirements:

- Sustain oscillation at the mechanical resonant frequency ω_d
- Maintain constant oscillation amplitude (velocity) despite temperature variations and aging
- Provide a clean, low-noise reference signal for demodulation
- Achieve these functions with low power consumption and minimal external components

3.37 Loop Gain in Oscillators

Before examining the specific circuit implementation, we review the fundamental condition for sustained oscillation.

3.37.1 Contrast with Negative Feedback

In negative feedback systems (e.g., operational amplifier circuits), we typically design for a large loop gain magnitude with $-180°$ phase shift to achieve precise closed-loop transfer functions. Stability is maintained by ensuring adequate phase margin.

In oscillators, we seek **positive feedback** to create a self-sustaining signal. The Barkhausen criterion states that for sustained oscillation, the loop gain must satisfy:

$$|H_{\text{loop}}| = 1 \quad \text{and} \quad \angle H_{\text{loop}} = 0° \text{ (or } 360°)$$

If $|H_{\text{loop}}| > 1$, the oscillation amplitude would grow without bound (until nonlinearities limit it). If $|H_{\text{loop}}| < 1$, the oscillation would decay. The ideal steady-state condition is exactly unity loop gain.

3.37.2 Applying to the Drive Resonator

As discussed in the previous class, the drive resonator at resonance behaves electrically as an equivalent resistance—it is a dissipative element. The mechanical damping removes energy from the system. To sustain oscillation, the electronics must inject exactly enough energy to compensate these losses.

The oscillator loop therefore consists of:

- The **mechanical resonator** (drive mode) with its transfer function from actuation force to displacement
- The **electrostatic actuator** that converts voltage to force
- The **capacitive displacement sensor** that converts motion to current
- An **electronic amplifier** that provides gain and phase shaping

The loop gain is the product of all these transfer functions around the loop. By designing the electronics to provide the correct gain and phase shift (ensuring $0°$ total phase at ω_d), and by incorporating amplitude control to maintain unity gain, a stable oscillation is established.

3.38 Loop Gain Analysis for Drive Oscillator

To design a self-sustaining drive oscillator, we must understand the loop gain conditions and how they are achieved in practice.

3.38.1 Barkhausen Criterion in the Context of the Drive Loop

In the simplified model of our drive loop, neglecting nonlinearities momentarily, the loop gain consists of two main components:

- The **electromechanical transfer function** of the drive resonator: from actuation voltage V_{in} to motion-induced current i_{out}. As derived previously, at resonance this behaves as an equivalent resistance R_m.
- The **electronic gain** A_{elec} of the amplifier and any additional stages.

The total loop gain is therefore $H_{\text{loop}} = A_{\text{elec}} \cdot (i_{\text{out}}/V_{\text{in}})$. For sustained oscillation, we require:

$$|H_{\text{loop}}| = 1 \quad \text{and} \quad \angle H_{\text{loop}} = 0^\circ$$

3.38.2 Start-Up Condition vs. Steady-State Condition

In practice, it is impossible to design a circuit that exactly satisfies $|H_{\text{loop}}| = 1$ at start-up due to uncertainties in:

- Mechanical parameters (quality factor, stiffness, mass) from fabrication tolerances
- Electronic component values (resistors, capacitors, transconductance)
- Temperature and aging effects

Therefore, the design strategy is to ensure that at **start-up**, the loop gain is significantly greater than unity—typically 10–20 dB (i.e., a factor of 3–10 in voltage gain). This guarantees that oscillation will begin to grow from noise.

The transition from start-up to steady-state oscillation is governed by **nonlinearity** somewhere in the loop. As the oscillation amplitude increases, some element (typically an amplifier) begins to saturate, reducing its effective gain. The system naturally settles at the amplitude where the average loop gain around the cycle is exactly unity, satisfying the Barkhausen criterion in steady state.

This is illustrated conceptually in Figure 3.11:

3.38.3 Oscillation Buildup in Time Domain

When the loop is powered on, there is no initial motion. However, electronic noise and thermomechanical noise (to be discussed in the next class) provide a broadband stimulus. Among all frequency components present, only the drive resonant frequency ω_d satisfies the Barkhausen phase condition ($\angle H_{\text{loop}} = 0^\circ$). At this frequency, the loop gain exceeds unity, so any component at ω_d is amplified each time it circulates around the loop, growing exponentially.

After many cycles, the amplitude increases until it is limited by the supply voltages of the electronic stages (e.g., ± 5 V). At this point, the amplifiers saturate, and the waveform becomes clipped (square-wave-like). The nonlinear saturation reduces the effective gain around the loop until the average loop gain drops to exactly unity, establishing a stable oscillation amplitude.

3.39 Circuit Implementation for Push-Pull Drive and Differential Sensing

We now examine a practical implementation of the drive oscillator electronics, suitable for the push-pull actuated, differentially sensed drive mode.

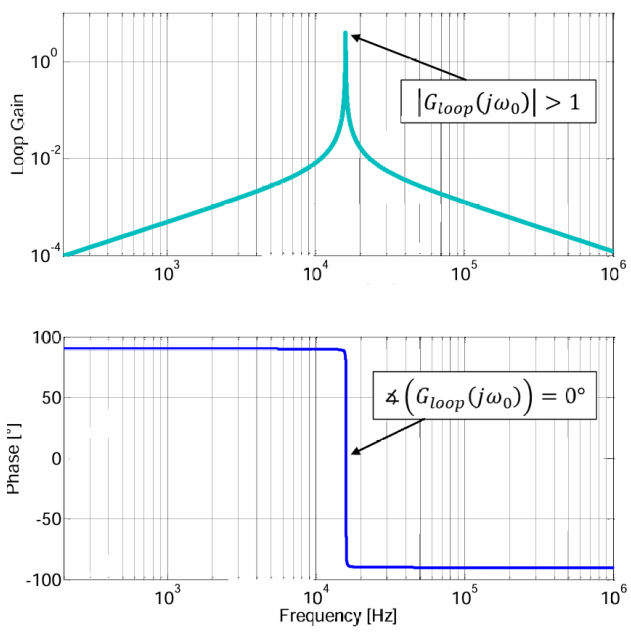

Figure 3.11: Nonlinear gain control: small-signal gain is high (> 1), but as amplitude increases, saturation reduces the effective gain until it reaches exactly 1 at the desired operating amplitude.

3.39.1 System Configuration

Figure 3.12 shows the complete drive oscillator system.

Key components:

- **Push-pull drive electrodes** (red and green): Actuated by anti-phase voltages $V_{\text{drive+}}$ and $V_{\text{drive-}}$.
- **Differential drive sense electrodes** (light blue and purple): Produce motion-induced currents i_{s1} and i_{s2}.
- **Fully differential charge amplifier**: Converts the differential current to a differential voltage.
- **90° phase shifter**: Corrects the phase relationship to satisfy Barkhausen criterion.
- **Amplitude control stage**: Provides adjustable drive amplitude and ensures stable oscillation.

3.39.2 Differential Charge Amplifier

The drive sense electrodes are biased with a DC voltage V_{DC} on the rotor, while the fixed sense electrodes are connected to the inputs of a fully differential amplifier. These inputs are maintained at virtual ground by the amplifier's feedback action.

The motion-induced currents from the two sense electrodes are equal in magnitude but opposite in sign:

$$i_{s1} = V_{\text{DC}} \frac{dC_1}{dt}, \quad i_{s2} = V_{\text{DC}} \frac{dC_2}{dt} = -i_{s1}$$

These currents flow into the amplifier through feedback impedances. In a **charge amplifier** configuration, the feedback path is dominated by capacitors C_f (with resistors R_f providing DC bias stability but having negligible effect at the drive frequency). The transfer function from input current to output voltage is:

$$V_{\text{out, diff}} = \frac{i_{s1} - i_{s2}}{sC_f} = \frac{2i_{s1}}{sC_f}$$

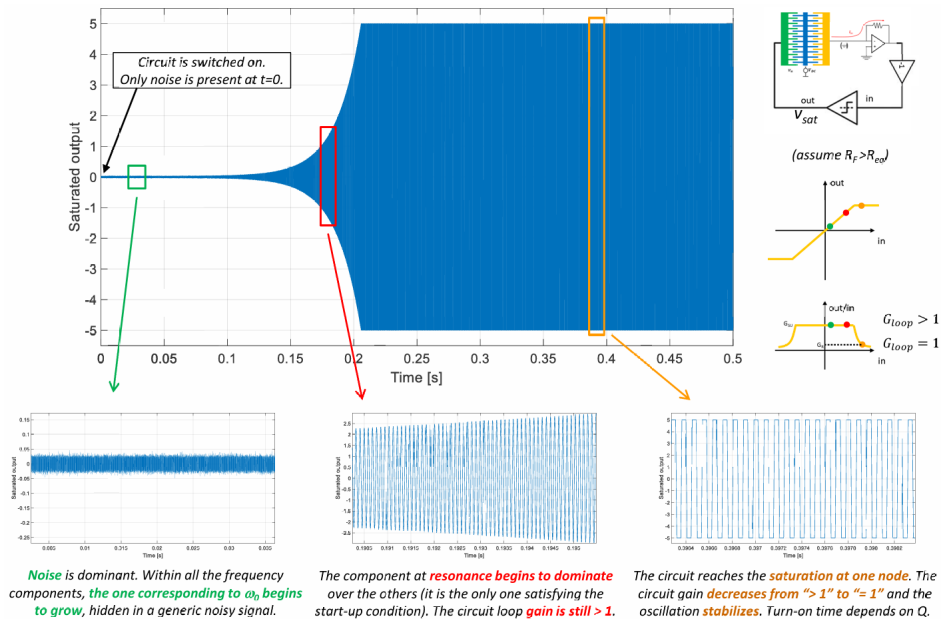

Figure 3.12: Complete drive oscillator circuit including push-pull actuation, differential sense, charge amplifier, phase shifter, and amplitude control.

The choice of capacitive feedback (rather than resistive) is motivated by noise considerations, which will be discussed in the next class. For now, note that the gain magnitude at frequency ω_d is $2/(\omega_d C_f)$.

3.39.3 Phase Analysis Around the Loop

To ensure the Barkhausen phase condition ($\angle H_{\text{loop}} = 0°$), we must carefully account for all phase shifts around the loop. Let us trace the signal path using the drive actuation voltage as phase reference.

1. **Drive actuation voltage**: Assume $V_{\text{drive+}}$ at 0° and $V_{\text{drive-}}$ at 180° (push-pull).

2. **Force to displacement**: At the drive resonance frequency ω_d, the mechanical transfer function from force to displacement has a $-90°$ phase shift. Therefore, displacement x lags the actuation force (and thus the actuation voltage) by 90°.

3. **Displacement to velocity**: Velocity is the time derivative of displacement, introducing a $+90°$ phase shift. Thus, velocity is in phase with the actuation voltage (0° and 180° for the two halves).

4. **Velocity to motion-induced current**: As derived, $i_s = V_{\text{DC}}(dC/dx)\dot{x}$. The current is proportional to velocity, so it shares the same phase: 0° for i_{s1} and 180° for i_{s2}.

5. **Charge amplifier**: The transfer function $1/(sC_f)$ introduces a $-90°$ phase shift (since $1/j = -j$). Therefore, the differential output voltage $V_{\text{out, diff}}$ lags the input current by 90°. Relative to our original reference, $V_{\text{out, diff}}$ is at $-90°$ and $+90°$ (or equivalently 270°).

At this point, the signal is in quadrature (90° phase shift) relative to the original actuation voltage. Connecting this directly back to the drive electrodes would not satisfy the Barkhausen phase condition (which requires 0° total phase shift).

3.39.4 90° Phase Shifter

To correct the phase, we insert a **90° phase shifter** after the charge amplifier. This can be implemented as:

- An **integrator** (providing $-90°$ shift)
- A **differentiator** (providing $+90°$ shift)

The circuit topology is similar in both cases: an operational amplifier with a capacitor and resistor in the feedback and input paths. By choosing component values appropriately, the stage can be designed to provide exactly $+90°$ or $-90°$ at the drive frequency ω_d. The choice depends on the polarity of connections; what matters is that the net phase shift from the charge amplifier output back to the drive electrodes becomes $0°$ (or $360°$).

After the phase shifter, the signals are again in phase with the original actuation voltage ($0°$ and $180°$), ready to close the loop.

3.39.5 Amplitude Control and Nonlinearity

The phase shifter output could theoretically be connected directly to the drive electrodes. However, several practical considerations motivate additional stages:

- **Controlled amplitude**: We need to set the drive displacement amplitude X_d to a precise value (e.g., 10 μm). This requires controlling the voltage amplitude applied to the drive electrodes.
- **Guaranteed nonlinearity**: While any amplifier will eventually saturate, providing a dedicated nonlinear stage gives predictable and stable amplitude control.
- **Start-up behavior**: The loop gain must be ¿1 for small signals but reduce to exactly 1 at the desired operating point.

A common implementation adds a **high-gain saturating amplifier** (often called a "limiter" or "H-bridge driver") after the phase shifter. This stage:

- Provides very high gain, ensuring that even small signals at its input produce full output swing.
- Saturates between fixed voltage levels, creating a square-wave output.
- The saturation amplitude can be controlled by adjusting its supply voltage (or by using a separate control voltage V_{ctrl}).

The square-wave drive is acceptable because the mechanical resonator acts as a high-Q bandpass filter, responding primarily to the fundamental Fourier component of the square wave. The effective drive amplitude is proportional to V_{ctrl}.

3.39.6 Complete Loop Operation

The full drive oscillator operates as follows:

1. At start-up, noise at ω_d circulates around the loop. The small-signal loop gain (including the high-gain saturating amplifier in its linear region) is > 1.
2. The signal grows exponentially.
3. Eventually, the saturating amplifier begins to clip, reducing its effective gain for the fundamental component.
4. A stable equilibrium is reached when the average loop gain around the cycle equals exactly 1. The oscillation amplitude at the drive electrodes is determined by V_{ctrl}.
5. The drive displacement amplitude X_d is proportional to this voltage, providing a means to precisely control the Coriolis force magnitude.

3.39.7 Summary of Drive Oscillator Design

The key design steps for the drive oscillator are:

1. Design the mechanical resonator (drive mode) with desired resonant frequency ω_d and quality factor Q_d.

2. Implement a differential charge amplifier to convert motion-induced currents to voltage with minimal added noise.

3. Analyze the phase shifts around the loop and insert a 90° phase shifter to satisfy the Barkhausen phase condition.

4. Include a saturating amplifier with controllable output swing to provide amplitude stabilization and adjustable drive level.

5. Ensure that the small-signal loop gain (before saturation) is significantly ¿1 (e.g., 10–20 dB) to guarantee reliable start-up.

This architecture provides a robust, self-sustaining oscillation at the mechanical resonant frequency, with well-controlled amplitude that can be tuned electronically. The clean reference signal derived from the drive oscillator (e.g., from the phase shifter output) will be used in the next class for synchronous demodulation of the sense signal.

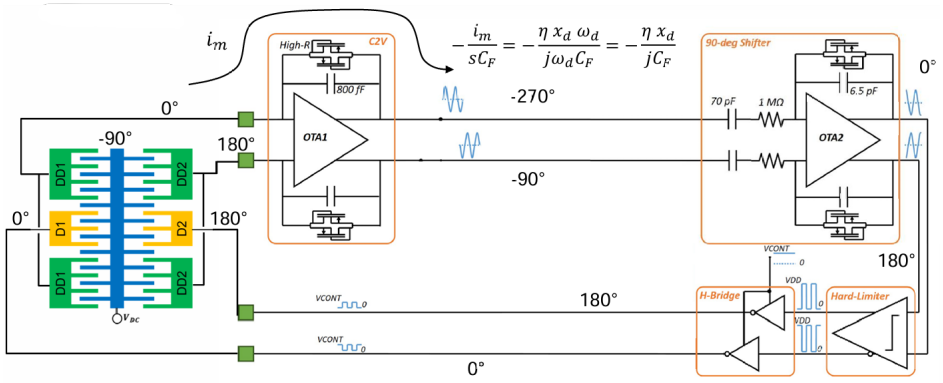

Figure 3.13: Integrated drive mode push-pull oscillator.

3.40 Why Control the Drive Voltage?

The control voltage V_{ctrl} serves a critical function: it directly determines the amplitude of the excitation signals applied to the drive electrodes, and consequently the amplitude of the drive displacement X_d. A larger control voltage produces a larger electrostatic force, resulting in a larger drive motion; a smaller control voltage reduces the motion. Therefore, V_{ctrl} provides a convenient electronic knob to set and maintain the desired drive displacement amplitude, ensuring consistent gyroscope sensitivity.

3.41 Overall Loop Gain Analysis

Figure 3.14 illustrates the transfer functions of individual stages and the resulting overall loop gain.

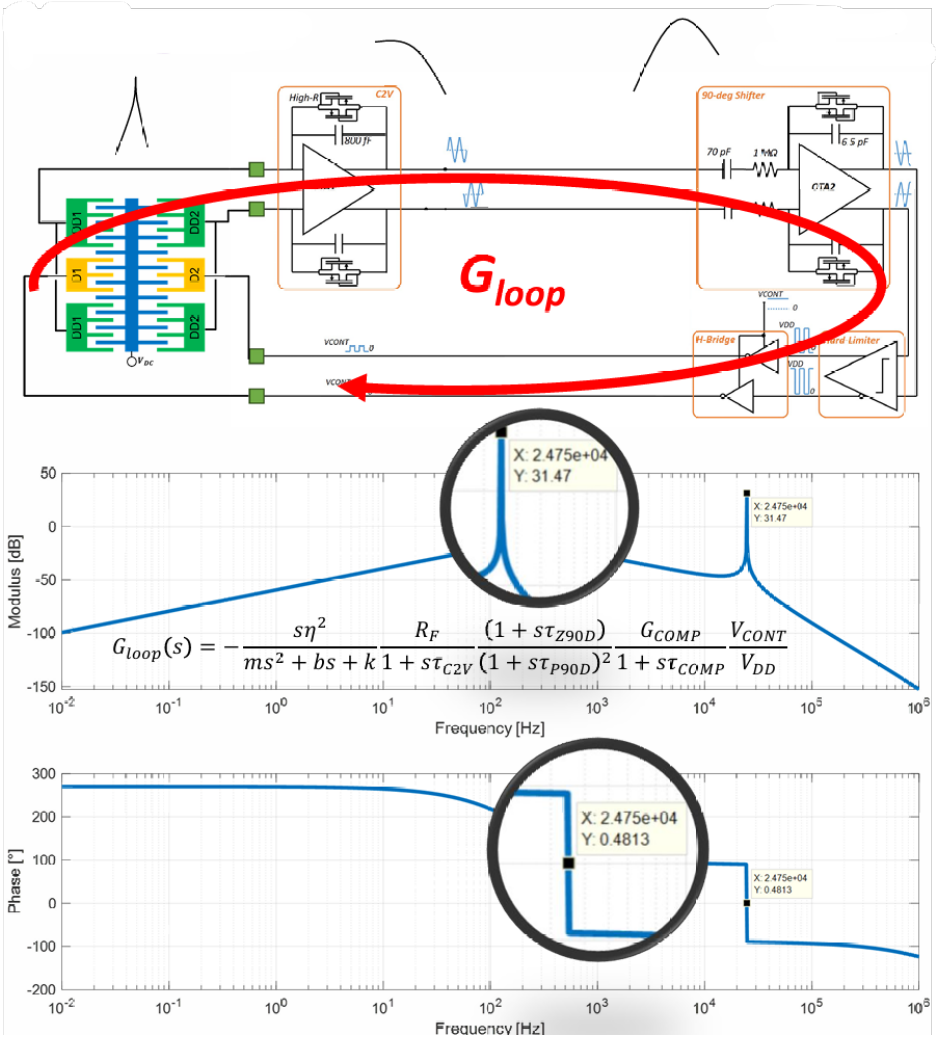

Figure 3.14: Bode plots of individual stages (top) and combined loop gain (bottom) showing the frequency where Barkhausen criteria are satisfied.

3.41.1 Stage Transfer Functions

- **MEMS resonator (drive mode)**: Exhibits a zero at DC (due to the velocity-to-current relationship), a pair of complex conjugate poles at the resonant frequency ω_d with high Q, and then rolls off at -20 dB/decade above resonance.
- **Charge amplifier (C-to-V)**: Configured as an integrator with a -20 dB/decade slope at frequencies above its corner (set by R_f and C_f). At the drive frequency, it provides a $-90°$ phase shift.
- **90° phase shifter**: Designed with a zero at the origin and two poles placed at the same frequency. This stage provides the necessary phase correction ($+90°$ or $-90°$ depending on implementation) to achieve zero net phase around the loop.

3.41.2 Combined Loop Gain

When these stages are cascaded, the overall loop gain $H_{\text{loop}}(s)$ has the following characteristics:

- Multiple zeros at DC (from MEMS and phase shifter) cause the low-frequency gain to rise.

- Poles from the charge amplifier and phase shifter shape the mid-band response.
- At the designed drive frequency ω_d, the loop gain magnitude is intentionally ¿1 (e.g., 31 dB in the example), and the phase shift is exactly 0° (or 360°).
- This unique frequency satisfies the Barkhausen criteria: sufficient gain and correct phase for sustained oscillation.

In the example shown, $\omega_d/(2\pi) = 24.75$ kHz, with a loop gain of approximately 31 dB and zero phase shift.

3.41.3 Design Rule: Pole Placement

To ensure accurate phase relationships, all poles in the electronic stages must be placed far from the operating frequency ω_d. A good rule of thumb is to place poles at least two decades away (either lower or higher) from ω_d. For example:

- If $\omega_d/(2\pi) = 24$ kHz, low-frequency poles (e.g., from the phase shifter) should be below 240 Hz to have negligible phase contribution at ω_d.
- High-frequency poles (e.g., from amplifier bandwidth limitations) should be above 2.4 MHz to avoid introducing additional phase lag.

Failure to follow this guideline results in phase errors that shift the oscillation frequency away from the intended ω_d or even prevent oscillation altogether.

3.42 Temperature Dependence of Quality Factor

The quality factor of the drive mode, particularly when dominated by fluid damping, is strongly temperature-dependent. Typical temperature coefficients are on the order of $-0.5\%/°\text{C}$ to $-1\%/°\text{C}$. Since the drive displacement at resonance is proportional to Q_d for a given actuation force, any variation in Q_d directly affects:

- Drive velocity amplitude $v_d = \omega_d X_d$
- Coriolis force magnitude $F_c = 2m\Omega v_d$
- Overall gyroscope sensitivity (scale factor)

For high-performance applications, scale factor stability requirements are extremely stringent—often better than 0.1% variation over the operating temperature range. Even the relatively modest ±1.5% specification shown in the example (typical for automotive/consumer grades) cannot be achieved without active stabilization.

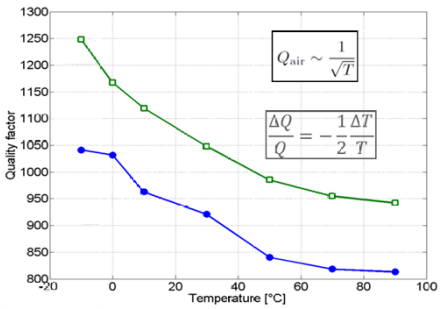

Figure 3.15: Q-factor over temperature.

3.43 Automatic Gain Control (AGC) Loop

To stabilize the drive displacement against temperature-induced Q_d variations, an additional negative feedback loop is incorporated: the **Automatic Gain Control (AGC)** loop.

3.43.1 Conceptual Operation

The AGC loop monitors the amplitude of the drive motion (indirectly through the charge amplifier output) and adjusts the control voltage V_{ctrl} to maintain constant amplitude. This is a classic negative feedback system superimposed on the positive oscillation loop.

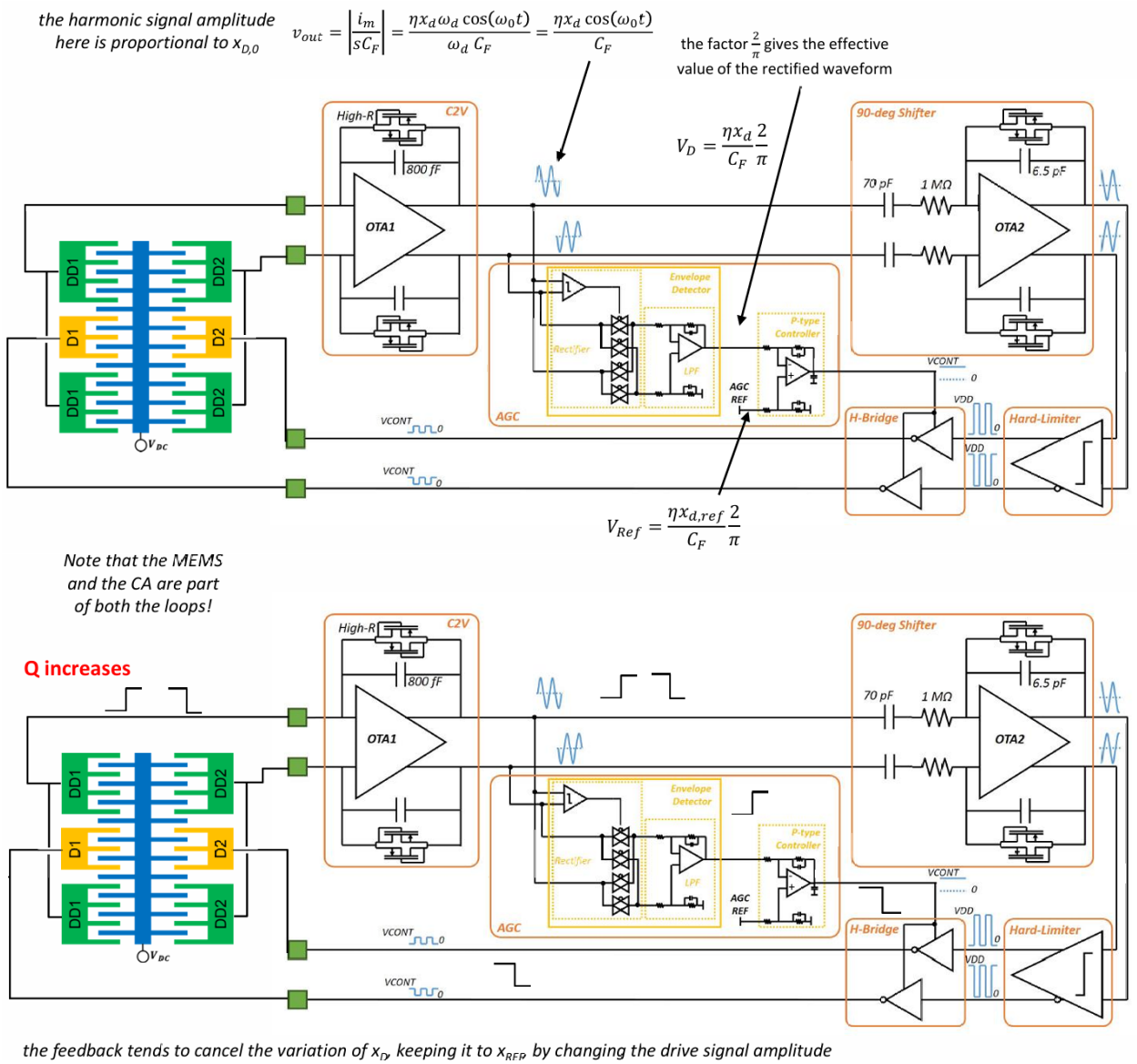

Figure 3.16: AGC loop within the drive oscillator: amplitude detection, comparison with reference, and control voltage adjustment.

3.43.2 Implementation Details

1. **Amplitude detection**: The differential output of the charge amplifier is fed to an envelope detector. This is typically implemented by:
   - Squaring the signals (e.g., using comparators to create square waves)
   - Synchronous demodulation using these square waves to rectify the signal
   - Low-pass filtering to extract the DC amplitude envelope
2. **Comparison**: The envelope detector output V_{env} (proportional to drive displacement amplitude) is compared to a stable reference voltage V_{ref}.
3. **Error amplification**: The difference $V_{\text{err}} = V_{\text{ref}} - V_{\text{env}}$ is amplified by an error amplifier. The output of this amplifier becomes the control voltage V_{ctrl}.
4. **Actuation**: V_{ctrl} sets the amplitude of the square waves applied to the drive electrodes (via the H-bridge driver), thereby controlling the electrostatic force amplitude.

3.43.3 Negative Feedback Action

Consider a temperature increase that reduces Q_d. Without AGC, the drive displacement would decrease. With AGC:

1. Reduced $Q_d \rightarrow$ reduced displacement $\rightarrow$ reduced charge amplifier output $\rightarrow$ reduced V_{env}
2. $V_{\text{env}} < V_{\text{ref}} \rightarrow$ error amplifier increases V_{ctrl}
3. Increased V_{ctrl} increases drive voltage amplitude $\rightarrow$ increased electrostatic force $\rightarrow$ displacement returns toward its original value

Conversely, if Q_d increases (e.g., temperature decrease), V_{env} would increase, causing V_{ctrl} to decrease and reduce the drive force, again stabilizing the displacement.

3.43.4 AGC Loop Stability

The AGC loop must be designed as a stable negative feedback system. Key considerations:

- The loop acts on the **envelope** of the oscillation, not on the instantaneous high-frequency signal.
- The plant (mechanical resonator + charge amplifier) has dynamics at the drive frequency, but these are removed by the envelope detection.
- The dominant time constants are those of the envelope detector's low-pass filter and the mechanical system's amplitude response.

A typical AGC loop bandwidth is in the range of 10–100 Hz (54 Hz in the example), which is sufficient to track temperature-induced variations (time scales of seconds to minutes) while providing adequate phase margin (e.g., 80° in the example) for stable operation.

3.43.5 Importance for Sensitivity Stability

Recall the general expression for sense displacement per unit angular rate (applicable to both mode-matched and split-mode operation):

$$\frac{y}{\Omega} \propto X_d$$

Stabilizing X_d through the AGC loop directly stabilizes the gyroscope's scale factor against temperature variations of Q_d. This is essential for meeting the tight stability specifications required in most applications.

3.44 Summary and Preview

In this lecture, we have covered:

- The complete drive oscillator electronics, including charge amplifier, phase shifter, and amplitude control stages.
- Loop gain analysis and the importance of pole placement for accurate phase conditions.
- Temperature dependence of quality factor and its impact on sensitivity.
- The AGC loop as a negative feedback system that stabilizes drive displacement amplitude.
- Design considerations for AGC loop bandwidth and stability.

In the next class, we will complete the gyroscope system by examining:

- Sense channel electronics (capacitive sensing, amplification)
- Demodulation and filtering to recover the angular rate signal
- Noise analysis and its impact on performance metrics
- Bandwidth considerations and trade-offs
- Advanced gyroscope architectures beyond the traditional designs discussed so far

3.45 Q&A: AGC Bandwidth Constraints

Question: Does the AGC loop have any constraints on its bandwidth? How is the bandwidth level determined?

Answer: Excellent question. The AGC loop bandwidth is intentionally designed to be narrow, typically in the range of 10–100 Hz. This choice is governed by several factors:

1. **Nature of disturbances**: The primary disturbances that AGC corrects are slow variations in quality factor due to temperature changes. Temperature changes occur over seconds to minutes, so a low bandwidth is sufficient.
2. **Separation from drive frequency**: The AGC loop acts on the **amplitude envelope** of the signal, not on the signal itself. By limiting the bandwidth to much less than the drive frequency (e.g., 54 Hz vs. 24.75 kHz), we ensure that the loop does not attempt to track or correct the instantaneous oscillations, which would cause instability.
3. **Stability**: A narrow bandwidth allows ample phase margin, ensuring that the negative feedback loop remains stable despite the presence of multiple poles in the system.
4. **Avoiding interaction with sense mode**: The AGC loop should not respond to signals at the drive frequency that might contain Coriolis information. By filtering down to baseband before the error amplifier, we ensure that only the DC amplitude (proportional to X_d) is controlled, not the modulated signal.

In summary, the AGC bandwidth is set just high enough to track thermal time constants but low enough to maintain stability and avoid interfering with the gyroscope's normal operation. The example shown (54 Hz bandwidth, 80° phase margin) represents a well-designed compromise.

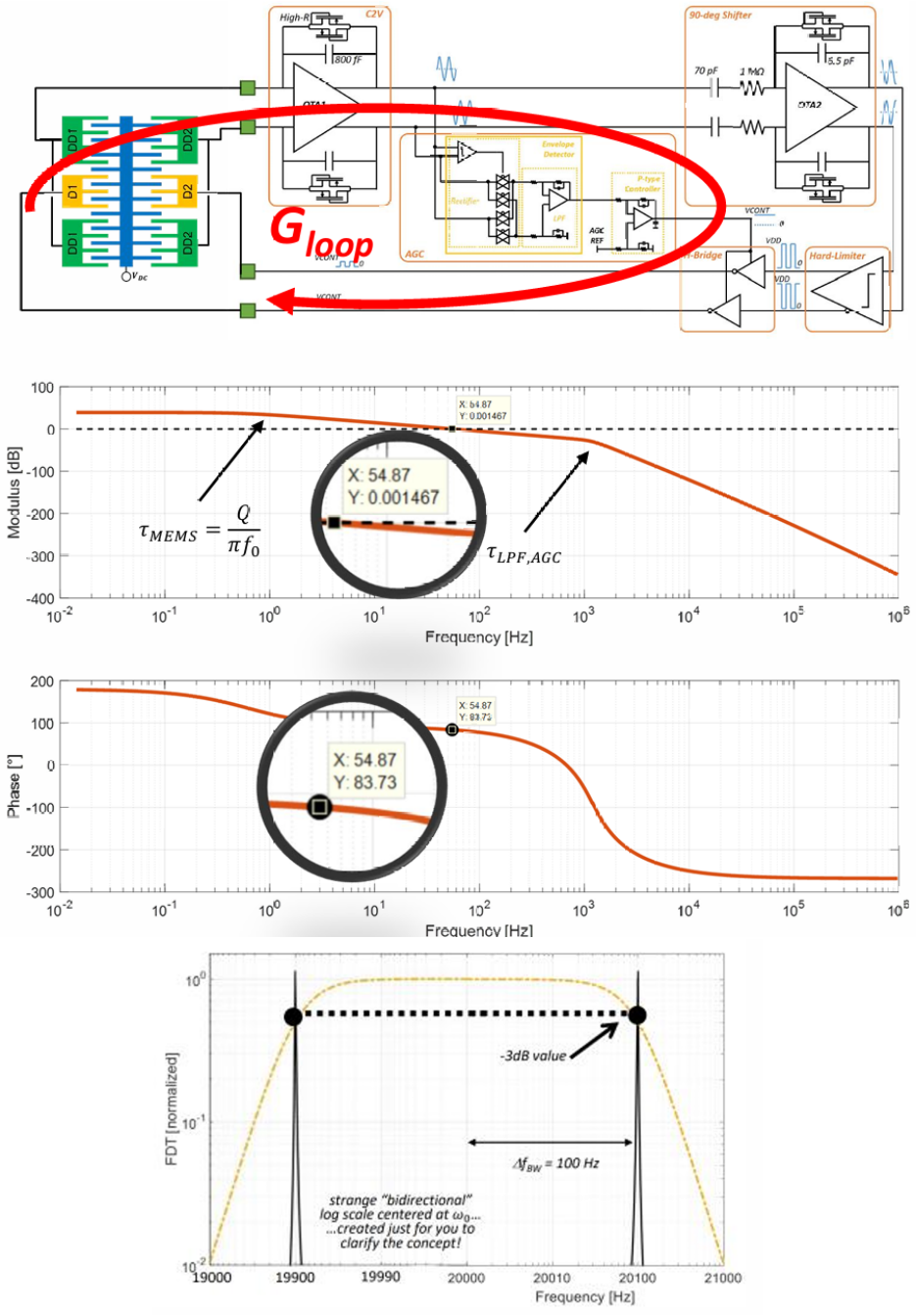

Figure 3.17: Typical P-type amplitude loop Bode diagrams.

Chapter 4

Noise and Nonidealities in AM Gyroscopes

4.1 Introduction to Parallel Plate Topology

The parallel plate configuration is often preferred in MEMS gyroscopes because it offers high transduction efficiency. This efficiency is maximized for small displacements, which is precisely the operating regime of these devices. While this topology introduces some inherent non-linearity, the resulting errors are typically negligible—on the order of 0.25%—for most applications. This can be verified by comparing the linearized capacitance expression with its full, non-linear form.

This discussion focuses on the push-pull, differential drive and detection scheme, commonly implemented with two anti-phase moving masses. This architecture, such as the four-gyro structure, is standard for both the drive and sense modes.

4.2 Sense Mode Capacitance Readout

The readout of the sense capacitance is straightforward. All moving parts (the rotor, including drive and sense frames) are biased at a fixed DC voltage, V_{DC}, typically in the range of 5–20 V. The differential sensing plates are connected to the virtual ground inputs of a fully differential amplifier—the same topology used for the drive mode readout.

When the proof mass moves in the sensing direction, the sense capacitances change. With a fixed bias voltage, the current generated by a time-varying capacitor is not $C\frac{dV}{dt}$, but rather the motion current, $i = V\frac{dC}{dt}$. This can be rewritten as $i = V\frac{dC}{dy}\frac{dy}{dt} = V\frac{dC}{dy}v_y$, where v_y is the velocity in the sense direction.

For a parallel plate capacitor with rest capacitance C_S and nominal gap g, the capacitive sensitivity is $\frac{dC}{dy} = \frac{C_S}{g}$. In a differential configuration, the two sense capacitors change in opposition, producing equal and opposite currents.

4.3 Differential Output Voltage

The differential output voltage, ΔV, is derived from the fully differential charge amplifier. The expression incorporates several key transduction stages:

$$\frac{\Delta V}{\Omega} = 2 \cdot \frac{V_{DC}}{C_f} \cdot \frac{C_S}{g} \cdot \frac{x_d}{\Delta\omega}$$

This scale factor can be understood as a product of gains:

- **2**: The factor from differential sensing.

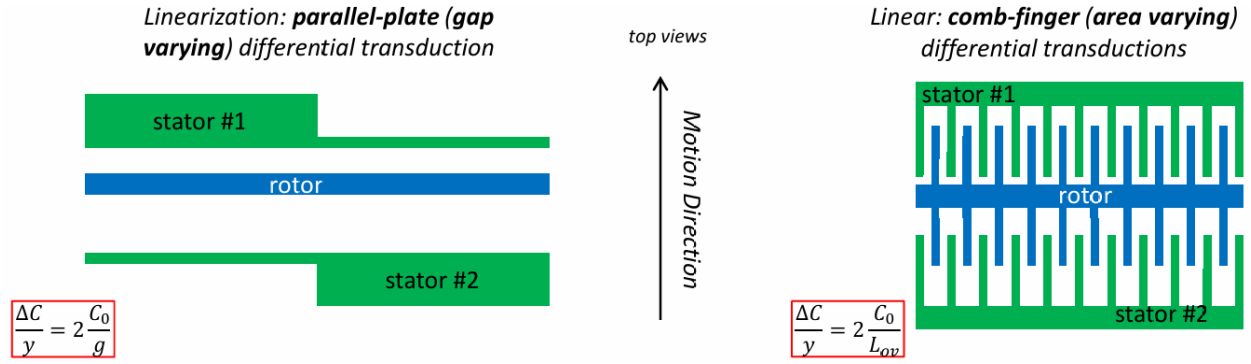

Figure 4.1: Gap and area varying capacitances.

- $\frac{V_{DC}}{C_f}$: The gain of the charge amplifier (capacitance-to-voltage conversion). At the drive frequency ω_0, the feedback capacitor C_f dominates, yielding a gain of $1/(\omega_0 C_f)$.
- $\frac{C_S}{g}$: The capacitive sensitivity (change in capacitance per unit displacement).
- $\frac{x_d}{\Delta\omega}$: The mechanical gain, where x_d is the drive displacement and $\Delta\omega$ is the frequency split between the drive and sense modes. This term relates the input angular rate Ω to the sense displacement y.

The output signal is an amplitude-modulated carrier: the Coriolis force, and thus the sense current, oscillates at the drive frequency. The amplitude of this carrier is proportional to the angular rate. Therefore, the raw output voltage must be demodulated to recover the baseband rate signal.

4.4 System-Level Architecture

The complete gyroscope system integrates the mechanical structure with control and readout electronics. The mechanically decoupled structure typically consists of three frames (drive, Coriolis, and sense) split into two anti-phase halves for common-mode rejection.

The drive loop maintains a sustained, constant-amplitude oscillation to provide the necessary velocity for the Coriolis force. The drive motion is sensed via a transimpedance amplifier. A phase shifter and gain stage satisfy the Barkhausen criteria for oscillation. An Automatic Gain Control (AGC) loop stabilizes the drive amplitude by adjusting the drive voltage, compensating for variations in the quality factor.

Angular rate input induces a sense displacement, generating a motion current in the sense electrodes. This current is converted to a voltage by an identical charge amplifier. The resulting amplitude-modulated signal is then synchronously demodulated using a reference signal from the drive loop. This demodulation, followed by low-pass filtering, recovers the baseband signal proportional to the input angular rate, which is then digitized and output.

4.5 Sensitivity as a Design Parameter

The scale factor $\frac{\Delta V}{\Omega}$ is not a figure of merit to be blindly maximized, but a design parameter to be precisely targeted. The design process begins with system-level specifications: the available output voltage swing (determined by the supply voltage and amplifier headroom) and the desired full-scale input rate range.

For example, with a 3 V supply and 0.5 V of headroom loss, the usable output swing is 2 V. For a full-scale range of ± 2000 dps, the target scale factor is $\frac{2\,V}{4000\,dps} = 0.5\,mV/^\circ/s$.

The physical and electrical parameters of the gyroscope—V_{DC}, C_f, C_S, g, x_d, and $\Delta\omega$—are then engineered to achieve this exact target. The scale factor is a deliberate outcome of this top-down design process.

True performance metrics, such as noise, bandwidth, and stability, are considered subsequently. Noise in MEMS gyroscopes originates from multiple physical domains. A primary source is mechanical-thermal

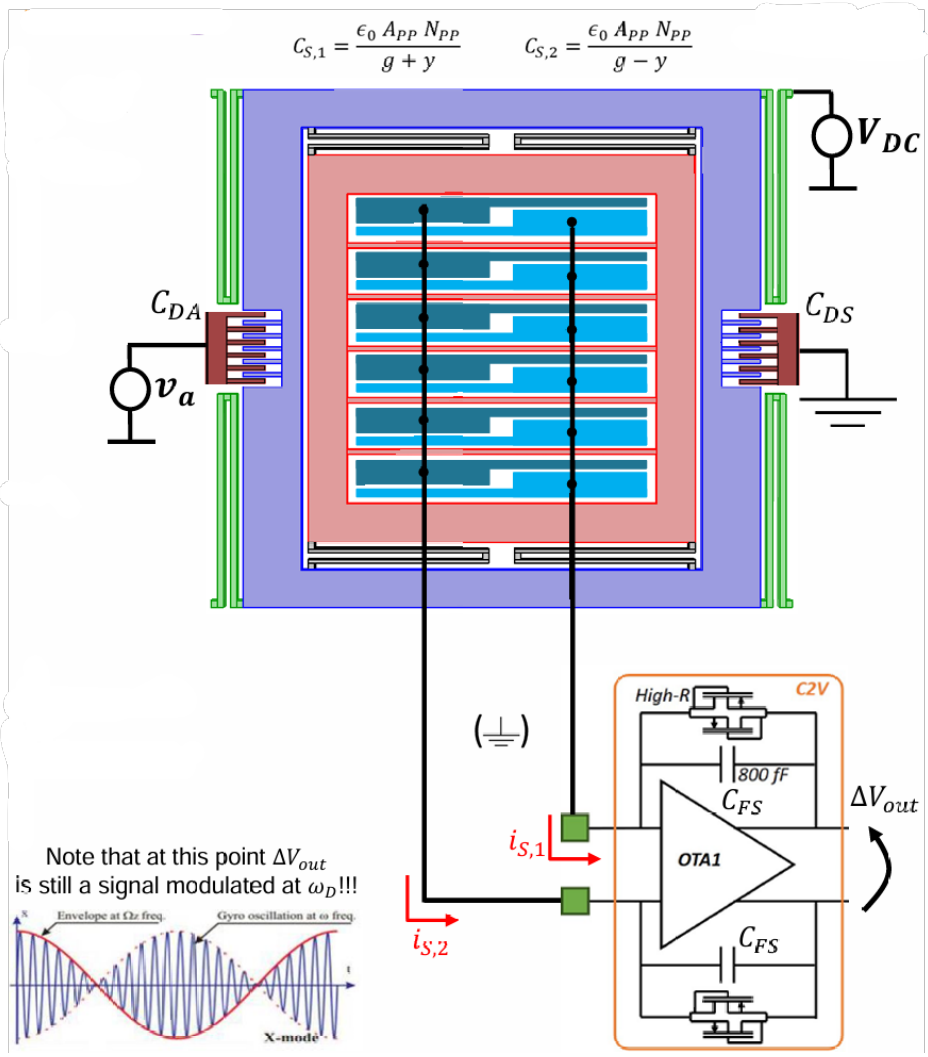

Figure 4.2: Parallel plate capacitive sensing overview.

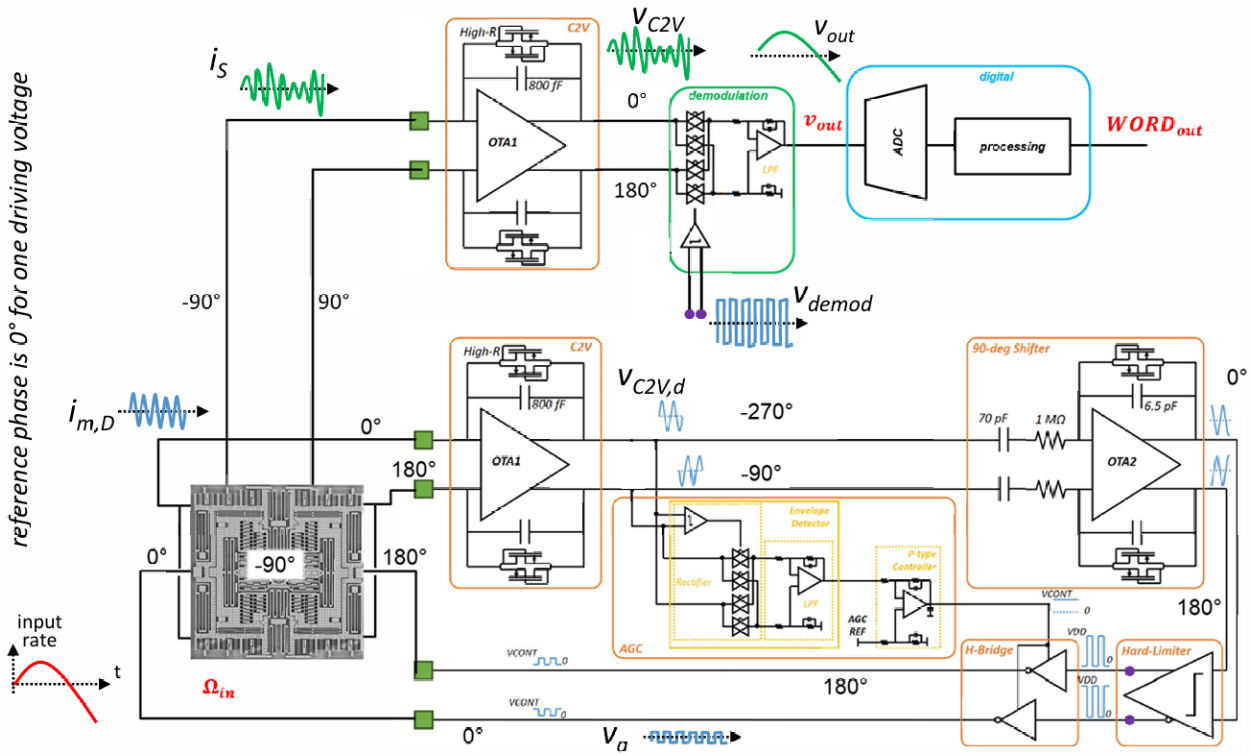

Figure 4.3: Overall system overview.

noise, which arises from the dissipation mechanisms (damping) in the system, analogous to Johnson-Nyquist noise in a resistor.

4.6 Noise Sources in MEMS Gyroscopes

The parallel between a mechanical spring-mass-damper system and an electrical RLC circuit extends to noise phenomena. In this analogy, mass corresponds to inductance, stiffness to capacitance, and damping to resistance. As the dissipative element, damping is the source of noise in the mechanical domain.

4.6.1 Mechanical-Thermal Noise

A intuitive picture helps understand the origin of mechanical noise. Consider a MEMS device packaged at a finite pressure (e.g., 0.1–1 mbar). The gas molecules in the package are not at absolute zero temperature; they possess thermal energy and move randomly. These agitated molecules continuously bombard the suspended proof mass. Even in the absence of external forces, this random bombardment excites the mass into motion. This phenomenon is described as a fluctuation force acting on the structure.

The power spectral density of this fluctuating force is given by:

$$S_F = 4k_B T b \quad [\mathrm{N}^2/\mathrm{Hz}]$$

where k_B is Boltzmann's constant, T is the absolute temperature, and b is the damping coefficient. This expression is the mechanical analog of Johnson-Nyquist noise in a resistor, $4k_BTR$. This relationship holds regardless of the specific physical mechanism responsible for damping—whether it is fluid damping, thermoelastic damping, or other dissipative processes. Any source of dissipation inherently introduces associated fluctuation forces.

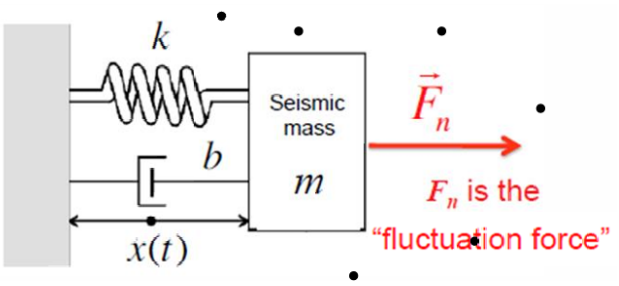

Figure 4.4: Mechanical noise origin.

4.6.2 Noise Equivalent Angular Rate Density

To determine the minimum measurable angular rate, we must translate this force noise into an equivalent angular rate noise. Because noise is a stochastic quantity, we work with power quantities. The power spectral density of the force noise can be referred to the input as an equivalent angular rate power spectral density:

$$S_\Omega = \frac{S_F}{(2m_s v_d)^2} = \frac{4k_B T b}{(2m_s \omega_d x_d)^2}$$

where:

- m_s is the sense mode mass
- $v_d = \omega_d x_d$ is the drive velocity
- ω_d is the drive resonance frequency
- x_d is the drive displacement amplitude

The square root of this quantity is the noise equivalent rate density (NERD), also known as angle random walk (ARW):

$$\text{NERD} = \sqrt{S_\Omega} = \frac{\sqrt{4k_B T b}}{2m_s \omega_d x_d} \quad [\text{rad/s}/\sqrt{\text{Hz}}]$$

For practical applications, we convert to degrees per second per square root of hertz using the factor $180/\pi$.

In a tuning fork configuration with two masses moving in anti-phase, the effective noise improves by a factor of $\sqrt{2}$ compared to a single mass of the same size, analogous to averaging two independent devices. This appears as a factor of 2 in the denominator of the expression.

4.6.3 Noise Optimization and Numerical Example

The NERD expression reveals key design trade-offs:

- Increasing the drive velocity ($\omega_d x_d$) improves noise by enhancing the Coriolis force for a given angular rate
- Increasing the sense mass m_s improves noise but typically requires larger area
- Reducing the damping coefficient b (e.g., by lowering package pressure) directly reduces noise

Consider a numerical example with representative parameters:

- Drive displacement: $x_d = 5$ μm
- Drive frequency: $f_d = 20$ kHz ($\omega_d = 2\pi f_d$)
- Sense mass per side: $m_s = 5$ μg (two masses in anti-phase)

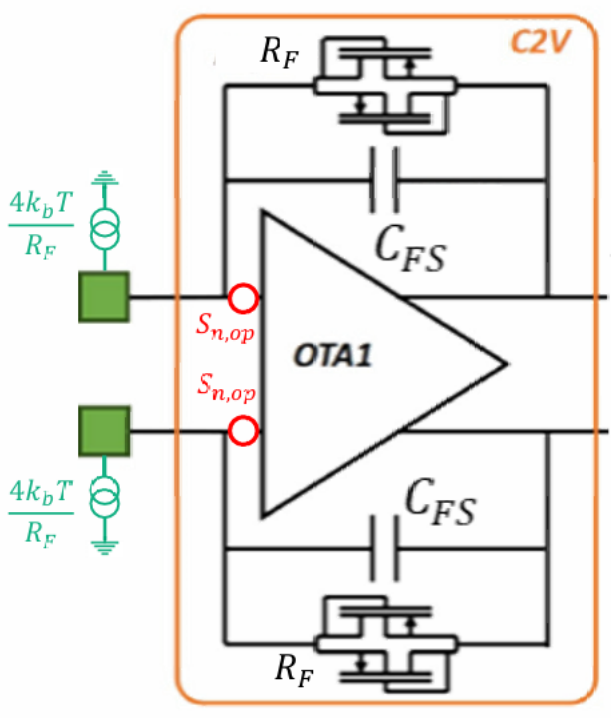

Figure 4.5: Electronic noise origin.

- Sense mode quality factor: $Q = 200$–300 (related to damping by $b = m_s\omega_s/Q$)

These parameters yield a NERD in the range of approximately 1 mdps/$\sqrt{\text{Hz}}$. Remarkably, this is within an order of magnitude of navigation-grade performance (approximately 0.1 mdps/$\sqrt{\text{Hz}}$ achieved by expensive hemispherical resonator gyroscopes or fiber optic gyroscopes). Achieving navigation-grade performance requires roughly a factor of two improvement in drive displacement, mass, and quality factor—a feasible objective from a purely mechanical-thermal noise perspective.

Consumer-grade devices typically operate at higher noise levels due to constraints on area, cost, and power consumption, illustrating the trade-offs in commercial applications.

4.7 Electronic Noise Sources

The sense chain electronics introduce additional noise contributions. The first stage (charge amplifier) dominates the electronic noise, as subsequent stages benefit from the gain already accumulated.

4.7.1 Noise Sources in the Sense Charge Amplifier

The differential charge amplifier includes several noise sources:

- Feedback resistors R_f (often implemented with off-state transistors), each generating current noise
- Input-referred voltage noise of the operational amplifier (dominated by the input pair)

These noise sources are uncorrelated and contribute independently to the output.

4.7.2 Input-Referred Angular Rate Noise from Electronics

A straightforward method evaluates electronic noise in terms of equivalent angular rate:

1. Calculate the noise voltage at the amplifier output (in V/$\sqrt{\text{Hz}}$)

2. Divide by the known scale factor $\frac{\Delta V}{\Omega}$ (in V/(rad/s) or V/(°/s))

Feedback Resistor Noise Contribution

Each feedback resistor R_f generates a current noise density:

$$i_{n,R_f}^2 = \frac{4k_B T}{R_f} \quad [\mathrm{A}^2/\mathrm{Hz}]$$

This current flows through the feedback impedance. At the drive frequency ω_d, the feedback impedance is dominated by the feedback capacitor C_f, giving a transimpedance of $1/(\omega_d C_f)$. For two uncorrelated sources, the output voltage noise density is:

$$v_{n,out,R_f} = \sqrt{2} \cdot \frac{1}{\omega_d C_f} \cdot \sqrt{\frac{4k_B T}{R_f}} \quad [\mathrm{V}/\sqrt{\mathrm{Hz}}]$$

The factor $\sqrt{2}$ accounts for the two independent noise sources.

Operational Amplifier Voltage Noise Contribution

The op-amp input voltage noise $v_{n,op}$ (in $\mathrm{V}/\sqrt{\mathrm{Hz}}$) appears at the output multiplied by the noise gain. At the drive frequency, this gain is:

$$G_n = 1 + \frac{C_0 + C_p}{C_f}$$

where C_0 is the MEMS rest capacitance and C_p represents parasitic capacitance at the input node (typically the dominant term). For a differential configuration with two uncorrelated op-amp noise sources, the total output noise density is:

$$v_{n,out,op} = \sqrt{2} \cdot v_{n,op} \cdot \left(1 + \frac{C_0 + C_p}{C_f}\right) \quad [\mathrm{V}/\sqrt{\mathrm{Hz}}]$$

4.7.3 Total Electronic Noise and Comparison

The total output noise voltage density is the root-sum-square combination of all contributions:

$$v_{n,out,total} = \sqrt{v_{n,out,R_f}^2 + v_{n,out,op}^2} \quad [\mathrm{V}/\sqrt{\mathrm{Hz}}]$$

To obtain the input-referred angular rate noise density, divide by the scale factor $SF = \Delta V/\Omega$:

$$\Omega_{n,electronics} = \frac{v_{n,out,total}}{SF} \quad [\mathrm{rad/s}/\sqrt{\mathrm{Hz}}]$$

This can be converted to degrees per second per square root of hertz using $180/\pi$.

4.7.4 Design Considerations

The feedback resistance R_f is typically designed to be large (using off-state transistors) for two reasons:

- It sets a low cutoff frequency for the charge amplifier, which is desirable
- Large R_f reduces its current noise contribution ($i_n \propto 1/\sqrt{R_f}$)

With careful design, electronic noise can be made negligible compared to mechanical-thermal noise, leaving the latter as the fundamental limit. The overall system noise is the combination:

$$\Omega_{n,total} = \sqrt{\Omega_{n,mechanical}^2 + \Omega_{n,electronics}^2}$$

This comprehensive noise model enables designers to allocate noise budgets across physical domains and optimize the gyroscope for target applications.

4.8 Dominance of Noise Terms in Applications

The feedback resistor noise contribution is typically straightforward to render negligible through proper design. The relative dominance between the remaining terms—op-amp voltage noise and mechanical-thermal noise—depends critically on the application domain and its associated constraints.

4.8.1 Application-Dependent Trade-offs

Consumer applications (e.g., mobile phones, wearables) operate under stringent power consumption limits. To make amplifier noise negligible relative to mechanical-thermal noise requires significant bias current, which is often prohibitive in these low-power contexts. In such devices, the electronic noise from the front-end amplifier typically dominates the overall noise budget.

High-end applications (e.g., space systems, autonomous vehicles, precision navigation) face less severe power constraints. Satellites have solar panels; autonomous driving systems have substantial battery capacity. In these domains, designers can afford the current necessary to reduce amplifier noise to the level of mechanical-thermal noise or below. A well-designed system achieves balance where multiple noise sources contribute comparably, representing an optimized allocation of the noise budget.

4.8.2 Noise Reduction Guidelines

Different strategies address different dominant noise sources:

- **Drive displacement** x_d: Uniquely benefits all noise sources. Increasing displacement enhances the Coriolis force at its origin, improving the signal before noise is added. This parameter should be maximized within mechanical constraints.
- **For electronic noise dominance**:
  - Design very large feedback resistors R_f (using off-state transistors) to minimize their current noise
  - Increase rotor voltage V_{DC} to boost transduction gain
  - **Minimize parasitic capacitance** C_p: This is arguably the most challenging task. Parasitics (typically 5–50 pF) amplify the effect of op-amp voltage noise through the noise gain factor $1 + (C_0 + C_p)/C_f$
- **For mechanical-thermal noise dominance**:
  - Reduce package pressure to lower damping coefficient b
  - Increase sense mass m_s (typically through larger area, though area is often constrained)

4.9 Numerical Example and Noise Budget Analysis

A comprehensive noise analysis illustrates the relative contributions of different sources. Consider a gyroscope with the following parameters:

- Feedback resistance: $R_f = 10$ GΩ (implemented with off-state transistors)
- Technology gap: $g = 2$ μm
- Rotor voltage: $V_{DC} = 15$ V
- Sense capacitance: $C_S = 80$ fF
- Drive frequency: $f_d = 20$–21 kHz
- Frequency split: $\Delta f = 1$ kHz
- Sense mass per side: $m_s = 5$ μg (two masses)

- Quality factor: $Q = 250$
- Drive displacement: $x_d = 5$ μm
- Op-amp input voltage noise: $e_n = 10$ nV/$\sqrt{\text{Hz}}$
- Parasitic capacitance: $C_p = 3$ pF
- Feedback capacitance: $C_f = 1$ pF

4.9.1 Achieving Low Op-Amp Noise

To achieve 10 nV/$\sqrt{\text{Hz}}$ input noise with minimal current consumption, the input transistors operate in sub-threshold (weak inversion). In this region, the transconductance is:

$$g_m = \frac{I_D}{nU_T}$$

where I_D is drain current, $U_T = k_B T/q$ is thermal voltage (≈ 25 mV at room temperature), and $n \approx 1.5$ is the sub-threshold slope factor. The input-referred voltage noise of a MOSFET is approximately:

$$e_n^2 \approx \frac{4k_B T\Gamma}{g_m}$$

with $\Gamma \approx 2/3$ in strong inversion and somewhat larger in weak inversion. For the target 10 nV/$\sqrt{\text{Hz}}$, the required transconductance is approximately 1.6 mS, corresponding to a bias current of roughly 4 μA per input transistor in sub-threshold operation.

4.9.2 Noise Budget Analysis

The total output noise voltage density is the root-sum-square of individual contributions:

$$v_{n,out,total} = \sqrt{v_{n,out,R_f}^2 + v_{n,out,op}^2 + v_{n,out,mech}^2}$$

where the mechanical-thermal noise is referred to the output through the scale factor. For this example:

- Thermo-mechanical noise: 1.3 mdps/$\sqrt{\text{Hz}}$
- Feedback resistor noise: negligible (but included in analysis)
- Op-amp noise contribution: dominant

The total noise equivalent rate density is approximately 3 mdps/$\sqrt{\text{Hz}}$. When comparing noise powers (variance), the contributions distribute as:

- Op-amp noise: $\approx 73\%$ of total variance
- Thermo-mechanical noise: $\approx 20\%$ of total variance
- Resistor noise: $\approx 7\%$ of total variance

This distribution is characteristic of consumer-grade gyroscopes, where amplifier noise dominates due to power constraints.

4.9.3 Current Required for Noise Matching

A key design question: what bias current makes electronic noise comparable to mechanical-thermal noise? Equating the two contributions:

$$\sqrt{\frac{4k_BT\Gamma}{g_m}} \cdot \left(1 + \frac{C_0 + C_p}{C_f}\right) \cdot \frac{1}{SF} = \text{NERD}_{\text{mech}}$$

Assuming sub-threshold operation ($g_m = I_D/(nU_T)$) and solving for current yields approximately 32 μA per input transistor. For a fully differential amplifier with two branches, this implies approximately 64 μA plus additional bias circuitry—roughly 70 μA total for the sense front-end.

In a state-of-the-art consumer three-axis gyroscope, total current consumption is approximately 500 μA. Allocating 70 μA to a single amplifier is impractical, confirming that consumer devices operate with electronic noise dominating the budget.

For high-end applications, this current is entirely acceptable. Achieving navigation-grade performance (< 0.1 mdps/$\sqrt{\text{Hz}}$) requires approximately another order of magnitude improvement. Pure current scaling is inefficient (noise $\propto 1/\sqrt{I}$), requiring 100× current for 10× noise reduction—impractical at 7 mA. However, combining modest current increases with optimized mechanical parameters (displacement, mass, quality factor) makes navigation-grade performance attainable, as suggested by the factor-of-two improvements discussed earlier.

4.10 Noise in the Drive Loop

The analysis has focused on sense-mode damping and sense electronics. Drive-loop noise sources have been temporarily neglected. These are typically not dominant provided a specific condition is satisfied. The next session will examine drive-loop noise and the condition under which it can be safely ignored.

4.11 Q&A Session: White Noise vs. Bias Instability

Question: For autonomous navigation applications, what is the interplay between white noise (angle random walk) and bias instability? Which is more critical?

Answer: This is an excellent question that touches on the time scales of navigation. Consider the power spectral density of gyroscope noise:

- White noise (angle random walk): flat spectrum
- $1/f$ noise: increases toward low frequencies
- Rate random walk: increases even more rapidly at low frequencies

The Allan variance (or Allan deviation) provides a complementary view in the time domain. When plotting Allan deviation of angular rate $\sigma(\tau)$ versus averaging time τ:

- White noise appears as a slope of $-1/2$ (decreasing with averaging)
- $1/f$ noise appears as a flat region (bias instability)
- Rate random walk appears as a slope of $+1/2$ (increasing with averaging)

For navigation, we typically care about angular position error—the integral of rate. When converting to the angle domain:

- White noise integrates to angle random walk: error grows as $\sqrt{t}$ (slope $+1/2$ in log-log plot)
- Bias instability integrates to error growing as t (slope $+1$)
- Rate random walk integrates to error growing as $t^{3/2}$ (slope $+3/2$)

The dominant error mechanism depends on navigation duration:

- For short navigation intervals (seconds to minutes), angle random walk (white noise) dominates
- For intermediate intervals (minutes to hours), bias instability ($1/f$ noise) may dominate
- For long intervals (hours to days), rate random walk and offset drifts dominate

In MEMS gyroscopes, modulation to a carrier frequency (typically 20–30 kHz) shifts signals above the $1/f$ corner frequency, making electronics $1/f$ noise negligible. The primary long-term degradation comes from environmental drift (temperature, humidity, aging) affecting the zero-rate output (ZRO). This drift, rather than white noise, limits performance for extended navigation times.

Thus, for applications requiring minutes of navigation without recalibration, angle random walk is critical. For applications requiring hours of navigation, bias stability (minimizing drift) becomes paramount. The optimal design balances both according to the intended mission profile and recalibration interval.

4.12 ADC and Bandwidth Requirements

The previous analysis considered noise from mechanical and electronic sources. In a complete system, the analog-to-digital converter (ADC) introduces quantization noise. Understanding bandwidth requirements is essential for selecting appropriate ADC resolution and architecture.

4.12.1 Bandwidth Determination in Mode-Split Operation

As with all design parameters, bandwidth is not an afterthought but a specification-driven choice. Given a target maximum system bandwidth, we design the gyroscope to accommodate it. The fundamental limit arises from the mode-split operation.

Recall that the Coriolis force is the product of angular rate Ω and drive velocity $v_d = \omega_d x_d \sin(\omega_d t)$:

$$F_c(t) = -2m_s v_d(t)\Omega(t)$$

The drive velocity is a sinusoidal signal at frequency ω_d. The angular rate $\Omega(t)$ contains frequency components of interest, typically from DC to some maximum frequency $f_{\Omega,\max}$ (e.g., 10–100 Hz for most applications). The product of two sinusoids generates components at sum and difference frequencies:

$$F_c(t) \propto \sin(\omega_d t) \cdot \sin(\omega_\Omega t) = \frac{1}{2}[\cos((\omega_d - \omega_\Omega)t) - \cos((\omega_d + \omega_\Omega)t)]$$

Thus, the Coriolis force contains two tones at $\omega_d \pm \omega_\Omega$, each with half the amplitude of the force that would result from a constant rate.

These force components excite the sense mode, which has a transfer function $H_s(\omega)$ with resonance at ω_s. In mode-split operation, ω_d is intentionally offset from ω_s by $\Delta\omega = |\omega_s - \omega_d|$.

For a constant rate ($\omega_\Omega = 0$), a single force tone at ω_d excites the sense mode at point A on the transfer function. As ω_Ω increases, the two tones separate symmetrically around ω_d. Provided ω_Ω is small compared to $\Delta\omega$, both tones lie within a region where the transfer function slope is approximately linear and symmetric. The gain increase for the upper tone equals the gain decrease for the lower tone, so the total response amplitude remains constant.

This explains the flat region in the system transfer function from angular rate to output voltage. As ω_Ω approaches $\Delta\omega$, one tone approaches the resonance peak ω_s, causing the response to increase. At $\omega_\Omega = \Delta\omega$, the upper tone exactly hits resonance, producing a peak in the transfer function.

4.12.2 Bandwidth Limitation and Filtering

The system bandwidth is therefore limited by the mode split $\Delta f = \Delta\omega/2\pi$. For frequencies beyond approximately $\Delta f/2$, the response becomes non-flat and eventually peaks. To obtain a well-behaved system with flat passband response, we must low-pass filter the output after demodulation.

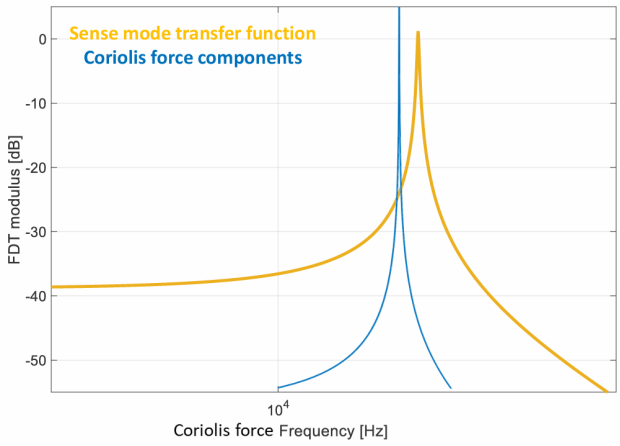

Figure 4.6: Sense mode transfer function with drive frequency offset by $\Delta\omega$

The filter (typically implemented after the synchronous demodulator) removes the rising portion of the response. With proper filter design, the usable bandwidth is approximately half the mode split:

$$\mathrm{BW}_{\max} \approx \frac{\Delta f}{2}$$

This is a key design relationship: the target bandwidth specification determines the minimum required mode split. For example, a 200 Hz bandwidth requires $\Delta f \geq 400$ Hz.

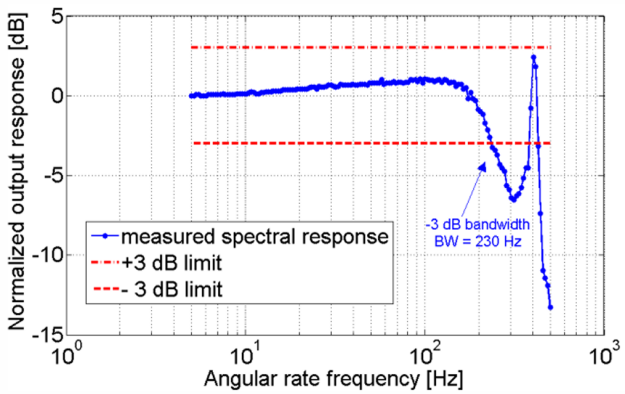

Figure 4.7: Example of a device with 400 Hz mode-split. The combination of the rising peak response and of the 200 Hz LPF brings the bandwidth to about 230 Hz, still lowering enough the undesired effect of the peak overshoot.

4.12.3 From Noise Density to RMS Noise

With bandwidth determined, we can convert noise density to RMS noise. For the consumer-grade example with 3 mdps/$\sqrt{\text{Hz}}$ total noise density and 200 Hz bandwidth:

$$\Omega_n = 3\,\text{mdps}/\sqrt{\text{Hz}} \times \sqrt{200\,\text{Hz}} = 42\,\text{mdps RMS}$$

4.12.4 Dynamic Range and ADC Resolution

Dynamic range (DR) is defined as the ratio of full-scale range to the minimum detectable signal (where SNR = 1). For a gyroscope with ±2000 dps full scale (4000 dps total range) and 42 mdps RMS noise:

$$\text{DR} = \frac{4000\,\text{dps}}{0.042\,\text{dps}} \approx 95,000$$

In decibels: $\text{DR}_{\text{dB}} = 20\log_{10}(95,000) \approx 100\,\text{dB}$.

The required ADC resolution to avoid dominating the noise budget must accommodate this dynamic range. The number of bits needed is:

$$N_{\text{bits}} = \lceil \log_2(\text{DR}) \rceil = \lceil \log_2(95,000) \rceil = 17\,\text{bits}$$

since $2^{16} = 65,536$ and $2^{17} = 131,072$.

This is remarkable: even consumer-grade MEMS gyroscopes require 17-bit effective resolution—a testament to the impressive dynamic range of these micromachined devices.

For navigation-grade applications targeting 0.3 mdps/$\sqrt{\text{Hz}}$ (10× lower noise), with similar full-scale range, the dynamic range increases by a factor of 10 to approximately 950,000, requiring 20–21 bits ($2^{20} = 1,048,576$). This explains why high-performance gyroscopes employ sigma-delta ADCs, which can achieve high effective resolution through oversampling and noise shaping.

4.12.5 Quantization Noise in the Overall Budget

Quantization noise from the ADC adds to the total noise budget. For an ADC with LSB size q, the quantization noise RMS is approximately $q/\sqrt{12}$ (assuming the quantization error is uniformly distributed).

For a 16-bit ADC with ±2000 dps full scale:

$$q = \frac{4000\,\text{dps}}{2^{16}} = 0.061\,\text{dps} = 61\,\text{mdps}$$

$$q_{\text{RMS}} = \frac{61\,\text{mdps}}{\sqrt{12}} \approx 17.6\,\text{mdps RMS}$$

In the previous noise budget example (total noise 42 mdps RMS from mechanical and electronic sources), adding 17.6 mdps RMS quantization noise increases total RMS noise to:

$$\sqrt{42^2 + 17.6^2} \approx 45.5\,\text{mdps RMS}$$

The quantization noise contributes approximately $(17.6/45.5)^2 \approx 15\%$ of the total noise power, making it non-negligible but not dominant. For a well-balanced design, quantization noise should be comparable to or slightly below other noise sources.

4.12.6 Programmable Full-Scale Range

Commercial MEMS gyroscopes often feature programmable full-scale range (e.g., ±250, ±500, ±1000, ±2000 dps). This flexibility allows the same device to serve different applications but has important implications:

- For a fixed ADC resolution, changing full-scale range changes LSB size proportionally
- Quantization noise scales with LSB size
- At lower full-scale settings, quantization noise becomes more significant relative to the signal
- At higher full-scale settings, quantization noise may become negligible, but resolution for small signals decreases

System designers must consider these trade-offs when selecting operating modes.

4.13 ADC Requirements and Oversampling

The previous analysis established that even consumer-grade MEMS gyroscopes require approximately 17 bits of effective resolution, while navigation-grade applications demand 20–21 bits. Conventional Nyquist-rate ADCs (successive approximation register, flash, pipeline) typically achieve 14–16 bits of effective resolution. Beyond this, they suffer from limitations due to component matching requirements: the accuracy of these converters relies on precise replication of elements (capacitors, resistors) across the chip area. Process variations introduce mismatch, leading to nonlinearity and reduced effective resolution.

4.13.1 Quantization Noise Fundamentals

An ADC samples the analog input at frequency f_s and quantizes it to discrete levels. The quantization error—the difference between actual analog value and its quantized representation—is deterministic given the input, but for unknown signals it behaves as an unpredictable noise source. The root-mean-square (RMS) value of this quantization noise is:

$$Q_{\text{RMS}} = \frac{\text{LSB}}{\sqrt{12}}$$

where LSB is the least significant bit size (full-scale range divided by 2^N for an N-bit ADC).

This quantization noise power is uniformly distributed across the Nyquist bandwidth, from DC to $f_s/2$. For a system with signal bandwidth B (e.g., 200 Hz), the Nyquist criterion requires $f_s \geq 2B$. Under these minimum conditions, all quantization noise falls within the signal band.

4.13.2 Oversampling Principle

Oversampling provides a straightforward method to reduce in-band quantization noise. By sampling at a rate much higher than the Nyquist minimum—by an oversampling ratio $K = f_s/(2B)$—the total quantization noise power $\text{LSB}^2/12$ is spread over a bandwidth K times larger. The in-band quantization noise power becomes:

$$Q_{\text{in-band}} = \frac{\text{LSB}^2/12}{K}$$

The RMS in-band quantization noise is therefore reduced by $\sqrt{K}$.

For example, with a 200 Hz bandwidth and 16-bit ADC ($\text{LSB} = 4000/2^{16} = 0.061$ dps), sampling at the Nyquist rate (400 Hz) yields:

$$Q_{\text{RMS}} = \frac{0.061}{\sqrt{12}} = 0.0176\,\text{dps RMS} = 17.6\,\text{mdps RMS}$$

If we oversample by $K = 100$ (sampling at 40 kHz), the in-band quantization noise becomes:

$$Q_{\text{RMS,in-band}} = \frac{0.0176}{\sqrt{100}} = 1.76\,\text{mdps RMS}$$

This dramatic reduction requires no change to the ADC hardware—only a faster sampling clock and subsequent digital filtering.

4.13.3 Decimation

After oversampling and digitization, the signal must be filtered to the target bandwidth and the sample rate reduced to the Nyquist rate for the signal of interest. This combination of filtering and downsampling is performed by a **decimator**. The decimator:

- Applies a low-pass digital filter with cutoff at the signal bandwidth
- Retains only every K-th sample, reducing the output data rate to $2B$

4.14 Noise Shaping and Sigma-Delta Converters

Oversampling alone can achieve significant quantization noise reduction, but requires very high oversampling ratios for large effective bit gains. **Noise shaping** provides additional suppression by modifying the spectral distribution of quantization noise.

4.14.1 Principle of Noise Shaping

In a feedback system, the transfer function from a noise source injected near the output to the output is the direct path divided by the loop gain. If we can inject quantization noise such that it experiences high loop gain at low frequencies, its contribution will be attenuated in the signal band.

This is precisely what a sigma-delta ($\Sigma\Delta$) converter accomplishes.

4.14.2 First-Order Sigma-Delta Architecture

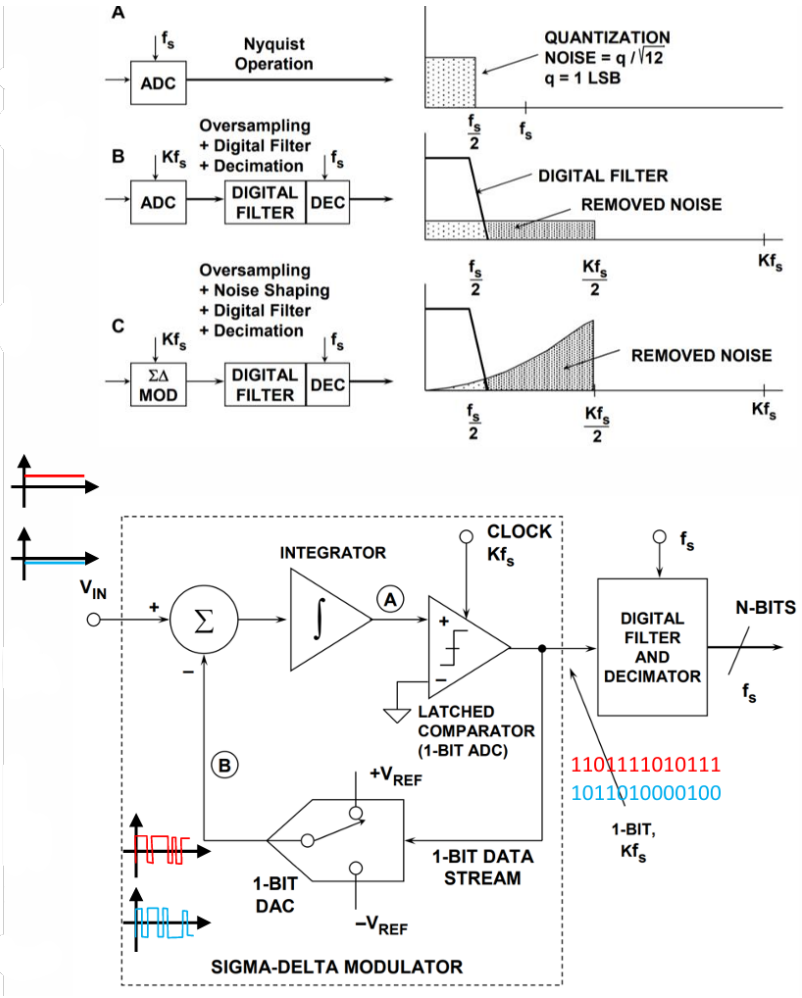

Figure 4.8: First-order sigma-delta modulator

The key elements are:

- **Summing junction** (Δ): computes the difference between input and feedback
- **Integrator** (Σ): accumulates the error

- **Quantizer**: typically a 1-bit ADC (comparator), producing a high-speed 1-bit stream
- **1-bit DAC**: converts the digital output back to analog for feedback

The system operates at a high oversampling clock frequency. The negative feedback forces the average of the feedback signal to equal the average input. Because the feedback signal can only take two values ($\pm V_{\text{ref}}$), it must switch rapidly between these levels such that its time-averaged value matches the input.

4.14.3 Operation and Waveform Interpretation

Consider a constant input voltage V_{in}. The output bitstream will contain a density of '1's proportional to V_{in}: if V_{in} is near $+V_{\text{ref}}$, most bits are '1'; if near $-V_{\text{ref}}$, most bits are '0'; if near zero, the bitstream alternates roughly equally between '1' and '0'. The analog feedback signal, reconstructed by the 1-bit DAC, is a two-level waveform whose average equals the input.

Thus, the information is encoded in the *density* of ones and zeros, not in the instantaneous values.

4.14.4 Noise Shaping Analysis

Quantization noise $e_q[n]$ is added by the quantizer. In a linearized model, the output $Y(z)$ is related to the input $U(z)$ and quantization noise $E(z)$ by:

$$Y(z) = \frac{H(z)}{1 + H(z)} U(z) + \frac{1}{1 + H(z)} E(z)$$

where $H(z)$ is the integrator transfer function ($H(z) = z^{-1}/(1 - z^{-1})$ for a discrete-time integrator). This gives:

$$Y(z) = z^{-1} U(z) + (1 - z^{-1}) E(z)$$

The signal transfer function is a pure delay (all-pass), while the noise transfer function is $1 - z^{-1}$—a first-order high-pass filter. Quantization noise is therefore *shaped* away from low frequencies toward high frequencies.

In the frequency domain, for frequencies much lower than the sampling rate, the noise transfer function magnitude is approximately:

$$|NTF(f)| \approx 2\pi f T_s = \frac{2\pi f}{f_s}$$

Thus, in-band quantization noise is attenuated by a factor proportional to frequency. Combined with oversampling, the in-band noise power for a first-order modulator is:

$$Q^2_{\text{in-band}} \approx \frac{\pi^2}{3} \cdot \frac{\text{LSB}^2/12}{K^3}$$

The $\pi^2/3$ factor comes from integrating the shaped noise over the signal band.

4.14.5 Higher-Order Modulators

Using multiple integrators in the loop filter yields higher-order noise shaping. An L-order modulator has noise transfer function $\approx (1 - z^{-1})^L$, giving in-band noise suppression proportional to $1/K^{2L+1}$ (for power) or $1/K^{L+0.5}$ for RMS. With modest oversampling ratios, high-order modulators can achieve 20+ bits of effective resolution using only a 1-bit quantizer.

4.14.6 Trade-offs and Applicability

Sigma-delta conversion trades speed for resolution: it requires high clock frequencies (MHz range) and significant digital filtering, increasing power consumption. This trade-off is favorable when the signal bandwidth is small—precisely the case for inertial sensors (200–500 Hz). For wideband applications (e.g., audio at 20 kHz, video at MHz), the achievable oversampling ratio is limited, making sigma-delta less effective.

For MEMS gyroscopes, the combination of:

- Small signal bandwidth (tens to hundreds of Hz)
- High dynamic range requirements (100+ dB)
- Feasibility of MHz-range clocks in modern CMOS

makes sigma-delta ADCs the natural choice.

4.15 Non-Idealities: Fabrication Process Variations

We now turn our attention to non-idealities that affect real devices. These fall into two categories:

1. **Part-to-part variations**: differences between nominally identical devices due to manufacturing tolerances
2. **Environmental dependencies**: changes in a single device's behavior with temperature, humidity, stress, etc.

4.15.1 Over-etch and Under-etch

Microfabrication processes involve etching steps to define structures. The exact dimensions of fabricated features depend on etch time, chemistry, temperature, and other factors. A structure designed with width w may emerge with actual width $w \pm \Delta w$, where Δw can be micrometers or fractions thereof.

This dimensional variation affects:

- **Spring constants**: stiffness $k \propto w^3$ for beams bending in-plane, so small width changes cause large stiffness variations
- **Resonance frequencies**: $\omega = \sqrt{k/m}$, so frequency varies as $\omega \propto w^{3/2}$
- **Gap spacing**: transduction gaps affect capacitance and electrostatic forces
- **Quality factors**: damping depends on geometry

4.15.2 Consequences for Calibration

Part-to-part variations necessitate calibration. Two scenarios:

1. **Stable devices with initial variation only**: If devices vary from part to part but each individual device's behavior is stable over time and temperature, one-time calibration suffices. Each device is tested and calibrated during manufacturing, then its calibration constants are stored (e.g., in on-chip memory).
2. **Devices with environmental sensitivity**: If device behavior changes with temperature, supply voltage, or aging, one-time calibration is insufficient. The device requires either:
   - On-board sensors (e.g., temperature sensor) and compensation algorithms
   - Periodic recalibration during operation
   - Design techniques that minimize environmental sensitivity

The magnitude of part-to-part variation determines calibration cost: small variations allow "trimming" of only a representative sample; large variations require individual calibration of every device, increasing test time and cost.

4.15.3 Design Strategies for Robustness

Minimizing the impact of process variations involves:

- Using symmetric, matched structures (common-centroid layouts)
- Designing for insensitivity to absolute dimensions (e.g., using ratioed quantities)
- Incorporating on-chip reference elements for self-calibration
- Electronic trimming (e.g., programmable capacitors, resistors)

The next session will delve deeper into mechanical non-idealities and their mitigation.

4.16 Calibration Strategies

Calibration addresses two distinct sources of error: part-to-part variations and environmental dependencies. The strategy differs based on the target application and performance requirements.

4.16.1 One-Time Calibration

For devices with stable behavior after manufacturing—where performance changes little with temperature, humidity, or aging—a single calibration at room temperature suffices. Each device is tested, its scale factor and offset are measured, and correction coefficients are stored (typically in on-chip memory). This approach is common in consumer applications where cost constraints limit test time.

4.16.2 Temperature Calibration

High-performance applications require stability across environmental conditions, particularly temperature. A MEMS gyroscope's behavior can change significantly from $-40°C$ to $+100°C$ due to:

- Temperature-dependent material properties (Young's modulus of silicon changes with temperature)
- Thermal expansion mismatches between materials
- Temperature sensitivity of electronics
- Stress induced by packaging

Temperature calibration involves characterizing the device across the full operating temperature range, storing a compensation model (e.g., polynomial coefficients), and using an on-chip temperature sensor to apply real-time correction during operation.

This calibration is expensive: temperature sweeps are slow (hours to days), require specialized equipment, and must often be repeated for multiple thermal cycles to verify hysteresis. For high-volume consumer devices, this cost is prohibitive. For automotive, industrial, and navigation applications, it is essential.

4.16.3 Design for Robustness

Regardless of calibration strategy, reducing inherent sensitivity to process and environmental variations benefits all applications. Lower sensitivity means:

- Tighter performance distribution before calibration
- Simpler calibration models
- Better long-term stability
- Reduced test time and cost

4.17 Part-to-Part Variations: Over-Etch and Under-Etch

The most significant source of part-to-part variation in MEMS is dimensional inaccuracy from the etching process, known as **critical dimension (CD) loss** or simply over-etch/under-etch.

4.17.1 Origin of CD Loss

Deep reactive ion etching (DRIE) is used to define high-aspect-ratio structures in silicon. The process involves:

1. Patterning a mask layer (e.g., photoresist or oxide)
2. Etching silicon through openings in the mask
3. The etch proceeds both vertically (desired) and laterally (undesired, but unavoidable)

The result is that features on the wafer differ from mask dimensions:

- **Holes and trenches**: become wider than designed
- **Beams and springs**: become narrower than designed

This is illustrated schematically: the mask defines a hole of width W_{mask}, but the etched hole has width $W_{\text{actual}} = W_{\text{mask}} + 2\Delta$, where Δ is the lateral etch under the mask. Similarly, a beam designed with width $W_{\text{beam,mask}}$ becomes $W_{\text{beam,actual}} = W_{\text{beam,mask}} - 2\Delta$.

4.17.2 Magnitude and Variability

Typical CD loss values depend on the process. For a mature MEMS process, one might have:

- Mean CD loss: $\mu_\Delta = 0.4\,\mu\text{m}$
- 3-sigma variation: $\sigma_\Delta = 0.2\,\mu\text{m}$ (i.e., 99.7% of devices have CD loss between $0.4 \pm 0.2\,\mu\text{m}$)

These numbers are significant compared to micrometric feature sizes. Consider a gap designed at $1.5\,\mu\text{m}$:

- After etching, mean gap $= 1.5 + 0.4 = 1.9\,\mu\text{m}$
- Part-to-part variation: gap ranges from 1.7 to $2.1\,\mu\text{m}$ (3-sigma)

A 20% variation in gap has substantial effects on performance.

4.17.3 Affected Parameters

CD loss impacts virtually all mechanical parameters:

- **Gaps**: wider gaps reduce capacitance and electrostatic forces
- **Beam widths**: narrower beams reduce stiffness ($k \propto w^3$ for bending beams)
- **Spring constants**: stiffness varies strongly with width
- **Resonance frequencies**: $\omega = \sqrt{k/m}$ changes with stiffness
- **Capacitance**: $C \propto A/g$ changes with both area and gap
- **Quality factors**: damping depends on geometry

4.17.4 Design Strategy

Two steps are essential:

1. **Compensate for mean CD loss**: design mask dimensions anticipating the known average etch bias, so fabricated dimensions match target values
2. **Minimize sensitivity to CD variation**: design structures whose performance metrics are inherently robust to dimensional fluctuations

4.18 Impact on Drive Loop and Scale Factor Stability

The drive loop's automatic gain control (AGC) stabilizes the drive displacement indirectly by regulating the output voltage of the charge amplifier. Understanding this relationship reveals how CD loss affects scale factor.

4.18.1 Drive Displacement vs. AGC Reference

The AGC compares the drive detection signal (after the charge amplifier) to a fixed reference voltage V_{ref}. The loop adjusts drive voltage to maintain this comparison. The resulting drive displacement x_d is related to V_{ref} by:

$$x_d = \frac{V_{\text{ref}}}{V_{\text{DC}}} \cdot \frac{C_{f,\text{drive}}}{dC/dx|_{\text{drive}}}$$

where $dC/dx|_{\text{drive}}$ is the capacitive sensitivity of the drive detection electrodes. For comb drives, $dC/dx = 2n\varepsilon_0 t/g_{\text{drive}}$ (with t = thickness, g_{drive} = gap, n = number of fingers).

The critical point: $dC/dx|_{\text{drive}}$ depends on geometry—particularly the gap g_{drive}—which varies with CD loss. Therefore, for a fixed V_{ref}, different devices will have different actual drive displacements x_d.

4.18.2 Scale Factor Sensitivity to Geometry

Recall the scale factor expression:

$$\frac{\Delta V}{\Omega} = 2 \cdot \frac{V_{\text{DC}}}{C_{f,\text{sense}}} \cdot \frac{C_S}{g_{\text{sense}}} \cdot \frac{x_d}{\Delta\omega}$$

Substituting x_d from above:

$$\frac{\Delta V}{\Omega} = 2 \cdot \frac{V_{\text{ref}}}{C_{f,\text{sense}}} \cdot \frac{C_S}{g_{\text{sense}}} \cdot \frac{1}{\Delta\omega} \cdot \frac{C_{f,\text{drive}}}{dC/dx|_{\text{drive}}}$$

Several important observations:

- V_{DC} **cancels**: scale factor is first-order independent of rotor voltage stability
- **Capacitance ratios matter**: if $C_{f,\text{drive}}$ and $C_{f,\text{sense}}$ are well-matched (same type, same layout), their ratio is stable despite absolute variations
- **Geometric terms remain**: C_S/g_{sense}, $1/\Delta\omega$, and $1/(dC/dx|_{\text{drive}})$ all depend on dimensions affected by CD loss

Expanding the geometric terms:

For sense parallel plates: $\dfrac{C_S}{g_{\text{sense}}} = \dfrac{\varepsilon_0 A_{\text{sense}}}{g_{\text{sense}}^2}$

For drive comb detection: $\left.\dfrac{dC}{dx}\right|_{\text{drive}} = \dfrac{2n\varepsilon_0 t}{g_{\text{drive}}}$

Thus:

$$\frac{\Delta V}{\Omega} \propto \frac{A_{\text{sense}}}{g_{\text{sense}}^2} \cdot \frac{g_{\text{drive}}}{1} \cdot \frac{1}{\Delta\omega} \cdot (\text{stable ratios})$$

4.18.3 Numerical Example of Variation

Assume nominal values:

- Sense gap $g_{\text{sense}} = 2\,\mu\text{m}$
- Drive comb gap $g_{\text{drive}} = 2\,\mu\text{m}$
- Sense plate area A_{sense} (length-dominated, minimally affected by CD loss)

With CD loss mean $0.4\,\mu\text{m}$ and 3-sigma variation $0.2\,\mu\text{m}$:

- Nominal ratio $g_{\text{drive}}/g_{\text{sense}}^2 = 2/(2^2) = 0.5\,\mu\text{m}^{-1}$
- If both gaps vary together (correlated): worst-case $g = 1.7\,\mu\text{m}$ or $2.1\,\mu\text{m}$
  - At $g = 1.7$: ratio $= 1.7/(1.7^2) = 1/1.7 = 0.588$ (+17.6%)
  - At $g = 2.1$: ratio $= 2.1/(2.1^2) = 1/2.1 = 0.476$ (-4.8%)
- This corresponds to approximately 8% variation around the mean of 0.5

If gaps vary independently (uncorrelated), worst-case variation can reach ±12%.

4.18.4 Design Freedom: Different Gaps

Crucially, nothing requires $g_{\text{drive}} = g_{\text{sense}}$. The designer can choose different gaps for the drive detection combs and sense parallel plates. This provides a degree of freedom to desensitize the scale factor to CD variations.

The geometric factor becomes:

$$F_{\text{geom}} = \frac{A_{\text{sense}}}{g_{\text{sense}}^2} \cdot g_{\text{drive}}$$

Treating A_{sense} as constant (length-dominated), we examine sensitivity to etching deviations δ from nominal gaps:

Let nominal gaps be g_{sense0}, g_{drive0}. After etching, actual gaps are $g_{\text{sense}} = g_{\text{sense0}} + \delta$, $g_{\text{drive}} = g_{\text{drive0}} + \delta$, assuming correlated variation (same etch bias everywhere).

The normalized geometric factor is:

$$\frac{F_{\text{geom}}(\delta)}{F_{\text{geom}}(0)} = \frac{g_{\text{sense0}}^2}{g_{\text{drive0}}} \cdot \frac{g_{\text{drive0}} + \delta}{(g_{\text{sense0}} + \delta)^2}$$

By choosing $g_{\text{sense0}} \neq g_{\text{drive0}}$, we can shape this function to be flatter around $\delta = 0$.

The plot demonstrates that with properly chosen unequal gaps, sensitivity variation over the expected $\pm 0.2\,\mu\text{m}$ etch range can be reduced from 8% to below 3%. This is a powerful design technique: trading nominal performance (the absolute scale factor value) for robustness against process variations.

4.18.5 Rule of Thumb

To minimize scale factor sensitivity to correlated CD variations, design such that:

$$\frac{g_{\text{drive}}}{g_{\text{sense}}} = 2$$

This equalizes the fractional change effects. In practice, optimization using process statistics yields the best result.

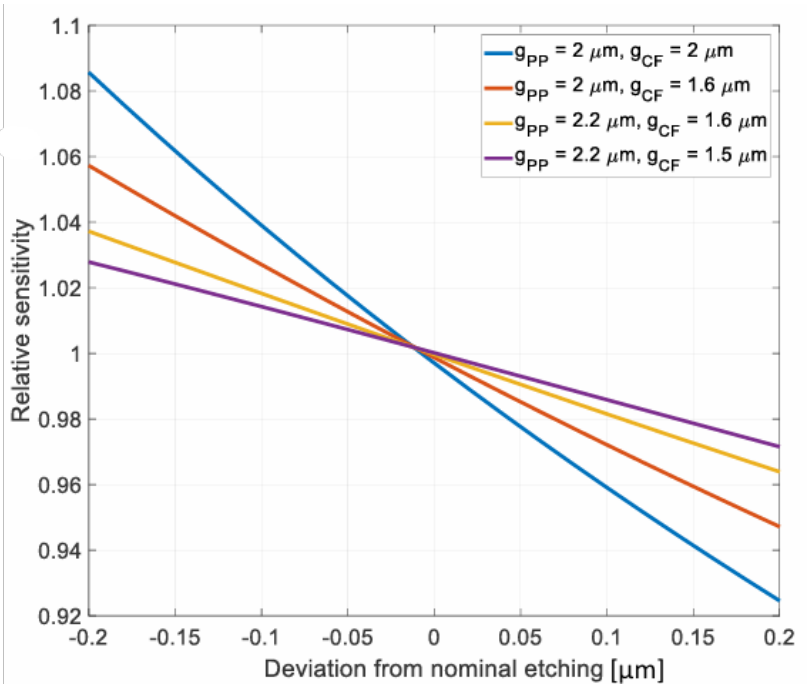

Figure 4.9: Normalized sensitivity vs. etch deviation for different gap choices. Blue curve: g_sense = g_drive = 2μm (steep slope, $\approx 8\%$ variation). Red curve: g_sense = 2.2μm, g_drive = 1.8μm (flatter, $< 3\%$ variation)

4.18.6 Additional Considerations

- The analysis assumed correlated variations across different structures on the same die—generally valid as etch conditions are uniform locally
- Uncorrelated variations (different parts of device experiencing different etch rates) are possible but less common; good layout practices minimize such effects
- The resonance frequency split $\Delta\omega$ also varies with CD loss, affecting scale factor; similar desensitization techniques apply

We will see this with an example in a few slides, but the principle is clear: by exploiting design degrees of freedom, we can make performance robust to the inevitable variations of microfabrication.

4.19 Trade-offs in Design for Robustness

The design freedom to choose different gaps for drive combs and sense plates does not come without cost. Two fundamental constraints limit this approach:

- **Minimum gap limitation**: Technology imposes a minimum feasible gap (e.g., 1.5–2 μm). Below this, etching becomes unreliable, stiction risk increases, and aspect ratio limitations appear.
- **Noise penalty**: Increasing the sense gap g_{sense} directly degrades mechanical-thermal noise. Recall the noise expression:

$$\text{NERD} \propto \frac{\sqrt{4k_BTb}}{m_s\omega_d x_d}$$

While gap does not appear explicitly, the transduction gain C_S/g_{sense} affects the signal level before electronic noise is added. A larger sense gap reduces C_S/g_{sense}, making electronic noise relatively more significant.

Thus, the fundamental trade-off is **repeatability vs. performance**. Larger sense gaps improve robustness to etching variations but worsen noise. The optimal design balances these competing requirements based on application priorities.

4.20 Frequency Split Stability

Even after desensitizing the geometric ratio $g_{\text{drive}}/g_{\text{sense}}^2$, the scale factor still depends on $\Delta\omega = \omega_s - \omega_d$. This frequency split also varies with CD loss because both resonance frequencies depend on spring dimensions.

4.20.1 Resonance Frequency Dependence on Geometry

The drive resonance frequency:

$$\omega_d = \sqrt{\frac{k_d}{m_d}} = \sqrt{\frac{EW_d^3\zeta_1}{m_d}}$$

where W_d is the drive spring width, ζ_1 encompasses other geometric parameters (length, number of folds, etc.), and E is Young's modulus.

Similarly, the sense resonance frequency:

$$\omega_s = \sqrt{\frac{k_s}{m_s}} = \sqrt{\frac{EW_s^3\zeta_2}{m_s}}$$

with W_s the sense spring width and ζ_2 a different geometric factor.

The participating masses differ: m_d includes drive frame plus Coriolis mass; m_s includes sense frame plus Coriolis mass. Thus $\zeta_1 \neq \zeta_2$.

4.20.2 Effect of CD Loss on Frequency Split

When etching varies, spring widths change by ΔW. The resulting variation in $\Delta\omega$ depends on the nominal widths W_d and W_s.

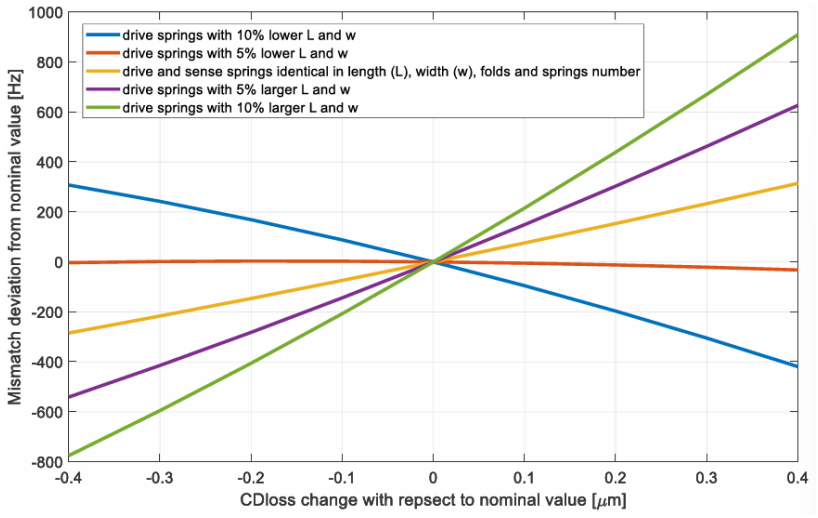

Figure 4.10: Normalized $\Delta\omega$ variation vs. etch deviation for different spring width ratios

If drive and sense springs are designed with identical widths, the frequency split variation can be substantial—e.g., nominal $\Delta f = 1\,\text{kHz}$ varying by $\pm 200\,\text{Hz}$ across process corners. This directly translates to $\pm 20\%$ scale factor variation.

By deliberately designing the springs with different widths, we can make $\Delta\omega$ nearly independent of CD loss. In the example shown, making drive springs 5% narrower than sense springs (in terms of width) flattens the response, reducing Δf variation to ± 10–$20\,\text{Hz}$.

4.20.3 Design Principle

The goal is to match the fractional sensitivities of ω_d and ω_s to width variations. Since $\omega \propto W^{3/2}$ (assuming mass unchanged), we have:

$$\frac{d\omega}{\omega} = \frac{3}{2}\frac{dW}{W}$$

For $\Delta\omega = \omega_s - \omega_d$, the condition for zero variation to first order is:

$$\frac{3}{2}\left(\frac{dW}{W_s} - \frac{dW}{W_d}\right) = 0 \quad \Rightarrow \quad W_s = W_d$$

But this ignores the different geometric factors ζ_1 and ζ_2. The full condition is more complex, requiring:

$$\frac{\partial\omega_s}{\partial W}\cdot\frac{1}{\omega_s} = \frac{\partial\omega_d}{\partial W}\cdot\frac{1}{\omega_d}$$

This generally leads to $W_d \neq W_s$. The optimal ratio depends on the specific spring designs and mass distributions, but the principle is clear: intentional mismatch in spring widths can desensitize $\Delta\omega$ to process variations.

4.21 Temperature Effects

Temperature influences MEMS gyroscope performance through multiple mechanisms. The two most significant are:

4.21.1 Young's Modulus Temperature Dependence

Silicon's Young's modulus decreases with temperature at approximately:

$$\frac{1}{E}\frac{dE}{dT} \approx -60\,\text{ppm/K}$$

Since resonance frequency $\omega \propto \sqrt{E}$, the temperature coefficient of frequency (TCF) is:

$$\text{TCF} = \frac{1}{\omega}\frac{d\omega}{dT} = \frac{1}{2}\frac{1}{E}\frac{dE}{dT} \approx -30\,\text{ppm/K}$$

Thus, a 20 kHz resonance frequency changes by $-0.6\,\text{Hz/K}$. Over a typical automotive temperature range ($-40°\text{C}$ to $+125°\text{C}$, $\Delta T = 165\,\text{K}$), this represents a $\pm 2.5\%$ variation.

4.21.2 Quality Factor Temperature Dependence

For devices where fluid damping dominates, the quality factor varies with temperature as:

$$Q \propto T^{-0.5}$$

This arises because:

- Damping coefficient $b \propto$ (gas density) $\times$ (mean molecular velocity)
- Gas density $\propto 1/T$ (ideal gas law at constant pressure)
- Mean velocity $\propto \sqrt{T}$
- Thus $b \propto \sqrt{T}$, and $Q = \omega m/b \propto 1/\sqrt{T}$

The temperature coefficient of quality factor (TCQ) is therefore -0.5 (i.e., $Q \propto T^{-0.5}$).

4.21.3 Impact on Scale Factor

Since scale factor $\propto 1/\Delta\omega$, and both ω_d and ω_s have the same TCF ($-30\,\text{ppm/K}$), their difference $\Delta\omega$ also varies with the same coefficient:

$$\frac{1}{\Delta\omega}\frac{d(\Delta\omega)}{dT} \approx -30\,\text{ppm/K}$$

Over a 165 K range, this yields approximately $\pm 2.5\%$ scale factor variation. For many consumer applications, this is acceptable without temperature compensation. For navigation-grade applications requiring $< 0.1\%$ stability, temperature calibration is essential.

4.21.4 Temperature Calibration Requirements

Achieving 0.1% scale factor stability over a 100°C range requires reducing the temperature coefficient from 30 ppm/K to approximately 1 ppm/K. This demands:

- Accurate temperature sensing (on-chip temperature sensor)
- Characterized temperature model (polynomial or lookup table)
- Real-time compensation in the digital domain
- Sufficient calibration time during manufacturing (temperature sweeps)

4.22 Mechanical Stress and Package Effects

The remaining slides (provided for reference) address additional non-idealities:

- **Substrate deformation**: Board bending, package stress from molding compounds, and thermal expansion mismatches induce stress in the MEMS structure.
- **Stress sensitivity**: Spring stiffness can change under stress, affecting resonance frequencies and thus scale factor.
- **Mitigation strategies**: Symmetric designs, stress-relief structures, and careful package selection (e.g., soft adhesives, stress-isolated packages).

These topics will be revisited in the next class.

4.23 Summary and Key Takeaways

This session covered the complete system-level view of MEMS gyroscopes:

1. **Noise sources**: Mechanical-thermal noise (fundamental limit) and electronic noise (amplifier, resistors). Trade-offs between power consumption and noise performance determine application suitability.
2. **Bandwidth and dynamic range**: Mode-split operation limits bandwidth to approximately half the frequency split. Dynamic range of consumer devices exceeds 100 dB, requiring 17+ bit ADC resolution.
3. **ADC requirements**: Sigma-delta converters with oversampling and noise shaping achieve the needed resolution by exploiting the small signal bandwidth of inertial sensors.
4. **Process variations**: CD loss (over-etch/under-etch) causes significant part-to-part variations in gaps, spring widths, and thus resonance frequencies and scale factor.
5. **Design for robustness**: Intentional mismatch of gaps and spring widths can desensitize scale factor to process variations, at the cost of noise or other performance metrics.
6. **Temperature effects**: Young's modulus temperature coefficient (-60 ppm/K) leads to -30 ppm/K frequency variation, causing scale factor changes that may require calibration for high-end applications.

The next class will continue with mechanical non-idealities, including stress effects and advanced mitigation techniques.

Chapter 5

Quadrature Error and Compensation in AM Gyroscopes

5.1 Resolution and Bandwidth in Consumer Applications

For current consumer applications, the required resolution is not extremely demanding. For instance, with a 4 kHz bandwidth, future applications may demand an order of magnitude improvement in resolution. However, let's examine current performance metrics. For an event rate of approximately 2000 or 800 Hz, a resolution of 50 mdps (50 millidegrees/sec) is achievable. This value represents the minimum measurable signal, corresponding to a very slow rotation.

A key design principle is that the operational bandwidth is typically set two to three times larger than the signal bandwidth to avoid attenuation. While this factor can be conservative, it is a crucial consideration in system-level calculations.

5.2 Displacement Scales: From Drive Motion to Earth's Rotation

The drive motion displaces the masses by approximately 10 µm through a constant anti-phase oscillation of the masses and their frames. For a gyroscope measuring the Earth's rotation frequency, the displacement in the sensing direction per unit angular rate is given by the well-known expression:

$$y = \frac{x_d}{\omega_y - \omega_x} \cdot \Omega \tag{5.1}$$

where x_d is the drive displacement, ω_x and ω_y are the resonant frequencies of the drive and sense modes, and Ω is the input angular rate.

Using this, we can calculate the sensor's physical displacement for two extreme cases:

- **Full-Scale Range (Maximum Rate):** For the maximum input rate the sensor can measure, the resulting sense-mode displacement is on the order of 80 nm. This is consistent with previous examples, where full-scale displacements were typically less than 100 nm.
- **Minimum Detectable Signal (Minimum Rate):** For the minimum resolvable signal, the displacement scales down dramatically. In this case, the required displacement resolution is in the range of picometers. Current consumer-grade devices (e.g., those in smartphones) already measure such small displacements. For future, high-performance applications, the required resolution must push even further, to tens of picometers or even hundreds of femtometers.

The key takeaway is the immense dynamic range required. While the drive motion is a robust 10 µm in the X-direction, the sense motion caused by the minimum angular rate is minuscule. The ratio of the sense displacement (y) to the drive displacement (x_d) is on the order of:

$$\frac{y}{x_d} \approx 10^{-6} \text{ to } 10^{-7} \tag{5.2}$$

This means that while the mass is undergoing a large oscillation in the drive direction, any spurious motion in the orthogonal sense direction must be suppressed to an extraordinary degree. A helpful analogy is to imagine continuously traveling back and forth between Milan and Boston (a distance of approximately 6000 km) without deviating from the path by more than one meter.

5.3 The Challenge of Spurious Motion: Quadrature Error

In an ideal gyroscope with no angular rate, the motion should be purely in the drive (X) direction. However, in practice, fabrication non-uniformities and defects (e.g., variations in etching on the order of a few nanometers) create asymmetries. These imperfections cause the drive motion to couple into the sense (Y) direction, producing a spurious signal.

This spurious motion can be decomposed into common-mode and differential-mode components. While common-mode signals are often rejected by a differential readout, differential-mode components are problematic. They appear as a displacement in the Y-direction, oscillating at the drive resonance frequency, and will be processed by the readout electronics.

5.3.1 Phase Relationship: The Key to Quadrature

The crucial difference between the desired Coriolis signal and this spurious signal lies in their phase relative to the drive motion.

- **Coriolis Force:** The Coriolis force $F_c = 2m\mathbf{\Omega} \times \mathbf{v}$ is proportional to the velocity of the drive mass. Since velocity is the time derivative of displacement, it is 90° out of phase with the drive displacement.

- **Quadrature Force:** The spurious force, which we call the **quadrature force**, is typically proportional to the drive *displacement* itself. This arises from mechanical asymmetries that create a direct X-to-Y coupling.

Consequently, the two forces, and the resulting motions they produce, are in *quadrature* (i.e., 90° out of phase). This phase difference is a critical property that allows for their separation in the signal processing chain, although non-idealities can still lead to output errors.

5.3.2 Sources of Quadrature Error

Several physical phenomena contribute to quadrature error:

1. **Mechanical Asymmetries:** Imperfections in the suspension springs are a primary source. In an ideal structure with four identical springs, motion is constrained to one axis. If one spring is slightly wider due to etching variations, it provides a different stiffness, causing the mass to move in an arc that has a component in the Y-direction.

2. **Electrostatic Asymmetries:** Non-uniform etching can also affect comb drives. If the moving fingers are not perfectly centered between the fixed fingers, the electrostatic forces in the X-direction can have a component in the Y-direction, or create a torque.

3. **Layout and Biasing:** In a complex device, biasing voltages and interconnections (e.g., routing traces passing underneath the structure) can generate spurious electrostatic forces or torques. Any asymmetry in the layout can lead to forces at the drive frequency.

5.4 Modeling Quadrature in the Signal Chain

The signal at the output of the front-end electronics (e.g., a charge amplifier or C2V converter), before demodulation, can be modeled as the sum of the in-phase and quadrature components.

The ideal signal due to an angular rate Ω is:

$$V_\Omega(t) = SF \cdot \Omega \cdot \cos(\omega_d t) \tag{5.3}$$

where SF is the scale factor (sensitivity) and ω_d is the drive frequency.

The quadrature error can be modeled as an equivalent, but fictitious, angular rate Ω_Q. This term is in quadrature with the Coriolis signal:

$$V_Q(t) = SF \cdot \Omega_Q \cdot \sin(\omega_d t) \tag{5.4}$$

The total signal before demodulation is therefore:

$$V_{out}(t) = SF\left[\Omega\cos(\omega_d t) + \Omega_Q \sin(\omega_d t)\right] \tag{5.5}$$

The quadrature term ($\Omega_Q \sin(\omega_d t)$) represents a significant spurious signal that is not caused by rotation. While it is in quadrature with the true rate signal, any phase error in the demodulation process will cause a portion of it to leak into the final output, appearing as a bias error. This necessitates careful system design, both mechanically to minimize Ω_Q at its source, and electronically to compensate for the remaining quadrature.

5.5 Synchronous Demodulation: The Ideal Case

To extract the angular rate signal from the modulated output, we employ **synchronous demodulation**. This process involves multiplying the signal from the front-end (Equation 5.5) by a reference signal that is perfectly in phase with the Coriolis component ($\cos(\omega_d t)$), and then applying a low-pass filter (LPF).

Mathematically, the process is as follows:

5.5.1 Demodulation of the Rate Signal

Multiplying the rate-induced term by the in-phase reference:

$$\begin{aligned} V_{out,\Omega}(t) \cdot \cos(\omega_d t) &= [SF \cdot \Omega \cdot \cos(\omega_d t)] \cdot \cos(\omega_d t) \\ &= SF \cdot \Omega \cdot \frac{1}{2}[1 + \cos(2\omega_d t)] \end{aligned}$$

After low-pass filtering, the high-frequency component at $2\omega_d t$ is removed, leaving:

$$V_{out,DC} = \frac{1}{2} SF \cdot \Omega \tag{5.6}$$

5.5.2 Rejection of the Quadrature Signal

Multiplying the quadrature term by the same in-phase reference:

$$\begin{aligned} V_{out,Q}(t) \cdot \cos(\omega_d t) &= [SF \cdot \Omega_Q \cdot \sin(\omega_d t)] \cdot \cos(\omega_d t) \\ &= SF \cdot \Omega_Q \cdot \frac{1}{2}\sin(2\omega_d t) \end{aligned}$$

This product contains only a high-frequency component at $2\omega_d t$. The low-pass filter completely attenuates this, resulting in zero output from the quadrature signal.

$$V_{out,DC} = 0 \tag{5.7}$$

This demonstrates the fundamental principle: **an ideal synchronous demodulation, with a perfectly phased reference, completely rejects quadrature error while preserving the true rate signal.**

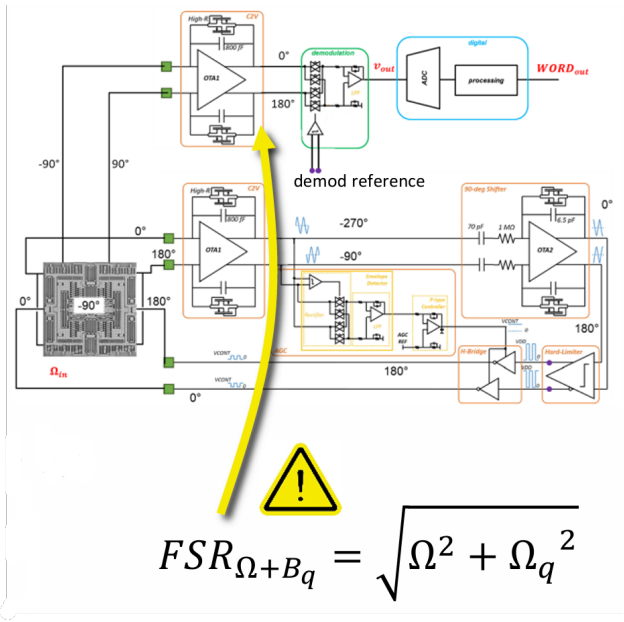

Figure 5.1: Quadrature error input referred modelling.

5.6 A Critical Caveat: Dynamic Range and Quadrature

Even with perfect rejection at DC, quadrature error poses a significant design challenge. The magnitude of the quadrature signal (Ω_Q), when referred to the input as an equivalent angular rate, can be orders of magnitude larger than the sensor's full-scale range (FSR).

The signal present before demodulation is the sum of the rate and quadrature components. Its peak amplitude is the quadrature sum of the two:

$$V_{out,peak} = SF \cdot \sqrt{\Omega^2 + \Omega_Q^2} \tag{5.8}$$

Since Ω_Q can be enormous (e.g., 1000x the FSR), the analog front-end (charge amplifier, variable gain amplifier, ADC) must be designed with sufficient dynamic range to handle this large signal without saturating. The gain of the stages before demodulation must be carefully chosen to accommodate the worst-case combination of the full-scale input rate and the unpredictable, but potentially huge, quadrature component.

5.7 Non-Ideal Demodulation: The Impact of Phase Errors

The perfect cancellation of quadrature relies on the demodulation reference being exactly $\cos(\omega_d t)$. In reality, the reference signal derived from the drive loop will have:

- **Phase Error (ϕ)**: A constant, systematic phase shift.
- **Phase Noise ($\delta\phi(t)$)**: Random fluctuations in the phase due to noise in the drive loop.

Our demodulation reference is therefore more accurately described as $\cos(\omega_d t + \phi + \delta\phi(t))$. Multiplying this by the total signal from Equation 5.5 and applying the low-pass filter yields a more complex result. Using small-angle approximations ($\cos(\phi) \approx 1$, $\sin(\phi) \approx \phi$), the output becomes:

$$\begin{aligned} V_{out,DC} &\approx \frac{1}{2} SF \left[\Omega \cdot \cos(\phi + \delta\phi) + \Omega_Q \cdot \sin(\phi + \delta\phi)\right] \\ &\approx \frac{1}{2} SF \left[\Omega + \Omega_Q \cdot (\phi + \delta\phi)\right] \end{aligned} \tag{5.9}$$

This result reveals two critical non-idealities:

1. **Scale Factor Error:** The rate signal Ω is multiplied by $\cos(\phi + \delta\phi) \approx 1$. The error introduced here is typically very small and is often neglected.
2. **Quadrature Feedthrough (Offset):** A portion of the quadrature signal now appears at the output. This is the product of the input-referred quadrature Ω_Q and the phase error ϕ. This creates a **DC offset** at the output, which is indistinguishable from a true angular rate. For a constant phase error, this is a fixed bias.
3. **Noise Contribution:** The phase noise $\delta\phi(t)$ modulates the quadrature signal, converting it into a low-frequency noise component at the output. While present, this term is often negligible in a well-designed high-Q drive loop.

The phase error term is the most harmful, as it directly transforms a large, spurious mechanical signal into a DC offset, corrupting the sensor's zero-rate output.

5.8 Visualizing Phase Error and Phase Noise

Figure 5.2 illustrates these concepts.

- The **ideal reference** (blue) is perfectly aligned with the Coriolis force.
- A **phase error** (orange) represents a constant time shift. The zero-crossing point is consistently delayed or advanced.
- **Phase noise** (yellow) causes the zero-crossing point to vary randomly around its mean position (jitter). This is a manifestation of noise in the drive loop.

In practical integrated implementations, demodulation is often performed using square waves rather than sine waves for simplicity and efficiency. A square wave is generated by passing the sine wave from the drive loop through a high-gain comparator. It is crucial to understand that this process does not eliminate phase noise or phase error. The zero-crossing jitter (phase noise) and the constant time shift (phase error) are preserved in the resulting square wave, and the detrimental effects described in Equation (5) remain fully applicable.

5.9 Electronic Implementation: The Passive Mixer Demodulator

From a circuit perspective, a synchronous demodulator can be implemented simply and effectively as a **passive mixer**. A typical topology using the differential outputs of a charge amplifier is shown in Figure 5.3.

5.9.1 Operating Principle

The differential outputs of the charge amplifier, which are still modulated at the drive frequency ω_d, serve as the inputs to the mixer. Each input line is connected to two branches, each containing a CMOS transmission gate (parallel NMOS and PMOS transistors) acting as a switch.

1. During the first half of the drive period (ϕ_1 high), the switches connect the positive input to the positive output node and the negative input to the negative output node.
2. During the second half of the period (ϕ_2 high), the switches toggle. The positive input is now connected to the negative output node, and the negative input is connected to the positive output node.

This switching action effectively performs a full-wave rectification. The positive output node always holds the positive portions of the differential signal, and the negative output node always holds the negative portions. After this stage, a simple low-pass filter extracts the DC envelope, resulting in a differential DC voltage proportional to the input signal amplitude. A following differential amplifier can then convert this to a single-ended signal for the ADC.

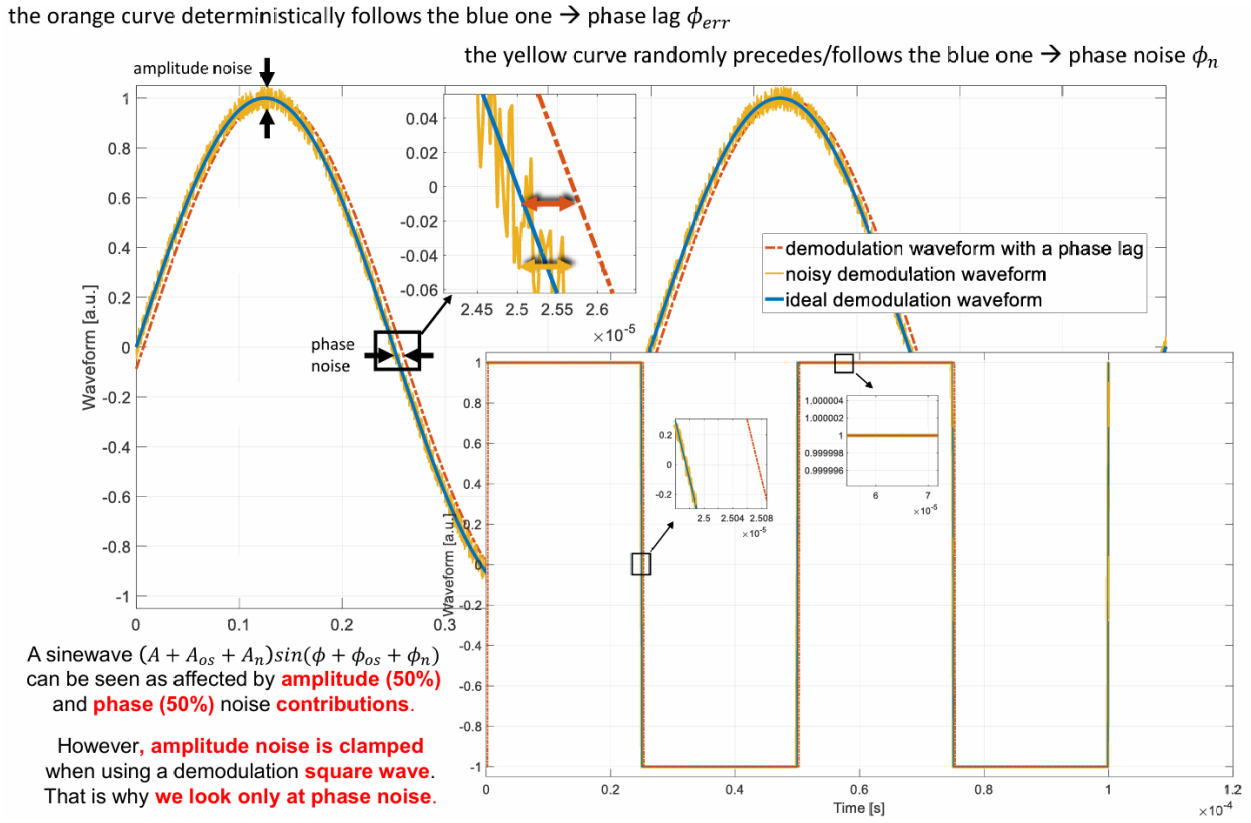

Figure 5.2: Illustration of an ideal reference sine wave (blue), one with a phase error (orange), and one with phase noise (yellow), highlighting the shift and jitter in zero-crossing points.

5.9.2 Key Design Considerations

- **Non-Overlapping Clocks:** The gate drive signals (ϕ_1, ϕ_2) must be non-overlapping to prevent momentarily shorting the two output nodes together, which would cause charge loss and signal corruption. A dedicated clock generator circuit is therefore required.

- **Demodulation Phase:** The phase of these switching clocks is critical. They are derived from the drive loop. To achieve perfect rejection of quadrature error, the demodulation must be synchronous with the Coriolis signal (velocity). The optimal point in the drive loop to derive this reference is **the output of the 90-degree phase shifter**, which aligns the reference with the velocity, and thus the Coriolis force.

5.10 Modeling the Origin of Quadrature: Cross-Axis Stiffness

While we have modeled quadrature as an equivalent input rate Ω_Q, its physical origin is a spurious force in the sense (Y) direction caused by displacement in the drive (X) direction. The natural way to model this force-displacement relationship is through an elastic coupling, or **cross-axis stiffness**, k_{xy}.

5.10.1 Equation of Motion in the Sense Axis

The full equation of motion for the sense mode, including the quadrature force, is:

$$m\ddot{y} + c_y\dot{y} + k_y y = F_{Coriolis} + F_{Quad} = 2m\Omega\dot{x} - k_{xy}x \tag{5.10}$$

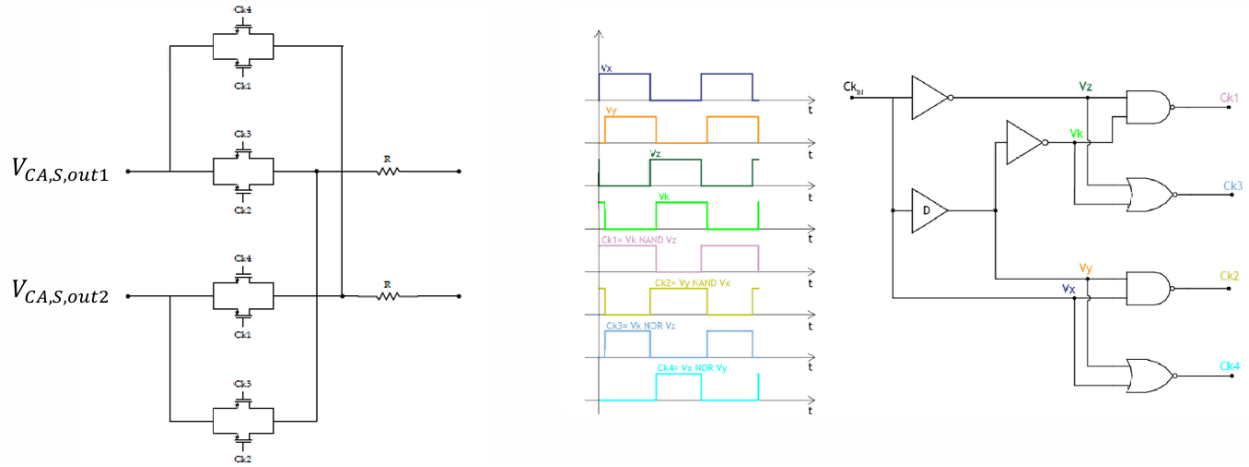

Figure 5.3: A passive mixer demodulator. The differential inputs from the charge amplifier are switched by non-overlapping clocks (ϕ_1, ϕ_2) to reconstruct the signal envelope.

The negative sign is a convention indicating the force opposes motion, but the key is the proportionality to x.

5.10.2 Why Neglect the Reverse Coupling?

The same cross-axis stiffness also generates a force in the drive direction proportional to the sense displacement ($-k_{xy}y$). This term is negligible in the drive-axis equation for two reasons:

1. The drive motion is dominated by the large electrostatic actuation force.
2. The sense displacement y is many orders of magnitude smaller than the drive displacement x (as established in Section 2).

Therefore, the effect on the drive loop is insignificant and can be safely ignored.

5.10.3 Input-Referred Quadrature Rate

To express this mechanical coupling as an equivalent input rate Ω_Q, we equate the two force terms on the right-hand side of Equation (1). In the Laplace domain, where $\dot{x} = sx$, and considering the system at the drive frequency ω_d, the Coriolis force is $2m\Omega(j\omega_d)X$. The quadrature force is $-k_{xy}X$. The factor of j confirms the 90° phase shift between the two.

By direct comparison, the input-referred quadrature rate is:

$$\boxed{\Omega_Q = \frac{k_{xy}}{2m\omega_d}} \tag{5.11}$$

This equation translates a physical stiffness coupling (k_{xy}) into an equivalent angular rate, which shares the same units as a true Ω but is in phase quadrature.

5.11 Relating Quadrature to a Physical Misalignment Angle

The cross-axis stiffness k_{xy} can be linked to a simple geometric imperfection: a misalignment of the primary drive motion. Instead of moving perfectly along the intended X-axis, fabrication tolerances cause the mass to move along a axis that is rotated by a small angle α, as shown in Figure 5.4.

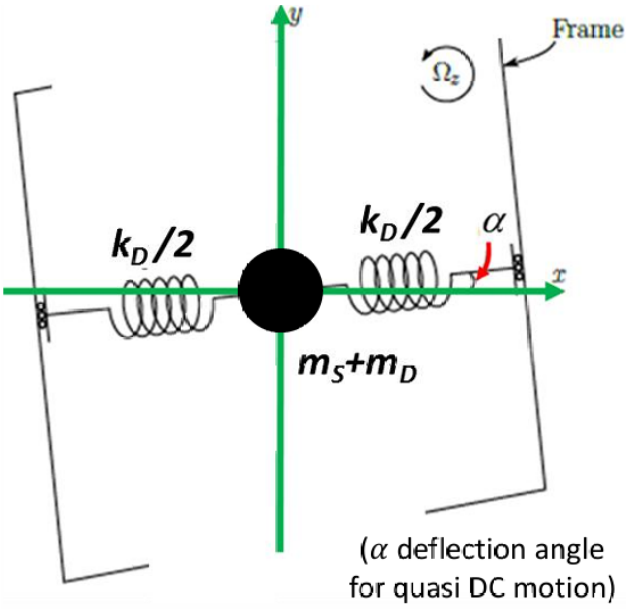

Figure 5.4: Illustration of the drive motion misalignment angle α, where the actual displacement deviates from the intended X-axis by a small angle.

For small angles, the displacement in the Y-direction due to the drive motion is $y \approx \alpha x$. The force required to produce this spurious Y-displacement is provided by the suspension stiffness in the Y-direction, k_y. The relationship between the cross-axis stiffness k_{xy} and the misalignment angle α is:

$$k_{xy} = \alpha k_y \tag{5.12}$$

5.11.1 A Simple Expression for Input-Referred Quadrature

Substituting Equation (3) into Equation (2) yields a remarkably simple and insightful expression for the input-referred quadrature error:

$$\boxed{\Omega_Q = \alpha \cdot \frac{\omega_d}{2}} \tag{5.13}$$

5.11.2 Interpretation and Units

- α is the misalignment angle (in radians), a dimensionless quantity representing the fabrication imperfection.
- ω_d is the drive frequency (in rad/s).

The product yields units of rad/s, which is consistent with an angular rate. This elegant result shows that the quadrature error, expressed as an equivalent input rate, is directly proportional to the small misalignment angle of the drive mode and the drive frequency. It provides a powerful tool for estimating the potential magnitude of quadrature based on expected fabrication tolerances.

5.12 The Frequency Trade-Off: Why Not Operate at Higher Frequencies?

As previously discussed, operating at a higher resonance frequency is desirable for several reasons:

- It places the operating frequency above typical environmental vibrations and disturbances.
- It reduces thermomechanical noise for a given displacement amplitude.
- It reduces the impact of amplifier noise.

This naturally raises the question: why are consumer gyroscopes typically designed with resonance frequencies in the 20 kHz to 30 kHz range, rather than, say, 200 kHz?

The fundamental limitation is **quadrature error**. As derived in Equation (4), the input-referred quadrature is directly proportional to the drive frequency:

$$\Omega_Q = \alpha \cdot \frac{\omega_d}{2} \tag{5.14}$$

Increasing the drive frequency ω_d would linearly increase the already problematic quadrature signal, making it even more difficult to manage within the electronic dynamic range and exacerbating offset errors due to phase mismatches. The choice of operating frequency is therefore a critical trade-off between noise/vibration immunity and quadrature-induced offset.

5.13 Physical Origins of the Misalignment Angle α

The misalignment angle α is a catch-all parameter representing how fabrication imperfections cause the intended drive motion to couple into the sense direction. Its origins can be broadly classified into two categories.

5.13.1 In-Plane Non-Uniformities: Spring Asymmetries

As discussed previously, the most dominant source for in-plane gyroscopes is the non-uniformity of the suspension springs. If the four suspension springs are not perfectly identical (e.g., due to lithography or etching variations), the stiffness matrix is no longer perfectly aligned with the drive and sense axes. This results in a principle axis of motion that is rotated by a small angle α, causing the drive motion to have a component in the sense direction.

5.13.2 Out-of-Plane Effects: Wafer Bow and Tilted Etching

For gyroscopes with out-of-plane sense modes (i.e., devices that measure in-plane angular rates by detecting vertical motion), a more insidious source of quadrature exists: **wafer bow during the Deep Reactive Ion Etching (DRIE) process**.

The fabrication process involves several steps with large temperature variations (e.g., DRIE, oxidation, wafer bonding). These temperature changes, combined with the stack of different materials (silicon, oxide, handle wafer) having different coefficients of thermal expansion, can cause the wafer to temporarily **bow or warp**.

Consider the scenario illustrated in Figure 5.5:

1. During the DRIE step, the wafer is bowed due to thermal stress.
2. The DRIE process etches vertically *with respect to the local surface normal*. On a bowed wafer, this means the etch is not vertical with respect to the global, flat wafer plane.
3. After the etching is complete and the wafer returns to room temperature, it flattens out again.

The result is that the etched structures, such as comb fingers and springs, are no longer perfectly vertical. They have a permanent, small-angle tilt relative to the substrate.

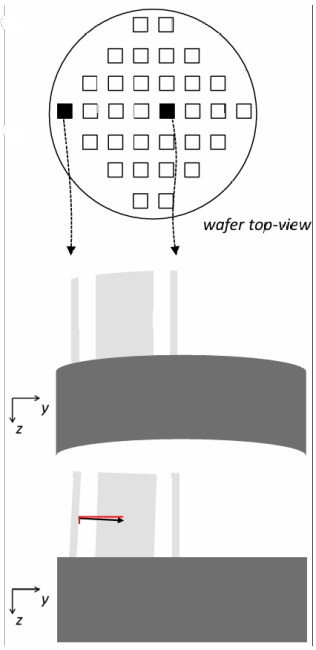

Figure 5.5: The effect of wafer bow during DRIE. Etching on a bowed wafer (left) creates structures that are tilted relative to the substrate. When the wafer is later flattened (right), features like comb fingers have non-uniform gaps and are no longer perfectly vertical.

Impact on Quadrature

For a gyroscope with an out-of-plane sense mode, this tilt is catastrophic. As shown in the right side of Figure 5.5, the gap between moving and fixed fingers becomes non-uniform along the vertical direction. When a drive voltage is applied to actuate the structure in the horizontal (X) direction, the electrostatic force is no longer purely horizontal. Due to the tilted geometry, the force vector has a small but significant vertical component. This vertical force, oscillating at the drive frequency, directly excites the out-of-plane sense mode, creating a massive quadrature signal.

5.14 Quantitative Impact of a Small Tilt

The magnitude of this effect is staggering. Let's consider a seemingly negligible misalignment angle of $\alpha = 0.5°$.

First, convert the angle to radians:

$$\alpha = 0.5° \times \frac{\pi}{180} \approx 0.008\,73\,\text{rad} \tag{5.15}$$

For a gyroscope with a drive frequency $f_d = 20\,\text{kHz}$ ($\omega_d = 2\pi f_d \approx 125\,663\,\text{rad}\,\text{s}^{-1}$), the input-referred quadrature rate is:

$$\begin{aligned}\Omega_Q &= \alpha \cdot \frac{\omega_d}{2} \\ &\approx 0.00873 \times \frac{125663}{2} \\ &\approx 548\,\text{rad}\,\text{s}^{-1}\end{aligned} \tag{5.16}$$

Converting to degrees per second:

$$\Omega_Q \approx 548 \times \frac{180}{\pi} \approx 31\,400\,^\circ\,\text{s}^{-1} \tag{5.17}$$

This is more than **10 times larger than the typical full-scale range** ($2000\,^\circ\,\mathrm{s}^{-1}$) of a consumer gyroscope. This calculation clearly demonstrates why quadrature is the dominant challenge for MEMS gyroscopes, particularly for pitch and roll (out-of-plane sensing) devices. For purely in-plane devices, the effect is less severe but still a primary concern.

5.15 Minimizing Quadrature: A Mechanical Design Perspective

Given the immense magnitude of potential quadrature, minimizing it at its source is paramount. Strategies can be categorized into basic process-related approaches and more advanced design techniques.

5.15.1 Basic Strategies

1. **Improve Process Uniformity:** This is primarily the responsibility of the process engineer. Tighter control over etching, lithography, and thermal budgets directly reduces the sources of asymmetry and wafer bow.

2. **Increase Beam Widths:** A designer can make the suspension beams wider. If the absolute variation in beam width due to etching (e.g., a 200 nm standard deviation) is constant, a wider beam experiences a much smaller *relative* variation, leading to more symmetric springs and a smaller α.

3. **Die Selection:** For high-end, cost-insensitive applications, only dies from the center of the wafer are used. These dies are less affected by the wafer bow and stress gradients that are more pronounced at the wafer edges.

5.15.2 Advanced Design Strategies

1. **Spring Topology Optimization:** The design of the suspension springs is critical. Multi-folded springs can be arranged in ways that are inherently less sensitive to certain fabrication errors. The choice of spring geometry directly impacts the native cross-axis stiffness (k_{xy}) of the suspension.

2. **Decoupling the Centers of Force Application:** This is a sophisticated technique. The Coriolis force is an inertial force that acts effectively at the **center of mass** of the sensing element. The quadrature force, however, originates from elastic coupling in the **springs**. By cleverly designing the mechanical structure, the points of application for these two forces can be made to differ. If the lever arms with which these forces act on the sense mode are different, their relative contribution to the overall sense-mode displacement can be balanced or minimized.

3. **Reduce Drive Frequency (with caution):** As Equation (4) shows, $\Omega_Q \propto \omega_d$. Lowering the drive frequency directly reduces quadrature. This is limited by the need to stay above the frequency of environmental vibrations and acoustic noise (e.g., the 20 kHz upper limit of human hearing is a common boundary for consumer devices to avoid interference from speakers).

4. **Increase Device Size:** A larger device allows for both a larger sense mass (improving SNR) and wider beams (reducing relative process variations). The trade-off is increased chip area and cost.

5.16 The Challenge of High-Performance Multi-Axis Gyroscopes

A key objective in advancing gyroscope technology is to achieve very high performance (approaching navigation grade) within a thin, compact form factor suitable for portable devices. The ultimate goal is to integrate pitch, roll, and yaw rate sensing on a single, flat MEMS die.

However, designing high-performance gyroscopes for **in-plane angular rates** (pitch and roll), which require an **out-of-plane sense mode**, is significantly more challenging than designing for out-of-plane rates (yaw), which use in-plane modes. The reasons for this are fundamental to MEMS fabrication and design:

Table 5.1: Summary of Quadrature Mitigation Strategies

Strategy	**Category**	**Trade-off**
Improve Process Uniformity	Process	Cost, Yield
Increase Beam Width	Design	Area, may increase mode matching complexity
Die Selection (Use Center Dies)	Manufacturing	Yield, Cost
Advanced Spring Topology	Design	Design Complexity
Separate Force Centers	Advanced Design	Extreme Design Complexity
Reduce Drive Frequency	System	Susceptibility to vibration/acoustic noise
Increase Device Size	System	Area, Cost

1. **Planar Process Nature:** MEMS fabrication is inherently planar. Designing structures that move precisely in-plane is much more straightforward than designing for precise out-of-plane motion.

2. **Asymmetric Capabilities:** It is difficult to create structures with perfect symmetry for out-of-plane motion. For example, the capacitive gaps for out-of-plane motion are often defined by different process steps (e.g., wafer bonding, layer thicknesses) than in-plane gaps (defined by DRIE), making them less uniform and controllable.

3. **Severe Quadrature from Sidewall Non-Orthogonality:** As derived in Section 5, the misalignment angle α caused by tilted sidewalls from DRIE (wafer bow) has a massive impact on out-of-plane sense modes, generating enormous quadrature signals.

4. **Dependence on Wafer Thickness:** The resonance frequency and stiffness of out-of-plane modes are highly dependent on the wafer thickness, a parameter with its own tolerances. In-plane modes are primarily defined by lithographic patterning, which can be more precise.

5. **Spurious Modes:** Suspensions designed to be compliant in the vertical direction often introduce many unwanted spurious modes at lower frequencies, which can be easily excited and interfere with proper operation.

Consequently, state-of-the-art, high-performance (e.g., navigation or space-grade) MEMS gyroscopes do not yet use a single flat chip. Instead, they are constructed by assembling three separate, high-performance **yaw** (in-plane sensing) gyroscopes orthogonally, as illustrated in Figure 5.6. While this approach yields a bulky device (on the order of centimeters), it bypasses the difficulties of designing a high-performance pitch/roll gyro. The ability to create a **flat, tri-axis, high-performance gyro** on a single chip remains a key research goal that would revolutionize the size and cost of such systems.

Figure 5.6: A typical high-performance multi-axis gyroscope implementation using three discrete yaw gyroscopes mounted orthogonally.

5.17 Electrostatic Sources of Quadrature: Comb-Drive Asymmetries

Beyond mechanical coupling from springs, quadrature error can also arise from **electrostatic** sources, specifically within the comb-drive actuators used for the drive mode. So far, we have assumed ideal, perfectly symmetrical comb fingers. However, fabrication non-uniformities can cause the gaps on either side of a moving finger to be unequal, as shown in Figure 5.7.

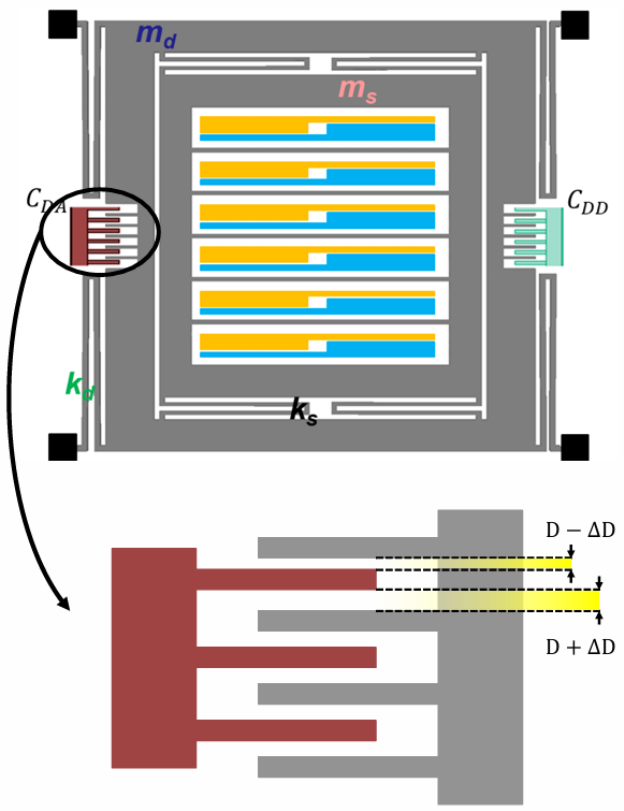

Figure 5.7: A moving finger with a gap asymmetry. The top gap is $d + \Delta d$, and the bottom gap is $d - \Delta d$, leading to unbalanced vertical electrostatic forces.

In an ideal finger, the gaps to the upper and lower fixed electrodes are equal (d). The electrostatic forces in the vertical (Y) direction from the top and bottom gaps are equal and opposite, thus canceling out. If a process imperfection causes the gaps to be different, e.g., $d + \Delta d$ on the top and $d - \Delta d$ on the bottom, the vertical forces no longer balance. The force from the smaller gap ($d - \Delta d$) will be larger, resulting in a **net vertical force** on the rotor at the drive frequency. This net force can excite the sense mode, contributing to the overall quadrature signal.

5.17.1 Modeling the Electrostatic Quadrature Force

To understand whether this force causes quadrature or an offset, we must model it. The electrostatic force in a particular direction is given by the general expression:

$$F = \frac{1}{2} V^2 \frac{\partial C}{\partial (\text{displacement})} \tag{5.18}$$

We are interested in the net force in the sense (Y) direction, F_y. Consider a single moving finger. Let's assume the rotor is biased with a DC voltage V_{dc}. For push-pull actuation, the stator on one side might be

driven with $-V_{ac}\cos(\omega_d t)$ and the other with $+V_{ac}\cos(\omega_d t)$. The voltage difference across a gap is therefore $(V_{dc} \pm V_{ac})$.

The capacitance for the top gap (with asymmetry) is a function of both the drive displacement $x(t) = x_0 \sin(\omega_d t)$ and the sense displacement y. The overlap area changes with x, and the gap changes with y and the imperfection Δd.

$$C_{top}(x, y) = \epsilon_0 t_f \frac{L_0 + x(t)}{d - \Delta d - y} \tag{5.19}$$

$$C_{bot}(x, y) = \epsilon_0 t_f \frac{L_0 + x(t)}{d + \Delta d + y} \tag{5.20}$$

where t_f is the finger thickness, L_0 is the initial overlap, and the signs for y and Δd are chosen so that a positive y reduces the top gap and increases the bottom gap.

The force in the Y-direction is the sum of the forces from the top and bottom:

$$F_y = \frac{1}{2}(V_{dc} - V_{ac})^2 \frac{\partial C_{top}}{\partial y} + \frac{1}{2}(V_{dc} + V_{ac})^2 \frac{\partial C_{bot}}{\partial y} \tag{5.21}$$

The derivatives are:

$$\frac{\partial C_{top}}{\partial y} = \epsilon_0 t_f \frac{L_0 + x(t)}{(d - \Delta d - y)^2} \tag{5.22}$$

$$\frac{\partial C_{bot}}{\partial y} = -\epsilon_0 t_f \frac{L_0 + x(t)}{(d + \Delta d + y)^2} \tag{5.23}$$

Substituting these, and considering that $x(t)$, $y(t)$, and Δd are all small compared to the nominal gap d, we can perform a Taylor expansion to find the significant frequency components of F_y.

The full expansion is lengthy, but the resulting expression for the force will be a product of terms:

- Constant terms and voltage squared terms (V_{dc}^2, V_{ac}^2, $2V_{dc}V_{ac}$).
- Terms containing the drive displacement $x(t) \propto \sin(\omega_d t)$.
- Terms containing the sense displacement $y(t) \propto \sin(\omega_d t)$ (which itself is a response to forces).
- Terms containing the asymmetry Δd.

Because the force expression involves the square of the voltage difference and products of these small terms, the total force F_y will contain multiple frequency components. The key is to identify which of these components are at the drive frequency ω_d and in-phase or in-quadrature with the drive velocity. This analysis, which involves expanding the expression and identifying the 12 resulting terms, allows us to determine whether this electrostatic imbalance contributes to the in-phase Coriolis signal (a rate offset) or the quadrature signal. We will examine the result of this expansion in the following section.

5.18 Electrostatic Sources of Quadrature and Offset: Summary

From the lengthy expansion of the electrostatic force in a comb-drive with gap asymmetries, only terms at the drive resonance frequency ω_d are significant, as they can efficiently excite the sense mode. The expansion yields three key terms at ω_d:

1. **Electrostatic Quadrature Term (Proportional to x):**

   $$F_{y,quad} \propto V_{dc}^2 \cdot \Delta d \cdot x(t)$$

   This is a force in the Y-direction proportional to the drive displacement $x(t)$. Since $x(t)$ is in phase with drive displacement, this force is in quadrature with the Coriolis force (which is proportional to velocity). This term is mathematically equivalent to the mechanical quadrature caused by spring asymmetries.

2. **Electrostatic Spring Softening Term (Proportional to y):**

$$F_{y,soft} \propto V_{dc}^2 \cdot y(t)$$

This is a force proportional to the sense displacement $y(t)$. It acts as an additional spring in the Y-direction, effectively modifying the sense-mode resonance frequency. This is known as the **electrostatic spring softening** effect. It does not directly cause an offset or quadrature but changes the sensor's dynamics.

3. **Direct Offset Term (Proportional to Velocity):**

$$F_{y,offset} \propto V_{dc} V_{ac} \cdot \Delta d \cdot L_0$$

This is a force in the Y-direction that is *not* proportional to x or y. It arises from the cross-product of the DC bias and the AC drive voltage. Crucially, the AC drive voltage $V_{ac}\cos(\omega_d t)$ is in phase with the drive *velocity* (due to the 90° phase shift at resonance between force and displacement). Therefore, this force is in phase with the Coriolis force and will pass through a synchronous demodulator *without* rejection, appearing as a **rate offset**.

5.18.1 Statistical Averaging and Typical Magnitudes

These electrostatic errors are random in sign and magnitude across different fingers, depending on local etching variations. For a system with N comb fingers, the net effect scales with $\sqrt{N}$ due to statistical averaging.

Using typical parameters ($V_{dc} = 20\,\text{V}$, $x_d = 10\,\mu\text{m}$, $d = 2\,\mu\text{m}$, $\Delta d_{rms} = 20\,\text{nm}$), the calculated values are:

- **Electrostatic Quadrature:** On the order of $50\,^\circ/\text{s}$ (input-referred). This is generally much smaller than mechanical quadrature (which can be $10\,000\,^\circ/\text{s}$ or more).
- **Direct Offset:** While smaller in magnitude than the quadrature terms, this component is critically important because it is *not* rejected by demodulation. It contributes directly to the sensor's zero-rate output (ZRO) offset.

5.19 The Path to Compensation: Understanding Phase Errors

The overall output of the gyroscope is corrupted by two main error types originating from the sources discussed:

- **Offset from Direct Electrostatic Forces:** This small signal passes directly through the demodulator.
- **Quadrature Feedthrough:** The enormous mechanical (and some electrostatic) quadrature signal, Ω_Q, is largely rejected by synchronous demodulation. However, any imperfection in the phase of the demodulation reference will cause a fraction of this large signal to leak into the output, creating a significant offset.

To compensate for these errors, we must first understand the origins of the **phase error** (ϕ) in the demodulation path. This phase error is the difference between the ideal demodulation phase (aligned with the Coriolis signal) and the actual phase of the reference signal applied to the mixer.

Referring to the system block diagram in Figure 5.8, the phase error accumulates from all elements between two points:

- **Point A:** The input to the demodulator in the sense chain (the modulated signal containing Ω and Ω_Q).
- **Point B:** The demodulation reference signal, derived from the drive loop (ideally after the 90° phase shifter).

Any phase lag introduced in the signal path from the drive motion to Point A, or in the reference path to Point B, will contribute to the overall phase error ϕ.

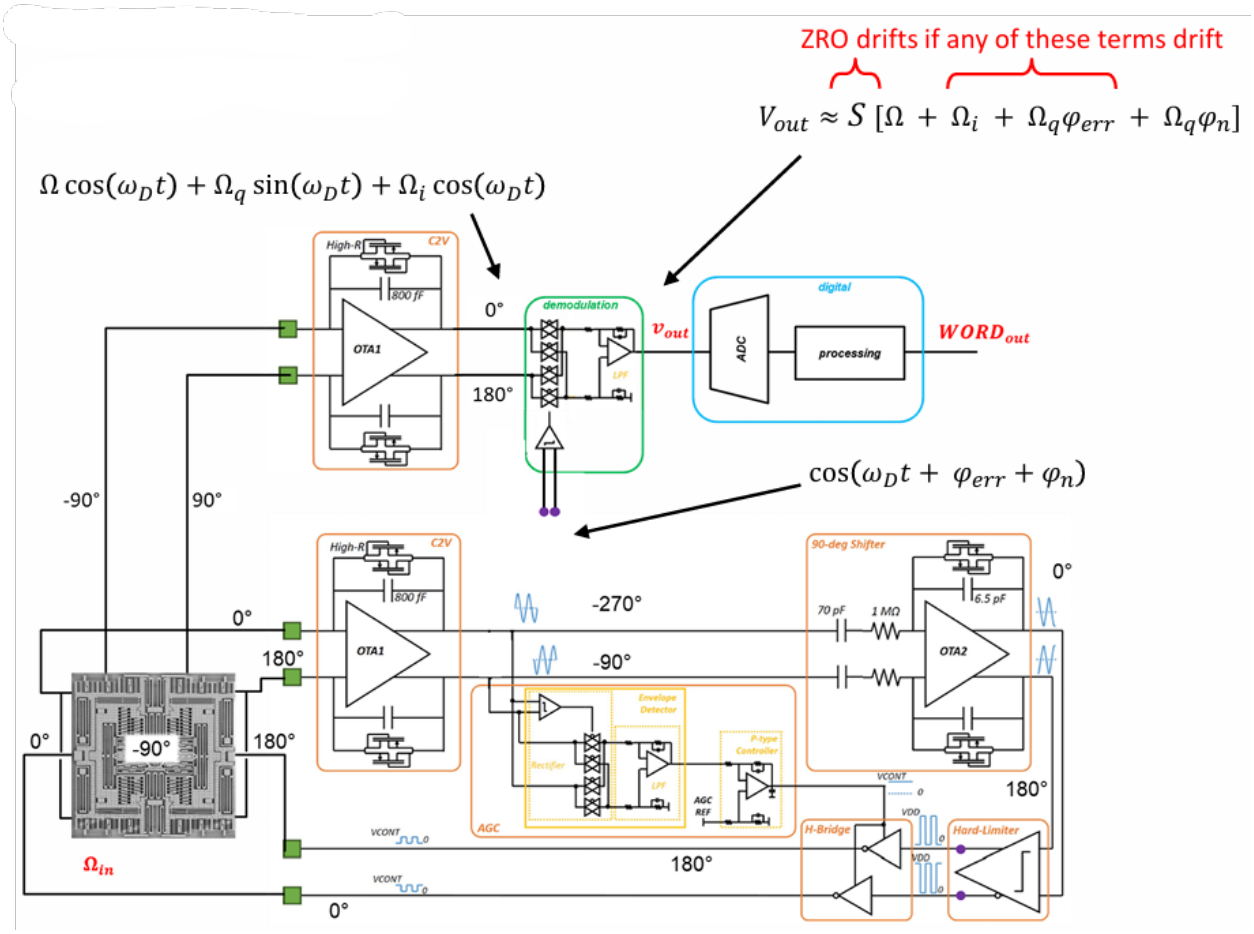

Figure 5.8: Simplified block diagram of a gyroscope system, highlighting the paths contributing to the demodulation phase error ϕ.

5.19.1 Sources of Phase Error

The phase error has two primary categories of origins: mechanical and electronic.

1. Mechanical Phase Error: The Sense Mode Transfer Function

This is a fundamental and often dominant source. The Coriolis force excites the sense mode at the drive frequency ω_d. In a mode-split gyroscope, ω_d is deliberately offset from the sense resonance frequency ω_y (typically by a few hundred Hz to a few kHz). The mechanical transfer function of a 2nd-order spring-mass-damper system from force to displacement is:

$$H_y(\omega) = \frac{1/m}{(\omega_y^2 - \omega^2) + j\frac{\omega_y}{Q_y}\omega} \tag{5.24}$$

This transfer function has a frequency-dependent phase shift $\phi_y(\omega)$, given by:

$$\phi_y(\omega) = -\arctan\left(\frac{\frac{\omega_y}{Q_y}\omega}{\omega_y^2 - \omega^2}\right) \tag{5.25}$$

Since $\omega_d \neq \omega_y$, the phase shift $\phi_y(\omega_d)$ is non-zero. This is the **first mechanical phase error**. The Coriolis force, which is perfectly in phase with drive velocity, produces a sense-mode displacement that is

phase-shifted by $\phi_y(\omega_d)$. The signal at the output of the sense front-end (Point A) therefore contains this mechanical phase shift.

2. Electronic Phase Errors

Several electronic components in both the sense and reference paths introduce additional phase lags:

- **Drive Loop Components:** The 90° phase shifter, comparators (for square wave generation), and any buffers all have finite bandwidth and propagation delays, introducing a phase lag $\phi_{drive,elec}$.
- **Sense Front-End:** The charge amplifier and any subsequent filtering or gain stages have a finite bandwidth, introducing a phase lag $\phi_{sense,elec}$ to the signal at Point A.
- **Passive Networks:** Any RC filtering or parasitic capacitances in the interconnects contribute to further phase shifts.

The total phase error ϕ is the sum of these contributions:

$$\phi = \phi_y(\omega_d) + \phi_{sense,elec} + \phi_{drive,elec} \tag{5.26}$$

5.19.2 Consequences of Phase Error

As derived earlier, the output after demodulation with a phase error ϕ becomes:

$$V_{out} \approx \frac{1}{2} SF\,[\Omega + \Omega_Q \cdot \phi] \tag{5.27}$$

where we have neglected the small $\cos(\phi)$ scaling of Ω and the phase noise.

This equation highlights the two critical challenges:

1. **Stability of Offset:** Even with a constant Ω_Q, any drift in ϕ (e.g., with temperature) will cause the offset ($\Omega_Q \cdot \phi$) to drift. This is often the dominant contributor to bias instability.
2. **Magnitude of Offset:** Given that Ω_Q can be on the order of 10 000 °/s, a phase error of just $\phi = 0.01$ rad (about 0.57°) results in an offset of 100 °/s, which is still 5% of a typical consumer FSR.

Therefore, the design goals are twofold: first, to minimize Ω_Q at its source (mechanical and electrostatic design), and second, to minimize and stabilize ϕ through careful electronic design and, as we will see, through active compensation techniques.

5.20 Mechanical Phase Error: A Quantitative Example

In a mode-split gyroscope, the drive frequency ω_d is intentionally offset from the sense resonance frequency ω_y. The transfer function of the sense mode introduces a phase shift $\phi_y(\omega_d)$ between the Coriolis force (which is proportional to drive velocity) and the resulting sense displacement.

Consider a typical example:

- Sense mode resonance frequency: $f_y = 20\,350$ Hz
- Drive mode frequency: $f_d = 20\,000$ Hz (a split of $\Delta f = 350$ Hz)
- Sense mode quality factor: $Q_y = 700$

As illustrated in Figure 5.9, even though the drive frequency is below the resonance peak, the phase shift is not zero. For this specific example, the phase error ϕ_{mech} is:

$$\phi_{mech} \approx 36\,\text{mrad} \tag{5.28}$$

This corresponds to approximately 2°.

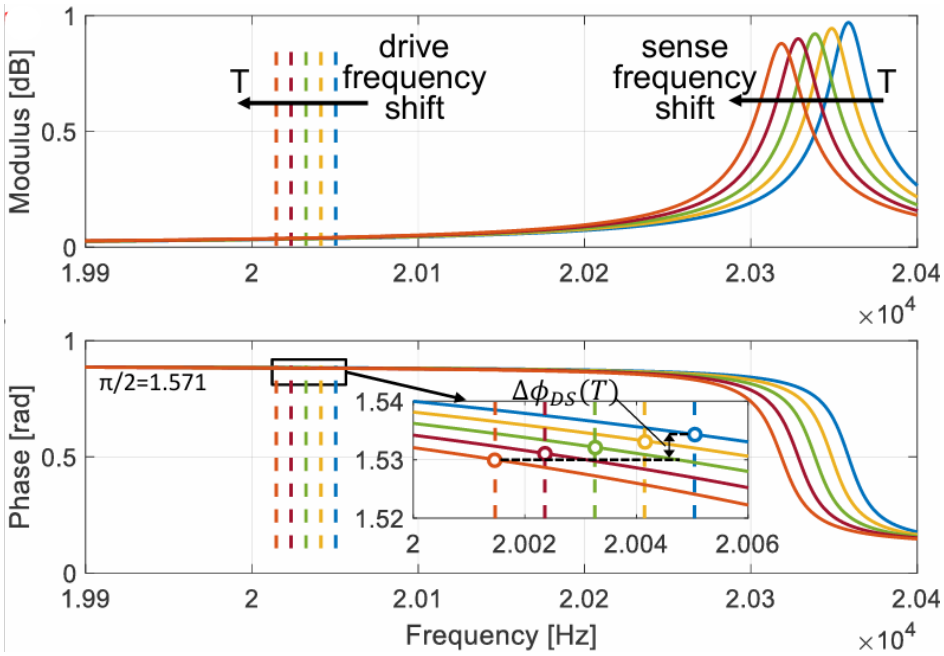

Figure 5.9: Phase response of the high-Q sense mode. Even at a drive frequency 350 Hz below resonance, the phase has already begun to deviate from its quasi-static value.

5.20.1 Impact on Output Offset

If the input-referred quadrature Ω_Q is significant, this mechanical phase error causes a portion of it to leak into the output. For an input-referred quadrature of $\Omega_Q = 1000\,^\circ/\text{s}$:

$$\text{Offset} = \Omega_Q \cdot \phi_{mech} = 1000 \times 0.036 = 36\,^\circ/\text{s} \tag{5.29}$$

This is a substantial offset, demonstrating that mechanical phase error alone can produce a significant zero-rate output, even with perfect electronics.

5.20.2 Temperature Dependence of Mechanical Phase Error

The situation worsens with temperature. Both the resonance frequencies and the quality factor are temperature-dependent:

- Frequencies drift at approximately -30 ppm/K. Since $f_y > f_d$, the *absolute* frequency shift of the sense mode is slightly larger than that of the drive mode. This causes the frequency split Δf to change with temperature.
- The quality factor Q typically decreases with increasing temperature.

Consider a temperature increase of 10 K:

- Both f_y and f_d decrease, but f_y decreases slightly more in absolute terms, reducing Δf.
- Q_y decreases, broadening the phase transition region.

The net effect is that the phase shift at the drive frequency changes. For the given example, a 10 K temperature change results in a variation of the phase error:

$$\Delta\phi_{mech} \approx 5\,\text{mrad} \tag{5.30}$$

This translates to a temperature coefficient of approximately:

$$TC_{\phi_{mech}} \approx 60\,\mu\text{rad/K} \tag{5.31}$$

The critical implication is that even if the phase error is calibrated at a single temperature (e.g., room temperature), it will drift with temperature, causing the output offset to drift as well.

5.21 Electronic Phase Errors

The electronic signal paths also contribute to the total phase error. Referring to the system block diagram (Figure 5.8), the key electronic blocks are:

- The charge amplifiers in both the sense and drive paths.
- The 90° phase shifter in the drive loop, which generates the demodulation reference.
- Any additional filters or buffers.

5.21.1 Matched Charge Amplifiers

A crucial design strategy is to make the charge amplifiers in the sense and drive paths **identical**. If they are identical, any phase lag introduced by these amplifiers (ϕ_{CA}) will affect both the signal path (to the mixer input) and the reference path (to the mixer reference port) equally. In an ideal mixer, these common-mode phase shifts cancel out, and they do not contribute to the net phase error ϕ. This is a powerful reason for using matched front-ends.

5.21.2 The 90° Phase Shifter and Amplifier Non-Idealities

The most critical electronic component for phase error is the 90° phase shifter. Any deviation from exactly 90° directly becomes part of the phase error ϕ. Furthermore, the operational amplifiers used in these stages are not ideal. They have a finite gain-bandwidth product and additional high-frequency poles.

Consider an amplifier with a high-frequency pole at f_p. This pole introduces an additional phase lag at the operating frequency f_d, given by:

$$\Delta\phi_{amp} \approx -\arctan\left(\frac{f_d}{f_p}\right) \tag{5.32}$$

Design Guideline: To keep this error small, the high-frequency poles of all amplifiers in the signal path should be placed at least two decades above the drive frequency.

- If $f_d = 20\,\text{kHz}$ and $f_p = 200\,\text{kHz}$ (one decade above), the additional phase lag is $\Delta\phi_{amp} \approx -\arctan(0.1) \approx -5.7°$ (100 mrad), which is unacceptably large.
- If $f_p = 2\,\text{MHz}$ (two decades above), the lag reduces to $\Delta\phi_{amp} \approx -\arctan(0.01) \approx -0.57°$ (10 mrad), which is much more manageable.

5.21.3 Temperature Dependence of Electronic Phase Error

The passive components (resistors, capacitors) and the amplifier characteristics that determine the pole frequency f_p also drift with temperature. A typical temperature coefficient for integrated circuit components is on the order of 200 ppm/K. This causes the pole frequency to shift, leading to a change in the phase lag $\Delta\phi_{amp}$.

For a well-designed system with $f_p \gg f_d$, the resulting temperature coefficient of the electronic phase error can be estimated. In the example provided, the drift is approximately:

$$TC_{\phi_{elec}} \approx 20\,\mu\text{rad/K} \tag{5.33}$$

5.22 Summary of Phase Error Contributions

The total phase error ϕ is the sum of its mechanical and electronic components:

$$\phi(T) = \phi_{mech}(T) + \phi_{elec}(T) \tag{5.34}$$

Both components are functions of temperature.

Table 5.2: Summary of Phase Error Magnitudes and Drifts (Example Values)

Source	**Absolute Error at RT**	**Temperature Drift**
Mechanical (Sense Mode)	$\sim 36\,\text{mrad}$ (2°)	$\sim 60\,\mu\text{rad/K}$
Electronic (Amplifier Poles)	$\sim 10\,\text{mrad}$ (0.57°)	$\sim 20\,\mu\text{rad/K}$

5.23 Meeting System-Level Stability Targets

Let's assess whether current designs can meet a challenging stability target, such as $2\,^\circ$/hour over a 100 K temperature range.

1. **Total Phase Drift:** From Table 5.2, the total phase drift is approximately $80\,\mu$rad/K. Over a 100 K range, the total change in phase error is:

$$\Delta\phi_{total} \approx 80 \times 10^{-6} \times 100 = 8\,\text{mrad} \tag{5.35}$$

2. **Resulting Offset Drift:** If the input-referred quadrature is $\Omega_Q = 1000\,^\circ$/s, the change in output offset due to this phase drift is:

$$\Delta\text{Offset} = \Omega_Q \cdot \Delta\phi_{total} = 1000 \times 0.008 = 8\,^\circ/\text{s} \tag{5.36}$$

   Converting to degrees per hour: $8\,^\circ/\text{s} \times 3600 = 28\,800\,^\circ$/hour. This is orders of magnitude larger than the $2\,^\circ$/hour target.

3. **Implication:** To meet the $2\,^\circ$/hour target, either the quadrature Ω_Q must be drastically reduced (through superior mechanical design) or the phase drift must be compensated (through electronic means). This calculation highlights why high-performance gyroscopes require both extremely low quadrature and stable, temperature-compensated electronics.

Key Takeaways

- Mechanical phase error from the sense mode transfer function is significant and temperature-dependent.
- Electronics should be designed with matched front-ends and all poles placed at least two decades away from the drive frequency to minimize their contribution.
- Even with careful design, the residual phase drift, when multiplied by a large quadrature signal, can produce offset drifts that far exceed high-performance requirements. Active compensation of quadrature and/or phase is therefore essential.

5.24 The Impossibility of Reaching High Performance without Quadrature Cancellation

Let us consider the implications of the phase drift calculated in the previous section. Assume we do not actively compensate for phase error. Using the numbers from our example:

- Input-referred quadrature: $\Omega_Q = 1000\,^\circ/\text{s}$ (this is a conservative estimate; native quadrature is often much larger).
- Mechanical phase error at room temperature: $\phi_{mech} \approx 36\,\text{mrad}$.

The resulting output offset due to quadrature feedthrough is:

$$\text{Offset} = \Omega_Q \cdot \phi_{mech} = 1000 \times 0.036 = 36\,^\circ/\text{s} \tag{5.37}$$

This $36\,^\circ/\text{s}$ offset is large, but in principle, it can be calibrated out at a single temperature. The real problem is drift. As calculated, the total phase drift (mechanical + electronic) is approximately $80\,\mu\text{rad/K}$. Over an automotive temperature range of 100 K, the phase drift is:

$$\Delta\phi_{total} = 80 \times 10^{-6} \times 100 = 8\,\text{mrad} \tag{5.38}$$

The resulting offset drift is:

$$\Delta\text{Offset} = \Omega_Q \cdot \Delta\phi_{total} = 1000 \times 0.008 = 8\,^\circ/\text{s} \tag{5.39}$$

Converting to degrees per hour: $8\,^\circ/\text{s} \times 3600 = 28\,800\,^\circ/\text{hour}$. This is *three orders of magnitude* larger than a target of $2\,^\circ/\text{hour}$, let alone navigation-grade performance ($0.1\,^\circ/\text{hour}$ or less).

5.24.1 Why Calibration and Trimming Are Insufficient

1. **Temperature Calibration:** One could characterize the phase error across temperature and apply a compensation lookup table. However, calibrating out a three-to-four order-of-magnitude drift is impractical due to accuracy limits and the enormous dynamic range required.
2. **Phase Trimming:** Another idea is to introduce a programmable phase delay in the demodulation path to cancel the phase error. To achieve the required stability (e.g., keeping residual phase drift below microradians), the trimming resolution would need to be extraordinarily fine. The minimum phase step is set by the clock frequency used to generate the delay:

$$\Delta\phi_{min} = 2\pi \frac{f_d}{f_{clock}} \tag{5.40}$$

To achieve $\Delta\phi_{min} < 1\,\mu\text{rad}$ with $f_d = 20\,\text{kHz}$, the required clock frequency would be:

$$f_{clock} > \frac{2\pi \times 20 \times 10^3}{1 \times 10^{-6}} \approx 125\,\text{GHz} \tag{5.41}$$

This is completely impractical for low-power, integrated systems.

Conclusion: For applications requiring stability on the order of degrees per hour or better, it is impossible to rely on phase calibration or trimming alone. The only viable path is to **remove quadrature at its origin**.

5.25 A Simple Partial Solution: Capacitive Quadrature Cancellation

Before discussing full electromechanical cancellation, it is worth noting a simpler technique often used in consumer and automotive applications to prevent saturation of the sense front-end.

As shown in Figure 5.10, a capacitor C_{comp} is connected between the output of the drive charge amplifier and the input of the sense charge amplifier. The current through this capacitor is:

$$I_{comp} = C_{comp} \frac{dV_{drive}}{dt} \tag{5.42}$$

Since V_{drive} is proportional to the drive displacement $x(t)$, its derivative is proportional to the drive velocity $\dot{x}(t)$, which is exactly in phase with the Coriolis force. However, recall that quadrature is proportional to $x(t)$, which is 90° out of phase with velocity. Therefore, $\dot{x}(t)$ is actually in phase with *quadrature*. By injecting this current into the sense node, we can cancel a portion of the quadrature signal.

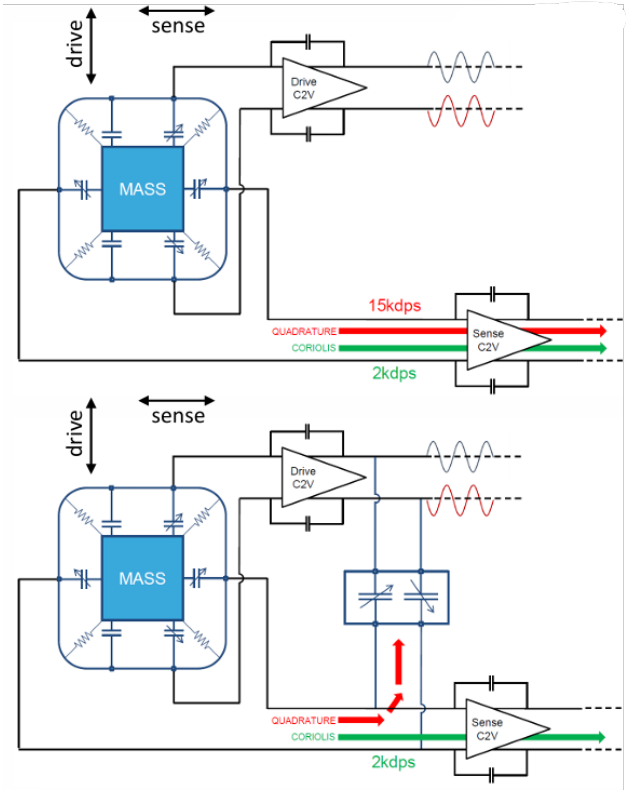

Figure 5.10: Capacitive feedforward for partial quadrature cancellation. A programmable capacitor C_{comp} diverts a current proportional to the drive velocity, which is in phase with the quadrature signal.

5.25.1 Practical Implementation

- A bank of programmable capacitors (e.g., binary-weighted: 1 pF, 2 pF, 4 pF, ...) is used.
- The sign of cancellation can be switched by routing the injection to either the positive or negative input of the sense charge amplifier, as the sign of quadrature is unknown a priori.
- At startup, a calibration routine finds the capacitor setting that minimizes the quadrature signal at the sense output.

5.25.2 Pros and Cons

Advantages	Disadvantages
Very low power	Only partial cancellation (limited by capacitor resolution)
Easy to implement	Does not compensate for phase drift
No MEMS design changes	Residual quadrature still causes temperature-dependent offset
Prevents front-end saturation	

This method is widely used in consumer devices (e.g., iPhones) to calibrate quadrature at the beginning of life, ensuring the sense chain does not saturate. However, it does not solve the temperature drift problem.

5.26 Electromechanical Quadrature Cancellation at the Origin

To achieve the high stability required for navigation-grade performance, quadrature must be canceled at its mechanical source. The goal is to design a structure that can generate a force in the sense direction that is:

- Proportional to the drive displacement $x(t)$ (to match the quadrature force).
- Programmable in both sign and amplitude (to counteract the unpredictable, fabrication-dependent quadrature).

5.26.1 Implementation for Yaw (In-Plane) Gyroscopes

Figure 5.11 shows a typical electrode arrangement for quadrature cancellation in a yaw gyroscope (in-plane sense mode). The red frame is the Coriolis frame (the mass that is driven in the x-direction and free to move in the y-direction). Within this frame, a set of dedicated quadrature cancellation electrodes is designed.

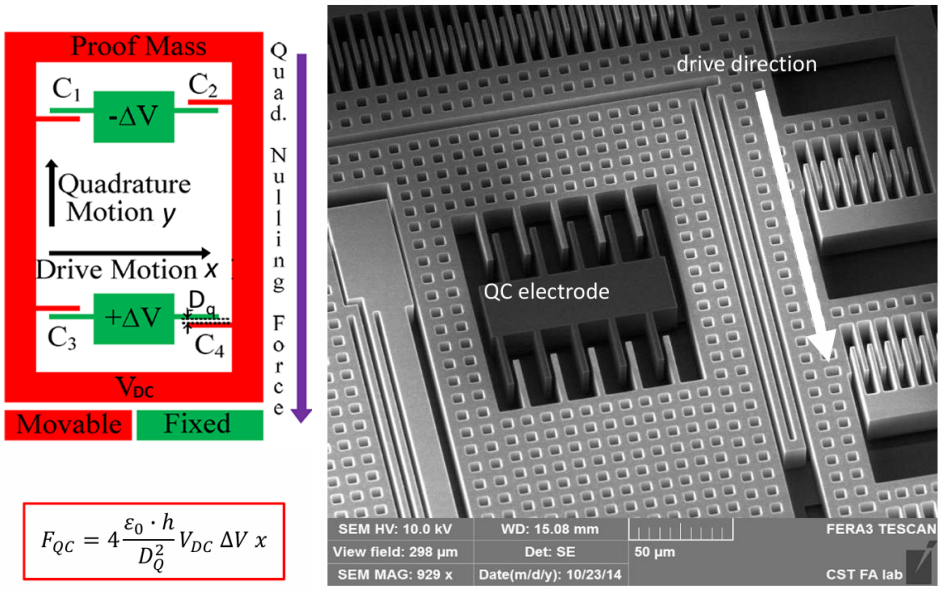

Figure 5.11: Quadrature cancellation electrodes integrated into the Coriolis frame of a yaw gyroscope.

Electrode Configuration and Biasing

- The rotor (Coriolis frame) is biased at a DC voltage V_R (e.g., 10 V to 20 V).
- Two sets of fixed electrodes are placed on opposite sides of the frame. One set is biased at $V_R + \Delta V$, the other at $V_R - \Delta V$, where ΔV is a small programmable voltage (e.g., 1 V).
- The electrodes are designed such that their capacitance varies with both the drive displacement x and the sense displacement y.

Force Derivation

The electrostatic force in the y-direction generated by these electrodes can be derived similarly to the comb-drive case. For a small displacement, the capacitance between the rotor and one electrode is approximately:

$$C(x, y) \approx \epsilon_0 \frac{A}{d_0} \left(1 + \frac{x}{L}\right) \left(1 + \frac{y}{d_0}\right) \tag{5.43}$$

where A is the nominal overlap area, d_0 is the nominal gap, and L is a characteristic length. The force is:

$$F_y = \frac{1}{2}(V_R - (V_R \pm \Delta V))^2 \frac{\partial C}{\partial y} \approx \frac{1}{2}(\mp \Delta V)^2 \frac{\partial C}{\partial y} \tag{5.44}$$

More importantly, the *difference* in forces between the two sides, which is what matters for a differential sense mode, yields a term proportional to $V_R \Delta V$ and to the drive displacement x. This is the desired effect:

$$F_{y,quad,cancel} \propto V_R \cdot \Delta V \cdot x(t) \tag{5.45}$$

Tunability

- **Amplitude:** The magnitude of the cancellation force is proportional to ΔV and V_R.
- **Sign:** The sign of the force can be reversed by swapping which electrode gets $+\Delta V$ and which gets $-\Delta V$. This is easily done with switches in the biasing circuitry.

By adjusting ΔV, the system can generate a force that exactly counteracts the mechanical quadrature force $k_{xy}x(t)$, effectively nulling the quadrature signal at its source. This is the most powerful method for eliminating quadrature-induced offset and its temperature drift.

5.26.2 Extension to Pitch/Roll (Out-of-Plane) Gyroscopes

A similar concept exists for gyroscopes with out-of-plane sense modes, though the electrode geometry is different (often involving electrodes above or below the moving mass). The principle remains the same: create a tunable electrostatic force proportional to drive displacement that acts in the sense direction.

5.27 Electromechanical Quadrature Cancellation: The Principle

To cancel quadrature at its origin, we need to generate a force in the sense direction that is:

- Proportional to the drive displacement $x(t)$.
- Tunable in both sign and amplitude.

This is achieved by designing dedicated electrodes within the Coriolis frame, as shown in Figure 5.12.

5.27.1 Force Derivation for In-Plane (Yaw) Gyroscopes

Consider the four electrodes labeled E1 to E4 in Figure 5.12. The rotor (red frame) is biased at a DC voltage V_{DC}. The fixed electrodes (green) are biased at $V_{DC} + \Delta V$ or $V_{DC} - \Delta V$, as indicated. We assume a small drive displacement $x(t)$ in the direction shown, and we are interested in the net electrostatic force in the y (sense) direction.

Force from Each Electrode

The electrostatic force is given by $F = \frac{1}{2}V^2 \frac{\partial C}{\partial y}$. For parallel-plate capacitors, the capacitance is $C = \epsilon_0 \frac{A}{d}$, where A is the overlapping area and d is the gap. The overlapping area for each electrode is a function of the drive displacement $x(t)$.

- **Electrode E1 (Top-Left):**
  - Overlap area increases with positive x: $A = A_0 + \alpha x$.
  - Voltage difference: $V_{DC} - (V_{DC} - \Delta V) = +\Delta V$.
  - Force direction: Attracts the rotor upward (positive y).
  - Force magnitude: $F_1 \propto +\Delta V^2 \cdot (A_0 + \alpha x)$.
- **Electrode E2 (Top-Right):**
  - Overlap area *decreases* with positive x: $A = A_0 - \alpha x$.
  - Voltage difference: $V_{DC} - (V_{DC} + \Delta V) = -\Delta V$.

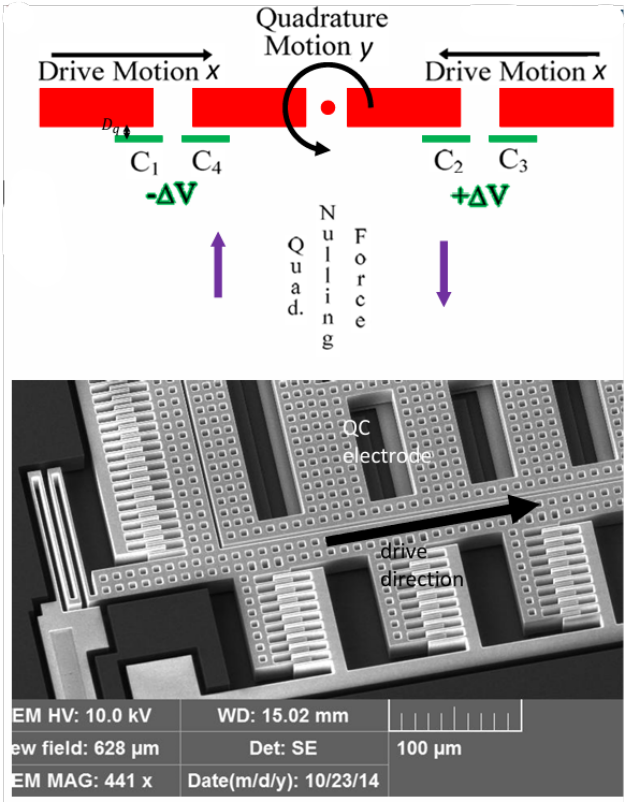

Figure 5.12: Quadrature cancellation electrodes integrated into the Coriolis frame. The rotor (red) is biased at V_{DC}, and fixed electrodes (green) are biased at $V_{DC} \pm \Delta V$. Quadrature cancellation for out-of-plane sense modes. Electrodes are placed beneath the moving structure, and their overlap area changes with drive displacement.

  - Force direction: Attracts the rotor downward (negative y).
  - Force magnitude: $F_2 \propto -(-\Delta V)^2 \cdot (A_0 - \alpha x) = -\Delta V^2 \cdot (A_0 - \alpha x)$.

- **Electrode E3 (Bottom-Left):**
  - Overlap area increases with positive x: $A = A_0 + \alpha x$.
  - Voltage difference: $V_{DC} - (V_{DC} + \Delta V) = -\Delta V$.
  - Force direction: Attracts the rotor downward (negative y).
  - Force magnitude: $F_3 \propto -(-\Delta V)^2 \cdot (A_0 + \alpha x) = -\Delta V^2 \cdot (A_0 + \alpha x)$.

- **Electrode E4 (Bottom-Right):**
  - Overlap area decreases with positive x: $A = A_0 - \alpha x$.
  - Voltage difference: $V_{DC} - (V_{DC} - \Delta V) = +\Delta V$.
  - Force direction: Attracts the rotor upward (positive y).
  - Force magnitude: $F_4 \propto +\Delta V^2 \cdot (A_0 - \alpha x)$.

Total Force

Summing all four forces:

$$\begin{aligned} F_{total} &= F_1 + F_2 + F_3 + F_4 \\ &\propto \Delta V^2(A_0 + \alpha x) \; - \; \Delta V^2(A_0 - \alpha x) \; - \; \Delta V^2(A_0 + \alpha x) \; + \; \Delta V^2(A_0 - \alpha x) \\ &= 0 \end{aligned} \tag{5.46}$$

All terms proportional to ΔV^2 cancel exactly. However, this analysis neglected the fact that the force is proportional to the square of the voltage difference. A more accurate derivation must include the full expression $(V_{DC} \pm \Delta V)^2$. Expanding these squares reveals additional terms.

$$(V_{DC} - \Delta V)^2 = V_{DC}^2 - 2V_{DC}\Delta V + \Delta V^2 \tag{5.47}$$

$$(V_{DC} + \Delta V)^2 = V_{DC}^2 + 2V_{DC}\Delta V + \Delta V^2 \tag{5.48}$$

When these expansions are used in the force expressions, the ΔV^2 terms still cancel as shown above. However, the cross-terms ($\mp 2V_{DC}\Delta V$) do not cancel in the same way. Recalculating with these cross-terms:

- For E1 (voltage diff $= -\Delta V$): Force $\propto (V_{DC}^2 - 2V_{DC}\Delta V + \Delta V^2)(A_0 + \alpha x)$.
- For E2 (voltage diff $= +\Delta V$): Force $\propto -(V_{DC}^2 + 2V_{DC}\Delta V + \Delta V^2)(A_0 - \alpha x)$.
- For E3 (voltage diff $= +\Delta V$): Force $\propto -(V_{DC}^2 + 2V_{DC}\Delta V + \Delta V^2)(A_0 + \alpha x)$.
- For E4 (voltage diff $= -\Delta V$): Force $\propto (V_{DC}^2 - 2V_{DC}\Delta V + \Delta V^2)(A_0 - \alpha x)$.

Summing these, all terms proportional to V_{DC}^2 and ΔV^2 cancel. The only terms that survive are those proportional to $2V_{DC}\Delta V$ and αx:

$$\begin{aligned} F_{total} &\propto [-2V_{DC}\Delta V(A_0 + \alpha x)] + [-2V_{DC}\Delta V(A_0 - \alpha x)] \\ &\quad + [-2V_{DC}\Delta V(A_0 + \alpha x)] + [-2V_{DC}\Delta V(A_0 - \alpha x)] \\ &\propto -8V_{DC}\Delta V \alpha x \end{aligned} \tag{5.49}$$

The result is a net force:

$$\boxed{F_{y,quad_cancel} \propto -8V_{DC}\Delta V \alpha x} \tag{5.50}$$

This force has exactly the desired properties:

- It is proportional to the drive displacement x.
- Its magnitude is tunable via ΔV.
- Its sign is tunable by swapping which electrodes receive $+\Delta V$ and $-\Delta V$.

5.27.2 Practical Implementation

- Multiple sets of such four-electrode groups can be used to increase the cancellation force (e.g., using multiple fingers).
- The electrodes are sized so that the maximum cancellation force can compensate for the worst-case expected quadrature (e.g., 5000 °/s input-referred).
- The sign is controlled by a simple switch network that interchanges the $\pm\Delta V$ connections.

5.28 Extension to Out-of-Plane (Pitch/Roll) Gyroscopes

A similar concept applies to gyroscopes with out-of-plane sense modes, where quadrature manifests as a vertical displacement during drive motion. Figure 5.12 shows the electrode configuration.

5.28.1 Operating Principle

- The drive motion is an anti-phase horizontal movement (tuning-fork mode).
- Electrodes are placed underneath the moving masses. As the mass moves horizontally, the overlap area between the mass and the underlying electrode changes.
- The vertical electrostatic force is proportional to the overlap area and the square of the voltage difference.

Electrode Configuration and Biasing

Consider four electrodes, E1 through E4, placed under the left and right masses. The biasing scheme is analogous to the in-plane case:

- Rotor (moving mass) biased at V_{DC}.
- Electrodes E1 and E4 biased at $V_{DC} - \Delta V$.
- Electrodes E2 and E3 biased at $V_{DC} + \Delta V$ (or vice versa, depending on desired sign).

For a positive drive motion (masses moving apart):

- Overlap of E1 (left mass, left electrode) *increases*.
- Overlap of E2 (left mass, right electrode) *decreases*.
- Overlap of E3 (right mass, left electrode) *increases* (depending on definition of positive motion).
- Overlap of E4 (right mass, right electrode) *decreases*.

Force Derivation

Writing the force for each electrode, with careful attention to the sign of the force direction (up/down) and the sign of the x-dependence, yields a result identical in form to the in-plane case:

$$\boxed{F_{z,quad_cancel} \propto V_{DC}\Delta V \cdot x} \tag{5.51}$$

Again, the force is proportional to x, tunable in amplitude via ΔV, and tunable in sign by swapping the $\pm\Delta V$ connections.

5.28.2 Layout Example

Figure 5.12 shows a typical layout implementation. Multiple electrodes (e.g., four fingers in this example) are used to achieve sufficient force magnitude. The electrodes occupy area on the chip, but this is a necessary trade-off for achieving high performance.

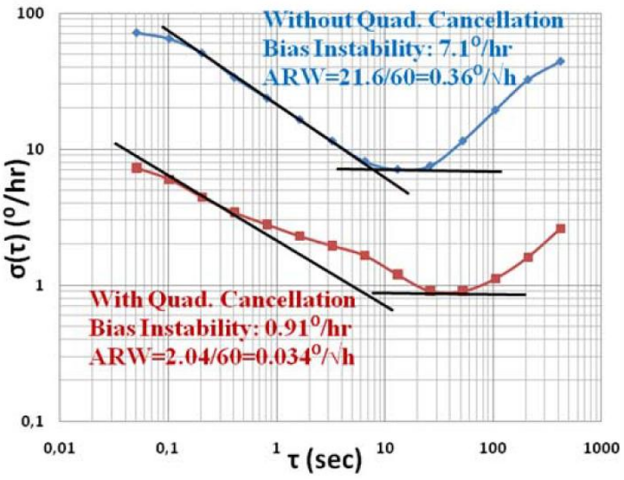

Figure 5.13: Allan deviation plot showing the dramatic improvement in bias instability with electromechanical quadrature cancellation.

5.29 Performance Impact: A Historical Example

The effectiveness of this technique is demonstrated by results published around 2010. Figure 5.13 shows the Allan deviation of a gyroscope with and without quadrature cancellation.

- **Without quadrature cancellation:** Bias instability $\approx 7\,^\circ$/hour.
- **With quadrature cancellation:** Bias instability $\approx 0.9\,^\circ$/hour.

This represents nearly an order of magnitude improvement. By cancelling quadrature at its origin, the dominant source of offset drift is removed, allowing the gyroscope to approach navigation-grade performance.

5.30 Summary

- Dedicated electrodes, biased with $V_{DC} \pm \Delta V$, generate a force proportional to $V_{DC}\Delta V \cdot x$ in the sense direction.
- This force can be tuned to exactly cancel the mechanical quadrature force.
- The sign is controlled by swapping the $\pm\Delta V$ connections.
- The concept works for both in-plane (yaw) and out-of-plane (pitch/roll) gyroscopes.
- Experimental results show dramatic improvements in bias stability, making this technique essential for high-performance applications.

5.31 Automatic Quadrature Cancellation: A Closed-Loop Approach

A one-time calibration of the quadrature compensation voltage ΔV can null the quadrature at a single temperature. However, as discussed, the mechanical quadrature Ω_Q (i.e., the misalignment angle α) itself can drift with temperature. Therefore, to maintain optimum performance over the device lifetime and across temperature, a more sophisticated approach is required: an **automatic quadrature cancellation loop**.

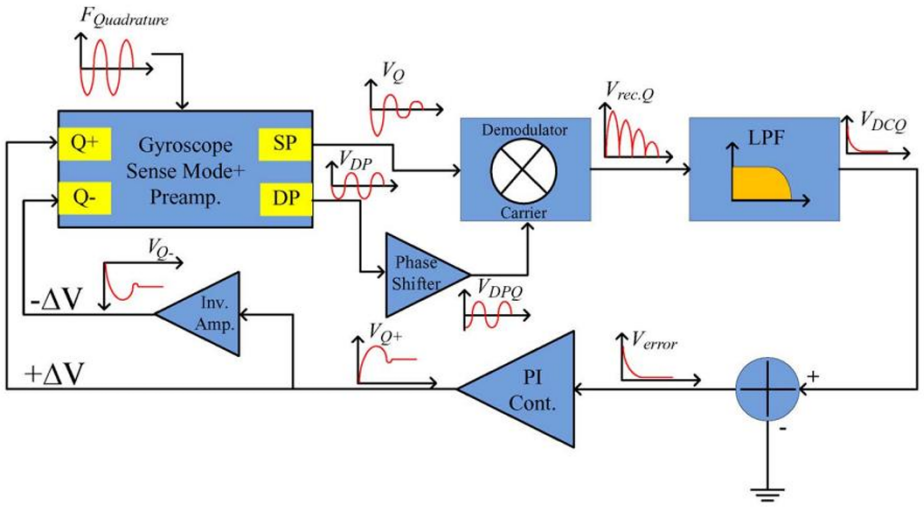

Figure 5.14: Automatic quadrature cancellation loop. The quadrature demodulator output is compared to zero, and a controller (e.g., PI) adjusts ΔV to continuously null the quadrature signal.

5.31.1 System Architecture

Figure 5.14 shows the complete sense chain with an additional feedback loop dedicated to quadrature nulling.
The key components are:

1. **In-Phase Demodulation Path:** The standard path that demodulates the signal with a reference in phase with the Coriolis signal (i.e., the output of the 90° phase shifter). This yields the angular rate output.
2. **Quadrature Demodulation Path:** A second demodulator that uses a reference shifted by an additional 90° (i.e., in phase with the drive displacement). The output of this path is proportional to the quadrature signal Ω_Q.
3. **Feedback Controller:** The quadrature output is compared to zero. The error signal is processed by a controller (e.g., a simple integrator or a proportional-integral (PI) compensator) to ensure loop stability.
4. **Quadrature Cancellation Electrodes:** The controller output generates the voltages $+\Delta V$ and $-\Delta V$ applied to the dedicated MEMS electrodes. This creates an electrostatic force that cancels the mechanical quadrature.

5.31.2 Operation

- The loop continuously monitors the quadrature output.
- If the quadrature signal changes (due to temperature drift or aging), the controller adjusts ΔV to bring the quadrature output back to zero.
- This creates a **third feedback loop** in the overall gyroscope system, joining the drive loop (amplitude control) and the main rate output loop.

5.31.3 Why Is This Only for High-End Applications?

The implementation of an automatic quadrature cancellation loop carries significant costs:

For these reasons, this technique is not typically used in cost-sensitive consumer applications. It is, however, **essential for achieving navigation-grade performance** (bias stability $< 1°$/hour) in high-end automotive, industrial, and aerospace applications.

Table 5.3: Trade-offs of Automatic Quadrature Cancellation

Aspect	Impact
MEMS Area	Requires additional area for quadrature compensation electrodes → increased die cost.
Electronics Complexity	Requires a second demodulator, a controller, and associated biasing circuits.
Power Consumption	Additional active electronics increase power draw.
Calibration	More complex system design and testing.

5.31.4 Interaction Between Loops

A common concern is whether the quadrature cancellation loop interferes with the rate output. As shown in Figure 5.14, the two paths are orthogonal in phase. Ideally, the quadrature loop acts only on the quadrature component. In practice, phase errors in the demodulation references can cause a small amount of coupling. However, if these phase errors are kept reasonably small (as discussed in previous sections), the interaction is negligible. Moreover, by nulling the quadrature at its source, the quadrature loop actually *reduces* the sensitivity of the rate output to any residual phase error, because the term $\Omega_Q \cdot \phi$ is minimized.

5.32 Summary of Quadrature Compensation Strategies

We have now seen three distinct approaches to dealing with quadrature, each with different trade-offs.

Table 5.4: Comparison of Quadrature Compensation Techniques

Technique	Principle	Pros	Cons
Passive Capacitive Feedforward	Inject current proportional to $\dot{x}$ at sense input	Low power, easy to implement, prevents saturation	Partial cancellation, no temperature tracking, residual quadrature remains
Open-Loop Electrostatic Cancellation	Dedicated electrodes with fixed ΔV calibrated at one temperature	Full cancellation at calibration point, no continuous power draw	Does not track temperature drift of quadrature
Closed-Loop Electrostatic Cancellation	Continuous measurement of quadrature and feedback to electrodes	Tracks temperature and aging, minimizes $\Omega_Q \cdot \phi$ term	High complexity, extra MEMS area, higher power and cost

5.33 Complete Gyroscope Structures: A Review

With all the elements now introduced, we can look at complete gyroscope layouts and recognize their components.

5.33.1 Yaw Gyroscope (In-Plane Sense Mode)

Figure 5.15 shows a typical yaw gyroscope with double decoupling.

Recognizable elements:

- **Drive Frame:** A relatively thin structure, driven by comb drives. It is designed to have low mass.

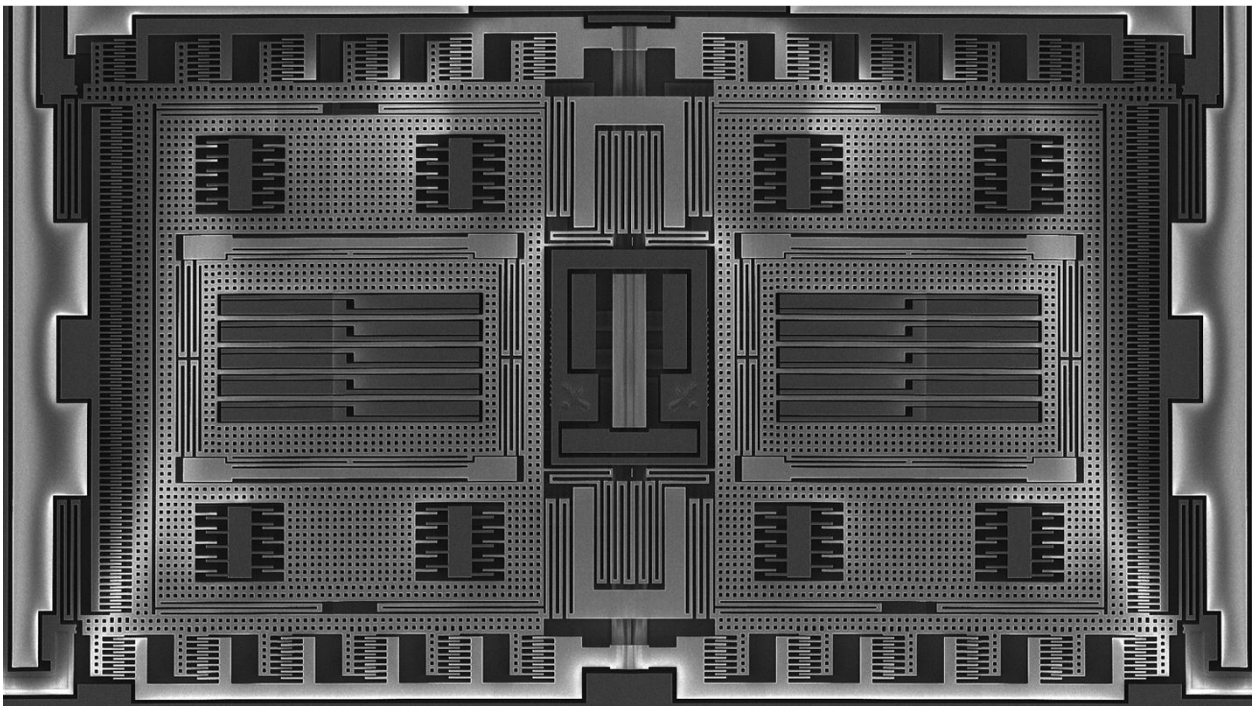

Figure 5.15: Complete yaw gyroscope layout. Key elements are labeled.

- **Drive Springs:** Anchor the drive frame and constrain its motion to the drive (X) direction.
- **Coriolis Frame:** A large, massive frame suspended from the drive frame by **decoupling springs**. These springs are stiff in the drive direction but compliant in the sense (Y) direction, allowing the Coriolis frame to be driven in X while remaining free to move in Y.
- **Sense Frame:** Attached to the Coriolis frame via springs that are stiff in the Y direction. Motion of the Coriolis frame in Y is transmitted to the sense frame.
- **Parallel Plate Sense Electrodes:** Used to capacitively detect the sense frame displacement.
- **Quadrature Compensation Electrodes:** Nested within the Coriolis frame, these are the "string" structures that generate the x-proportional force in the Y direction.
- **Tuning Fork Springs:** Couple the two halves of the structure to ensure anti-phase motion.

5.33.2 Pitch/Roll Gyroscope (Out-of-Plane Sense Mode)

Figure 5.16 shows a gyroscope designed to sense in-plane rates via an out-of-plane sense mode.

Recognizable elements:

- **Drive Frame and Comb Drives:** Similar to the yaw gyroscope, providing the anti-phase drive motion.
- **Coriolis Frame:** Suspended such that it can move vertically (out-of-plane) in response to a Coriolis force.
- **Quadrature Compensation Electrodes:** Nested within the Coriolis frame. These are electrodes that create a vertical force proportional to the horizontal drive displacement, as derived in Section 8.
- **Sense Electrodes:** Typically located underneath the moving structure to detect vertical displacement via capacitance change.

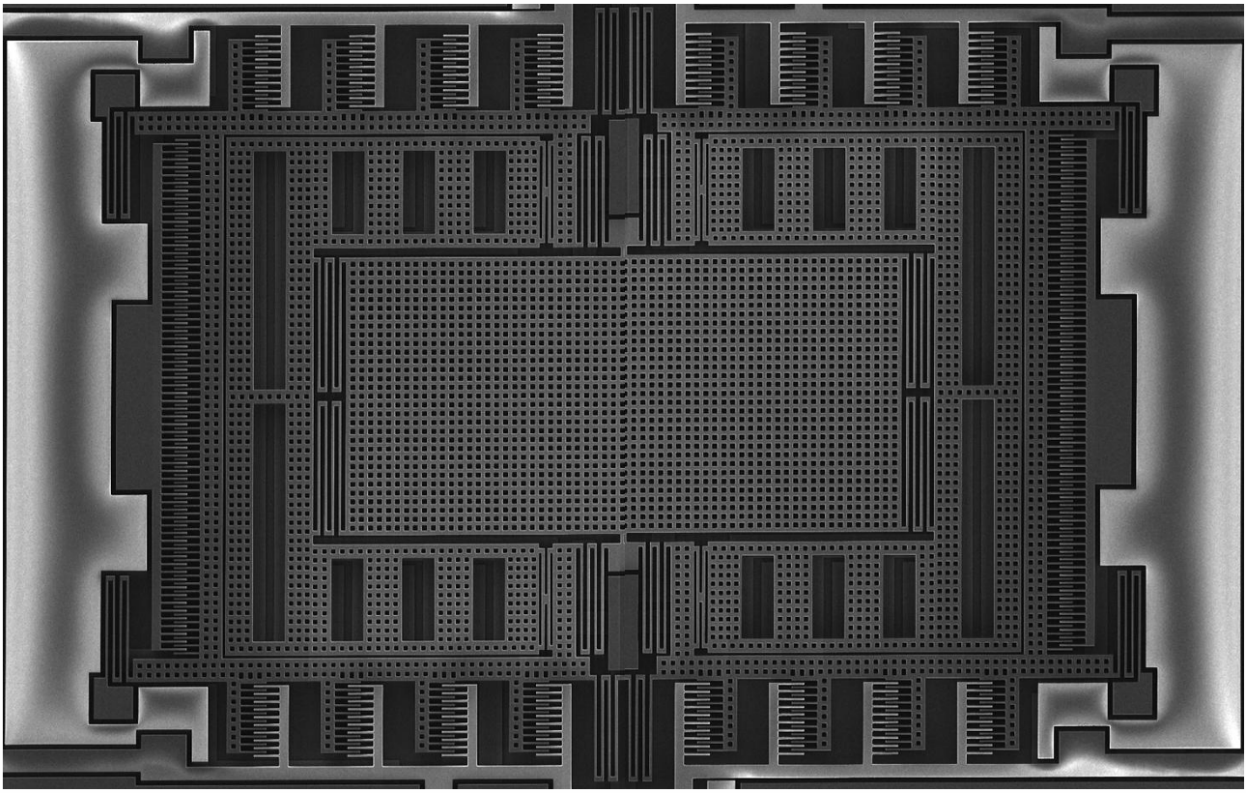

Figure 5.16: Complete pitch/roll gyroscope layout. Quadrature compensation electrodes are nested within the structure.

5.34 Conclusion

This class has covered the most complex and critical issue in real MEMS gyroscopes: **offset induced by quadrature error**.

- Quadrature originates from mechanical asymmetries (spring non-uniformities, tilted sidewalls) and electrostatic asymmetries (unequal comb gaps).
- It manifests as a sense-mode displacement proportional to the drive displacement, i.e., 90° out of phase with the Coriolis signal.
- Ideal synchronous demodulation rejects quadrature, but **phase errors** (mechanical and electronic) cause a fraction of the large quadrature signal to leak into the output, creating a temperature-dependent offset.
- Simple capacitive feedforward can prevent front-end saturation but does not solve the temperature drift problem.
- **Electromechanical quadrature cancellation** using dedicated electrodes and bias voltages ($\pm\Delta V$) can generate an opposing force proportional to x, nulling the quadrature at its source.
- For the highest performance, a **closed-loop automatic quadrature cancellation system** continuously nulls the quadrature by measuring it (via a separate demodulator) and adjusting ΔV in real-time.
- This adds complexity, area, and cost, making it suitable only for high-end navigation-grade applications.

A final note: in addition to quadrature, there is also a small **in-phase offset** arising from comb-drive defects (the term proportional to $V_{DC}V_{ac}\Delta dL_0$ derived earlier). This term is in phase with the Coriolis signal and cannot be rejected by demodulation. It must be calibrated at the system level. However, its magnitude is typically much smaller than the quadrature-induced offset, making quadrature the dominant concern.

This concludes the discussion on amplitude-modulated (AM) gyroscopes.

Chapter 6

AM Resistive NEMS Gyroscopes

6.1 Introduction to the Alternative Technology

The class this afternoon introduces an alternative technology to capacitive sensing. This approach is resistive-based and exploits nano-electromechanical systems (NEMS) structures that will be discussed in depth and thoroughly throughout the course.

The general idea is to quickly review the key points highlighted in previous classes about advanced amplitude-modulated capacitive gyroscopes and then introduce possible alternative approaches for achieving high-performance gyroscopes.

6.2 Quick Review of Capacitive Gyroscopes

Capacitive gyroscopes are based on comb-finger actuation for the drive motion and capacitive readout of the Coriolis motion. However, the sensing transduction element is typically a parallel-plate capacitor rather than a comb finger (in most modern solutions).

These devices are amplitude-modulated: the Coriolis force modulates the amplitude of a carrier signal oscillating at the drive-mode frequency. We have examined several examples of advanced electronic designs and have also addressed part-to-part repeatability, design strategies to mitigate temperature-induced drift, quadrature errors, and the combined effects of non-idealities in both the electronics and the mechanical domain under temperature variations.

Guidelines for overall system optimization were derived, leading to the conclusion that certain conceptual limitations (or at least significant challenges) remain unavoidable.

6.2.1 Identified Limitations and Trade-offs

- To achieve good performance in a small footprint, it is advantageous to employ different sensing topologies for the drive and sense modes.
- A large drive displacement (a few microns up to 10 µm) is desirable; this is easily obtained with comb fingers.
- The same comb fingers, however, provide relatively poor transduction for the Coriolis motion. Unless a very large area is used, parallel-plate capacitors are preferred for sensing.
- This choice introduces part-to-part repeatability issues due to the mismatch between the two different sensing mechanisms.
- Capacitive coupling (and its associated drift) can perturb or even corrupt the measurement. Temperature-dependent drift of these couplings may inject unwanted contributions from the drive loop into the sense chain.

- In the Coriolis frame (the only frame where a large proof mass is desired), adding quadrature-compensation electrodes limits the available area. The sense frame cannot be made arbitrarily small, which forces part of the area outside the Coriolis frame for the implementation of parallel-plate sensors.
- Equivalent performance could be obtained with fewer parallel plates by accepting higher power consumption, but fundamental trade-offs persist.

6.3 Alternative Approaches

Several alternative strategies can be envisioned:

1. **Change the technology while keeping the working principle** (amplitude-modulated gyroscopes operating in mode-split conditions). This may involve new actuation or sensing mechanisms.
   - Piezoelectric actuation was discussed earlier (a student from Brescia recently completed a PhD on piezoelectric MEMS actuation, although not focused on gyroscopes).
   - Today we focus on a variant in which the actuation remains electrostatic (comb drive) but the sensing principle is changed to *piezoresistive* (instead of capacitive).
2. **Keep the same technology** (identical process steps, layer thickness, gaps, etc.) but adopt a completely different working principle, e.g., frequency modulation instead of amplitude modulation. This topic will be covered in the next class.
3. **Combine both changes** or explore emerging technologies (3D printing, micro-glass-blowing, etc.). These will not be treated in dedicated lectures but may appear in student presentations during the final mini-conference (the day before the final exam). An invited seminar on nonlinear effects in MEMS gyroscopes (by a colleague from the Structural Engineering Department at Politecnico di Milano) will also be given that morning.

6.4 Today's Focus: Piezoresistive NEMS Sensing

Today's lecture first presents the alternative fabrication process (which is not radically different and therefore does not introduce major new process challenges). The key advantage is the possibility of embedding miniaturized piezoresistive gauges (NEMS structures) directly into the MEMS device.

We will then examine:

- Optimized gyroscope design based on piezoresistive sensing,
- The associated electronic readout architecture and scale-factor calculation,
- Noise sources and the achievable performance improvement,
- Reliability aspects of the miniaturized NEMS elements and overall stability considerations.

6.5 The Fabrication Process

The process (provided by CELT) starts from a non-standard silicon wafer. Instead of the usual polysilicon structural layer on a standard substrate, the starting material consists of:

- A silicon substrate,
- A thin buried oxide layer,
- A thin monocrystalline silicon layer (typically 200 nm thick).

This monocrystalline layer is heavily doped, conferring peculiar electrical and mechanical properties that are highly beneficial for sensing.

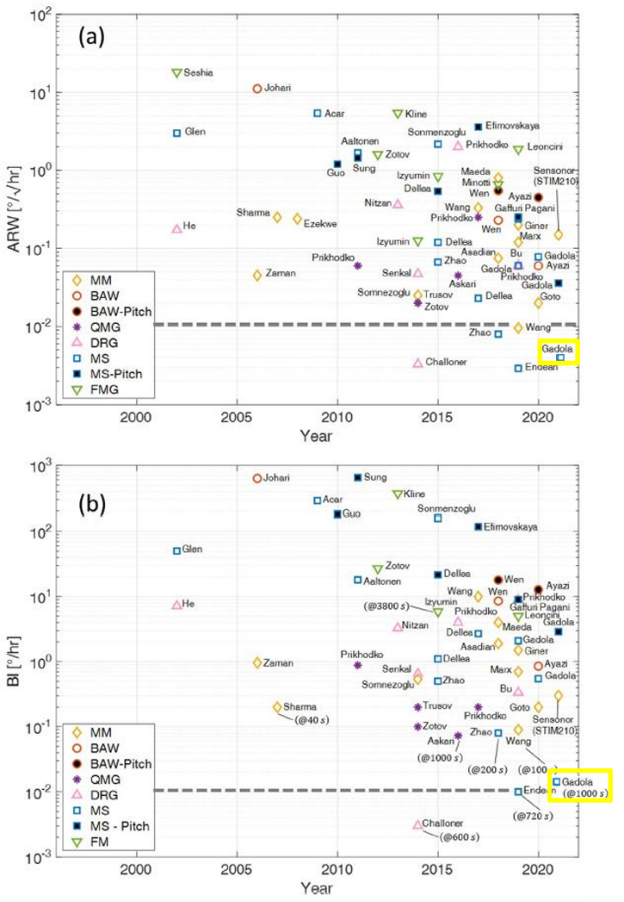

Figure 6.1: Survey data on MEMS-based processes.

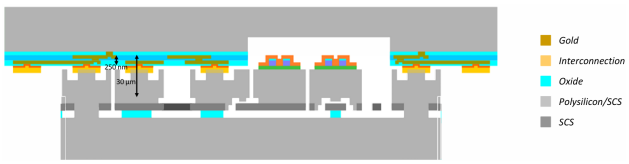

Figure 6.2: NEMS fabrication process.

6.5.1 Key Process Steps (Simplified)

1. **Patterning of the NEMS layer:** The 200 nm monocrystalline silicon layer is patterned to form piezoresistive gauges with typical cross-sections of 200 nm × 200 nm. The out-of-plane dimension can be chosen freely according to design needs.

2. **Isolation:** An additional oxide layer is deposited to electrically and mechanically isolate the NEMS gauges from the forthcoming thick structural layer.

3. **Growth of the MEMS structural layer:** A thick silicon (or polysilicon) layer is grown epitaxially on top of the isolated NEMS structures and is then patterned exactly as in standard surface-micromachining flows.

4. **Interconnections and barriers:** The same isolation oxide can also serve as a barrier for routing interconnections, analogous to previous processes.

5. **Wafer-level encapsulation:** The MEMS wafer is bonded to a cap wafer using *thermal compression bonding*. The cap wafer contains several metallization layers (similar to a CMOS back-end). The bottom-most metal layer of the (flipped) cap wafer is brought into contact with designated regions on the MEMS wafer. Simultaneous application of pressure and heat (approximately 400 °C to 500 °C) causes the deposited alloy to form a eutectic bond with silicon, thereby:

   - Hermetically sealing the device,
   - Providing electrical contacts from the MEMS structures to the external pads.

   This bonding scheme also grants greater freedom in the number of interconnection layers compared with standard anodic or glass-frit bonding.

The resulting structure embeds piezoresistive NEMS gauges inside a conventional thick MEMS resonator, combining the advantages of monocrystalline silicon sensing elements with the well-established MEMS fabrication platform.

6.6 Key Advantages of the Process

Apart from the capping/bonding scheme (which is conceptually similar to the previous process), the fundamental difference lies in the ability to create suspended nano-electromechanical structures with a thickness – and in general a cross-section – on the order of 200 nm.

When the structure is flipped upside-down, the input/output pads remain accessible on both sides of the device, exactly as in the former process. The final MEMS chip therefore consists of:

- A thick MEMS structural layer (the original MEMS wafer),
- A cap wafer bonded on top,
- Two rows of pads for electrical connections.

This process simultaneously provides:

- Relatively thick structural layers (20 µm in the example) – essential for inertial sensors because a large proof-mass is required from several design perspectives,

- Extremely thin (215 nm in the example) monocrystalline silicon NEMS elements obtained directly from the native device layer of the SOI wafer.

In other words, a single SOI starting wafer yields both a high-mass MEMS resonator and sub-micrometric NEMS sensing elements.

6.7 Piezoresistive Effect in Heavily Doped Monocrystalline Silicon

The thin monocrystalline silicon layer is heavily doped. This material exhibits pronounced *piezoresistive* properties: the electrical resistance of the element changes when it is elongated or compressed.

The resistance of a resistor is given by

$$R = \rho \frac{L}{A},$$

where ρ is resistivity, L is length and A is cross-sectional area.

The relative change in resistance $\Delta R/R$ is *not* dominated by the geometric terms ($\Delta L/L$ and $\Delta A/A$), which are negligible. Instead, it originates almost entirely from a change in the resistivity ρ.

This effect is quantified by the *gauge factor* (GF):

$$\mathrm{GF} = \frac{\Delta R/R}{\Delta L/L}.$$

Thus, elongation (or compression) of the resistor produces a proportional change in resistance. The sign of the change (increase or decrease) depends on the doping type (n- or p-type) and orientation.

Using the generalized Hooke's law,

$$\varepsilon = \frac{\sigma}{Y},$$

where $\varepsilon = \Delta L/L$ is the axial strain, σ is the axial stress (force per unit area) and Y is Young's modulus, we obtain

$$\frac{\Delta R}{R} = \mathrm{GF} \cdot \frac{\sigma}{Y}.$$

For the technology under discussion, GF ≈ 50. Because the NEMS cross-section is extremely small ($\approx$ 200 nm $\times$ 200 nm), even a modest force produces a very large stress σ, and consequently a large relative resistance change. This is one of the key reasons for the high sensitivity of these gauges.

6.7.1 Simplified Physical Origin of Piezoresistivity

When stress σ is applied, the material undergoes strain ε. The interatomic distance between silicon atoms changes, which modifies the energy-band structure. This alters the number of free carriers (electrons or holes), thereby changing the resistivity ρ and, ultimately, the resistance R.

6.8 Design Considerations for Scale Factor and Linearity

To size the gyroscope parameters and achieve a target scale factor, the maximum stress applied to the NEMS gauges must remain well below the rupture stress of silicon (typically $2--3$GPa, process-dependent).

A more practical limit is set by linearity: the piezoresistive response should stay within $\approx 1\,\%$ error. For this technology, the corresponding maximum allowable stress is approximately 500 MPa.

6.8.1 Example Gauge Parameters

Consider a nanogauge with:

- Length $L = 3\,\mu$m,
- Square cross-section 250 nm $\times$ 250 nm.

At $\sigma = 500\,\text{MPa}$, the maximum axial elongation is

$$x_{\max} = \varepsilon_{\max} \cdot L = \frac{\sigma_{\max}}{Y} \cdot L \approx 15\,\text{nm}.$$

The axial stiffness (force-displacement relationship) follows the usual Hooke's law for axial loading:

$$k = \frac{F}{x} = \frac{\sigma \cdot A}{x} = \frac{Y \cdot A}{L} \approx 3000\,\text{N/m}.$$

This stiffness is far too high for a gyroscope operating at typical drive frequencies ($\approx 20\,\text{kHz}$) with realistic proof-mass values. The required modal stiffness for such devices is usually in the range $50 - -200\text{N/m}$. Consequently, the gauges *cannot* be attached directly to the Coriolis frame or sense frame; a mechanical leverage or amplification structure is mandatory (to be discussed in subsequent slides).

6.9 Actuation and Drive-Mode Detection Strategy

Piezoresistivity has no reverse effect: it is impossible to generate mechanical deformation by applying a voltage or current (unlike capacitive or piezoelectric transduction). Therefore:

- Actuation remains electrostatic, using comb fingers exactly as in conventional amplitude-modulated capacitive gyroscopes.

For drive-mode detection we now have a choice:

- Classical comb-finger capacitive sensing,
- NEMS piezoresistive gauges.

Because the Coriolis (sense) mode is already read out with NEMS gauges, it is highly advantageous to use identical NEMS gauges also for drive-mode detection. In this way:

- Drive and sense modes are transduced by exactly the same physical principle and identical structures,
- Part-to-part statistical spread, temperature coefficients, and long-term drift are inherently self-compensated,
- Repeatability and stability are dramatically improved (as will be quantified in later slides).

(Note: the lecturer occasionally pronounces "gauges" with a French-influenced "gauches"; both terms refer to the same NEMS piezoresistive elements.)

6.10 Incorporation of NEMS Gauges into the Mechanical Structure

Direct attachment of the NEMS gauges to a moving Coriolis frame or sense frame is impractical because of their extremely high axial stiffness ($\approx 3000\,\text{N/m}$).

The solution consists in mechanically de-amplifying the displacement (which simultaneously amplifies the force) by means of a lever system. Two simultaneous amplification mechanisms are exploited:

- **Displacement de-amplification** (lever ratio),
- **Stress amplification** due to the drastic reduction in cross-sectional area (from $20\,\mu\text{m}$ thick MEMS layer to $200\,\text{nm} \times 200\,\text{nm}$ NEMS gauge).

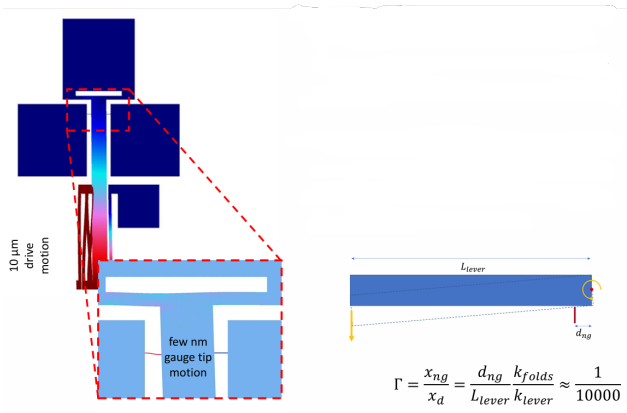

Figure 6.3: Differential sensing NEMS for in-plane sensing.

6.10.1 In-Plane Lever Configuration (for Small Displacements)

The basic building block is a rotational hinge with an attached lever. The Coriolis frame (or sense frame) displaces by Y and rotates the lever around the hinge. The NEMS gauge is placed close to the rotation centre at a short distance L_{NG}, while the lever length is L_{lever}.

The displacement de-amplification factor (often called γ) is

$$\gamma = \frac{L_{\mathrm{NG}}}{L_{\mathrm{lever}}}.$$

Hence the gauge-tip displacement is

$$Y_{\mathrm{NG}} = \gamma \cdot Y.$$

A scanning-electron-microscope image of a typical implementation shows the gauge suspended between a fixed anchor (square region) and a suspended lever/mass. The gauge experiences pure axial elongation/compression.

Example Numbers (Sense Mode)

Typical dimensions:

- Distance of gauge from rotation centre: 20 µm,
- Overall lever length: 200 µm,
- $\Rightarrow \gamma = 0.1$.

For a Coriolis displacement $Y \approx 15\,\mathrm{nm}$ (typical value for a drive amplitude of $\approx 10\,\mu\mathrm{m}$ and mode split of $\approx 1\,\mathrm{kHz}$):

$$Y_{\mathrm{NG}} \approx 1.5\,\mathrm{nm} \quad \text{(well below the 15 nm linearity limit)}.$$

Corresponding stress:

$$\sigma \approx 150\,\mathrm{MPa} \quad \text{(well below the 500 MPa limit)}.$$

This lever geometry is ideal for sensing small in-plane motions and is the standard configuration for Coriolis (sense-mode) readout in NEMS-based gyroscopes.

6.10.2 Drive-Mode Detection (Large Displacements)

Drive displacements are much larger (5 µm to 10 µm). A pure lever with $\gamma \approx 10^{-3}$ would require an impractically long lever arm.

The solution merges the previous lever with a set of folded springs:

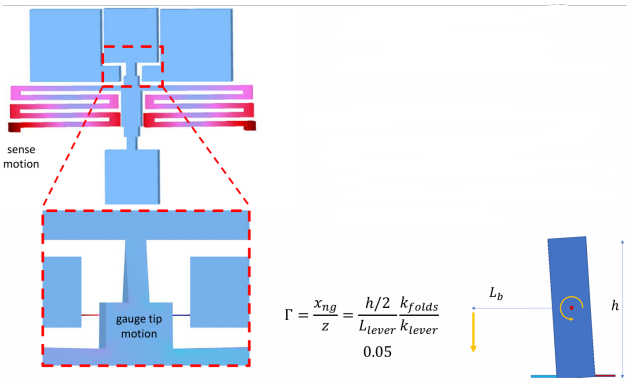

Figure 6.4: Differential sensing NEMS for out-of-plane sensing.

- The drive frame is attached to the folded springs, which absorb the bulk of the large displacement.
- Only a residual displacement is transmitted to the lever.
- The lever further de-amplifies the motion at the gauge tip.

The effective de-amplification factor becomes

$$\gamma_{\text{eff}} = \gamma_{\text{lever}} \times \frac{k_{\text{folded}}}{k_{\text{lever+gauges+hinge}}},$$

where k_{folded} is the stiffness of the folded springs and $k_{\text{lever+gauges+hinge}}$ is the combined stiffness of the lever mechanism. Values around $\gamma_{\text{eff}} \approx 10^{-3}$ are routinely obtained.

Example: drive displacement 10 µm ⇒ gauge elongation ≈ 10 nm (still within linearity limits).

The gauges are placed in a **differential configuration** (always used, exactly as in capacitive sensing) to cancel common-mode effects.

6.10.3 Out-of-Plane Sensing (Pitch/Roll Gyroscopes)

For large out-of-plane (rotational) displacements the same principle applies, but the geometry changes slightly:

- An external set of folded springs (or direct attachment) converts the out-of-plane rotation into a smaller rotation of a central lever element.
- Differential stress appears on the NEMS gauges placed on opposite sides.

In this case the de-amplification factor is lower because:

- The effective lever length is limited by the Coriolis-frame geometry.
- The numerator (gauge distance) is fixed at half the structural thickness (≈ 10 µm).

Typical $\gamma \approx 0.01$–0.05, resulting in smaller gauge elongations than in yaw devices. Consequently, pitch/roll gyroscopes are more challenging to design for high sensitivity, even though the ultra-sensitive NEMS gauges still provide a net advantage.

6.11 Overall Gyroscope Architecture

The complete device retains the same topology as conventional amplitude-modulated capacitive gyroscopes, with only the sensing elements changed:

- **Actuation**: electrostatic comb fingers (single-ended or push-pull) driving the drive frame in anti-phase.

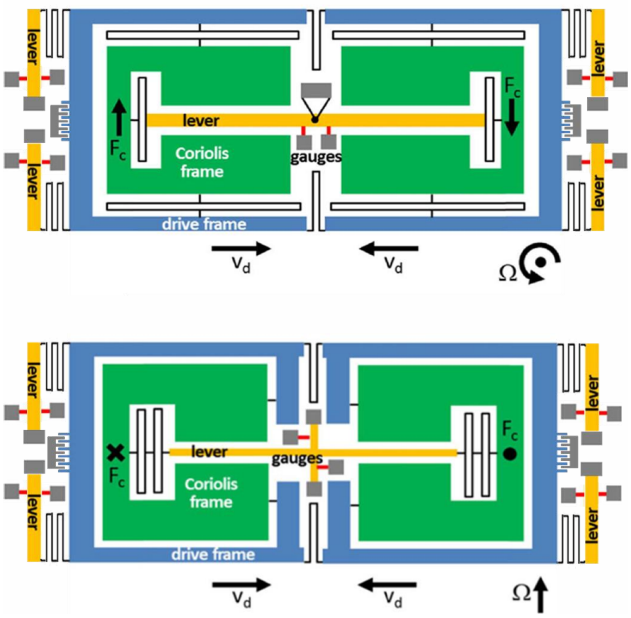

Figure 6.5: Complete NEMS for in- and out-of-plane sensing.

- **Drive-mode detection**: NEMS gauges + lever + folded springs ($\gamma \approx 1000$), converting $\approx 10\,\mu$m drive motion into $\approx 10\,$nm gauge elongation.
- **Coriolis (sense) detection**: identical NEMS lever system ($\gamma \approx 0.1$) attached to the Coriolis frame.

Because drive and sense modes are read out by exactly the same physical structures (NEMS piezoresistive gauges), statistical spread, temperature drift and long-term ageing are inherently matched and self-compensated.

6.12 Complete Gyroscope Layout and Symmetry Considerations

To maximise structural symmetry, the same lever+gauges configuration is placed at all four corners of the device. Only two gauges are actively read out; the other two can be left as *dummy* gauges (to save power consumption in the readout electronics – see later slides).

The drive mode (horizontal motion of the blue drive frame) drags the green Coriolis frame along with it. Decoupling springs allow the Coriolis frame to move independently in the sense direction when a Coriolis force appears. The working principle is therefore identical to that of conventional amplitude-modulated capacitive gyroscopes.

The main difference is that there is *almost no sense frame*: the sense function is performed by a very small, nearly massless lever that de-amplifies the Coriolis displacement Y into the gauge-tip displacement Y_{NG}.

When the lever tilts, one gauge is stretched while the opposite gauge is compressed – a differential sensing scheme, exactly as in capacitive designs.

Summary of the elements:

- Blue: drive frame,
- Green: Coriolis frame,
- Central lever system: senses Coriolis motion,
- Outer lever systems + folded springs: detect drive displacement,

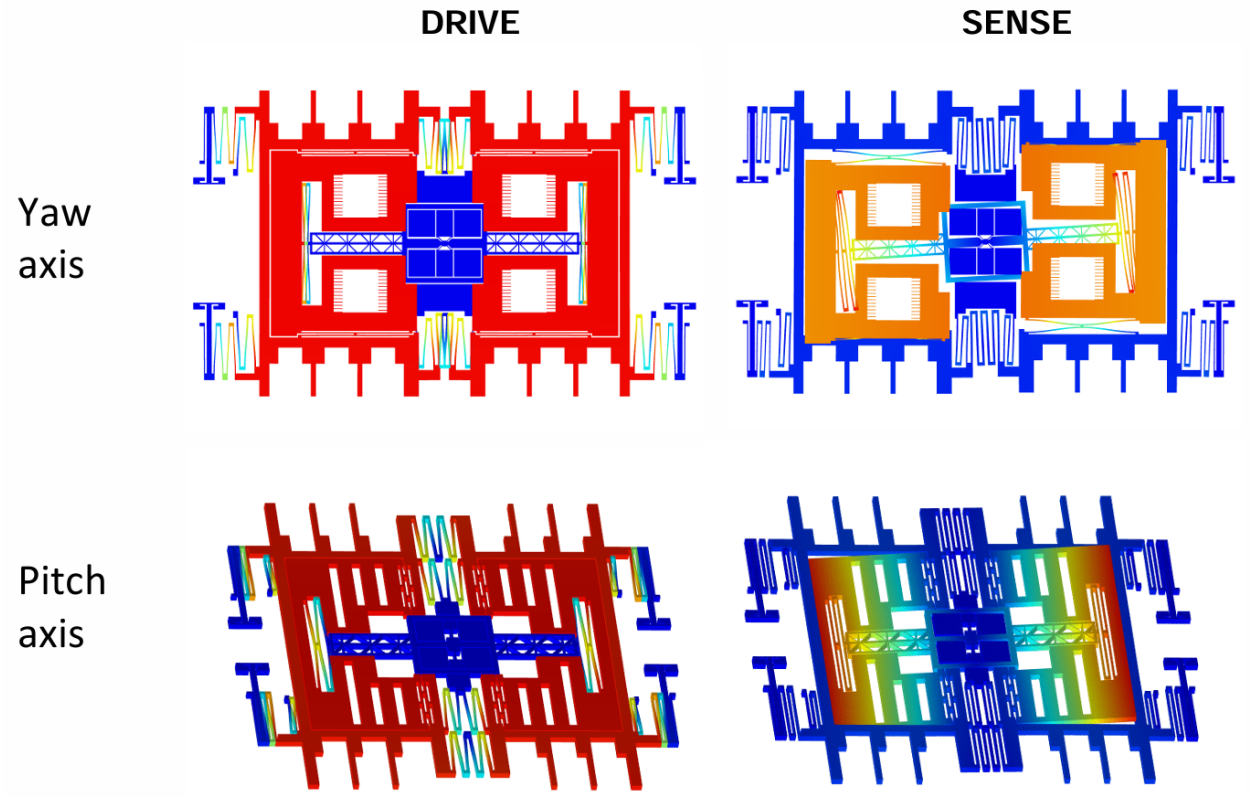

Figure 6.6: Yaw and Pitch NEMS gyroscopes.

- Both drive and sense modes use identical NEMS piezoresistive gauges ⇒ resistive sensing electronics required.

6.12.1 Pitch / Roll Gyroscopes

The drive section (fingers, drive frame, drive detection) remains unchanged. The Coriolis force now applies opposite out-of-plane forces on the two halves of the Coriolis frame, producing an out-of-plane rotation of the central lever. This induces opposite stresses in the two NEMS gauges.

Again, four symmetric gauges can be implemented (two active + two dummy) for perfect mechanical balance.

6.13 Examples of Realised Devices

The first fabricated yaw gyroscope clearly shows:

- Folded springs + lever for drive detection,
- Central lever system for Coriolis detection.

(The NEMS gauges are false-coloured for visibility; in reality they are invisible at this scale.)

The pitch gyroscope follows exactly the same drive architecture; only the sense lever responds to the out-of-plane Coriolis torque.

6.14 Mechanical Advantages

A striking feature is visible in the layout: virtually the entire area allocated to the Coriolis + sense subsystem is occupied by the Coriolis frame. There is no large parallel-plate sense frame stealing area.

Recall the earlier guideline: we want a **large mass on the Coriolis frame** and **very low mass on the sense frame** (and drive frame). This topology achieves that goal perfectly because the sensing elements are miniaturised NEMS levers that do not remove area from the proof mass.

For pitch/roll devices the same benefit exists (sensing plates are placed underneath the structure, as in capacitive implementations).

Finite-element simulations confirm:

- Holes in the Coriolis frame are only for quadrature-compensation electrodes (no parallel plates),
- Sense lever is intentionally perforated to minimise its mass,
- Drive mode: in-plane anti-phase motion,
- Sense mode: perpendicular displacement (yaw) or rotation (pitch/roll),
- NEMS gauges are located near the rotation centres (not visible in the colour maps).

6.15 Process Summary and Sensitivity Promise

We have now seen how NEMS piezoresistive gauges can be embedded inside a process that is essentially identical to standard epitaxial-growth + deep-reactive-ion-etching surface micromachining, with only the addition of a thin monocrystalline SOI device layer.

The resulting sensing elements combine:

- Large gauge factor (heavily doped single-crystal silicon),
- Extreme stress concentration (tiny cross-section),
- Mechanical de-amplification levers.

This promises a very large sensitivity (scale factor). However, as repeatedly emphasised in previous classes, high sensitivity alone means little. The real figures of merit are **noise** and **stability**. These will be analysed in the following slides.

6.16 Question from the Audience

Gabriele Gattere: "Since the sense configuration is based on several mechanical anchors and the lever amplification factor, how does the system behave in terms of external mechanical stress rejection?"

Lecturer's reply: This is an excellent and advanced point. In earlier slides we showed that placing anchors very close to each other greatly improves rejection of wafer bending / packaging stress (the differential displacement between nearby anchors is negligible). In the present designs the anchors are relatively far apart (four corners), but the difference with respect to conventional capacitive gyroscopes is not dramatic. The lever amplification does not appear to degrade stress rejection significantly. We have not yet optimised the anchor placement for maximum stress immunity in these first prototypes. Suggestions for better configurations are welcome – especially useful for students who may design devices with this technology next semester.

6.17 Scale-Factor Derivation (Resistive Topology)

We can reuse the mechanical analysis already derived for amplitude-modulated capacitive gyroscopes. Once the drive displacement X_d is stabilised by an automatic gain control (AGC) loop (discussed later), the Coriolis-induced sense displacement per unit angular rate Ω is identical:

$$\frac{Y}{\Omega} = \frac{2\,X_d\,Q_s}{\omega_s} \qquad \text{(same expression as before).}$$

What changes is the electrical transduction: we now measure resistance instead of capacitance.

From the piezoresistive equation:

$$\frac{\Delta R}{R} = \mathrm{GF}\cdot\varepsilon = \mathrm{GF}\cdot\frac{Y_{\mathrm{NG}}}{L_{\mathrm{NG}}},$$

and the lever relation:

$$Y_{\mathrm{NG}} = \gamma\cdot Y,$$

substituting yields the relative resistance change per unit angular rate (single gauge, before differential readout):

$$\frac{\Delta R/R}{\Omega} = \mathrm{GF}\cdot\gamma\cdot\frac{1}{L_{\mathrm{NG}}}\cdot\frac{Y}{\Omega}.$$

Inserting the mechanical term and converting from rad/s to °/s gives the final scale-factor expression (in Ω^{-1}):

$$\frac{\Delta R/R}{\Omega} = \mathrm{GF}\cdot\gamma\cdot\frac{1}{L_{\mathrm{NG}}}\cdot\frac{2\,X_d\,Q_s}{\omega_s}\times\frac{180}{\pi}.$$

Before converting this into an output voltage, let us first examine a numerical example of the typical resistance change we can expect.

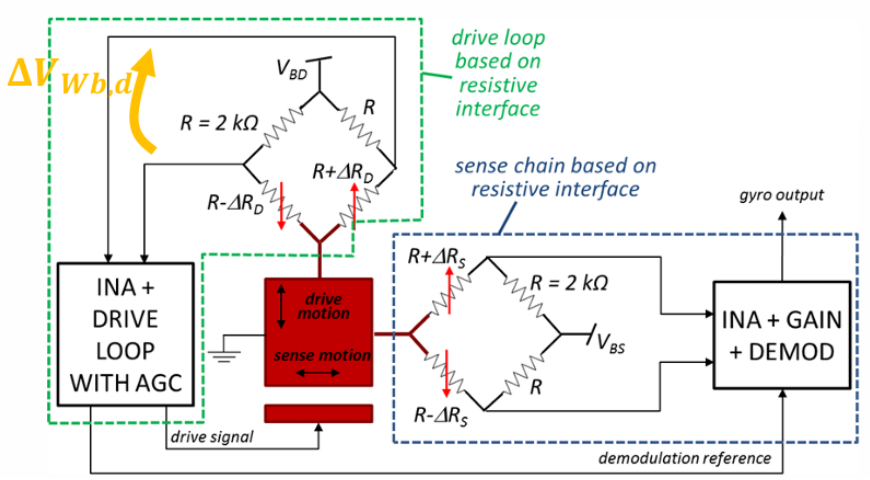

Figure 6.7: Automatic Gain Control system.

6.18 Numerical Evaluation of the Scale Factor

Typical values for the technology:

- Gauge factor $\mathrm{GF} \approx 50$,
- De-amplification factor $\gamma \approx 0.1$ (yaw gyroscopes); $\gamma \approx 1/2$–$1/3$ for pitch/roll devices,
- Nano-gauge length $L_{\mathrm{NG}} \approx 3\,\mu\mathrm{m}$ to $5\,\mu\mathrm{m}$,
- Drive displacement $X_d = 10\,\mu\mathrm{m}$,

- Frequency split $\Delta f \approx 1\,\text{kHz}$.

The resistance of a typical gauge (5 µm long, 250 nm × 250 nm cross-section, at the doping level of the monocrystalline layer) is in the range $R \approx 2 - -3k\Omega$.

Substituting into the expression derived in the previous section yields a relative resistance change per unit angular rate of

$$\frac{\Delta R/R}{\Omega} \approx 28\,\text{ppm/dps}.$$

Examples:

- At $\Omega = 100\,\text{mdps}$ (typical minimum measurable signal): $\Delta R/R \approx 2.8\,\text{ppm}$,
- At full-scale range $\Omega = 2000\,\text{dps}$: $\Delta R/R \approx 5.6\,\%$ (a very significant fraction of the nominal resistance).

Thus the piezoresistive transduction produces a consistent, easily measurable resistance variation even at modest angular rates.

6.19 Readout Electronics Architecture

The overall system is conceptually identical to the amplitude-modulated capacitive gyroscope, the only difference being the sensing elements.

- **Drive section**: electrostatic comb fingers (single-ended or push-pull) + H-bridge driven by an automatic-gain-control (AGC) loop that stabilises the drive amplitude.
- **Drive detection**: Wheatstone bridge formed by two active NEMS gauges (changing as $+\Delta R$ and $-\Delta R$) plus two dummy gauges (unstressed). The rotor is grounded.
- **Sense (Coriolis) detection**: identical Wheatstone bridge + dummy gauges, biased by the same DC voltage.

The differential voltage that appears across each bridge is amplified by an instrumentation amplifier (INA). The drive signal is phase-adjusted and fed back to the AGC. The sense signal is demodulated (synchronous demodulation at the drive frequency), low-pass filtered, digitised by an ADC, and processed exactly as in the capacitive case.

The system remains an amplitude-modulated gyroscope; only the transduction principle has changed from capacitive to resistive.

6.19.1 Wheatstone Bridge Output (for non-electronics students)

With the standard quarter-bridge configuration (two active arms, two dummies):

$$\Delta V_{\text{WB}} = \frac{V_{\text{bias}}}{2} \cdot \frac{\Delta R}{R},$$

where V_{bias} is the DC voltage applied across the bridge.

Inserting the relative-resistance scale factor derived earlier gives the overall voltage scale factor:

$$\frac{\Delta V_{\text{out}}}{\Omega} = \left(\frac{2\,X_d\,Q_s}{\omega_s}\right) \cdot \text{GF} \cdot \gamma \cdot \frac{1}{L_{\text{NG}}} \cdot \frac{V_{\text{bias}}}{2} \times \frac{180}{\pi}.$$

The bridge bias V_{bias} plays exactly the same role as the charge-amplifier gain in the capacitive architecture.

All system-level considerations previously derived for capacitive gyroscopes (loop stability, demodulation, ADC requirements, etc.) apply directly.

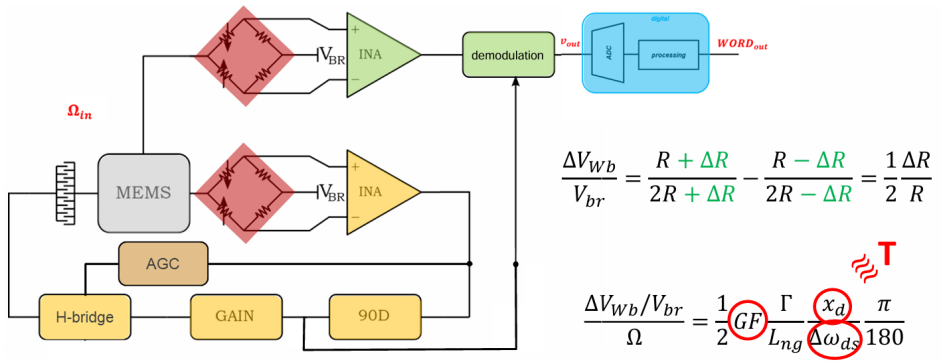

Figure 6.8: Wheatstone bridge whole system.

6.20 Temperature Sensitivity of the Scale Factor

The elements that may vary with temperature are:

- Drive displacement X_d: compensated by the AGC (exactly as in capacitive devices),
- Frequency split $\Delta\omega = \omega_s - \omega_d$: same expression and target ($< 0.25\,\%$ variation over temperature) as before,
- Gauge factor GF: material property that depends on temperature (will be compensated later),
- De-amplification factor γ: in principle could change if temperature-induced wafer bending alters lever geometry; however, for a well-designed device this is a purely geometrical factor and can be considered temperature-stable to first order.

6.21 First Experimental Results

Early prototypes confirm the working principle:

- Linearity error $< 0.2\,\%$ (actually within measurement noise up to $\pm 3000\,\mathrm{dps}$),
- Scale factor in the usual range $0.5 - -1 mV/dps$ (for $2\,\mathrm{V}$ supply and $\pm 2000\,\mathrm{dps}$ full scale),
- Bandwidth set by the low-pass filter after demodulation (identical to capacitive implementations).

6.22 Noise Sources

Because we are now performing resistive readout we are free from capacitive coupling issues, but the gauges are dissipative elements and therefore introduce their own thermal (Johnson) noise.

The total noise-equivalent rate density is still composed of three main contributions (exactly as in capacitive gyroscopes):

1. Thermo-mechanical (Brownian) noise of the mechanical structure,
2. Electronic noise of the sensing chain,
3. Phase-noise contribution folded through quadrature (from the drive loop).

We will neglect the third term (as before). The thermo-mechanical noise expression remains unchanged because the mechanical topology and operating principle are identical. Only a couple of minor differences appear in the electronic-noise term (to be detailed next).

6.23 Thermo-Mechanical Noise

Neither yaw nor pitch/roll devices suffer from small sensing gaps that would induce squeezed-film damping. Consequently, for an otherwise identical structure, the sense-mode damping coefficient is lower than in a parallel-plate capacitive design. In addition (at least for yaw gyroscopes), no large battery of parallel plates needs to be embedded, so the effective Coriolis mass can be larger.

Therefore, everything else being equal, the intrinsic thermo-mechanical (Brownian) noise-equivalent rate density is slightly lower. A realistic improvement factor is in the range 1.5–2 (possibly more). This is a welcome by-product, but not the dominant advantage of the technology.

An interesting secondary benefit is that, for the same resonance frequencies, a larger mass allows the use of thicker springs. Thicker springs are more tolerant to process non-uniformities, etch variations, and temperature-induced stress gradients.

6.24 Electronic Noise Contributions

The total noise-equivalent rate density still comprises the same three terms as in amplitude-modulated capacitive gyroscopes:

1. Thermo-mechanical noise (slightly lower, as discussed above),
2. Amplifier noise (input-referred noise of the instrumentation amplifier),
3. Thermal (Johnson) noise of the NEMS resistors.

(The quadrature contribution from the drive loop is again neglected.)

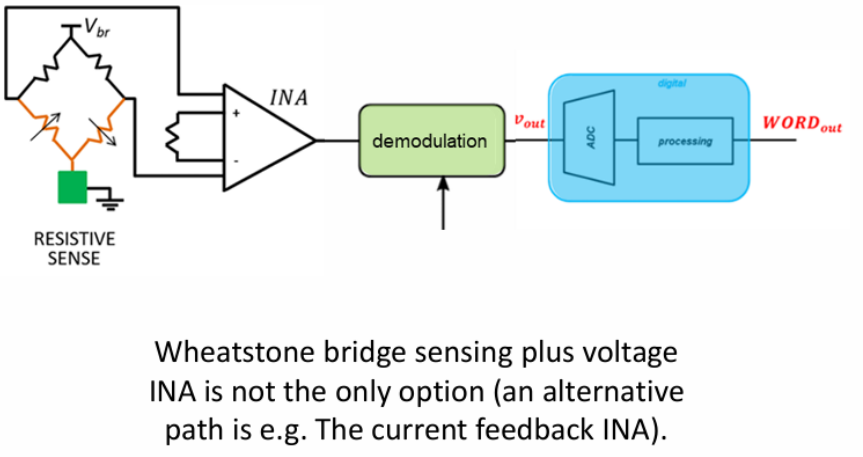

Figure 6.9: Electrical noise model.

6.24.1 Amplifier Noise

Because the readout is purely resistive, there is *no* noise amplification by parasitic capacitances between the MEMS and the electronics. Consequently the amplifier noise contribution can be made significantly lower than in a conventional capacitive architecture.

6.24.2 Resistor Thermal Noise (New Term)

The gauges themselves are dissipative elements and therefore add voltage noise density at the input of the instrumentation amplifier:

$$v_{n,R} = \sqrt{4\,k_B\,T\,R},$$

where R is the gauge resistance. This noise is referred back to angular rate by dividing by the scale factor derived earlier.

Both electrical contributions are evaluated at the same differential point (across the Wheatstone bridge) and then divided by the overall scale factor to obtain noise in $^\circ/s/\sqrt{\text{Hz}}$.

6.25 Alternative Low-Power Readout Topology

An alternative architecture (investigated in a previous European project and recently resumed for the next-generation integrated electronics) replaces the instrumentation amplifier with a simple degenerated MOS buffer. The same bias current that sets the amplifier operating point is reused to bias the gauges themselves (current-reuse technique).

Advantages:

- Lower overall power consumption,
- The bias current simultaneously reduces both amplifier noise and resistor noise.

The classical Wheatstone-bridge + INA solution still requires separate currents for the bridge and for the amplifier, leading to higher consumption.

6.26 Pros, Cons and Power-Noise Trade-off

Advantages of resistive sensing:

- Complete immunity to parasitic capacitances (noise, feed-through, drive-to-sense coupling),
- Inherent self-compensation of temperature drift and process spread (identical drive and sense transducers).

Disadvantage:

- Extra DC current must flow through all gauge branches (four branches in the minimum configuration: two for drive detection, two for Coriolis detection).

A useful comparison with the capacitive case (where the sensing chain also draws bias current) shows that increasing the gauge bias current reduces *both* noise terms *linearly* (the resistor-noise term scales as $1/I$ under the square root). In contrast, increasing amplifier current only improves noise as $1/\sqrt{I}$. Therefore, once the amplifier is biased at a minimum acceptable current, any additional available current should preferentially be allocated to the gauges.

For low-power consumer applications this may be a drawback. For "consumer-plus" or high-end miniature applications that can tolerate a few hundred microamps, the technology becomes very attractive because the extra current dramatically improves noise performance.

6.27 Numerical Noise Example

All mechanical parameters (frequencies, mode split, drive amplitude) are taken from the reference capacitive design shown in previous classes. Only the following changes are applied:

- Higher sense-mode quality factor (no squeezed-film damping),
- Slightly larger Coriolis mass,
- Yaw-gyro geometry ($\gamma \approx 0.1$),
- Gauge resistance $R \approx 2.5\,\text{k}\Omega$.

Bias current per gauge: $60\,\mu\text{A}$ (total extra current $\approx 240\,\mu\text{A}$).

Resulting noise contributions:

- Amplifier noise: $\approx 600\,\mu\text{dps}/\sqrt{\text{Hz}}$,
- Resistor (Johnson) noise: $\approx 533\,\mu\text{dps}/\sqrt{\text{Hz}}$,
- Thermo-mechanical noise: $\approx 585\,\mu\text{dps}/\sqrt{\text{Hz}}$.

Total noise-equivalent rate density:

$$\approx 990\,\mu\text{dps}/\sqrt{\text{Hz}} \quad (\text{almost } 1\,\text{mdps}/\sqrt{\text{Hz}}).$$

Doubling the gauge current halves the electrical contributions (linear dependence), while in the capacitive case a four-fold current increase was needed for only a modest improvement. This demonstrates that the piezoresistive approach is particularly advantageous whenever a modest increase in supply current is acceptable.

The lecturer concludes that the technology is especially promising for miniaturised high-end or "consumer-plus" gyroscopes that demand excellent noise and stability while tolerating a few hundred microamps of extra consumption.

```
kb = 1.38e-23;
T = 300;

g = 2e-6;
Vb = 3;

fs = 22e3;
ws = 2*pi*fs;
fd = 21000;
wd = 2*pi*fd;
fds = fs-fd;
wds = 2*pi*fds;
ms = 7e-9;
Qs = 1800;
bs = ws*ms/Qs;

xd =10e-6;
gamma_leva = 0.1;
rho = 2.3e-5;
GF = 55;
Ang = (230e-9)^2;
R = 2200;
ir = Vb/2/R;

gamma = 2/3;
n = 1.5;
is = 4.5e-6;
Vth = 25.8e-3;
```

$$\sqrt{S_{\Omega n,tot}} = \frac{180}{\pi}\frac{1}{x_d}\sqrt{\underbrace{\frac{\frac{2k_BT\,\gamma\,nV_{th}}{i_s}}{\left(\frac{i_r\,\rho}{2\,A_{ng}}\frac{GF\,\Gamma}{\Delta\omega_{ds}}\right)^2}}_{303\ \mu dps/\sqrt{Hz}} + \underbrace{\frac{4k_BTR}{\left(\frac{i_r\,\rho}{2\,A_{ng}}\frac{GF\,\Gamma}{\Delta\omega_{ds}}\right)^2}}_{265\ \mu dps/\sqrt{Hz}} + \underbrace{\frac{k_BT\,b_S}{(m_S\,\omega_s)^2}}_{292\ \mu dps/\sqrt{Hz}}}$$

$$495\ \mu dps/\sqrt{Hz}$$

```
Sn_R = 180/pi/xd * sqrt((4*kb*T*R)/(ir*rho/2/Ang*GF*gamma_leva/wds)^2)
Sn_INA = 180/pi/xd * sqrt((2*kb*T*gamma*n*Vth/is)/(ir*rho/2/Ang*GF*gamma_leva/wds)^
Sn_M = 180/pi/xd * sqrt(kb*T*bs/(ms*wd)^2)

Sn_tot = sqrt(Sn_R^2+Sn_INA^2+Sn_M^2)
```

Figure 6.10: Numerical example of noise calculation.

6.28 Reliability Considerations (Opening)

At this point many people raise the question of reliability: are these extremely thin (200 nm) monocrystalline silicon NEMS elements robust enough to survive shocks, vibrations, and long-term operation without breaking?

6.29 Reliability Assessment

The next part of the lecture briefly presents reliability experiments performed in collaboration with STMicroelectronics (STM) and CEA-Leti. Additional campaigns were conducted directly by the CEA-Leti team.

6.29.1 Why the NEMS Gauges Do Not Compromise Reliability

A key conceptual point is that the NEMS gauges contribute negligibly to the modal stiffness of either the drive or sense mode. In the drive-mode configuration (lever + folded springs), the main stiffness is provided by the folded springs and the drive-frame suspension. Removing the gauges changes the total stiffness by only $\approx 5\,\%$.

Consequently:

- Shocks and accelerations are absorbed primarily by the main springs,
- Mechanical stoppers limit the displacement exactly as in any conventional gyroscope,
- No special reliability issues arise from the presence of the sub-micrometric NEMS elements.

6.29.2 Standard Reliability Test Procedures

The following characterisation steps are standard for any MEMS gyroscope (the same protocol is used for capacitive devices).

1. **Shock-machine testing**
   A rigid heavy block is lifted and released onto the device under test. Typical profiles:
   - Peak acceleration: up to 10 000 g (example shown: 3400 g),
   - Pulse duration: few hundred microseconds (Gaussian-like shape).

   The test is repeated along six orientations (three principal axes + cross-axes).

2. **Drop (free-fall) testing**
   The packaged device is dropped from a height of 1.2 m onto a rigid marble block. The experiment is repeated 10–50 times; the random impact angle produces different stress conditions.

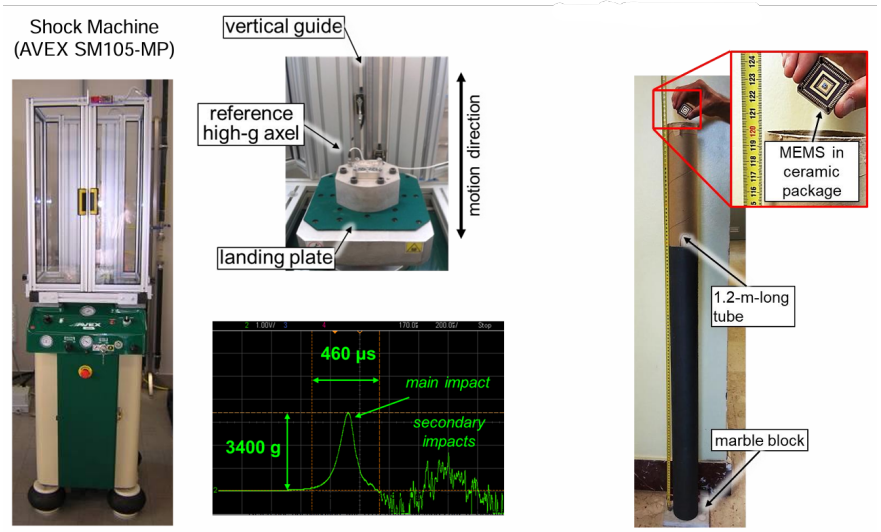

Figure 6.11: Shock test setup conforming to MIL-STD standards.

Survivability is verified by measuring scale factor and offset *before* and *after* the tests.

6.29.3 Experimental Results

- **100 % survivability** was obtained for all tested devices (both shock-machine and drop tests).
- Scale factor and offset showed *no measurable change* after the stress campaign.

The lecturer emphasises that reliability is therefore *not* a concern, even for 215-nm-thick, few-micron-long monocrystalline silicon elements.

Audience interaction (Stefan): Devices have been tested up to 10 000 g (the conventional level guaranteed in consumer datasheets). Higher values have also been explored, although exact numbers are not recalled.

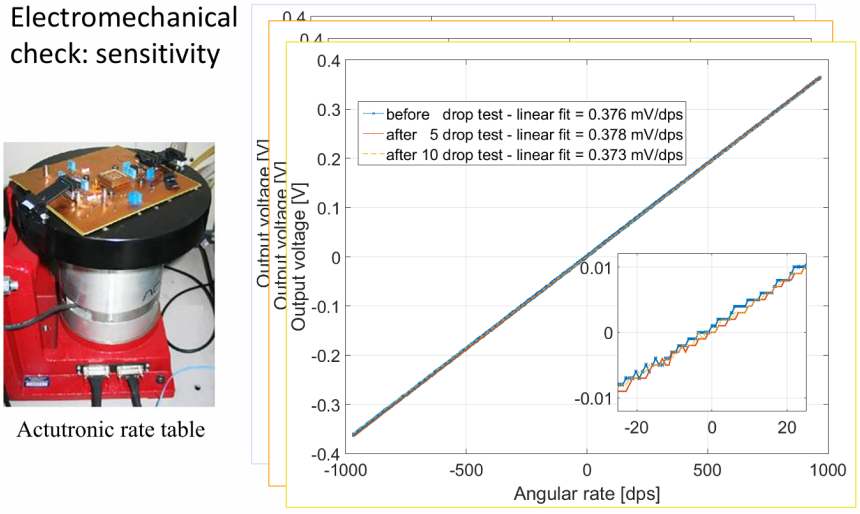

Figure 6.12: Results of shock tests and electromechanical verification.

6.30 Transition to Stability Analysis

The most interesting part of the lecture (from the lecturer's perspective) concerns long-term and temperature stability – the key figure of merit for next-generation applications.

The scale-factor expression is now re-examined including the drive AGC loop. The device schematic shows:

- Drive motion (horizontal),
- Sense (Coriolis) motion (vertical),
- Resistive drive loop with AGC,
- Identical resistive sense chain.

6.30.1 Biasing Strategy and Voltage Levels

The rotor is kept at ground. To obtain $\approx 50\,\mu\text{A}$ per gauge branch, the Wheatstone-bridge bias is only $\approx 0.2\,\text{V}$ (DC).

Typical measured quality factors:

- $Q \approx 20000$ at package pressure slightly below 1 mbar.

Drive actuation (no DC voltage on the rotor):

- Overlap DC + AC on the drive electrodes,
- Push-pull configuration: 1.5 V DC + 0.3 V AC,
- Achieves $\approx 5\,\mu m$ to $15\,\mu m$ drive displacement (example calculated with $Q = 15000$).

The sense chain can be biased at a standard integrated-circuit supply (1.8 V).

Major system-level advantage: The entire gyroscope (drive + sense electronics) can operate from a single 1.8 V supply. No high-voltage charge pump is required for a large rotor DC bias (as in conventional capacitive gyroscopes).

While charge-pump power consumption is modest, the generated high-voltage rail often contains ripples/spikes that can propagate through the system and create disturbances. The NEMS piezoresistive approach completely eliminates this issue.

(The PCB demonstrator used 5 V only because of discrete-component limitations; an integrated implementation can stay at 1.8 V.)

6.31 Low-Voltage Operation Advantage

The absence of a high-voltage rotor bias (and therefore of any charge-pump stage) is a clear system-level benefit. All voltages in the gyroscope remain very low:

- Wheatstone-bridge bias $\approx 0.2 - -0.3V$,
- Drive actuation: 1.5 V DC + 0.3 V AC (push-pull),
- Sense-chain bias: standard 1.8 V supply.

The entire system can therefore operate from a single 1.8 V rail with no voltage boosting.

6.32 Scale-Factor Derivation with Drive AGC

We now rewrite the scale factor assuming the drive amplitude is stabilised by the AGC loop.

6.32.1 Drive Wheatstone-Bridge Output

The voltage at the drive-bridge output depends only on the drive displacement:

$$\Delta V_{\mathrm{WB,drive}} = \frac{V_{\mathrm{bias,drive}}}{2} \cdot \mathrm{GF} \cdot \gamma_d \cdot \frac{X_d}{L_{\mathrm{NG}}},$$

where γ_d is the drive-mode lever ratio.

The AGC forces this voltage to a fixed reference value V_{ref}. Solving for the drive displacement gives

$$X_d = \frac{2\,V_{\mathrm{ref}}}{V_{\mathrm{bias,drive}} \cdot \mathrm{GF} \cdot \gamma_d} \cdot L_{\mathrm{NG}}.$$

6.32.2 Final Scale Factor

Inserting the expression for X_d into the sense-mode scale factor yields

$$\frac{\Delta V_{\mathrm{out,sense}}}{\Omega} = V_{\mathrm{ref}} \cdot \frac{\gamma_s}{\gamma_d} \cdot \frac{1}{\Delta\omega} \times \frac{180}{\pi},$$

where $\Delta\omega = \omega_s - \omega_d$ is the frequency split and γ_s is the sense-mode lever ratio.

Remarkable Cancellations

Almost every parameter disappears:

- Gauge factor GF (drive and sense gauges are on the same die, $\approx 200\,\mu m$ apart $\Rightarrow$ process and temperature variations are perfectly correlated),
- Bridge bias voltages $V_{\text{bias,drive}}$ and $V_{\text{bias,sense}}$ (both derived from the same supply),
- Gauge length L_{NG} (identical for drive and sense gauges),
- Drive displacement X_d (controlled by the AGC).

The only temperature-dependent term that remains is the frequency split $\Delta\omega$, whose variation is limited to $\approx 0.25\,\%$ over the full temperature range (as derived in previous classes).

The same simplifications hold for pitch/roll gyroscopes (only the numerical values of γ_s and γ_d change).

This is a dramatic improvement over amplitude-modulated capacitive gyroscopes, where drive (comb) and sense (parallel-plate) transducers introduce uncorrelated process and temperature dependencies.

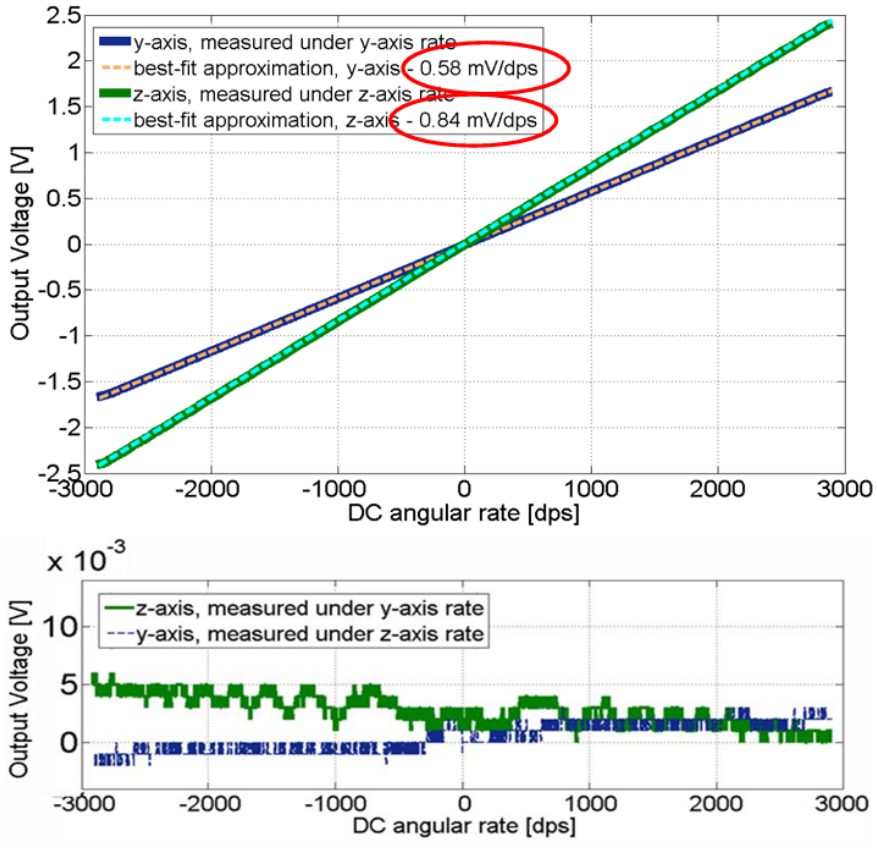

Figure 6.13: Cross-axis rejection representation.

6.33 Feed-Through Capacitance and Parasitic Immunity

A well-known issue in capacitive gyroscopes is the parasitic feed-through capacitance C_{ft} between drive and sense ports.

The mechanical resonator can be represented by its RLC equivalent circuit. Without C_{ft} the admittance is the red curve (pure resonance peak). Adding C_{ft} modifies the transfer function to the yellow curve: an additional term $\propto s\,C_{\text{ft}}$ appears, producing a rising contribution at high frequencies.

This is especially problematic in single-ended drive:

- It forces placement of filter poles far from resonance to avoid phase errors,
- The parasitic term grows unbounded and can trigger spurious oscillations or amplify noise.

Push-pull drive mitigates the effect by symmetry, but perfect cancellation is impossible in real devices.

In the resistive gyroscope the situation is radically different. The NEMS gauges provide a large intrinsic amplification (gauge factor $\times$ stress concentration). Consequently:

- The electrical equivalent model is resistive (flat low-frequency response),
- Even a parasitic feed-through capacitance *ten times larger* than in the capacitive case produces a negligible contribution.

Numerical example (same resonance frequency):

- At $\omega = 10\,\omega_0$: feed-through effect is still three orders of magnitude below the resonance peak,
- In the capacitive case the parasitic term already equals the peak value.

Thus the resistive approach is inherently tolerant to parasitic capacitances. Any drive-to-sense coupling (and its temperature drift) has a dramatically smaller impact on offset stability and demodulation accuracy.

6.34 Measured Noise and Stability Performance

We now compare real experimental results obtained on the two families of devices (capacitive and piezoresistive) using the same PCB-based discrete-electronics setup.

6.34.1 Capacitive Reference Gyroscope

The yaw and pitch gyroscopes shown in earlier classes (with coloured functional blocks) were tested with open-loop quadrature compensation (Tatar scheme) and a drive amplitude of $\approx 5\,\mu m$ to $6\,\mu m$.

Allan-deviation plot (green curve):

- Noise: $\approx 3\,\mathrm{mdps}/\sqrt{\mathrm{Hz}}$ (typical of high-end consumer products),
- Bias instability: $\approx 10\,^\circ/\mathrm{h}$ (with rate random walk appearing after $\approx 3\,\mathrm{s}$ observation time).

These values are representative of state-of-the-art capacitive devices of similar size ($< 1\,\mathrm{mm}^2$).

6.34.2 Resistive NEMS Gyroscope

The same measurement campaign was repeated on a piezoresistive device biased at $120\,\mu A$ per gauge branch (total extra current $\approx 480\,\mu A$ for the four branches).

Results (same PCB setup, no integrated circuit):

- Noise: $\approx 550\,\mu\mathrm{dps}/\sqrt{\mathrm{Hz}}$ (almost an order of magnitude better),
- Bias instability: $\approx 1.8\,^\circ/\mathrm{h}$ at $\approx 200\,\mathrm{s}$ observation time.

This noise figure is among the lowest ever reported for a MEMS gyroscope smaller than $1\,\mathrm{mm}^2$.

The extra current consumption is acknowledged: an integrated implementation will still draw this current (absent in capacitive designs). However, the dramatic improvement in noise and especially stability fully justifies the modest power penalty for high-performance miniature applications.

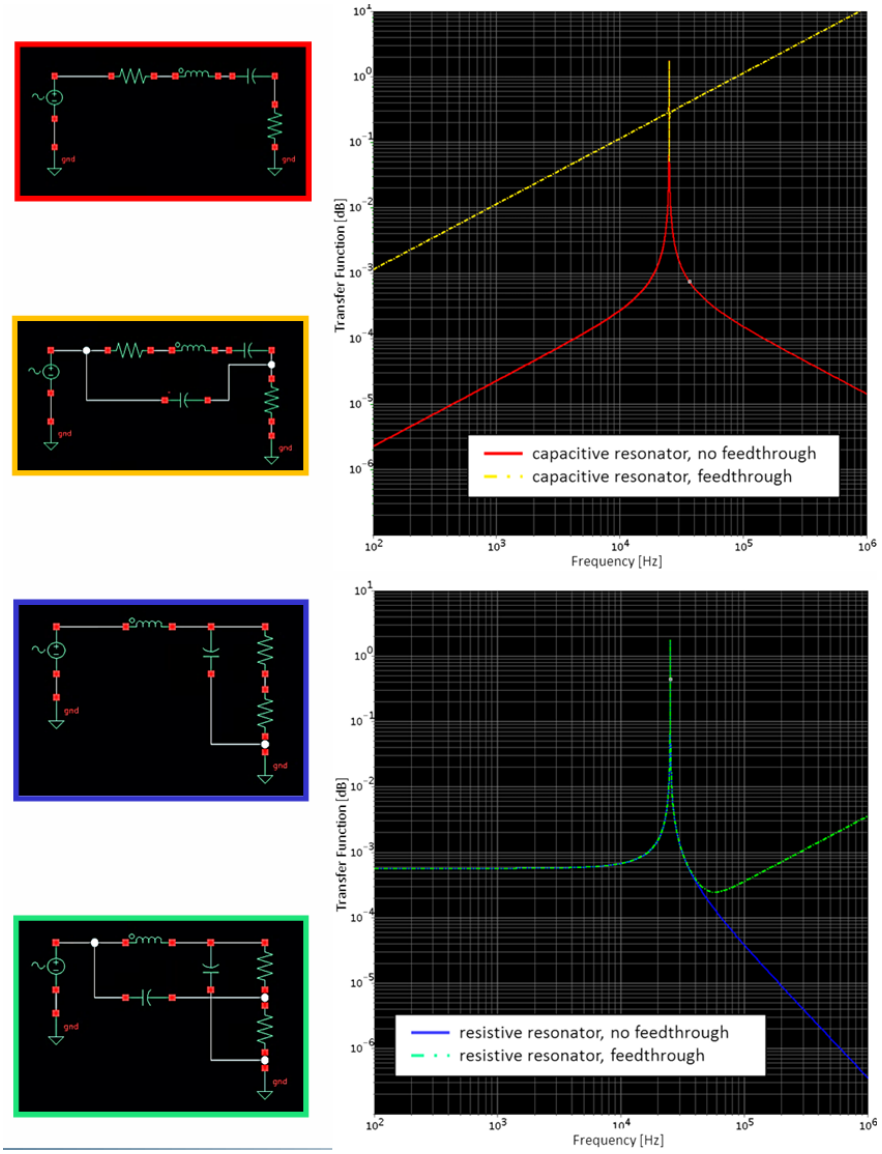

Figure 6.14: Comparison of electrical equivalent models for capacitive and resistive readout.

6.35 Measuring Earth's Rotation with MEMS Gyrocompassing

The planet Earth rotates at 15 °/h (one full turn per day), corresponding to a constant angular rate of

$$4.16\,\text{mdps}.$$

A high-stability gyroscope can in principle detect this rate. However, a constant (DC) signal is easily confused with offset or offset drift.

To discriminate Earth's rotation from offset, the technique known as *carouseling* (or MEMS gyrocompassing) is used.

6.35.1 Experimental Setup

A rate table rotates the gyroscope (but *not* to apply a test rotation). The gyroscope sensing axis is mounted *orthogonal* to the table's rotation axis.

The device is tilted by an angle equal to the local latitude λ ($\approx 45°$ in Milan/Grenoble/Brescia/Pavia). Four phases during one table revolution:

- Sensing axis points toward celestial north $\Rightarrow$ maximum Earth-rate signal (positive),
- Sensing axis orthogonal $\Rightarrow$ zero sensitivity,
- Sensing axis points toward celestial south $\Rightarrow$ maximum Earth-rate signal (negative),
- Sensing axis orthogonal again $\Rightarrow$ zero sensitivity.

The perceived Earth rate is therefore modulated at the table rotation frequency.

A control experiment is performed with opposite tilt ($-\lambda$): the sensing axis remains always orthogonal to the celestial-north direction $\Rightarrow$ *no* modulated Earth-rate signal appears.

6.35.2 Results

The orange curve clearly shows the modulated 4.16 mdps Earth-rate signal. Any residual oscillation due to table misalignment (small cross-axis sensitivity) is distinguishable and does not mask the Earth-rate component.

This measurement was performed on a twin device with slightly worse electronics (non-optimal phase alignment), producing an earlier rate random walk. The best resistive gyroscope (with perfectly optimised phase) achieves even higher stability thanks to the inherent advantages of resistive sensing (immunity to parasitic couplings and temperature drifts of feed-through).

The lecturer notes that this experiment demonstrates the outstanding stability reachable with the NEMS-piezoresistive technology and confirms that the resistive approach makes critical phase-optimisation issues much easier to manage than in capacitive designs.

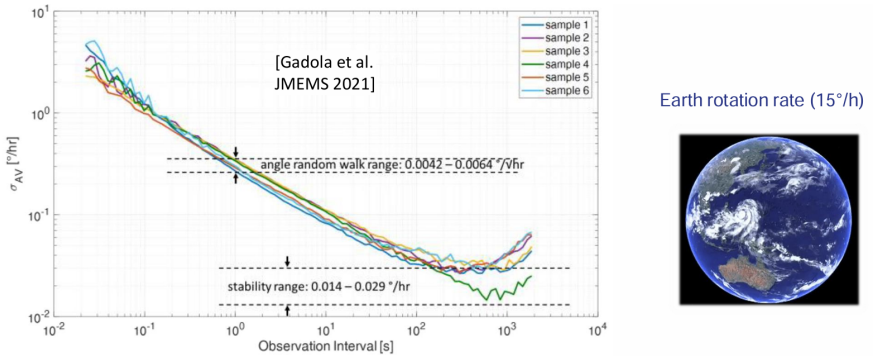

Figure 6.15: Achieved noise and stability performance compared to state-of-the-art.

6.36 Temperature Stability of the Scale Factor

The ideal target for scale-factor variation (after AGC stabilisation) remains $\pm 0.25\,\%$ over the full industrial temperature range ($-40\,°\mathrm{C}$ to $80\,°\mathrm{C}$).

Measurements were performed over a reduced range, so the theoretical limit would be $\approx \pm 0.15\,\%$. The actual result (blue curve, no post-processing compensation, no initial calibration) is

$$\pm 0.45\,\%.$$

This is already a very good value, but still slightly above the ultimate limit. The residual variation is most probably linked to temperature-induced wafer bending affecting the lever ratios γ_s and γ_d (exactly the point raised earlier by Gabriele).

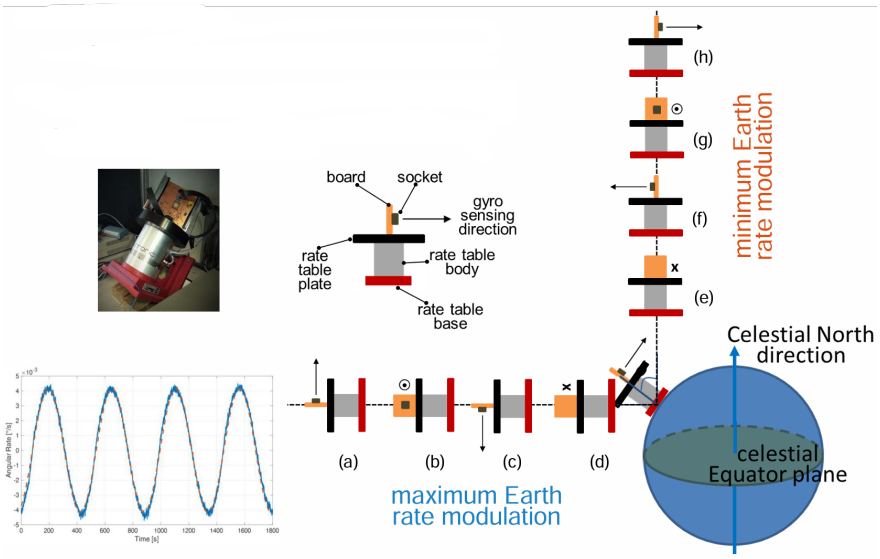

Figure 6.16: Demonstration of Earth rate detection and carouseling technique.

6.37 Multi-Axis Monolithic Gyroscope Concept

Although the main focus of the course is on single-axis devices, the same process allows fully monolithic three-axis gyroscopes (conceptually similar to the capacitive triple-axis device shown earlier).

Preliminary design (currently under fabrication):

- Single drive mode (in-plane anti-phase),
- Three orthogonal sense modes (in-plane yaw + out-of-plane pitch/roll),
- NEMS gauges embedded directly inside the central structure (the inner frame itself acts as the lever),
- ≈ 20 gauges in total (to prove reliability with a larger number of elements).

When an angular rate appears, the appropriate lever tilts and the gauges experience differential stress.

This layout has never been fabricated before but demonstrates that the NEMS approach scales naturally to multi-axis devices while preserving the mechanical advantages (large Coriolis mass, negligible sense-frame mass).

6.38 Conclusions and Commercial Perspective

The lecture ends with a broader reflection on technology transitions.

Sometimes promising new approaches appear in the literature, yet the commercial market is reluctant to adopt them because:

- Changing the fabrication process implies new equipment, higher costs and unknown risks,
- The performance improvement must be *significantly* better to justify the change.

In the present case the comparison is fair (same lab, same supervisors, two PhD students, identical PCB electronics):

- Noise improved by almost one order of magnitude ($550\,\mu\text{dps}/\sqrt{\text{Hz}}$ vs. $3\,\text{mdps}/\sqrt{\text{Hz}}$),
- Bias instability improved by several times ($1.8\,^\circ/\text{h}$ vs. $10\,^\circ/\text{h}$),
- Device area: $\approx 800\,\mu\text{m} \times 500\,\mu\text{m}$ (well below $1\,\text{mm}^2$).

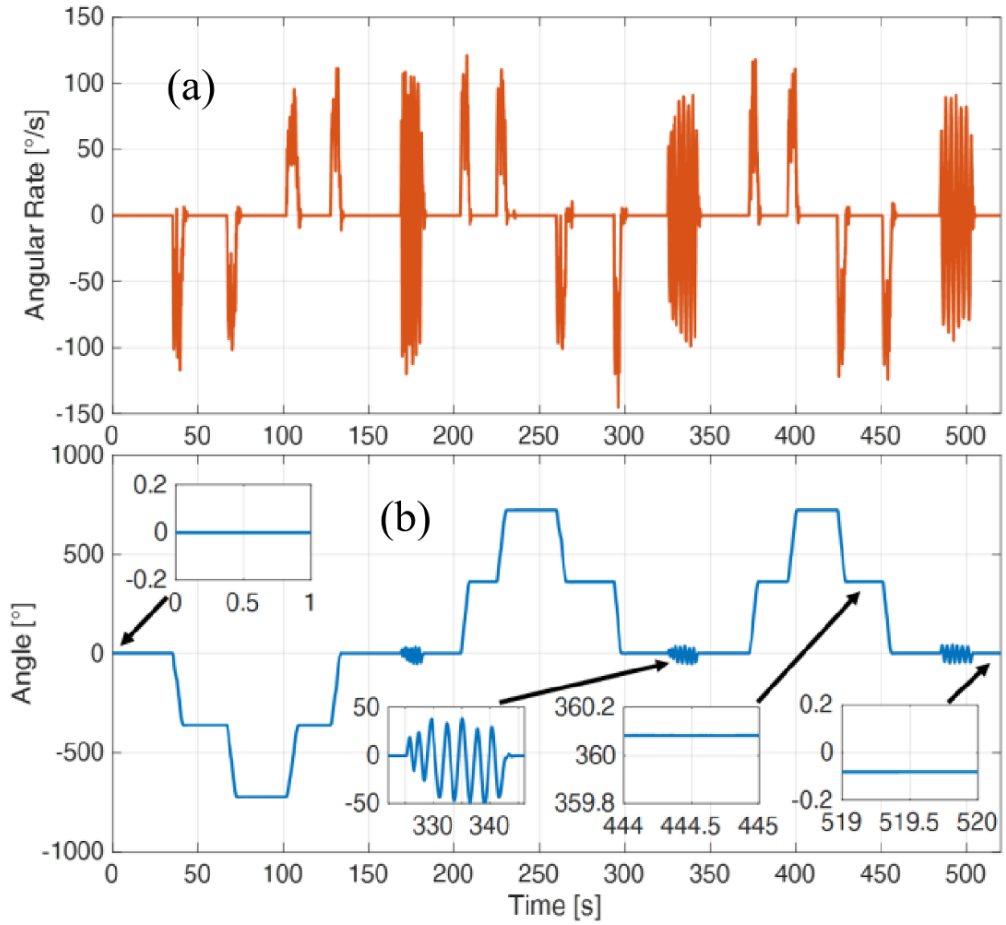

Figure 6.17: Long-term angle estimation performance without external aiding.

This is probably the best noise + stability performance ever reported for a MEMS gyroscope of this size, obtained purely through high-sensitivity piezoresistive NEMS sensing.

Pros

- Dramatic noise and stability improvement,
- Self-compensation of most non-idealities via identical drive/sense transduction,
- Low-voltage operation (no charge pump),
- Immunity to parasitic capacitances and feed-through,
- Possibility of higher resonance frequencies without sensitivity loss.

Cons

- Extra current consumption (≈ few hundred microamps for the gauge branches),
- Starting wafer (SOI) slightly more expensive than standard polysilicon.

The lecturer (who has worked on this technology for nearly a decade) presents it as an interesting and practically viable alternative for high-end miniature gyroscopes.

6.39 Overview of the Remaining Course

- **Next Thursday afternoon** (one week): frequency-modulation gyroscopes (same polysilicon technology, completely different working principle) – a promising route to inherently stable scale factor and offset.
- **Two weeks from now – Thursday morning**: invited seminar by Prof. Valentina Zega (Structural Engineering Dept.) on nonlinear effects and damping modelling in MEMS gyroscopes.
- **Two weeks from now – Thursday afternoon**: final exams (12–14 students, ≈ 10 min each, fits in one afternoon).

The lecturer invites questions on the presented material, on the exam format, or on how to prepare the student presentations.

6.40 Question & Answer – Vibration Sensitivity (Thierry)

Thierry: "You did not mention sensitivity to vibrations. I think the advantage of piezoresistive detection is related to vibration rejection."

Lecturer's answer:

Vibrations must be considered at three levels:

1. Quasi-static board bending (already discussed earlier),
2. Low-frequency broad-spectrum vibrations (up to $\approx 20\,\text{kHz}$),
3. High-frequency vibrations near the resonance frequencies.

For high-frequency vibrations (near-mode excitation) a well-balanced structure is required in *both* capacitive and piezoresistive designs; no major conceptual difference exists.

For moderate vibrations (well below resonance) the piezoresistive approach offers a clear advantage:

- In capacitive gyroscopes, vibration-induced displacement of the proof mass can generate nonlinear effects (moderate but present),
- In NEMS-gauge gyroscopes these nonlinearities are essentially absent thanks to the inherently linear piezoresistive response.

Published works and videos demonstrate this improved vibration tolerance.

Additional by-product: In capacitive devices large drive/sense displacements are mandatory to obtain sufficient signal. In the NEMS design the Coriolis displacement is deliberately de-amplified by the lever; therefore the same output signal can be obtained with a much smaller physical displacement of the Coriolis frame. This allows:

- Higher resonance frequencies (one PhD student is currently working on $\approx 50\,\text{kHz}$ devices),
- Larger stiffnesses (better process robustness),
- Larger frequency split (if needed),

without any loss of scale factor. The lever ratio can simply be adjusted to keep the gauge elongation at the optimum value.

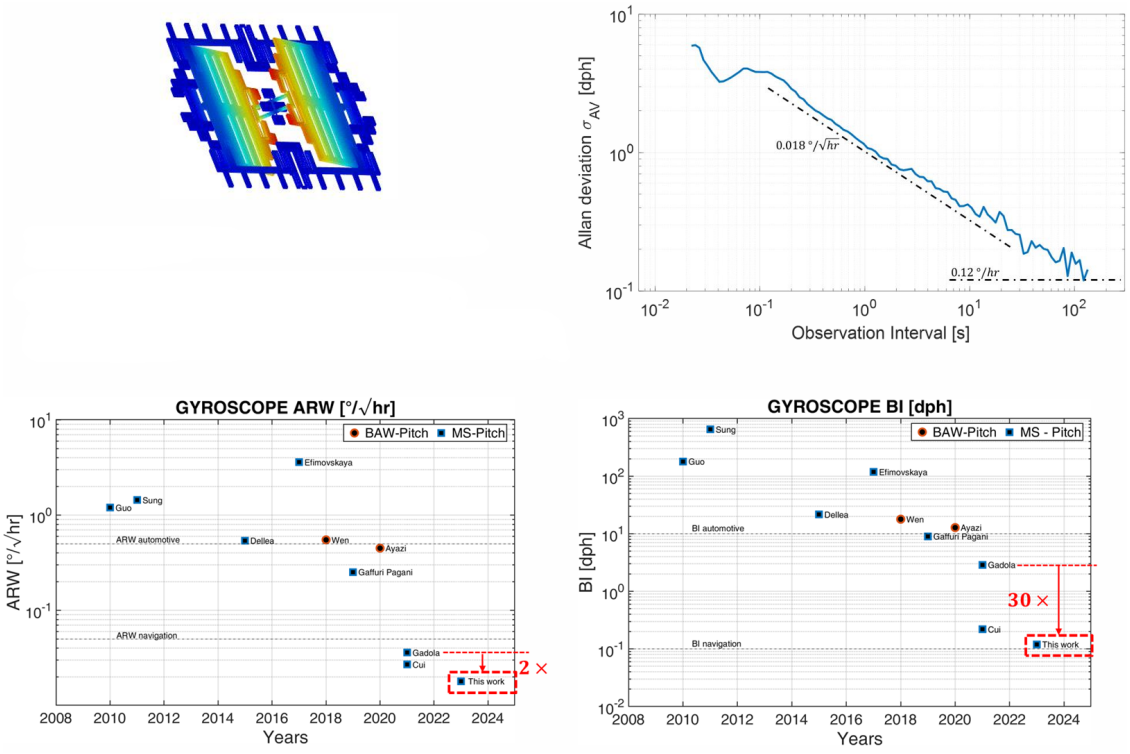

Figure 6.18: Announcement of the highest performance MEMS pitch/roll gyroscope.

Chapter 7

FM and RIG MEMS Gyroscopes

7.1 Introduction

In the last class I talked about new working principles for new technologies. We have seen an example of a new technology. Today we will have a look at two examples of alternative working principles.

This means we can use *any* technology—we can rely on conventional epitaxial poly-silicon capacitive processes—but change the way we deal with our system. I immediately told you that we will *not* be changing the force we rely on: we will still use the Coriolis force behind this working principle. We are only governing the system in a different way, which will lead to different responses of our gyroscope.

7.2 Review of the Conventional Amplitude-Modulated Capacitive Working Principle

Before starting, let us briefly review the motivations by recalling the conventional amplitude-modulated (AM) capacitive working principle we studied up to the previous class.

We have a drive mode, typically operated in anti-phase. In the example shown for the x-axis we drive the proof mass along the x-direction. If an angular rate Ω_z is present along the z-axis, the Coriolis force appears orthogonal to both the drive and the angular-rate directions (in this case along the y-axis):

$$\mathbf{F}_{\text{Coriolis}} = -2m\,\boldsymbol{\Omega} \times \mathbf{v}_{\text{drive}}.$$

We operate in the so-called *weak-coupling* condition: the sense-mode transfer function is excited *slightly off-resonance*. We therefore do not require automatic mode matching; the system remains stable against fabrication and operational variations.

We reviewed the expressions for the **scale factor** S and the **zero-rate offset** (ZRO), two key parameters that must be stabilised for future navigation-grade applications.

7.2.1 Temperature Dependence of the Scale Factor

The scale factor depends on temperature because the drive-mode amplitude is affected by the quality factor Q:

$$S \propto \frac{Q_{\text{drive}}}{m\,\omega_0}.$$

However, the automatic gain control (AGC) circuit rejects these variations extremely well. With an AGC loop gain of at least 100 ($\approx 40\,\text{dB}$), the residual scale-factor variation stays within $0.1\,\%$ over the entire temperature range. The higher the loop gain, the better the rejection of Q-induced drive-amplitude changes.

The resonant frequencies themselves are set by the structural thickness and stiffness. The stiffness depends on Young's modulus E, which varies with temperature. Calculations performed in previous classes (both for resistive and capacitive gyroscopes) show that the mode frequencies shift by approximately $0.25\,\%$ over the full temperature range—a figure compatible with most applications.

The capacitive transduction gain and the electronic gain must also be considered. The electronic gain is predictable from the integrated-circuit environment and usually exhibits a repeatable temperature dependence that can be calibrated once and for all.

The parallel-plate capacitance itself is temperature-dependent because package deformations can warp the MEMS substrate. Fixed electrodes then move differently from the suspended parts, altering parasitic capacitances. These effects are difficult to quantify analytically because they depend on the adhesive, soldering process, and package materials; we cannot rely on simple equations.

Overall, even after careful temperature calibration, a residual scale-factor drift with temperature remains.

7.2.2 Zero-Rate Offset (ZRO) Behaviour

When we look at the gyroscope output at zero angular rate, the observed offset is influenced by quadrature error, phase errors, and crosstalk. These errors propagate through the scale factor, producing three distinct temperature-dependent contributions to the output.

Assume the quadrature error is stable (does not change with time or temperature) and that the scale factor has already been calibrated to minimise its own temperature dependence. If the *phase error* varies with temperature, a temperature-dependent offset appears.

If instead we calibrate the phase, the *quadrature* itself may still change with temperature. Quadrature arises from mechanical and electromechanical imperfections. Package deformations alter gaps and stiffnesses, modifying both the mechanical quadrature and the electrical (capacitive) quadrature; the latter introduces an additional bias component.

A further "crosstalk" term exists: parasitic coupling from the drive signal (or any oscillating node) to the sense output through substrate parasitics. Because the AGC changes the drive amplitude when Q varies with temperature, this crosstalk also becomes temperature-dependent.

The system is therefore a complex interplay of electronics, device, package, and environment. Predicting ZRO behaviour over temperature is extremely challenging. Factory calibration (MEMS + circuit + package) helps, but once the customer solders the device onto a board the thermal boundary conditions change. Package deformation under temperature therefore differs from customer to customer. Temperature calibration performed in the fab is consequently only valid up to a certain point.

7.3 Motivation for Alternative Working Principles

The ideal solution would be a working principle inherently more immune to temperature, environmental changes, and mechanical deformations—both for scale factor and for zero-rate offset.

In this class we will examine two such principles:

1. **Frequency-modulated (FM) gyroscope.**
   The information is no longer encoded in amplitude (as in AM capacitive gyroscopes). Instead, the output is a frequency:
   $$f_{\text{out}} = f_0 + S_{\text{FM}} \cdot \Omega,$$
   where S_{FM} converts angular rate directly into hertz variations. The scale factor therefore appears in units of Hz/(deg/s).

2. **Rate-integrating (angle-output) gyroscope.**
   Offsets are particularly harmful in navigation because the angular rate must be integrated to obtain heading; any constant offset produces a linearly growing angle error. What if the working principle itself integrates the rate inside the mechanical domain? The sensor would then deliver an angle directly at the output, still based on the Coriolis effect, without requiring post-processing integration in electronics.

These are two radically different approaches:

- The first still provides an angular-rate output but with greatly improved stability of scale factor and offset.
- The second bypasses the rate domain entirely and outputs angle.

We will start with frequency modulation (the discussion of the rate-integrating principle will follow after a short break).

7.4 First Comments on Frequency-Modulated Gyroscopes

The class will be rather equation-heavy; I will try to keep the derivations as intuitive as possible.

Typical statements found in the literature on frequency-modulated sensors (not only gyroscopes) are:

- "We only need to read out the frequency of an oscillator instead of the amplitude."
- "FM gives an extremely large full-scale range because the principle is super-linear."

Let us examine these claims.

First, what does "reading frequency" mean? The sensor is an oscillator whose frequency of oscillation changes as a function of angular rate Ω. We simply measure the frequency of (for example) a square wave that toggles between 0 and 1. No amplitude demodulation is required.

The claim of an "extremely large full-scale range because of super-linearity" is *not entirely true*. The underlying sensing physics remains non-linear. Recall the basic relation for a resonator:

$$\omega = \sqrt{\frac{k}{m}}.$$

The Coriolis force does *not* change frequency directly. It modifies the *apparent stiffness* k_{app} seen by the mode; the frequency shift then follows from the square-root dependence above. The non-linearity of the stiffness modulation will be analysed in detail in the next slides.

7.4.1 Linearity and Full-Scale Range

As these frequency shifts are very small, it is basically like linearising our working principle around a specific operating point and working there. We still have some residual non-linearities.

It is true that the full-scale range becomes typically large, but that is only because we move by a very small amount around the operating point—a direct consequence of the fact that the scale factor is not that large.

It is exactly analogous to a capacitive gyroscope: if you deliberately choose a small scale factor, you move only mildly around the working point. We know the capacitive plates are non-linear, but if we linearise around the operating point (as we did a few classes ago), we accept the linearity. The situation is precisely the same here—except that the scale factor in frequency-modulated sensors (as we will see) is dictated purely by geometry and not by electronics. This is advantageous because it is extremely stable, but it is also very small, which is precisely why the linearity is excellent.

7.4.2 Temperature Immunity Claim

Another typical statement is that frequency modulation is immune to temperature changes because you bypass all temperature drifts of the electronic stages.

That is partially true. If you have an amplifier that simply squares up a frequency signal (0 to 1), any offset in the amplifier, any drift of that offset, or any change in gain does not matter—the output still switches cleanly from 0 to 1.

However, the frequency you actually have to read out still contains a common-mode frequency shift (as we will see later). That common-mode component *does* drift with temperature, and we will have to find ways to bypass or cancel it.

7.4.3 "Quasi-Digital" Output Claim

A further fascinating statement often found in the literature is that frequency modulation has a "quasi-digital" output (0, 1, 1, 0, 0, 1, 1, 0), so it should be very stable against temperature changes because "we are already in the digital domain".

That is absolutely not true. It is correct that the signal switches between two levels, but we do not care about the amplitude—only about the *zero-crossing times* (the instants when it goes from 0 to 1 and from 1 to 0). Those instants can vary continuously in time. The information is therefore still fundamentally analogue: it is a continuous time interval. The signal is not yet quantised.

It is true that you can process this signal with purely digital circuits (counters, timers, etc.), but that does not mean the raw sensor information is already digitised.

7.4.4 Intuitive Cartoon Comparison: AM vs. FM Gyroscopes

Once these caveats are given, let us visualise the intuitive difference between AM and FM gyroscopes in cartoon style.

Amplitude-Modulated (AM) Case (Review)

We apply a force along one axis (the drive axis) and keep the proof mass in oscillation, producing a velocity $\mathbf{v}_x$. With an angular rate Ω_z (out-of-plane), the Coriolis force appears along the orthogonal (y) axis.

Neglecting damping for a moment and assuming the drive electrostatic force perfectly cancels the damping term (exactly what a closed-loop drive does), the force-balance equation along the drive axis reduces to the stiffness term only:

$$m\ddot{x} + kx \approx 0.$$

Hence the resonance frequency along the drive axis is independent of Ω. The same holds for the sense axis.

We already derived that the sense displacement is

$$x_y \propto \frac{x_{\text{drive}}}{\Delta\omega},$$

where $\Delta\omega$ is the frequency split between drive and sense, plus electronic gains. Operation is classical: one oscillator (with AGC) on the drive axis, capacitive readout on the sense axis, demodulation, etc.

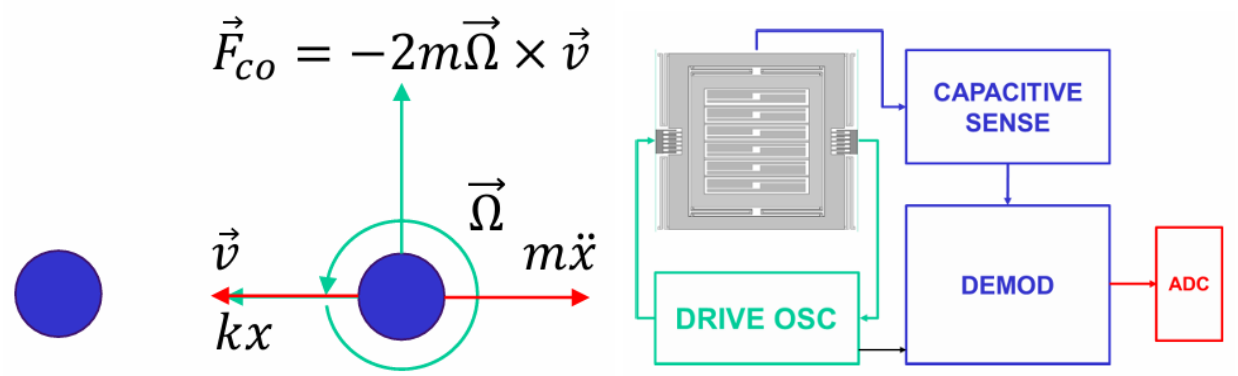

Figure 7.1: AM setup.

Frequency-Modulated (FM) Case

Now consider an FM gyroscope. In this first example we assume two identical modes (perfectly frequency-matched) driven in quadrature.

"Quadrature drive" means: when displacement along x is maximum, displacement along y is zero, and vice versa. The proof mass therefore traces a circular trajectory in the x-y plane. At every instant the velocity is tangential to the circle.

With an angular rate Ω_z, the Coriolis force now acts radially (inward or outward depending on the instantaneous velocity direction). The force-balance equation along each axis now contains the stiffness term, the inertial term, *and* the Coriolis term.

Linearising (because the Coriolis contribution is much smaller than $\sqrt{k/m}$), the resonance frequency of each mode acquires a shift proportional to Ω:

$$\omega_{x,y} = \omega_0 \pm S_{\mathrm{FM}} \cdot \Omega.$$

The sign and exact form depend on the relative phase between the two axes.

Viewed another way: in an AM gyroscope the angular-rate information appears as a change in displacement amplitude; in an FM gyroscope the same information appears as a change in resonance frequency of at least one (actually both) axes. The scale factor is now

$$S_{\mathrm{FM}} \propto \frac{x_{\mathrm{drive}}}{x_{\mathrm{sense}}} \times (\text{mechanical gain}),$$

with *no* electronic gains involved. The scale factor is therefore set purely by geometry and is extremely stable.

Operation requires two self-sustained oscillators (one per axis) plus a phase-control loop (discussed in the next slide). Each oscillator produces a square wave (0 to supply). We simply measure the two frequencies digitally; their difference (or suitable combination) is proportional to Ω.

That is the cartoon view. We will now move to the actual equations. The class will be equation-heavy, but I do not expect you to memorise every term. My goal is only to highlight the challenges, design guidelines, benefits, and drawbacks compared with classical AM operation.

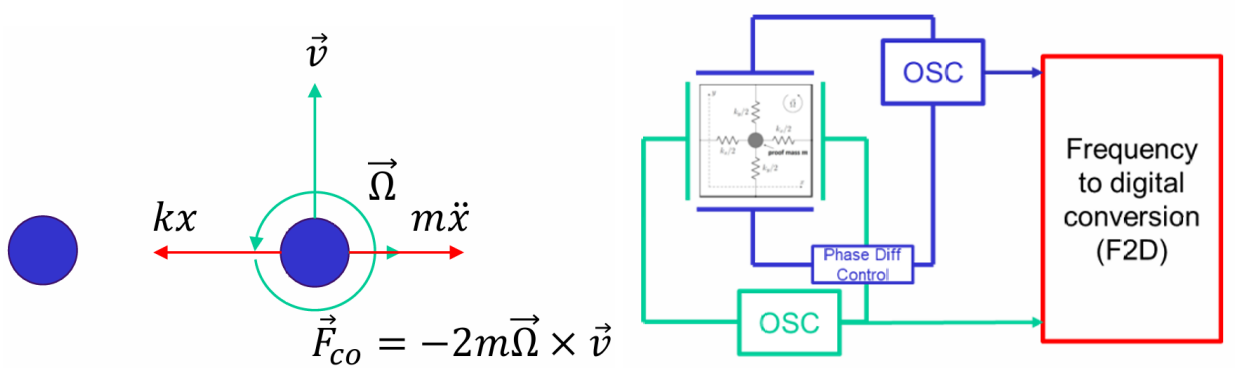

Figure 7.2: FM setup.

7.5 Equations for the Frequency-Modulated Gyroscope

According to the new view of the system, we no longer have a "drive mode" versus a "sense mode". Forget that distinction. The two modes are identical from our perspective—one along x, the other along y—and we keep *both* in self-sustained oscillation.

We apply an electrostatic force that sustains the x-axis oscillation and another that sustains the y-axis oscillation. The Coriolis forces now act on *both* axes simultaneously.

7.5.1 Physics of the Sign Convention

From the vector cross-product physics, if your angular rate is along the $+z$-axis (the usual z-axis gyroscope assumption), the Coriolis force direction depends on the velocity:

- If velocity is positive along $+x$, Coriolis force is negative along y.
- If velocity is positive along $+y$, Coriolis force is positive along x.

That is why the signs differ in the two equations.

7.5.2 Equations of Motion

We now keep *both* axes in sustained oscillation. The analysis is symmetric, so we develop the equation for the x-axis; the y-axis equation is identical except for the sign of the Coriolis term.

The coupled system reads (assuming identical mass m, stiffness k, damping c):

$$m\ddot{x} + c\dot{x} + kx = F_{\text{elec},x} - 2m\Omega\dot{y}, \tag{7.1}$$

$$m\ddot{y} + c\dot{y} + ky = F_{\text{elec},y} + 2m\Omega\dot{x}. \tag{7.2}$$

7.5.3 Phasor Analysis (High-Q Approximation)

High-Q oscillators produce nearly pure sinusoidal motion. We therefore use phasor notation. All quantities (x, $\dot{x}$, $\ddot{x}$, forces) are represented as complex amplitudes (the $e^{j\omega t}$ time dependence is implicit):

$$x = X_a\, e^{j\phi_x}, \quad y = Y_a\, e^{j\phi_y}.$$

Differentiation rules in the phasor domain:

$$\dot{x} \to j\omega X, \qquad \ddot{x} \to -\omega^2 X.$$

The electrostatic drive force is phased to exactly cancel damping losses, so it lies along the imaginary axis (in phase with velocity).

The Coriolis term couples the two equations: it contains the phasor of the orthogonal axis multiplied by $2m\Omega j\omega$. You cannot solve the two equations independently; the relative phase $\phi_y - \phi_x$ becomes critical.

7.5.4 Phasor Diagram

We plot the x-equation in the complex plane (real part horizontal, imaginary part vertical). The mass term ($-m\omega^2 X$) and stiffness term (kX) lie on the real axis. Damping and electrostatic forces lie on the imaginary axis.

The Coriolis contribution from the y-phasor appears as an additional vector whose direction depends on the relative phase $\phi_y - \phi_x$. This vector can point anywhere in the plane. Its projection onto the real axis modifies the apparent stiffness (hence the resonance frequency); its projection onto the imaginary axis modifies damping.

The frequency modulation (apparent stiffness change) is therefore strongly phase-dependent. If the y-phasor is aligned such that its Coriolis contribution is purely real, the frequency shift is maximum. If it is purely imaginary, there is no frequency shift.

A constant relative phase is possible only if the two natural frequencies are perfectly matched. Otherwise the phase difference drifts continuously in time.

7.5.5 In-Phase Matched Modes ($\phi_y - \phi_x = 0°$)

This first FM principle (studied in laboratories at the University of California and later in Japan) keeps the two axes perfectly frequency-matched and drives them with *zero phase difference*.

When displacement along x is maximum, displacement along y is also maximum. The trajectory is therefore a straight diagonal line in the x-y plane.

After linearisation the resonance frequencies become

$$\omega_x \approx \omega_0 - O\Omega, \qquad \omega_y \approx \omega_0 + O\Omega,$$

where O is a dimensionless mechanical coupling factor ($O = 1$ in this ideal configuration). The difference $\omega_y - \omega_x = 2O\Omega$ cancels the common-mode ω_0 and yields a pure rate signal.

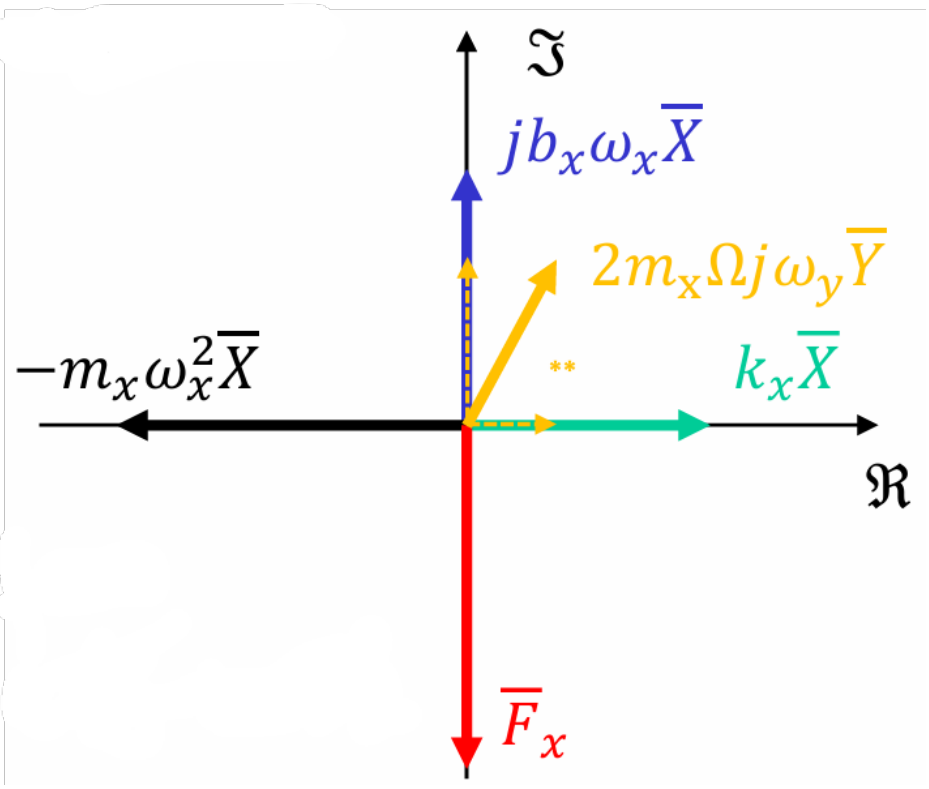

Figure 7.3: Phasor representation.

7.5.6 Scale Factor and Full-Scale Range in Practice

Typical navigation-grade full-scale ranges discussed in previous classes are around 2000°/s. For a resonance frequency of 20–30 kHz (chosen to stay far from acoustic and vibration disturbances), this produces only 5–6 Hz of frequency shift per axis.

The differential output is therefore

$$\Delta f = 2O\Omega,$$

with O a mechanical parameter that represents how much of the effective mass participates in the Coriolis action. O cannot exceed 1 and typically lies between 0.5 and 1 for well-designed MEMS structures (pictures will be shown later).

The relative frequency change with respect to the operating frequency remains extremely small, confirming the excellent linearity and huge practical full-scale range.

7.5.7 Pros and Cons of In-Phase Operation

- **Advantage**: common-mode resonance frequency ω_0 is perfectly rejected by subtraction.
- **Drawback**: requires continuous mode-matching ($\omega_x = \omega_y$), which is difficult to achieve by fabrication alone and usually needs auxiliary tuning circuits.

7.5.8 Quadrature Operation ($\phi_y - \phi_x = \pm 90°$)

This is exactly the circular-trajectory case shown in the cartoon.

With a 90° phase shift the ratio of the phasors becomes $\pm j$. Consequently both frequencies shift in the *same* direction:

$$\omega_x \approx \omega_0 - O\Omega, \qquad \omega_y \approx \omega_0 - O\Omega.$$

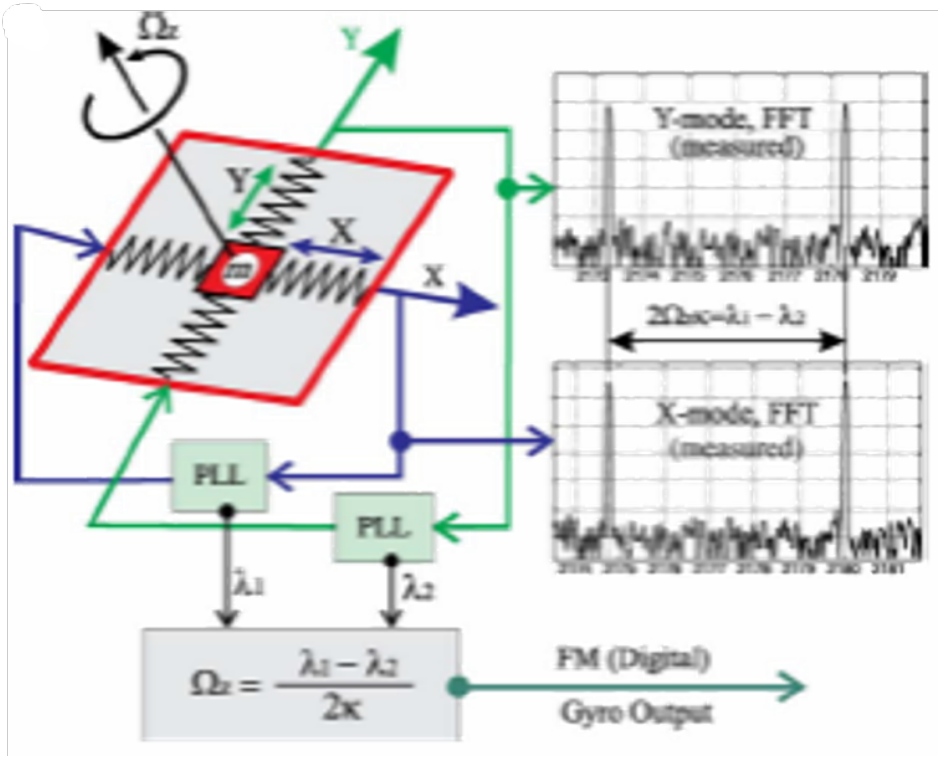

Figure 7.4: Mode matched with phase control at 0° case.

The output must be taken as the *sum* of the two frequencies:

$$f_{\text{out}} = \omega_x + \omega_y \approx 2\omega_0 - 2O\Omega.$$

The gyroscope remains perfectly mode-matched at all times (a major advantage over the in-phase case). However, the large common-mode term $2\omega_0$ drifts with temperature (Young's modulus temperature coefficient $\approx -30\,\text{ppm/K}$), appearing directly as offset.

To cancel this drift we fabricate *two identical gyroscopes*:

- Gyro 1: $+90°$ phase shift (counter-clockwise circular motion) $\rightarrow$ sum $\approx 2\omega_0 - 2O\Omega$.
- Gyro 2: $-90°$ phase shift (clockwise circular motion) $\rightarrow$ sum $\approx 2\omega_0 + 2O\Omega$.

The difference of the two sums cancels the $4\omega_0$ term and yields a signal proportional only to Ω. This architecture (called *quadrature frequency-modulated gyroscope*) requires four oscillating axes in total; the natural frequencies of the two separate gyroscopes must also be matched to each other.

Additional robustness: the sum operation automatically rejects small amplitude mismatches between X_a and Y_a. The effective coefficient becomes

$$\left(\frac{Y_a}{X_a} + \frac{X_a}{Y_a}\right) \approx 2$$

even when $Y_a \neq X_a$ (the expression is stationary at equality). Amplitude-control imperfections are therefore strongly suppressed.

7.5.9 Lissajous (Split-Frequency Modulated) Frequency-Modulated Gyroscope

So far we assumed the two modes were frequency-matched.

What if we deliberately introduce a small frequency split $\Delta\omega$ (typically 100–200 Hz) while keeping the operating frequency at 20–30 kHz?

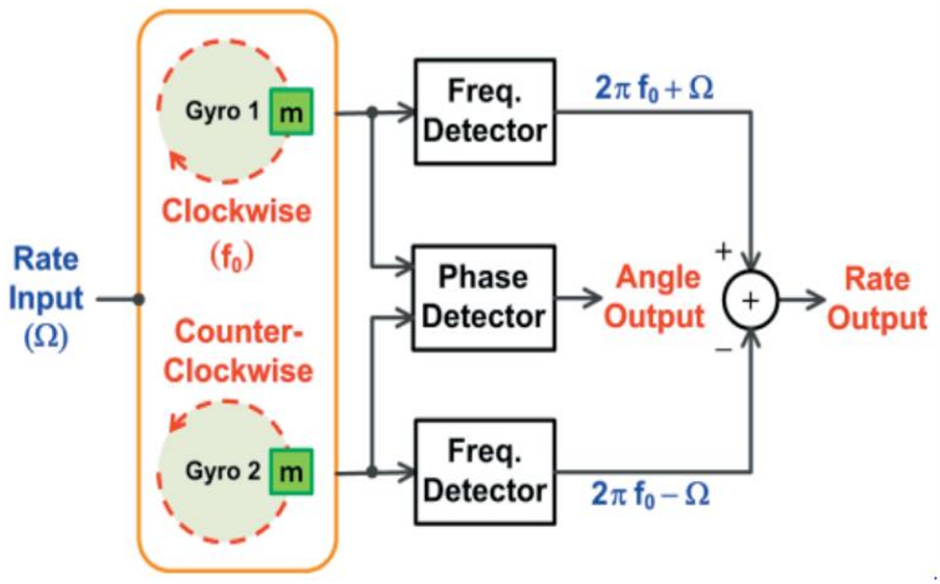

Figure 7.5: Mode matched with phase control at 90° case.

The relative phase now evolves continuously:

$$\phi_y - \phi_x = \Delta\omega\, t \pmod{2\pi}.$$

It increases linearly with time and wraps every 2π. The Coriolis phasor (coming from the y-axis) therefore rotates steadily in the complex plane at frequency $\Delta\omega$.

Phasor Diagram with Rotating Coriolis Term

In the x-axis phasor diagram we still have:

- Mass term: negative real axis,
- Stiffness term: positive real axis,
- Damping term: positive imaginary axis,
- Electrostatic drive: negative imaginary axis.

The Coriolis contribution is proportional to the y-phasor multiplied by $2m\Omega j\omega$. Because the y-phasor rotates at $\Delta\omega$, this vector continuously sweeps all directions (aligned with real axis $\rightarrow$ imaginary axis $\rightarrow$ negative real axis $\rightarrow$ negative imaginary axis, and repeats).

At each instant the force-balance equation must still be satisfied on both axes. Along the y-axis the drive force automatically adjusts its amplitude to compensate (amplitude modulation). Along the x-axis the resonance frequency itself modulates to keep the equation satisfied (frequency modulation).

After linearisation the instantaneous resonance frequency of each axis acquires a term that oscillates at exactly $\Delta\omega$:

$$\omega_x(t) \approx \omega_0 + O\Omega \cdot \sin(\Delta\omega\, t + \varphi).$$

(The exact form and phase φ depend on the chosen reference.)

The angular-rate information is therefore *frequency-modulated* (information carried in frequency deviation) but also *amplitude-modulated* onto a carrier at $\Delta\omega$. Slow temperature drifts (which affect the carrier ω_0) can now be separated from the rate signal (sidebands around $\Delta\omega$) by ordinary modulation-demodulation techniques. This completely bypasses $1/f$ noise and slow environmental drifts.

Implementation and Lissajous Trajectory

This architecture is realised with a specific MEMS layout using four shuttles (two for the x-drive and two for the y-drive) that push a central proof mass. The shuttles move only along one axis each, while the central mass can move in both directions.

Because the relative phase between the two drive forces changes continuously at $\Delta\omega$, the trajectory of the centre of mass periodically evolves:

- In-phase ($\phi = 0°$): diagonal line,
- Quadrature ($\phi = +90°$): counter-clockwise circle,
- Opposition ($\phi = 180°$): anti-diagonal line,
- Opposite quadrature ($\phi = -90°$): clockwise circle.

The path repeats every $1/\Delta\omega$ seconds. This is exactly the *Lissajous figure*, which is why the principle is called the **Lissajous frequency-modulated gyroscope**.

The frequency-modulation effect is strongest precisely when the trajectory is circular (quadrature instants) and vanishes when it is linear (in-phase instants). The modulation is therefore inherently periodic at the beat frequency $\Delta\omega$.

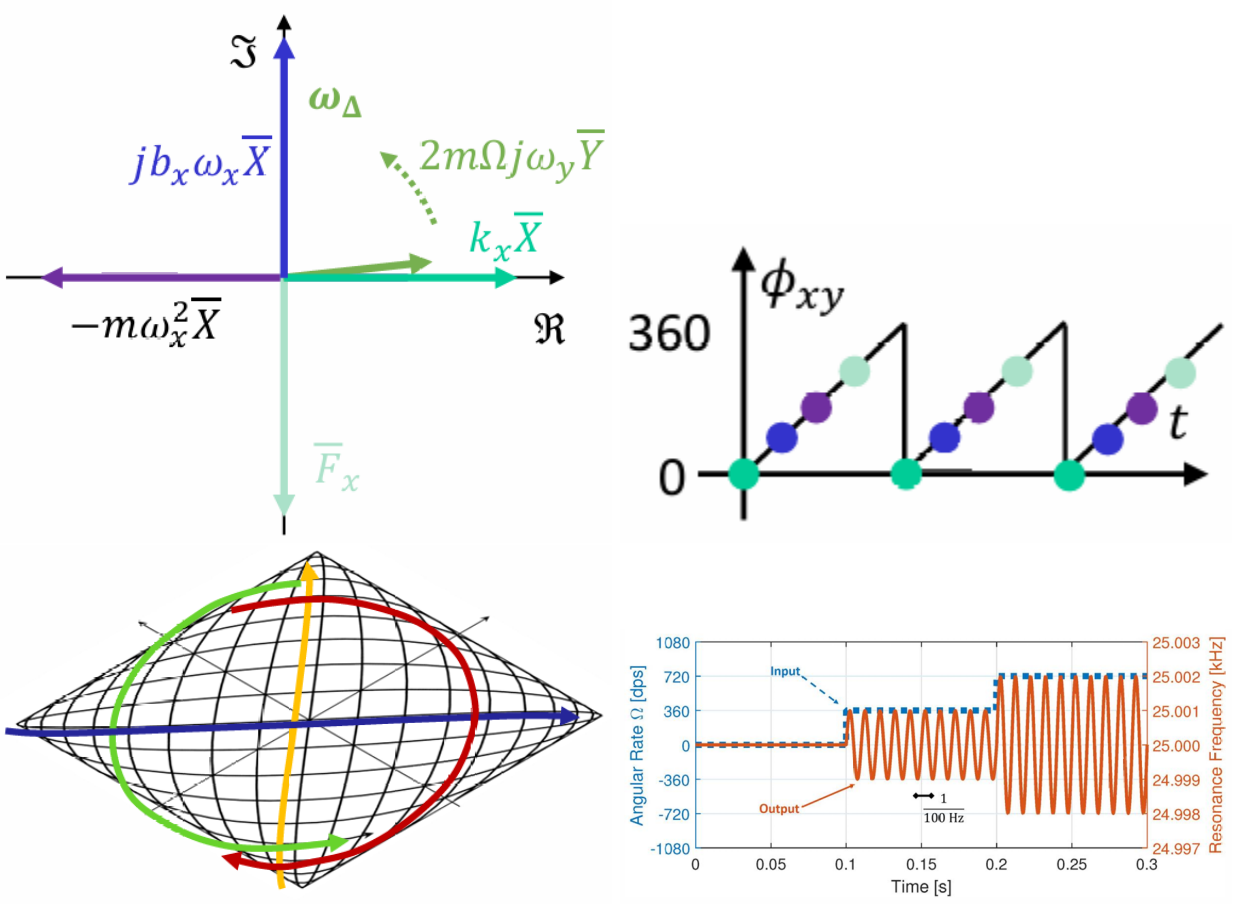

Figure 7.6: Pattern of Lissajous frequency modulation.

Time-Domain Frequency Output

Consider a gyroscope designed to resonate at 25 kHz with $\Delta\omega = 100$ Hz:

- At zero angular rate ($\Omega = 0$): the resonance frequency stays constant at 25 kHz (blue line).
- At $\Omega = 360°/\text{s}$ ($\equiv 1$ Hz in the earlier convention): the frequency deviates by ± 1 Hz, but this deviation is modulated at 100 Hz (carrier appears as sidebands).
- At $\Omega = 720°/\text{s}$: deviation becomes ± 2 Hz, still modulated at the same carrier.

The modulation depth (amplitude of the frequency excursion) is directly proportional to Ω. Demodulation at $\Delta\omega$ recovers a clean rate signal immune to temperature-induced carrier drifts.

7.6 Geometrical Factor, Real Implementation and Performance

7.6.1 Geometrical Factor O

Only the inner proof mass participates in the Coriolis effect. The outer shuttles are designed to be very stiff in the orthogonal direction, so they contribute negligibly to the motion. Consequently the effective mechanical coupling factor is

$$O = \frac{m_{\text{inner}}}{m_{\text{total}}}.$$

In the example structures shown, $O \approx 0.8$. The frequency shift for $360°/\text{s}$ therefore becomes ≈ 0.8 Hz instead of 1 Hz—a modest reduction in scale factor.

7.6.2 Real Device Example

The structure developed approximately 10 years ago is exactly the four-shuttle layout described above. The two axes are intentionally split by $\Delta f \approx 80$–200 Hz. When operated, the centre-of-mass trajectory visibly cycles through the full Lissajous sequence: diagonal → counter-clockwise circle → anti-diagonal → clockwise circle, repeating at the beat frequency.

7.6.3 Extension to Pitch and Roll (Out-of-Plane) Axes

Implementing the same principle for the other two sensing axes is significantly more difficult. For yaw (in-plane) the structure is almost perfectly axis-symmetric; only minor mass adjustments on the shuttles are needed to create the intentional frequency split. For pitch or roll the required oscillations are out-of-plane: comb-finger actuation is impossible, only parallel-plate electrodes can be used, and out-of-plane damping is higher. Maintaining the required symmetry between the two modes becomes extremely challenging, which penalises performance. Consequently, commercial 3-axis gyroscopes based on Lissajous FM are not straightforward.

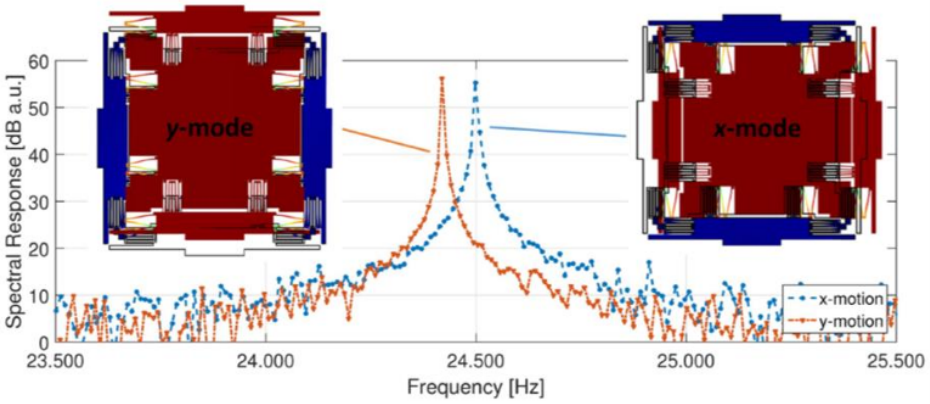

Figure 7.7: Spectral response of the modes in yaw devices.

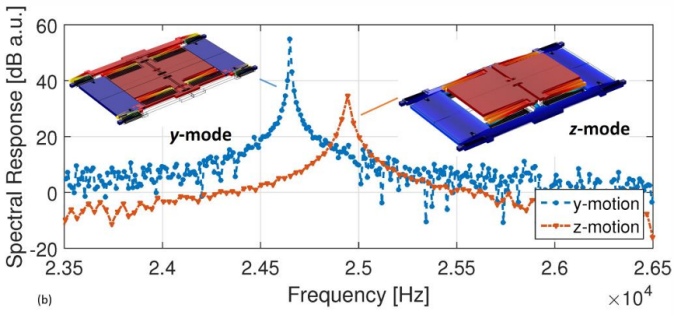

Figure 7.8: Spectral response of the modes in pitch/roll devices.

7.6.4 Experimental Results

Measurements on several fabricated samples (different Δf from 80 to 200 Hz) show:

- Individual axis frequency shifts vary slightly from part to part (one axis more sensitive than the other).
- The extracted scale factor (average of the two normalised shifts) remains extremely stable across parts and across different frequency splits.
- The effective coefficient $(X_a/Y_a + Y_a/X_a) \approx 2$ is robust against amplitude mismatches, as theoretically predicted.

Compared with classical AM gyroscopes (where scale factor depends linearly on frequency split and therefore drifts with any mismatch), the Lissajous FM scale factor is far more repeatable: part-to-part variation is in the range of a few parts per thousand (much better than the typical ppm-level repeatability of AM devices). No individual calibration or screening is required.

The same excellent repeatability was confirmed on yaw-rate sensors.

7.6.5 Electronics Architecture

The electronics for each axis are almost identical to the drive loop of a classical AM gyroscope:

- Differential capacitive readout,
- Phase shifter (90° for quadrature drive in some variants),
- AGC-controlled push-pull actuation,
- Self-sustained oscillator.

Two independent oscillators run in parallel. The square-wave outputs of both are fed to a digital frequency counter (or PLL-based frequency-to-digital converter). The frequency difference (or suitable demodulated combination) is extracted digitally and yields the angular-rate output.

7.6.6 Noise and Long-Term Stability

In AM gyroscopes we worried mainly about amplitude noise. In FM gyroscopes only phase (or frequency) noise matters, because the information is carried in zero-crossing timing. The inherently small scale factor (few hertz change around tens of kilohertz) makes the system sensitive to oscillator phase noise. Consequently the angle random walk (ARW) is typically a couple of orders of magnitude worse than navigation-grade requirements.

However, the Allan-deviation plot shows excellent long-term stability: the bias instability remains flat (no drift) even after thousands of seconds.

A simple rotation-table experiment (similar to the resistive-gyro test shown in a previous class) over approximately 5 minutes yielded a heading error of a few degrees—larger than the resistive-gyro result, but

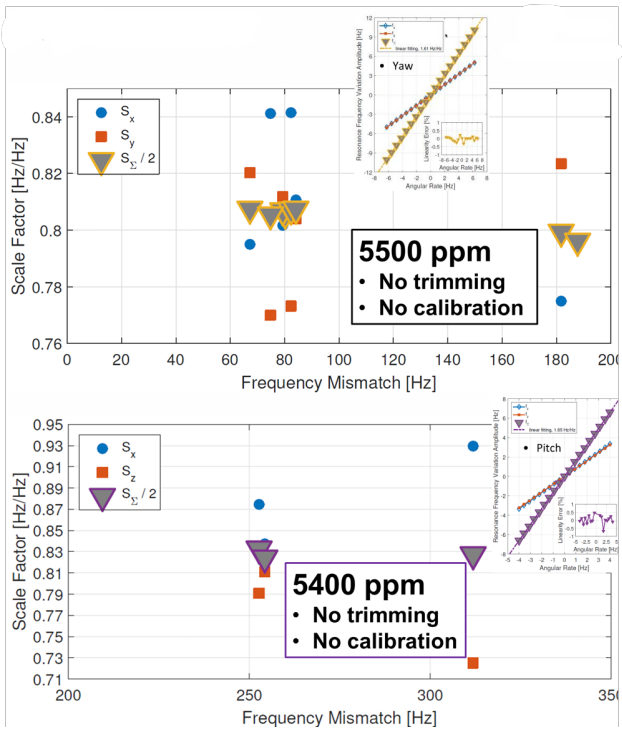

Figure 7.9: Scale factor part-to-part repeatability.

still in the degrees-per-hour range. The device was not pushed to its limits (very stable table, low-vibration environment).

If the noise floor can be reduced (better oscillators, higher Q, optimised electronics), the principle offers outstanding long-term stability for navigation applications.

7.7 Rate-Integrating (Whole-Angle / Full-Angle) Gyroscopes

It will become even worse if you look at the final working principle. Analysing it in full mathematical detail would be extremely heavy and we simply do not have time, but we can at least get a glimpse of the operation of what we call **rate-integrating gyroscopes**.

The strange name simply means that the rate is integrated *inside* the working principle and not in external electronics. They are also known as **whole-angle** or **full-angle** gyroscopes because they directly output an angle (degrees of rotation) at the output.

7.7.1 Summary of All Architectures Studied

We can operate classical amplitude-modulated gyroscopes in:

- mode-split (sense chain open-loop),
- force-rebalance closed-loop (sense mode kept at zero displacement),
- mode-matched,
- frequency-modulated approaches (in-phase, quadrature, or Lissajous).

In all these cases the output is still an angular *rate* (Ω in deg/s).

Rate-integrating gyroscopes, on the contrary, operate exclusively in perfectly mode-matched conditions and behave as full-angle devices: the output is directly the *angle* θ (degrees of rotation), not the rate.

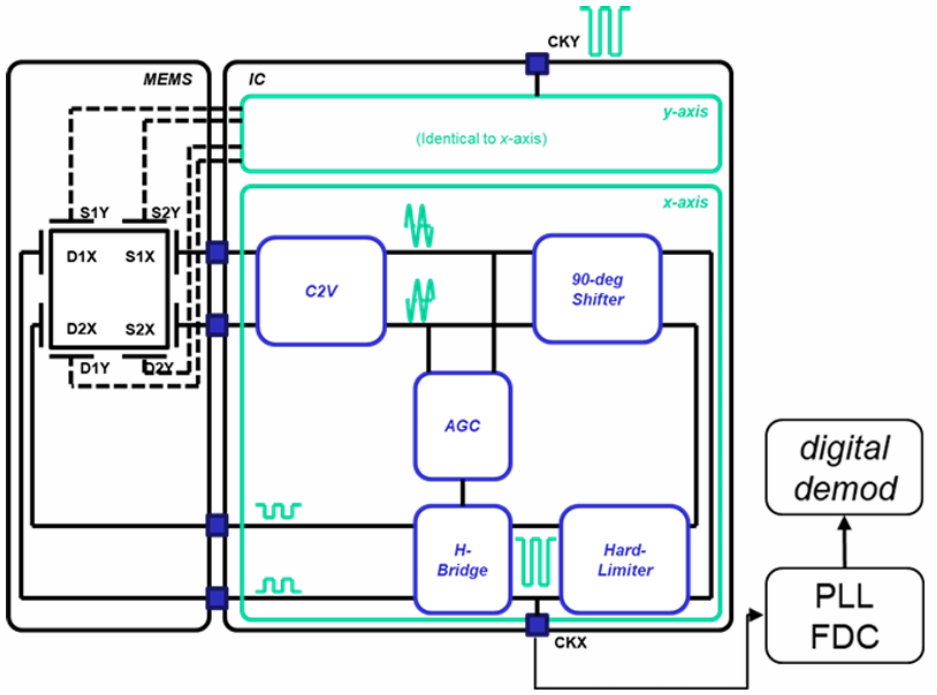

Figure 7.10: Oscillator topology.

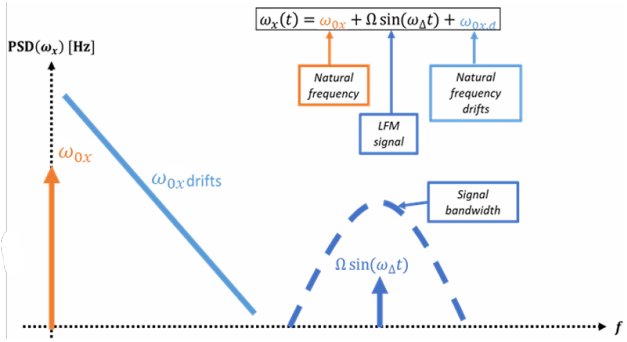

Figure 7.11: Phase noise representation.

7.7.2 Motivation in Inertial Navigation

In inertial navigation we are ultimately interested in the *attitude* (heading) at any instant, not the instantaneous rate. We use accelerometers and gyroscopes to track position by integrating velocity and angle.

If we integrate the output of an ordinary rate gyroscope we also integrate noise and offset. Large noise accumulates statistically; any residual offset (even after calibration) produces a linearly diverging angle error.

If the integration of the rate occurs *mechanically*—ideally without added noise or offset—the stability can be inherently superior.

7.7.3 Historical Example: Hemispherical Resonator Gyroscope (HRG)

We already saw this principle in the first class with the hemispherical resonator gyroscope (wine-glass or mushroom shape). When the platform rotates, the positions of the nodal and anti-nodal lines of the vibration pattern precess. The difference between the geometric rotation and the nodal-line rotation is a function *only* of the material properties (not geometry). The output is therefore a direct angle.

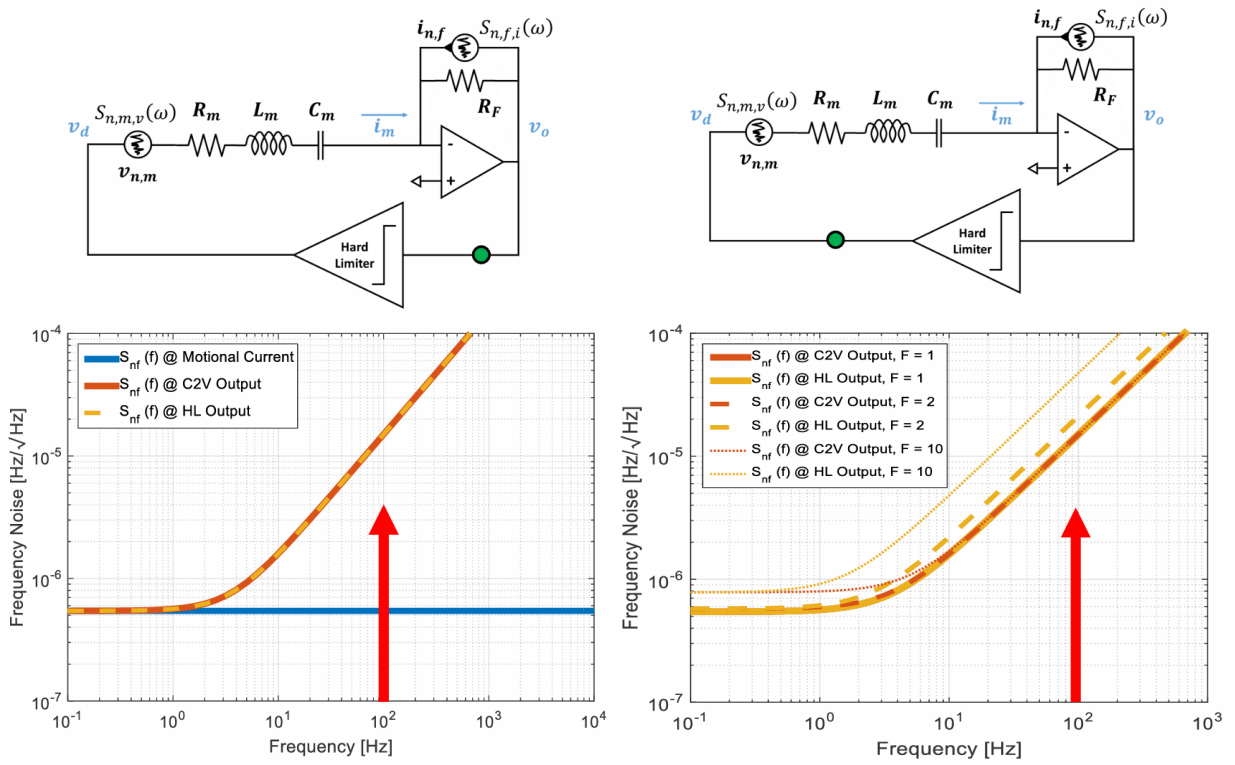

Figure 7.12: Frequency noise representation.

These structures are traditionally fabricated in quartz (extremely low losses, Q up to tens of millions). However quartz is non-conductive: the resonator must be metallised, which introduces losses and reduces Q to around one million. Electrodes must also be manually aligned—far from the beauty of a self-aligned batch MEMS process.

7.7.4 Goal: Silicon MEMS Implementation

The challenge is to realise the same rate-integrating principle in standard 2-D flat silicon MEMS technology while preserving the essential features: two perfectly degenerate (frequency-matched) modes and very high Q.

7.7.5 Requirements and Possible Designs

Any rate-integrating gyroscope requires *two degenerate modes*—i.e., two oscillation directions that are perfectly symmetric and occur at exactly the same frequency.

Several geometries achieve this:

Ring Resonator

A central disk (or hub) surrounded by concentric rings linked by thin spokes, with electrodes placed all around the perimeter. Two degenerate elliptical modes exist:

- one along the x-y axes,
- the other along the 45° axes.

Higher-order radial modes (e.g., with electrodes at 120° intervals) are also possible.

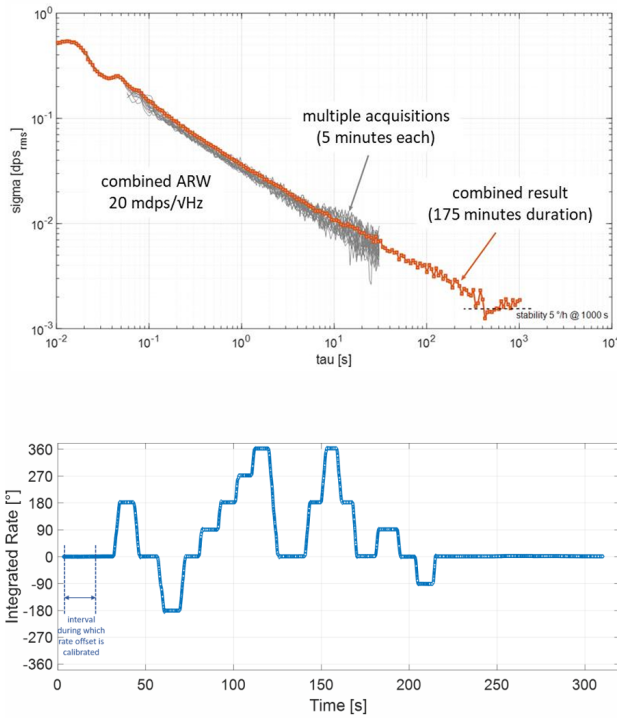

Figure 7.13: Measured Allan variance.

Lantern-Element (Quadrature Mass) Design

Very similar in principle to the structure we used for Lissajous FM gyroscopes. Four masses oscillate in anti-phase: when two move outward along x, the other two move inward, and vice versa (same behaviour along y). Perfect symmetry is obtained by design.

Dual-Mass / Dual-Frame (Balanced) Design

An inner proof mass and an outer frame move in opposite directions:

- inner mass to the right $\rightarrow$ outer frame to the left,
- inner mass down $\rightarrow$ outer frame up.

The centre of mass remains stationary during oscillation, providing excellent linear acceleration rejection. This is the design most groups have explored for full-angle operation.

In all cases the two modes must be kept perfectly frequency-matched (difficult but achievable with tuning electrodes or electrostatic softening).

7.7.6 Equations of Motion and Ideal Behaviour

We are discussing systems with even higher quality factors than the gyroscopes seen so far. We therefore operate at even lower pressure and achieve Q in the range 100 000–1 000 000. Initially we completely neglect forcing and damping terms and are left with the neutral equations of motion.

To be even more precise, recalling the kinematics of relative motion in inertial and non-inertial frames, the full equations also contain angular-acceleration (Euler) and centrifugal terms:

$$m\ddot{x} + kx - 2m\Omega\dot{y} - m\dot{\Omega}y - m\Omega^2 x = 0$$

(and the symmetric equation for y).

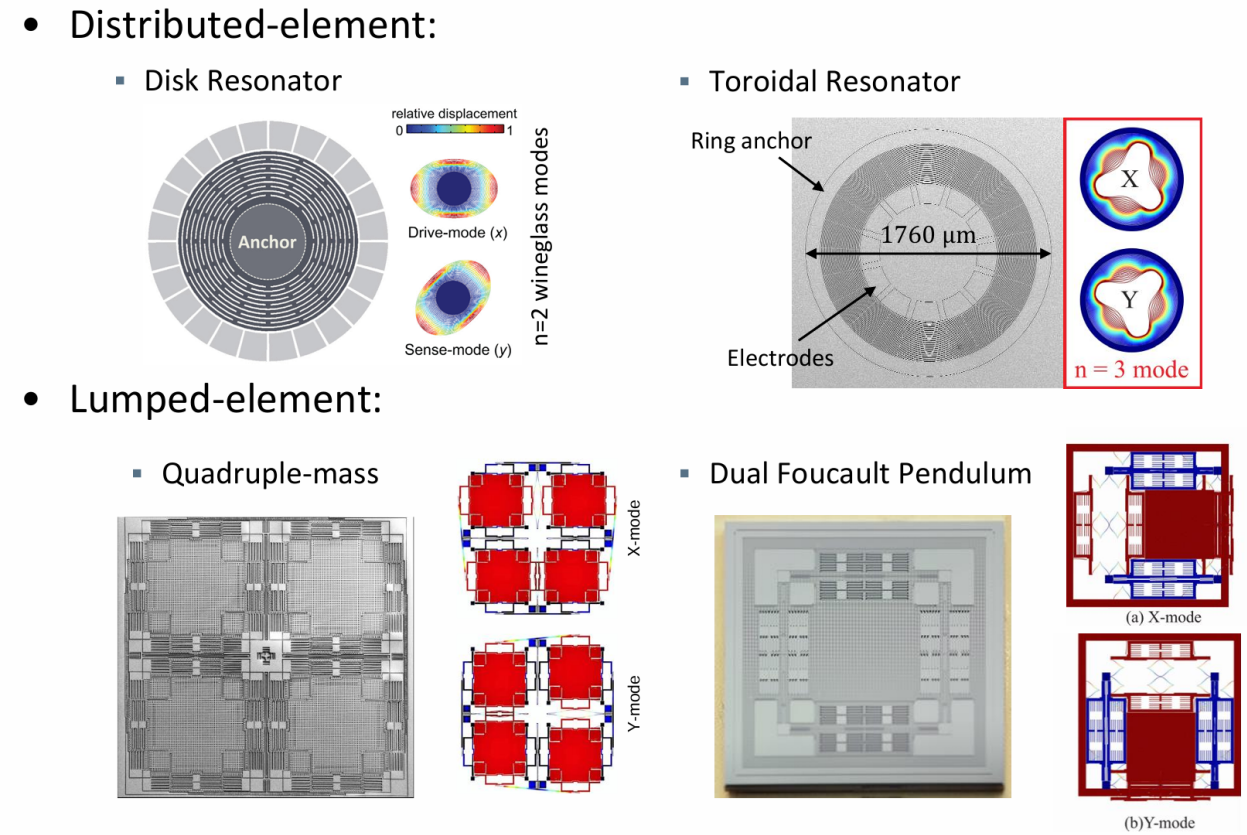

Figure 7.14: Structure designs.

In an ideal undamped system (electrostatic forces exactly cancel the tiny residual damping), the generic solution is an *elliptical* trajectory in the x-y plane. The major/minor axes and orientation depend only on initial conditions; once set, the trajectory remains constant.

If the reference frame now rotates with angular velocity Ω, the oscillation pattern precesses relative to the body axes. In the rotating (non-inertial) frame the observed precession angle is (ideally) exactly equal to the integrated platform rotation.

This is precisely the operating principle of the famous Foucault pendulum in the Pantheon in Paris: a large mass on a long wire oscillates with near-zero ellipticity; as Earth rotates, the oscillation plane appears to precess.

In a MEMS version the same physics applies: keep a tiny proof-mass structure in free oscillation; the vibration pattern precesses at a rate proportional to the input angular velocity. Electrodes fixed to the substrate read out the precession angle directly.

7.7.7 Non-Idealities in Real MEMS

Even with very high Q, several non-idealities remain:

- Residual damping (isotropic + cross-axis),
- Anisotropic stiffness (slight frequency mismatch between x and y modes due to thickness or etch variations),
- Quadrature (cross-stiffness coupling: force in x proportional to y-displacement),
- Anisotropic damping (different damping coefficients along the two axes).

All these terms must be actively nulled by electrostatic forces.

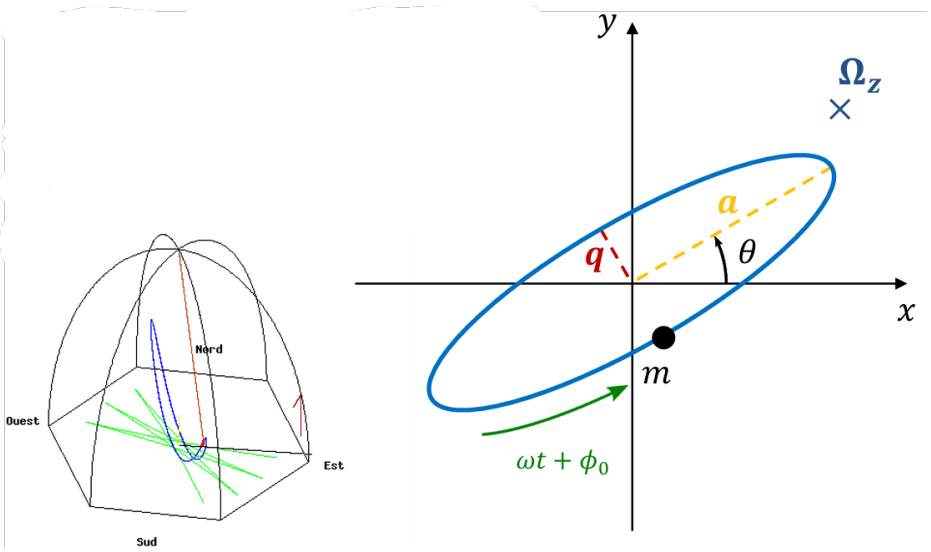

Figure 7.15: Elliptical trajectory example.

$$\begin{cases} m\ddot{x} + b_x\dot{x} + k_x x + b_{xy}\dot{y} + k_{xy} y = F_x + 2m\gamma\Omega\dot{y} \\ m\ddot{y} + b_y\dot{y} + k_y y + b_{yx}\dot{x} + k_{yx} x = F_y - 2m\gamma\Omega\dot{x} \end{cases}$$

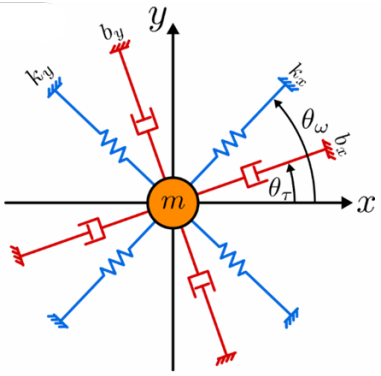

Figure 7.16: Nonidealities representation.

7.7.8 Control Architecture

In classical AM or FM gyroscopes the electrostatic forces only sustain oscillation (compensate damping). Here the forces must simultaneously compensate damping (isotropic and anisotropic), stiffness mismatch (keep perfect mode matching), quadrature, and cross-damping.

This requires multiple nested closed-loop controllers:

- **Energy loop**: keeps total oscillation energy constant ($A_x^2 + A_y^2 = \text{const}$),
- **Quadrature loop**: drives ellipticity $q \to 0$ (forces the trajectory to remain linear),
- **Phase / frequency loop**: locks the oscillation frequency (analogous to the phase-locked loop in a classical drive).

Additional loops handle precession readout and higher-order compensation. The resulting controller is complex (multiple interacting loops), but it enables direct mechanical integration of the rate and an inherently stable angle output.

This set of controllers is, in principle, the same as what we have in a conventional single-axis gyroscope, but now applied to two axes simultaneously. The goal is to keep the linear trajectory continuously aligned, even as the system precesses due to rotation. Because the two modes are matched, a single phase controller

suffices. The system, therefore, requires two end-to-end controllers (for energy and quadrature) and one phase controller. Additional loops handle precession readout and higher-order compensation. The resulting controller is complex (multiple interacting loops), but it enables direct mechanical integration of the rate and an inherently stable angle output.

To understand this better, let's look at a block diagram of the control scheme. Starting from the MEMS device, we have the displacement outputs along the X and Y axes. These signals are first read out using conventional techniques (e.g., capacitive sensing). The next critical step is to demodulate these displacements. The purpose of this demodulation is to extract, at any point in time, the complete state of the oscillating mass: its position, the trajectory's orientation, amplitude, and phase.

By knowing the X and Y displacement, we can calculate the instantaneous angle of the precession pattern. The demodulated signals also provide us with the oscillation amplitude and phase. This information is fed into a digital control system. On one side, it directly calculates the output angle. On the other side, it provides the necessary quantities to close the control loops mentioned above. These controllers, typically Proportional-Integral (PI) controllers, work to drive the energy to a desired setpoint, the quadrature component to zero, and the phase to match a reference.

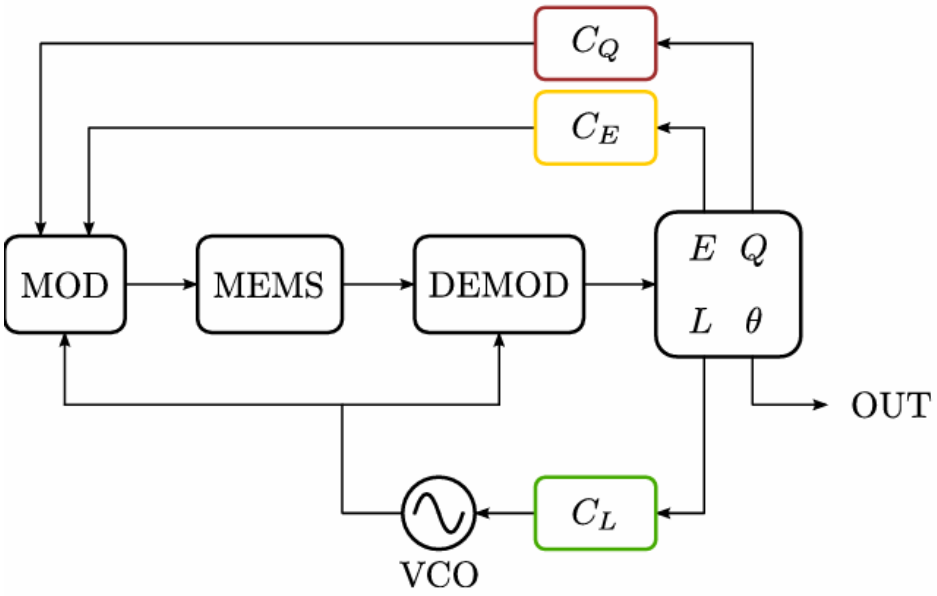

Figure 7.17: Lynch's control scheme.

The Critical Issue: Frequency Matching and a Self-Tuning Method

The most demanding requirement for a rate-integrating gyroscope is perfect mode-matching ($\omega_x = \omega_y$). If the stiffnesses are not perfectly matched ($K_x \neq K_y$), it introduces a significant error. Let's analyze this effect.

Assume ω_0 is the mean frequency of the two axes. If the two frequencies are not matched, the frequency generated by your oscillator (e.g., a PLL within the phase controller) will not remain constant, even under a constant input rotation. Instead, it will oscillate at a rate corresponding to the mismatch between the axes. In other words, the phase controller will generate a frequency that is time-varying.

This effect provides a valuable built-in diagnostic. The peak-to-peak oscillation of this frequency within the phase controller is a direct measure of the frequency mismatch. This observable quantity can be used to create an additional negative feedback loop. This new loop reads the peak-to-peak frequency oscillation and applies compensation forces (via specific tuning electrodes or by adjusting the forcers F_x and F_y) to minimize it. This creates a self-tuning system that actively keeps the modes matched.

Experimental results demonstrate the power of this method. Starting from an initial frequency difference of a few hertz, an additional control loop can measure the resulting angle oscillation in the phase controller

and react to tuning voltages, reducing the mismatch to something on the order of tens of millihertz (e.g., 20-60 mHz).

This raises a key question: How well matched do we need to be? Is 60 mHz of mismatch on a 30 kHz resonance a good value, or is it still too large?

The answer lies in the width of the resonant peak. Every resonator has a -3 dB bandwidth given by $\Delta f = f_0/Q$. The higher the quality factor (Q), the narrower this peak. For a high-Q MEMS resonator with $f_0 = 30$ kHz and $Q = 200{,}000$, the -3 dB bandwidth is approximately 30,000/200,000 = 0.15 Hz, or 150 mHz. As long as the frequency difference between the two modes is smaller than this -3 dB bandwidth, the system is considered well-matched. Therefore, controlling the mismatch to within a few tens of millihertz is more than sufficient to place the system in a quasi-ideal operating condition.

By implementing this self-tuning scheme, the scale factor of the gyroscope—which defines the relationship between the actual angle of rotation and the measured precession angle of the vibration pattern—becomes extremely stable and repeatable, both from measurement to measurement on the same device and from device to device. This high level of performance is what makes the complexity of the control architecture worthwhile for navigation-grade applications.

The Challenge of Anisodamping (Damping Asymmetry)

While the self-tuning method for stiffness mismatch is highly effective, it addresses only one half of the problem. The key challenge—and the open research question that currently limits the widespread adoption of rate-integrating gyroscopes—is anisodamping, or asymmetry in the damping coefficients along the two axes.

To understand this, we must introduce the concept of a time constant, τ, associated with each axis. This time constant is inversely proportional to the damping coefficient: a lower damping coefficient means a longer ring-down time, and thus a larger τ. In an ideal, perfectly symmetric device, the time constants for the X and Y modes are equal ($\tau_x = \tau_y$). However, due to fabrication imperfections, material properties, or even the way the resonator is anchored, these time constants can differ. This difference is defined as:

$$\Delta\left(\frac{1}{\tau}\right) = \frac{1}{\tau_x} - \frac{1}{\tau_y}$$

This quantity, $\Delta(1/\tau)$, is a direct measure of the damping asymmetry. A non-zero $\Delta(1/\tau)$ has two profoundly detrimental effects on the output of a rate-integrating gyroscope.

First, it introduces a sinusoidal error term into the output angle. Recall the ideal JavaScript-like equations of motion we considered earlier. When we solve these equations including anisodamping, the output angle θ is no longer a pure integral of the input rate. Instead, if the gyroscope is subjected to a constant angular rate, the output will be the desired, linearly increasing angle with a sinusoidal oscillation superimposed upon it. The frequency of this oscillation is related to the rotation itself, creating a complex and undesirable modulation of the output.

The second, and even more critical, effect is a phenomenon known as locking or quadrature nulling. In the absence of any input rotation ($\Omega = 0$), the vibration pattern should, ideally, remain fixed at its initial orientation. However, with anisodamping, this is not the case. The system will exhibit a slow, spontaneous precession towards a stable equilibrium point, which corresponds to the axis with the minimum damping (i.e., the largest time constant τ). If you start the oscillation at any other angle, it will gradually drift and eventually lock onto this "preferred" axis. This behavior is fundamentally similar to a "lock-in" phenomenon and represents a major source of drift and bias instability.

Minimizing $\Delta(1/\tau)$ is, therefore, the holy grail of rate-integrating gyroscope research. If this can be achieved, the performance potential is enormous. Researchers have been able to compensate for anisodamping effects for short time intervals—long enough to capture an Allan variance plot. When the angle output is numerically differentiated to compare it to a conventional rate gyroscope, the Allan variance shows no sign of rate random walk or bias instability. The plot does not exhibit the characteristic upward slope at longer averaging times; it is dominated only by angle random walk. In other words, the drift is eliminated.

The difficulty lies in doing this reliably and in real-time. Unlike stiffness, which can be tuned electrostatically, damping is a more fundamental dissipation mechanism that is harder to counteract with forces.

Finding a robust, practical method to compensate for anisodamping in real-time remains a critical and active area of research. Solving this would represent a giant leap toward realizing navigation-grade MEMS gyroscopes that directly output a stable, drift-free angle.

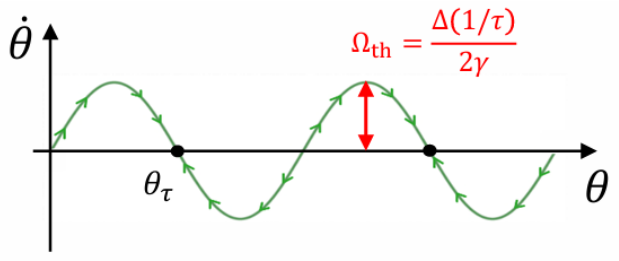

Figure 7.18: Angle locking behaviour.

7.7.9 Conclusion: Two Paths Beyond Amplitude Modulation

In this discussion, we have explored two alternative working principles for MEMS gyroscopes that move beyond the classical amplitude-modulated (AM) approach. Both are motivated by the same goal: to overcome the fundamental limitations of AM gyroscopes, namely the instability of the scale factor and the drift caused by integrating offset and noise.

- **Frequency-Modulated (FM) Gyroscopes** exploit the properties of a resonant element whose frequency is modulated by rotation. By operating in modes like Lissajous FM, they achieve a scale factor that depends only on the resonant frequency and a mechanical gain factor, which can be extremely stable. The output is a quasi-digital frequency signal, inherently robust to amplitude variations and many electronic offsets.
- **Rate-Integrating (Whole-Angle) Gyroscopes** take a different approach. They aim to create a near-perfect mechanical oscillator that is left free to precess under the influence of the Coriolis force, mimicking a Foucault pendulum. The output is the direct angle of rotation, mechanically integrated within the structure. This is achieved not by driving the motion, but by using complex control loops to cancel out all non-idealities—damping, stiffness asymmetry, and quadrature—leaving the system to behave ideally.

Both approaches trade the relative simplicity of an AM gyroscope for significantly increased control complexity. However, this complexity is the price for dramatically improved stability. The FM approach offers an ultra-stable scale factor, while the rate-integrating approach promises an inherently drift-free angle output. The primary remaining hurdle for the latter is the real-time compensation of anisodamping ($\Delta(1/\tau)$). If this challenge can be overcome, it would unlock a new class of navigation-grade MEMS gyroscopes with performance rivaling much larger and more expensive technologies.